Lectures

on

“Philosophical Propaedeutics”

from the Academic Year 1860–61

translated from the Danish by H. Halvorson

R. Nielsen

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Contents

Translator’s Note on Terminology

Nielsen lectured in 1860–61 in a philosophical Danish that had been formed on German models, and much of his vocabulary is a set of Danish calques on Kantian and Hegelian terms. A few of these words have modern English cognates that mean something quite different, and a reader who takes them at face value will be misled. The renderings below are used consistently throughout; a dagger footnote marks the first occurrence of each in the text.

***den sunde Forstand* — “sound understanding”**

Literally “the healthy understanding”; the standing Danish equivalent of German *der gesunde Menschenverstand*, and close to English “common sense.” But *Forstand* is *Verstand*, the understanding as opposed to reason, and Nielsen’s whole argument turns on its being a *faculty* that can be shown to contradict itself — not a stock of shared opinion. He almost always sets it in quotation marks, and those are preserved. The fuller form *den sunde Menneskeforstand* is given as “sound human understanding.”

***Anskuelse* — “intuition”**

Anschauung: the immediate presentation of an object to sense, in Kant’s technical sense. Never “opinion” or “view,” which is what the cognate suggests in modern Danish.

***Forestilling* — “representation”**

Vorstellung: the mental image or presentation, standing between sensation and concept. Nielsen’s argument at pp. 10–11 depends on the contrast between what can be *thought* and what can be *represented* — a round square can be thought, he says, but not represented.

***sandselig, Sandsning, Sandseindtryk* — “sensuous, sensation, sense-impression”**

sinnlich: pertaining to the senses. “Sensuous” here carries none of the modern connotation of “sensual.”

***Skarpsindighed* / *Spidsfindighed* — "acumen" / "over-subtlety"**

A matched pair in Danish (*skarp* = sharp, *spids* = pointed) that English cannot match morphologically. The first is the virtue of exact analysis; the second is its vice — subtlety that dazzles by distinctions only half true. Distinguishing them is the business of Part I.A, and Nielsen insists the same utterance may be either.

***Menen* / *Viden* — "opining" / "knowing"**

The Platonic *δόξα*/*ἐπιστήμη* distinction. Nielsen puns on it at p. 22 ("the question is not of opining but of knowing — still, what is your opinion?").

***Ophæve, Ophævelse* — "sublate, sublation"**

aufheben: at once to cancel, to preserve and to raise up. Where the context is plainly non-technical the ordinary "cancel" or "annul" is used instead.

***Tvetydig* / *Tvesidig* — "ambiguous" / "two-sided"**

Another pair the English breaks: literally "two-meaninged" and "two-sided." Nielsen's definition of dialectic at p. 15 is that it is the science which discovers the two-*sided* in the two-*meaninged.*

***Grund* / *Følge* — "ground" / "consequent"**

Grund/*Folge*, the logical pair, not cause and effect.

***dialektisere* — "dialecticize"**

Nielsen's own verb, and deliberately ungainly in Danish too: to derive one thing from another by dialectic. It is kept rather than paraphrased, since the dialogue of pp. 24–28 turns on the question of what can and cannot be so derived.

***Grundsætning* / *Sætning* — "fundamental principle" / "proposition"**

Sætning does duty for both "sentence," "proposition" and "theorem"; where Nielsen means one of the laws of thought, "principle" is used.

diversi respectus

Left in Latin, as Nielsen leaves it: the scholastic maxim *diversi respectus tollunt omnem contradictionem* — different respects remove every contradiction. It is the hinge of the first dialogue, and translating it would blunt the joke that B. has never troubled to study what a *respectus* is.

Nielsen cites three of his own earlier books by short title, without further identification: *Propæd.* or *Propædeut.* is *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og*

Dialektik. These are left in Danish, as he gives them. All page references in such citations are to the Danish originals.

Nielsen's own footnotes are marked, as in the print, *) and **); translator's notes are marked with a dagger and open with the words *Translator's note.*

Preface

Should the present work attract any attention, the interest taken in it will doubtless be of two kinds. With the question of the possible worth of its contents, most readers, keeping to the letter of the title, will surely join a second and more practical question: *on what principles, and to what extent, was the philosophical propaedeutic here delivered to the students?*

As regards this latter question, it will not be enough to remark merely that the motives, leading ideas and whole design of the present exposition correspond to the oral lectures, but that in matters of detail a good deal has been altered, something added, something else omitted, and so forth; the reader will think himself entitled to a clearer account.

It is therefore expressly noted that, apart from a few less considerable editorial changes, the whole of the first part, pp. 1–30, is a straightforward reproduction of the lectures as delivered. The same holds of the introductory piece of the second part, *General Remarks*, pp. 30–39; of Philosophy's Relation to Mathematics, pp. 39–63; of Philosophy's Relation to Chemistry, pp. 81–100 and pp. 135–177; and of Philosophy's Relation to Physiology, pp. 177–180, pp. 199–288, pp. 386–437, pp. 475–482.

In the presentation of the remaining portions, by contrast, considerable changes have been made. The section on the psychophysical investigations was not touched upon at all in the oral lectures; sharper and further-reaching analyses have been introduced here and there; and it goes without saying that by far the greater part of the contents of the notes was added in the course of the revision.

Since the effort to give the treatment of the physiological problems in particular that fullness which investigations of so intricate a nature can hardly do without brought with it a considerable expansion of the second part, the third part — Philosophy and Academic Study — has been confined to a bare statement of its moments, and is appended as a separate part only in order to exhibit the course of the whole.

11 June 1862.

R. Nielsen.

I.

The Concept of Philosophy.

A.

The Doctrine of Contradiction.

What is philosophy? "A science full of contradictions!" However much this definition may seem, viewed from one side, designed to pass sentence on philosophy altogether, it is nonetheless well founded, both in the history of the science and in the nature of the case. For any other science such an admission would be annihilating; for philosophy it furnishes an essential starting point, a dependable fulcrum, a firm and unshakable foundation. That philosophy must contain contradictions is a matter of course, since it is precisely the science which everywhere discovers and exhibits contradictions — in existence itself no less than in sound understanding's[†] grasp of existence (Propæd. pp. 40–44)[†] — and it is just in this respect that philosophy is to be understood as a *general doctrine of contradiction.*

As a doctrine of contradiction, however, philosophy must not merely discover and exhibit contradictions; it must also test them, for there are differences among contradictions. Some are superficial, unnecessary, easily avoided; others are insistent and not to be waved aside. Some creep in through thoughtlessness, others are manufactured by over-subtlety, others again brought to light by acumen.[†] Some contradictions play about

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[†] *Translator's note.* Nielsen's short titles for his own earlier books, cited throughout without further identification: *Propæd.* (also *Propædeut.*) is his *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og Dialektik*. Page numbers refer to the Danish originals.

[†] *Translator's note. Spidsfindighed* and *Skarpsindighed* form a matched pair in Danish (*spids* = pointed, *skarp* = sharp) that English cannot match. The first names the vice of subtlety that dazzles by half-true

as in a game of thought; others mark essential knots, difficult problems not easily untied.

The following examples may serve to show how necessary it is, where contradictions are concerned, to distinguish the superficial from the essential, the acute from the over-subtle.

The Liar (ψευδόμενος). This contradiction is put as follows: when someone says that he is lying, is he lying in that very statement, or is he telling the truth? The answer demanded is either yes or no. "Yes, he is lying!" — And he says so himself, calls the thing by its right name, says what is true, and hence is not lying. "No, he is not lying, he is speaking the truth!" — Then one must believe what he says; but what he says is precisely that he is lying; and if it is true that he is lying, then he is not speaking the truth.

That there is a contradiction here, sound understanding can readily grasp; but what sort of contradiction is it, and where does the fault lie?

Equal Magnitudes. That a magnitude is equal to itself, and that two, three .. magnitudes are equal to one another — in this sound understanding sees no contradiction. But if it were asserted that two magnitudes cannot possibly be equal to one and the same third, and hence equal to each other, because that would be a contradiction, sound understanding would feel itself affronted. Is there then really any contradiction whatever in the proposition just cited? To see what is actually being spoken of here, one must expressly note that the talk is not of things that have magnitude but of magnitudes. If the talk were not of magnitudes themselves but of things that have magnitude — an apple, say, a triangle, an hour, and so on — there would certainly not be a trace of contradiction. That two triangles, each of which is equal to one and the same third, are equal to one another, goes without saying. So long as we have to do with concrete magnitudes, difference and plurality can coexist perfectly well with equality; but when it is pure magnitudes that are in question, the matter stands otherwise. A plurality of magnitudes necessarily presupposes that the one can be distinguished from the other: without mutual difference, no plurality. But by what should pure magnitudes be distinguished from one another if not by size? If the magnitudes are equal to one another, then every glimmer of difference has vanished, and with it the plurality: to say that several magnitudes are equal to one another is therefore to say, in other words, that they are not several but one and the same magnitude, which, however many times it be repeated, is not thereby multiplied but remains the very same.

Should "sound understanding" wish to reject this consequence because there is something offensive about it, that can hardly be done out of hand by a mere dictum of

distinctions; the second the virtue of exact analysis. Distinguishing them is the business of this section, and Nielsen insists that one and the same utterance may be either.

authority; a sensible rejection must set about showing wherein the difficulty consists and how it is to be removed. If one and the same magnitude can, by being named twenty times, become twenty magnitudes equal to one another, then one and the same species must likewise be able, by being named twenty times, to become twenty species; but if it is not enough to repeat the same magnitude — if in repeating it a difference must also be thought — then where is that difference to come from?

The Instant.† If the old question — when did Socrates die? — is answered by pointing to the year of his death, sound understanding will hardly suspect any contradiction in that. Unless indeed it be that Socrates has no year of death at all, inasmuch as he did not take a year over dying: he was hale and healthy through the whole of that last year, until he drank from the cup. But if, in accordance with a well-known and warranted usage, one means by his year of death the year in which the day of his death fell, then there is no contradiction here. If it be objected that he did not take a whole day over dying either — well then, let one think of the hour of his death; and if one will go to the very limit, of the instant of his death. So then: when did Socrates die? "At the instant he ceased to live!" But this answer, which tells us only that he died when he died, is no answer at all. The question concerns the point of time in Socrates' existence at which his death occurred. To specify this point of time, four cases may be conceived: either he died α) while he was still living, or β) when he was already dead, or γ) when he was partly living and partly dead, or δ) when he was neither living nor dead. But α) to die while one is still living, to be a living dead man, is surely a contradiction. And to die when β) one is already dead is surely a contradiction too. The instant at which Socrates died must therefore have been the instant when, lying on the border between life and death, he was in such a condition that one part of him — his feet, say — was already dead while another part was still alive. By this explanation, however, the question is not answered but only pushed further back; for the question then is: when did the already dead part die? If the answer is that the already dead part died when it was still partly alive and partly dead, then the contradiction concealed in case γ) becomes obvious. And to conceive Socrates in a condition in which he was neither living nor dead is, finally, likewise to conceive a contradiction (δ); for how long does a living thing live? Until it dies!

"A philosophy that stands in open contradiction with 'sound understanding' is false; a philosophy that conforms itself immediately to sound understanding is superficial. Philosophy does not wish arbitrarily to contradict the natural understanding, but only to show

† *Translator's note. Øieblikket*: literally "the eye-blink," the smallest unit of time. "Instant" is used rather than "moment" throughout, to keep the word clear of the Hegelian *Moment* that Nielsen also employs.

that so-called sound sense, without having any inkling of it, incessantly contradicts itself." (Propæd. p. 41).

In so far as contradictions of these various kinds develop in the course of thinking, and their origin is viewed from the subjective side, the psychological share that reflection has in the contradictions offensive to "sound sense" must be determined more closely. The aberrations belonging here may be briefly characterized as the *eristic*, the *over-subtle* and the *sophistical*.

The striking mark of eristic is a one-sided bent for quarrel and strife, a delight in shining everywhere by objections, in raising difficulties, in tearing down what is generally accepted and stubbornly defending one's own divergent claims. As for ancient philosophy, it is above all the Megarian school, akin to the Eleatic, that has earned the name of eristic; but neither can it be denied that every distinctively developed form of philosophy has its polemical element, and that eristic, for all its dark sides, is to that extent of unmistakable importance in the development of the science. For insight and knowledge it is not enough that one form what is commonly called a reasonable opinion about things, and, by combining sense-impressions with natural judgment, be led to distinguish correct from incorrect, acceptable from objectionable opinions; by means of opinions one attains only the probable, not the true. Whoever would rise from mere opining to knowing[†] must arm himself against the deceptions of sense, let thought test what has been accepted, and acknowledge as true only what admits of being thought — declaring untrue, without regard to the multitude's otherwise unimpeachable opinions, everything that does not admit of being thought. That philosophy, on these terms, must come into manifold conflict with sound human understanding is evident. (Cf. Philos. og Mathem. p. 12, note; Mathem. og Dialektik pp. 34–35, pp. 116–117, pp. 131–32, etc.) Rigorous analysis of concepts demands acumen; but acumen that is one-sided and unfruitful, dazzling by distinctions which are subtle yet only half true or even untrue, yields over-subtlety. Easy as it is to state the difference between acumen and over-subtlety so long as one keeps to generalities, it is just as impossible to lay down a rule valid for every particular case; for one and the same utterance may, seen from one side, look over-subtle, and yet, seen from another, bear the stamp of sound and genuine acumen. The contradiction set up in "the Liar" may serve as an example. If the task is to drive home the truth that one should not undertake off-hand to answer every question, since there are questions whose apparently clear and easily grasped sense proves on closer inspection to be senseless, then "the Liar" may stand as a model of

[†] *Translator's note. Menen* and *Viden*, the Platonic pair δόξα/ἐπιστήμη. Nielsen returns to the pun at p. 22.

genuine acumen. If, on the other hand, one guilelessly takes up the answer demanded — and does so in such a way that, watching with astonishment as the affirmation turns into a denial and the denial back into an affirmation, one ends by imagining oneself before some deep riddle, some exceedingly intricate knot whose solution may perhaps shed a hitherto unknown light upon "the eternal truths" — then over-subtlety has outwitted sound understanding and caught it in its web.

In secret league with over-subtlety and hair-splitting we find sophistry. The sophist is not content to exhibit contradictions; he manufactures contradictions, plays with contradictions, draws inferences, and dazzles by false inferences. For it is one thing to bring contradictions into relief, another to misuse them in the service of sophistic; how great the difference is may be seen, for instance, from what was said above about "the Instant." No sensible person can regard it as sophistry to call attention to the contradictions upon which thought must necessarily strike when it tries to give an account of the meaning of the instant; only when it is a question of the use made of the contradictions does sophistry become recognizable, by what is false, untenable and dazzling in the inferences. Thus if anyone were to infer, from the contradictions contained in the thought of the instant of death, the unthinkability of death, and from the unthinkability of death its impossibility, and were then to build upon that a new proof of the immortality of the soul, fitted out with a peculiar acumen, we should have to consider him a sophist.

B.

The Doctrine of Possibility.

If philosophy as a doctrine of contradiction is not to dissolve into over-subtlety and sophistry, it must always find itself able, from one side or another, to master the contradictions. It must then be able to determine their relation both to things and to our reflections upon things, both to objective being and to subjective thinking; in seeing through the essence of contradiction it must discover the marks of what is in itself thinkable and what is in itself unthinkable, and thereby, so to speak, wring from contradiction a dependable measure of the acceptable and the objectionable, the reasonable and the absurd, the possible and the impossible: a thoroughgoing doctrine of contradiction is inseparable from a *general doctrine of possibility.*

What, then, is philosophy? "A doctrine of the possible." But since philosophy cannot possibly be a doctrine of everything possible, this Wolffian definition is satisfactory only on condition that the concept of possibility — its extension, its content, and its relation to contradiction — be exactly determined. The principle that everything is possible which does not contain a contradiction (*possibile est, quod non implicat contradictionem*) — which in turn presupposes that everything containing a contradiction is impossible — will, on the strength of what has gone before, hardly furnish a dependable guiding line, whether dogmatic or critical.

In the first place it is self-evident that the principle as stated specifies possibility only formally, and is therefore unfit to orient us with respect to the real possibilities. Even if for the time being we declare everything that contains a contradiction impossible, and everything that does not contain a contradiction possible, the concept of possibility so given will still be so negative, so empty and unsettled, that it cannot be brought into any positive connection whatever with the actual. To determine the possibilities that hold in actuality, it is necessary to know actual things together with their conditions and laws; and knowledge of actual things is reached only by the road of experience. Without the correctives of experience, dogmatizing thought with its empty concepts will be as readily tempted to set the real possibilities too narrow bounds as to allow them too wide a play. Or who, back in the days when Wolff was fixing the bounds and defining the possibilities, would have seen the possibility that ships should be able to go both against wind and against current; that carriages without horses should be able to run — not like horses, but swift as flying birds; or the possibility that a man in London should be able to correspond with a man in Calcutta as quickly and easily as if they were talking together? So long as philosophy, with its formal concepts and definitions, attempts to exhaust existence and to set itself up as judge over what is possible and impossible in actuality, so long will sound human understanding have manifold occasion for the triumphant application of Hamlet’s words: *There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy.*

But even if philosophy gives up trying to fathom the positive, empirical content of the real possibilities, it may still contribute essentially to purifying and heightening our knowledge of the possible, provided only that it can secure for us the negative side of possibility by determining the impossible. Now if the impossible can be known only as what is in itself unthinkable, and if the unthinkable must always be recognizable by contradiction, then the fundamental question presses itself upon us of its own accord: *how, then, does the contradictory stand to the thinkable?* Here we are at a decisive turning point.

In presupposing without more ado the unity of thinking and being, dogmatic philosophy proceeds at the same stroke from the assumption that the contradictory is unconditionally an unthinkable, and the unthinkable an impossible. But is the contradictory really unthinkable in every respect?

"What does contradiction mean? An existing nothing is a contradiction; a round square is a contradiction; a flying horse is a contradiction too. And yet these contradictions are of different kinds." (Propæd. p. 43).

If what is contradictory in itself is straightway unthinkable, how then does thought come to apprehend this unthinkable? How does it happen that we not only understand what is meant by the word "contradiction," but also see clearly what lies in the essence of contradiction? If we set, say, "something" against "nothing," "that which is" against "that which is not," thought finds it just as easy to grasp the negative as to grasp the positive; and if we set the proposition "that which is, is" against the proposition "that which is, is not," one notices at once that a conflict between subject and predicate is as evident to thought as an agreement. But if thought apprehends the conflict as readily as the agreement, sees through an existing nothing as clearly as an existing something, and so far finds the former as thinkable as the latter — how then can what is in itself conflicting and contradictory show itself to thought as an unthinkable, and this unthinkable coincide with the impossible?

To extricate oneself from the difficulty, let one try letting representation come to the support of thinking.† Pure, abstract thinking, detached from all intuition and representation, does indeed grasp the negative as readily as the positive, since the negative is precisely abstraction's own abstract product; but when the light of intuition dawns, when representation joins in, the opposition between the positive and the negative, between something and nothing, becomes conspicuous. For something admits of being intuited, whereas nothing, pure nothing, does not; to the positive there answers a representation, but nothing — the negative — repels every representation, and is for that reason in turn repelled and rejected by representation.

How thinking, by being combined with representation, comes to apprehend the contradictory as the unthinkable, and the unthinkable as the impossible, may be seen from the example of the round square. The proposition "the square is round" can indeed be thought, in so far as the predicate's conflict with the subject can be thought; but the round square cannot be represented, since the representation of the round displaces

† *Translator's note. Forestilling* is *Vorstellung*, the mental presentation or image standing between sensation and concept; *Anskuelse*, just below, is *Anschauung*, intuition in Kant's technical sense — never "opinion" or "view." The whole argument of this page turns on the contrast between what can be *thought* and what can be *represented.*

and effaces the representation of the square. Thought, seeing that the round square makes a demand upon representation which representation must reject, now brands the contradiction as something objectionable, declares the round square an unthinkable, and this unthinkable an impossible. That it is precisely intuition and representation which in such cases must help thought to discover the unthinkable is seen not only from propositions in whose components representation manifestly rules — "black is not white," "thick is not thin" — but also from examples that do not directly display the contradiction in their words and manner of expression, as when one asks: is a regular hexahedron possible?[†] is it possible for a line parallel to an ellipse to be an ellipse? and so on.

If only we could rely on intuition and representation to furnish, always and in every case, infallible marks of the thinkable and the unthinkable, we should at least have something firm to hold to here; but now look at the example of the flying horse! Natural sense surely feels that under the name "flying horse" a contradiction lies hidden. But to whom is sound human understanding now to turn to have the contradiction cleared up? To thinking! That is no help: pure thinking thinks everything, the unthinkable included. To intuition! No help either: intuition lives with the poet, and the poet rides Pegasus, and Pegasus reconciles intuition with every winged horse. To a union of thinking and intuition! But how can a union of two unreliables yield anything reliable? To sense-impressions! Sense-impressions can indeed, in union with thinking, teach us something about the possible, in so far as they teach us something about the actual, of course; but what do they teach us about the merely possible, and what could they possibly teach us about what is in itself contradictory and impossible?

To determine the relation of possibility to contradiction, it is above all necessary to grasp *the principle of contradiction* — its form, its validity, its application.

In formal logic we find the propositions of identity and of contradiction set out so separately that we are thereby supplied not merely with two fundamental principles but, into the bargain, with two independent principles. If there is really anything in this duality of principles, we should expect some account of the difference of principle between them. But wherein is the difference supposed to consist? To explain the proposition of identity — A is what A is — one slips in underneath it the proposition of contradiction: that A is what A is precisely because A is not other, and cannot be other, than what A is. Now if the principle of contradiction has had to do service in setting

[†] *Translator's note.* The example is puzzling as it stands, since the regular hexahedron is simply the cube. Nielsen's point is presumably that these are questions whose answer cannot be read off the words — unlike "black is not white" — so that intuition must be consulted before one can say whether a contradiction is present at all.

up pure identity — and that, in the logical sense, before it was even born — then it is no more than fair that the principle of identity, once born and the logical birth-pangs happily over, should in its turn help the principle of contradiction to due recognition. Why, then, is no thing what it is not? Answer: because every thing is what it is. And why is every thing what it is? Answer: because no thing is what it is not. In other words: when identity is asked after, one answers with contradiction; and when contradiction is asked after, one answers with identity. On closer inspection it must undeniably strike sound human understanding that there are here not really two fundamental principles but only one.

Yet suppose that, in order so far as possible to avoid conflict with traditional forms, we let formal logic keep its two fundamental principles undiminished — so that, for the sake of "certainty" and "truth" and "determination," "statuation," "whatness," "decision," "unanimity" and "concordance,"† — we held identity literally apart for itself and contradiction apart for itself: would such a holding-apart then perhaps help sound understanding to a clear and distinct insight into the relation between the contradictory and the impossible? At first glance it seems that some difficulty still remains. For if every *A*, provided only that it is what it is, has by the principle of identity a claim to "a valid is," then it looks almost as though the senseless and the thoughtless could likewise make a claim — a valid claim — to "a valid is"; if every thing, provided only that it is not what it is not, can satisfy the principle of unanimity and concordance, why should the conflicting, the absurd, the self-contradictory and the foolish not be able to satisfy the principle of unanimity and concordance as well?

But to overlook the principle of contradiction, still less to reject it, will not do at all; on the contrary, when thought attempts to deny the validity of the principle of contradiction, it is led by the very denial to confirm it. To posit "black is white" instead of "black is not white" is not only possible but reasonable and necessary if the corresponding contradiction is to be made properly plain; yet in being branded in this way as a contradiction, the contradiction vindicates the principle. The unthinkable, in this sense, is not a something that thought simply cannot think; the unthinkable is the thought which thinking itself condemns, in view of a purpose given within a given context — a thought that comes into being only to vanish, a thought in whose sublation†

† *Translator's note.* The list is ironic: Nielsen is parading the scholastic-sounding coinages under which formal logic defends its two principles. *Hvadhed* is a Danish calque of *quidditas*, "whatness"; *Statuering* is the laying down of something as established; *Eenstemmighed* and *Samstemmighed* are "unanimity" and "concordance."

† *Translator's note. Ophævelse*, Hegel's *Aufhebung*: at once cancelling, preserving and raising up. Where the context is plainly non-technical, the ordinary "cancel" is used instead.

thinking recognizes its own law and satisfies itself. Only when the unthinkable is thus determinately set against the thinkable can the latter be called the logical mark of the possible and the former that of the impossible. Contradiction, then, logically regarded, cannot be abolished outright. Could the principle of contradiction abolish contradiction once and for all, it would at the same stroke render itself superfluous. But since thought cannot stir without contradiction — its sting and its spur — stirring too, the principle stands in force because the cooperation of contradiction stands in force; in order to determine the thinkable, sift the possible and dismiss the impossible, the principle of contradiction must incessantly contradict contradiction, must in resolving contradiction uphold contradiction.

In the application of the principle of contradiction the difference between critical and dogmatic philosophy comes into view. Agreeing with dogmatism in setting contradiction outwardly hostile against thought and the movement of thought, critical philosophy then diverges from the dogmatic in this: that in the unthinkable it sees only the subjective side of the impossible, and not its objective side as well. When Kant attempts to think the origin of the world, he strikes, quite rightly, upon contradiction; the same recurs when he tries to think the essence of substantiality, of necessity and of freedom objectively. The acute thinker, consistent from his own point of view, is here so far from being rid of the contradictory that on the contrary, seeing contradiction tie its knots and settle into antinomies that cannot be waved aside (Propæd. pp. 29–30; Mathem. og Dialekt. pp. 125–126), he declares it insuperable. The objective essence of the world belongs therefore, according to Kant, to what is for us unthinkable; but what does Kant infer from this? By no means that the reality of an objective world should on that ground be held impossible — that is an inference which first arises with Fichte (Propæd. p. 33) — no, the author of "Kritik der reinen Vernunft" warns precisely against precipitate inferences from the subjectively unthinkable to the objectively impossible. The separation of thinking and being, the original characteristic mark of critical philosophy, rests precisely on the presupposition that for the human understanding there are essential, necessary, insoluble contradictions.

But are the Kantian antinomies really insoluble? That depends on whether one chooses to grant contradiction a merely subjective significance, or a significance at once subjective and objective. Such a choice cannot of course be arbitrary; it must follow from the nature of the case. If, then, it turns out that by denying the objective validity of contradiction, holding fast to the critical separation of thinking and being, and so rejecting every attempt to know "*the thing in itself*," one by no means escapes contradiction but rather entangles oneself in new and greater contradictions — what then?

C.
Dialectic.

What, then, is philosophy? "A doctrine of ambiguity, a science of the ambiguous." The correctness of this definition seems at least to find ample confirmation in what has gone before. For although we did distinguish between acumen and over-subtlety, we had to admit that one and the same thing could be taken both as over-subtle and as not over-subtle; when it was a matter of separating the thinkable from the unthinkable, what came out was that the unthinkable was something that both could and yet could not be thought; and when it finally came to a resolution of contradiction, the upshot was that contradiction both could and could not be resolved. Should sound understanding find such a predilection for the ambiguous suspicious, let it be noted that the expression "doctrine of ambiguity" is itself ambiguous. While higher instruction in making the clear obscure, the distinct veiled and the intelligible unintelligible is always to be sought among the sophists, philosophy must contrive to set the ambiguous in another light — a light that makes the ambiguity vanish. As the science which everywhere discovers the two-sided in the ambiguous,[†] and by means of contradiction clarifies the doubling, philosophy is *dialectic*[*)].

The truth that philosophy by its higher nature is and must be dialectic was categorically stated by Plato. (Cf. Propædeut. pp. 124–129; p. 150 and elsewhere.) In Plato the presentation of the dialectical (διαλεκτική from διαλέγεσθαι) has, as is well known, taken the form of conversation; and connected with this is the fact that the dialectician is characterized precisely as the one who in conversation knows how to ask and to answer

[†] *Translator's note. Tvetydig* and *tvesidig*: literally "two-meaninged" and "two-sided." The definition depends on the Danish pair, which English cannot reproduce.

[*)] That the form of philosophy is in its essence dialectical must not be misunderstood as though the development of concepts had everywhere to display immediately the formal antinomies by which logical transitions are motivated; on the contrary! Philosophical exposition can and should also take up reporting, descriptive and critical components. The only decisive point is that the elements communicated in the form of report, description, critical discussion and the like should everywhere show themselves, by the design of the whole and the prevailing current of thought, to serve the dialectical. There is in the Hegelian method a formal overstrain, a logical over-exertion, brought on chiefly by the endeavour to have the exposition sparkle at every point with dialectical flashes of lightning. It is with dialectic's flashes of lightning, and with the elements in which they are prepared and out of which they develop, as it is with natural lightning: the lightning-fire must as a rule presuppose clouds, but the clouds are not on that account lightning.

rightly (Τὸν δὲ ἐρωτᾶν καὶ ἀποκρίνεσθαι ἐπιστάμενον ἄλλό τι σὺ καλεῖς ἢ διαλεκτικόν; *Krat.* 390 c.). That what essentially matters in dialectic is not the conversation but the development of concepts conditioned by "the art of language" goes without saying; but that the conversational form, when everything irrelevant is kept out, is nonetheless a very natural form for the beginning dialectician, may readily be shown by a few examples.

Something and Other. A. If the categories "something" and "other" simply excluded one another, there would be no dialectic; if they coincided without more ado, there would be no dialectic either. The dialectical shows itself precisely in this, that the two propositions "something is what it is, not an other" and "something is what it is not, hence an other" not only cancel but also mutually presuppose and ground one another. B. If the dialectical rests on so slippery a ground, then all dialectic is false. A. How so? B. The claim you have set up stands in open contradiction with sound human understanding, and *a philosophy that stands in open contradiction with 'sound understanding' is false.* A. If what you here call *open contradiction* is to be dismissed out of hand by an immediate verdict, philosophy can attach no weight to the ruling; for *a philosophy that conforms itself immediately to 'sound understanding' is superficial.* But if "sound understanding" will condescend to an inquiry, the inquiry can easily be got under way. B. It would have to be the second proposition, under whose negative form some ambiguity or other has hidden itself, and which might therefore become an object of inquiry; in the first proposition, the proposition of identity, the pure proposition of certainty (*principium certitudinis*), there cannot be a trace of ambiguity. A. But suppose it is after all the first proposition in which the ambiguity, the indeterminacy, and so far the uncertainty, lies hidden — what then? B. Then it must be capable of proof. A. Is it altogether forbidden in logic to say anything else of something than that it is something? B. By no means; for how, by merely repeating "something is something," should one ever determine *what* something is? Logic, however, which has no fondness for ambiguities, draws a definite distinction, as easy to grasp as it is unmistakable, between determining something by means of an other and saying of something something other than what it is. A. So it is permitted to determine what something is by determining what it is not? B. Not only permitted but necessary. A. Then it is necessary that in the determination of something a determination of an other be contained. B. That too may be granted — but on the express condition that something, precisely by that determination, excludes the other. A. Something, then, is —? B. A thing different from an other. A. And if now, on that condition, we exclude the other, something is —? B. A different thing.† A. But now the

† *Translator's note. Et Forskjelligt*, the substantivized neuter adjective: "a differing one," something whose whole character is to be different. English has no natural form for it; "a different thing" is used

other, on its side, is —? B. A thing different from something. A. And when, likewise on that condition, we let the other exclude something, we say that the other is —? B. A different thing. A. So then: something, when determined, is a different thing; the other, when determined, is a different thing; something is therefore what the other is, namely a different thing. B. This ambiguity is easy enough to see through. Certainly something is a different thing, just as the other is; but different and different are two things — something is a different thing precisely by being different from the other, and the other a different thing by being different from something. Something and other are therefore still, as before, different from one another. A. Certainly something and other are different from one another; only a sophist would take it into his head to deny their difference. What was to be proved here was only this: that their difference is grounded in their identity. B. What is different is not identical, and what is identical is not different: only by pure categories can one secure oneself both against hidden ambiguities and against open contradictions.

A. But if I am not mistaken, you have already yourself, by your definite and lucid answers, made valuable contributions toward sharpening our eye for the dialectical unity of something and other. B. I should very much doubt it. A. But you did stress that something can be determined only as a different thing, and, for clarity's sake, as a thing different from an other? B. Yes, that I did stress. A. Then you have at the same stroke expressly required that something, in virtue of that difference, must be taken as an other than that from which it is different. B. Of course. A. So on your view something is other than the other, in opposition to the other — and hence, after all, an other. B. Your play on words leads nowhere; for when something is taken as an other, it becomes another other — namely the other that something itself is, as against all the other that something is not. A. That I can understand; now you are talking like a true dialectician. So you have now seen for yourself that, with respect to the determination contained in something, there are two kinds of other: an other that something itself is, as against another other that something is not — in other words, that something can exclude the other only by itself being an other. B. Yes — so long as one keeps to expressions as general as "something" and "other," one can no doubt practise without much difficulty the game of thought of turning something into other and other into something. But let us try it with something determinate! Show me in definite examples that something really is identical with other, and I shall grant you that there is "sound understanding" in the dialectical. A. You take a leaf in a book to be something determinate? B. Naturally. A. And the two sides of a leaf

throughout the exchange, and the reader should hear the emphasis on *different* rather than on *thing*.

are surely so opposed to one another that if we call the front a something, we must call the back an other. B. Excellent! Now surely it will occur to no one to identify the front with the back, and then maintain that the front is what the back is, because both are sides, or because the front becomes the back when the leaf is turned, and the back the front. A. But on this occasion too the identity is surely as conspicuous as the difference? B. I can see no identity. A. What do you think, then, about the leaf? B. That the leaf which has the two sides consists precisely in the difference and opposition of the sides! For if the two sides were turned into one side, we should get only a surface, not a leaf. A. And if the two sides were separated in such a way that each came to belong to a body of its own, we should get two surfaces but no leaf. B. Yet it makes an essential difference whether I say "in the leaf the front is united with the back" or "the front is what the back is, the front is back." The first I can grant, the second not. A. If the leaf that unites the two sides were a third thing subsisting on its own over against the sides, we should certainly not be entitled to take it as the existing unity of the sides; but since the leaf, taken in separation from the sides, does not exist at all, the one side must in the leaf show itself identical with the other. B. Are we then perhaps to derive from this that the front itself is both front and back, and the back in turn both back and front? A. That is a consequence you will hardly reject. If we call the two sides α and β, it is evident that α, which in and for itself is neither front nor back, must by its opposition to β be front in so far as β is back, and conversely; α is therefore, in virtue of its identity with β, both front and back. B. That is a peculiar extension of the concept of identity. What I have to object against the concept of identity so extended I would rather hold back, however, until you have produced one more example of a different sort.

A. What do you say to calling the greater a something and the lesser an other? B. That example I find very suitable. A. And what is it, then, that we are to prove? B. What else than that something is both a greater and a lesser, the other both a lesser and a greater, and one and the same magnitude itself both lesser and greater. And in truth nothing is easier than to prove all this. A cursory comparison of three magnitudes, $\alpha > \beta > \gamma$, tells us at once that α is a greater in so far as β is a lesser, and β a greater in so far as γ is a lesser. But what follows from this? That β is in one respect a lesser and in another respect a greater — in that there is no contradiction; but if anyone maintains that the magnitude β is at one and the same time and in one and the same respect both greater and lesser than itself, he does fall into contradiction, and not only with the natural understanding but with true logic. That one and the same side of the same leaf can at one instant be front and at another back, and at one and the same instant — when, mark you, it is regarded from two opposite points of view — both front and back, contains no contradiction, but a confirmation of the old rule *diversi respectus tollunt omnem contradictionem.* If, on the other hand, one ignores the *diversi respectus*, overlooking what both Plato and Aristotle, for all the difference of their intellectual bent, so strictly inculcated — that in such inquiries what is asked about must each time be seen to be τὸ αὐτό, and regarded κατὰ τὸ αὐτό, πρὸς τὸ αὐτό, ἐν τῷ αὐτῷ χρόνῳ — then one may indeed dazzle the inexperienced for a moment with piquant turns of identity and surprising contradictions, but one will never convince the expert, never satisfy the sensible thinking that inquires after truth. A. Weighty words, and important! So now your "sound understanding" has almost got the better of contradiction and finished off dialectic; I say *almost*, for one small point still remains — a point we simply must have cleared up, and one you will surely be able to clear up easily: it concerns *diversi respectus*, and you have of course studied *diversi respectus.* B. Studied? It has never occurred to me that there was anything further to study in logic's *diversi respectus*; I have always supposed it enough to remark in a general way that there are *diversi respectus.* A. Then it is surely necessary to distinguish between *respectus*, since there are, after all, *diversi respectus.* Of the two sides of the same leaf — to return to one of our examples — it undeniably holds in one respect that they are the same, namely that both are sides, and in another respect that they differ, namely that one is front and the other back; in one respect the same, that both are, say, written on, in another respect different, that one is, say, white and the other red, and so forth. Each of these different respects may in turn give occasion for new ones; for although it holds of both sides in one respect that both are written on, yet on this very point it also holds in another respect that they differ, namely that the one is written in black ink and the other in red. These doublings of respect can be continued indefinitely. If thinking is not to lose itself in an unsurveyable tangle of all manner of regards to all manner of respects and remarks and considerations and reflections, logic must necessarily distinguish not only between essential and inessential respects but also between respects that fall indifferently outside one another and respects that are contained in one another and therefore follow from one another with inner necessity. That one side of a leaf is written on does not of course entail that the other must be so too; one can take account of the former without taking account of the latter; the former can be without the latter's therefore being, and no one but sophists would think of positing the latter as identical with the former. It stands otherwise, however, with being front and being back: here it is impossible to take account of the former without taking account of the latter, for these two modes of existence are contained in one and the same existent in such a way that the regard to the former actually coincides with the regard to the latter — the former is, despite the difference, indeed in virtue of the difference, *identical* with the latter. True insight into the essence of identity, difference and contradiction depends on analysing the *diversi respectus* sufficiently.

B. But can any well-founded objection really be made against the proposition *A* is what *A* is? A. Among others, the objection that the proposition lacks a ground. B. But when the proposition is one with the principle — the principle of identity, the principle of truth, the principle of certainty, *principium certitudinis* — and is of course self-evident in every way, it needs no ground. A. Is it then immediately evident from the proposition *that A* actually is? B. No, not *that A* actually is, but *what A* is. A. Is it immediately evident *what A* actually is? B. Not exactly *what A* actually is, but that, *if A* is at all, *then A* is *what A* is. A. But suppose *A* is not at all? B. Then it is still what it is. A. So in order to be what it is, *A* does not even need to be? B. You understand very well what I mean. A. The question here is not of opining but of knowing; still — what is your opinion? B. That even if *A* is a non-being, it is still what it is, namely a non-being. A. But non-being is surely the same as what we otherwise call nothing, no thing? B. Naturally. A. So then, whether I say "no thing is what it is" or "every thing is what it is," it must at once be evident to sound understanding — by the principle of identity, of course, the principle of truth, the principle of certainty, *principium certitudinis* — that with both propositions I mean one and the same thing!

B. But can anything at all be adduced in defence of so self-contradictory a proposition as "something is what it is not"? A. Among other things this: that a proposition which by its open contradiction so boldly challenges thinking must awaken reflection and call for grounds, in that it demands a refutation. B. But how should one refute by grounds a proposition that immediately refutes itself? A. The proposition "something is what it is not" by no means refutes itself immediately. B. If what is unthinkable, absurd and senseless in this proposition is not self-evident, then what is self-evident, and what is sound human understanding? A. Sound human understanding by no means sees anything absolutely absurd in something's being able to be what it is not. B. So sound human understanding is supposed to be able to conceive that *something* — mark you, at a given time and in a quite definite sense — *is what it* — at the same time and in the same sense — *is not*; no, that really is too much. A. Yes, that really is too much; but so absurd a demand is made by the proposition *something is* — mark you, when its existence is grounded — *what it is not* only when the proposition is held one-sidedly in superficial, abstract immediacy. B. Well, that is another matter, if your defence rests on an interpretation. A. And another matter too, presumably, if your refutation rests on an interpretation. B. Then prove to us, by means of your dialectical interpretations, that one and the same something can at one and the same time and in one and the same respect be both what it is and what it is not, and that one and the same magnitude can at once be both greater and lesser than itself. A. Such proofs belong to sophistic, not to dialectic. B. But to prove that something can in one respect be what in another respect it is not, or that one and the same magnitude can in one relation be a lesser and in another a greater, nothing is needed but *diversi respectus*, no dialectic at all. A. But to prove that something, though different from the other, is nevertheless essentially one with its other, because the being of both has the same source, the same root; to prove that the same magnitude is on one and the same ground both a greater and a lesser, since the two relations, the two respects, have one and the same point at which they meet, namely being a magnitude; to prove that the identity of something and other, of the greater and the lesser, contains precisely the ground of something's being in existence different from the other, and the greater different from the lesser — for that, dialectic is needed; for this dialectic can prove, but dialectic alone.

Concept and Existence. C. A dialectical development of the inner connection of the *a priori* concepts, and of their transition into one another, is at the present standpoint of the science no doubt to be acknowledged on the whole; but it is another question whether, because the dialectical plays a considerable part in philosophy, one is on that ground entitled not merely to reproach formal logic with not being dialectical, but even — on the strength of a Platonic expression which can always be taken in more than one sense — to transform the whole of philosophy into sheer dialectic. A. Had you followed the preceding inquiry attentively, you would, as to the first point, easily have convinced yourself that there was no thought here of delivering a detailed critique of logical formalism, but only of rebuffing the self-sufficiency with which one not seldom allows oneself, invoking "sound human understanding," to consign every logical objection to the immediate validity of immediate first principles to the realm of fancy and sophistry; as to the second point, your misgivings could do with a clearer statement. C. *A priori* concepts — such as something and other, essence and semblance, ground and consequent, and the like — can easily be produced out of one another by the application of contradiction. Whoever posits the ground already implies the consequent; the ground is therefore, at bottom, both itself and the consequent.[†] But since the ground becomes a ground proper

[†] *Translator's note.* Danish *Grund* carries the whole weight here: it is at once the logical ground (*Grund* as against *Folge*, consequent), the "bottom" of a thing, and the "reason why." Nielsen plays on all three, and the English can only follow at a distance.

only on account of the consequent, the ground is consequent and the consequent ground. The same thing is not merely ground in one respect and consequent in another; the two respects also coincide in a single respect: to dialecticize[†] the concepts out of one another is, if I may use the expression, the same as to dialecticize *diversi respectus* into one another. As for the *a posteriori* concepts, the concepts of experience, it is impossible to dialecticize existence out of the concept; or who would try to dialecticize the actual horse out of the concept of a horse, or the actual snail out of the concept of a snail? But if it is impossible to dialecticize the existent out of its concept, then presumably it is for the same reason impossible to dialecticize one concept of existence out of another — the concept of a horse, say, out of the concept of a fly, and this in turn out of the concept of a snail. In the world of experience *diversi respectus* fall outside one another, and where *diversi respectus* fall outside one another dialectic plays a very subordinate part. A. But if you could not succeed in dialecticizing existence out of the concept (cf. Propæd. p. 31), have you never tried going the opposite way — dialecticizing the concept out of its existence, as for instance dialecticizing your concepts of snails out of their corresponding snail-existence? C. One does not dialecticize concepts of experience out of existing objects; one *abstracts* them. A. So there is nothing one might call a dialectic of abstraction? C. Not so far as I know. A. But the concepts abstracted from the objects must surely be, not only in our consciousness, but also in the objects themselves? C. Naturally; for if the concepts were not in the objects, they could not be abstracted from the objects. A. But the object from which the concept is abstracted is surely not itself a concept? C. By no means. A. Yet the concepts, in so far as they are in the objects, may be said to exist? C. That the concepts of experience exist in the existing objects, no one can well doubt. A. So in one and the same existence there are two existents — the existing object, and the concept existing in the object? C. No, it cannot be taken that way; it is not the object itself but its material that is opposed to the concept. A. Then presumably we get three existents: the existing material, the existing concept, and that existence of the object in which both exist? C. No, one cannot take it that way either. A. How, then, is it to be taken? C. The material by itself is no separate existent, any more than the concept is; the material and the concept are two sides — two sides of existence — of the same existing one, the object. A. The existing one is, then, over against and in opposition to the two sides of existence, the existent in and for itself? C. No, one cannot take it that way either; for if we let abstraction remove the two sides of existence, the

[†] *Translator's note. Dialektisere*, Nielsen's own verb and deliberately ungainly in Danish too: to derive one thing out of another by dialectic. It is kept rather than paraphrased, since the whole dialogue turns on the question of what can and cannot be so derived.

existing unity will itself become an abstraction, a side of existence that does not exist. A. But how, then, is it to be taken? C. So that the existing unity is seen to exist in the two sides of existence, which precisely thereby come to exist in inseparable unity. A. Then however many parts, qualities and sides the existing object may otherwise have, we may now, on your account, assume that these parts, with their sides, qualities and mutual relations, are all equally material by virtue of the material and intellectual by virtue of the concept; every side of existence thus becomes an existential unity of two sides, the material and the intellectual. C. That we may quite safely assume. A. But what is material in the object is surely something sensuous,† which as sensuous cannot be thought, and what is intellectual in the object something non-sensuous, something thinkable that cannot be sensed? C. Perfectly correct. A. Then in the existing unity the unthinkable is after all a thinkable, and the non-sensuous a sensuous. C. That I do not understand! A. Then neither do you understand the existing unity; for how should the thinkable in an existing unity be able to penetrate the sensuous without the sensuous thereby becoming, by virtue of the thinkable, itself thinkable, and the thinkable, by virtue of the sensuous, sensuous? C. It almost looks as though there were something in what you say. A. The concept of existence, then, is a system not only of intellectual but also of material determinations — a system of intellectual sense-determinations, of sensuous intellectual determinations. C. Yes, so it is. A. But then the concept of existence betrays a most striking contradiction. C. What contradiction? A. That of demanding existence by setting itself against existence. C. What does that mean? A. If the whole system of sensuous intellectual determinations is posited in the intellectual form, then the sensuous vanishes as a moment of reflection sublated in the thinkable — that is to say, the existent transforms itself into the concept of existence, but existence goes out. If conversely the whole system of intellectual sense-determinations is posited in the sensuous form, answering to immediate sense-impressions, then the object steps into existence, but the concept of existence goes out.... C. No, the concept of existence does not go out; it is completed in existence: the existent is, by what we have said, nothing other than the concept of existence complete in every respect. A. But was it not you who, by what we have said, granted that the object from which the concept is abstracted was not itself the concept? C. Yes, but I was thinking then of the abstract and so far incomplete concept, not of the concept concrete and complete in itself. A. So it is after all possible to dialecticize existence out of the concept, once one has learned to dialecticize the concept out of existence? C. To prove that in deed would here amount to showing it

† *Translator's note. Sandselig = sinnlich*, pertaining to the senses; it carries none of the modern connotation of "sensual."

in definite examples.

A. Sound human understanding tells us that the horse is an animal, that the fly is an animal, and the snail likewise; but how do such judgments actually stand to "the principle of identity and of contradiction"? C. The matter is plain enough. A. Yes, the matter is that if we let formal identity hold for the predicate, we deny it in each subject; whereas if we let it hold for each subject, we deny it in the predicate. C. How so? A. If we posit unconditionally "animal is animal," then the horse becomes a fly and the fly a snail; if we posit unconditionally "the horse is horse, the fly is fly, the snail is snail," then animal (horse) is not animal (fly) — that is, the common predicate contradicts itself and must fall away. C. This difficulty can be overcome well enough. A. With dialectic or without? C. Without any dialectic at all! One need only bethink oneself of the factual characteristic marks of the concepts of existence, and of the relation of the disjunctive judgment to *diversi respectus.* A. That we must certainly hear! C. If we compare animals with plants, it is seen that all existing animal individuals have one thing in common: being animal. If on the other hand we compare the animal existences with one another, then one animal series shows itself, by given characteristic marks, different from another. If we compare the existences in the one series with the existences in the other, then the individuals within each series are alike; but if instead we compare the existences belonging to a given series with one another, particular marks will show how one class is distinguished from another, and so forth. The animal, then, in opposition to the plant, is what the animal is; but by general and constant marks of difference the animal, in opposition to itself, is either mollusc or articulate or vertebrate.[†] In opposition to articulates and vertebrates the mollusc is always mollusc; but in opposition to itself the mollusc is either a mollusc proper or a radiate. Now if one simply goes on in this way with the divisions (*divisiones* and *subdivisiones*) through every series, every class, every order, every family, guided by the marks established by experience, until by way of genera and species one has reached individuals — what contradiction is there in that, and what have such classifications to do with dialectic? A. Unless it be that dialectic finds a good deal to do with *diversi respectus.* C. There is surely no danger of that. A. Then tell us, on the strength of your completed classification: what actually is a horse? C. A horse (*Equus Caballus*) is, by the category of its kingdom, an animal; by the category of its series, a vertebrate; by the category of its class, a mammal; by the category of its order, a pachyderm; by its

[†] *Translator's note. Bløddyr*, *Leddyr*, *Hvirveldyr*, and *Straaledyr* just below, are the Danish names of Cuvier's four *embranchements*: Mollusca, Articulata, Vertebrata and Radiata. The classification was standard in 1861 and has since been abandoned; "radiate" covered what we should now distribute among the echinoderms, cnidarians and others.

family category, akin to asses and mules; and a species-category of its own within a genus-category of its own. A. A fine collection of categories, then, all of which had to combine here to yield the horse's concrete category of existence! If only one could now, with the help of *diversi respectus*, get it properly explained how these many different categories have actually grown together — not to say, been bundled together — into a single one! C. They of course enclose one another in such a way that the more general always encompasses the more special. A. Surely not, though, in the way one envelope encloses another, or one ring encircles another? C. No, they rest upon one another, the more special having in each case settled upon

the more general. A. The more general a category is, the more abstract it is; the more abstract it is, the further it is from existing: so the existent rests upon the non-existent? C. No, I meant to say: they rest upon one another in such a way that the more general is always borne by a more special, this by one more special still, until we come to the most special of all, which is one with the existent itself. A. So it is not being a horse that depends on being an animal, but being an animal that depends on being a horse? C. No, that is not how I conceived it either; indeed, at this moment I cannot, to speak plainly, say how I actually did conceive it.

II.
Philosophy and the Particular Sciences.

General Remarks.

"With respect to the a posteriori, the advance of knowledge will doubtless depend on new information's being constantly gathered, new matters of fact tested, new discoveries made, and so on; but is the matter thereby settled? If the advance of knowledge in relation to the a posteriori is regarded from this side alone, it is easy to dismiss philosophy. If it cannot so much as exhaust the a priori — seeing that mathematics owes it no thanks — how much less can it meddle with positive matters of fact! Let the historian lay a problem before it: it cannot answer, for a historical problem is answered only historically. Let a chemist apply to it: it has no knack for experiment and makes no discoveries in chemistry. Let the astronomer want help with an observation: it scarcely understands him. Let the physician ask it for advice: it is ignorant. Even what it has speculated out on its own account concerning the relation of soul and body the physiologist may unfortunately be unable to use." (Propædeut. pp. 99–103.)

The extent to which it holds that philosophy is dialectic is seen from its relation to the subject-sciences,† for that relation is dialectical in the strictest sense. If one overlooks the dialectical, the relation becomes ambiguous and doubtful; for while the subject-sciences are by nature inclined to dismiss philosophy, philosophy in its turn, when it conceives its vocation dogmatically, is tempted from its side to exalt itself at the expense of the other sciences. Schelling's words are characteristic in this regard: "Speculation is everything;

† *Translator's note. Fagvidenskab*, literally the science of a *Fag* — a subject, a trade, a department. "Subject-science" is used here, as in the translations of Nielsen's other works, and its practitioner, the *Fagmand*, is "the specialist." These are to be kept distinct from *de særskilte Videnskaber*, "the particular sciences," of this part's title.

it is to behold in God, to contemplate that which is. Science itself has worth only in so far as it is speculative, that is to say, contemplation of God as he is." (Propædeut. § 17, pp. 81–83).

What has chiefly contributed in recent times to confusing the issue is the circumstance that Hegel — whose penetrating grasp of contradiction, of the negative, and of the immanent method of development (Propæd. pp. 150–158) must place him so high in the ranks of the dialecticians — has nevertheless, in his account of philosophy's relation to the subject-sciences, overlooked the dialectic of the relation. Instead of acknowledging the scientific independence of the particular sciences, Hegel closes his propaedeutic survey of the encyclopaedia of the sciences (Propæd. pp. 84–85) with the declaration that the encyclopaedia is properly "an exposition of the general content of philosophy; for whatever in the sciences is founded upon reason depends upon philosophy, whereas whatever there is in them of what rests on arbitrary and external determinations, or, as it is called, is positive and statutory, lies outside it." But to let philosophy's superiority rest in this way directly upon the subjection of the subject-sciences is to take the matter undialectically; to see philosophy's independence grounded in its dependence upon the independent subject-sciences is, by contrast, to grasp the relation dialectically.

The independence of the subject-sciences over against philosophy is evident from the course of their development; for the methods and discoveries that mark the great advances of these sciences are so far from being attributable to philosophy that "the history of the inductive sciences" (Whewell) shows, on the contrary, to what degree independent research and a procedure answering to the nature of the case have depended on the researcher's being able to free himself from the one-sided influence of philosophical speculation (Propæd. pp. 167–68). What in the subject-sciences "is founded upon reason" by no means depends upon philosophy in particular. In the first place it is not reason that owes its origin to philosophy, but on the contrary philosophy that owes its origin to reason. And next — and in this connection this is decisive — it must be emphasized in the strongest terms, not only for the sake of the subject-sciences but above all for philosophy's own sake (Propæd. § 23), that there is a knowledge which is scientific in the strictest sense although it is not philosophical; indeed a thinking whose scientific energy depends precisely on its not being philosophical. For that in the empirical knowledge of the empirical there is a certainty of sensation, a many-sidedness of representation, an added gift of intuition which no philosophical glimpse of the Idea can replace, is surely as undeniable as it is undeniable that there is an exact and critical thinking not to be exchanged for the dialectical (Mathem. og Dialekt. p. 3). The rational distinctiveness of the particular sciences is seen, too, from the fact that each one severally presupposes a scientific talent of its own answering to it.

But if one now sets up the proposition that *the subject-sciences are independent of philosophy, whereas philosophy — particularly as regards the appropriation of positive content — is dependent on the subject-sciences*, then this presupposition, true as it is in itself, may easily occasion a false inference. If philosophy, it is said, is so dependent on the exact and positive sciences in everything that concerns the exact, the empirical and the positive, then this dependence must plainly be a sign of its lack of independence. Every time philosophy comes into contact with some determinate positive subject matter, be its content mathematical, mechanical, chemical or historical, the philosopher must of course put up with being not merely set right about the facts but even duly schoolmastered by the specialist in the development of the concepts themselves. It looks, then, as though the concepts were perfectly trustworthy, clear, distinct and secure in all their consequences so long as they are preserved in the form in which the subject uses them, but abandoned to distortion as soon as they are made an object of philosophical reflection; it looks as though the philosopher had to borrow everything from the specialist, while the specialist for his part may regard the philosophical as a matter of complete indifference. This way of seeing things rests on a misunderstanding. Philosophy's dependence on the subject-sciences is, essentially regarded, a sign of its independence, of its ideal self-validity.

In contrast to the material objects of the surrounding world all sciences are of course ideal, all have ideality; but in contrast to philosophy, as the representative of ideality, the subject-sciences must be said to stand for reality. If the opposition is determined categorically, the real in its finitude will be recognized by its asserting itself through excluding its other, whereas the ideal proves its reality precisely by taking the other up into itself and relating itself to itself through the other. That mathematics as mathematics, chemistry as chemistry, natural history as natural history each keep to their own[*)], and, even when contributions are fetched from other subjects, still have first and directly to do each with its own, is precisely a mark of these sciences' finite reality. That philosophy, by contrast, must always have its ideal wealth in real poverty, its positive in its negative, was illustrated by Plato in an ingenious myth: at Agathon's banquet Plato has his Socrates, as is well known, present Eros, the ideal philosopher, as the son of Poros and Penia.

[*)]"There is a cardinal rule which can never be violated with impunity, and which is nevertheless constantly violated in our groping advance toward the light. It is: *never to attempt the solution of the problems of one science by the order of ideas belonging to another.* There is one class of conceptions belonging to physics, another belonging to chemistry, a third belonging to physiology, a fourth belonging to psychology, a fifth belonging to social science. While all these sciences are closely connected, each has its own independent province, which must be respected." G. H. Lewes: Hverdagslivets Physiologie, Danish translation, part 1, p. 66.[†]

The realistic science, once founded, may in the course of time need manifold reshaping and undergo many changes, yet its supports and foundations will through it all remain unshaken. If the idealistic science, on the other hand, is to remodel its building, a *de omnibus dubitandum* will gradually shake every support; the whole building must then be pulled down to the ground, that it may be raised in its new style from the ground up. Each time such a critical juncture arrives it does indeed look for the moment as though reality were at stake, as though it were now all over with philosophy; but history confirms what also lies in the nature of the case, that such pervasive crises mark precisely the turning points in philosophy's rebirth.

"The subject-sciences are solid; they always have steady work, ample means and secure recognition in the republic of letters. Philosophy, by comparison, is like a traveller without settled provision, a hovering shape that sinks and rises, dies away and lives again, as the spirit will." (Propæd. p. 102). — Yet it is only from a wholly external, realistic standpoint that the ideal is taken as merely the vague, the indeterminate, the in itself indeterminable; the dialectical form is a form determinate and determinable in itself, and philosophy's relation to the subject-sciences is therefore to be stated categorically.

If philosophy is to be able to place itself in an inner and essential relation to a particular science, it must discover in that science a problem which it can also call its own problem. In order to show how such a *common problem* is chosen and treated, and so as not to scatter our attention over too many points, we shall in what follows confine ourselves to examples drawn from mathematics, chemistry and physiology; and we have thus, in mathematics, *the problem of infinity*; in chemistry, *the problem of change*; in physiology, *the problem of life*.

But even if such common problems are characteristic enough when viewed from the side of ideality, it is by no means thereby settled that they show themselves so when they are grasped realistically. Because the philosopher has noted some pervasive problem in one or another determinate subject-science, it is not yet said that the specialist will acknowledge the same problem in a corresponding way. A chemist who wished absolutely to resist a philosophical conception might, for example, reason thus: "There is in chemistry properly nothing at all that one may call a problem of change, but rather a concept of change, which arises and comes to be applied in so far as chemistry investigates the natural laws according to which the inner changes of bodies take place. But it is one thing to investigate the laws according to which the changes take place, and another to want to fathom the essence of change, to speculate about its possibility and impossibility — a task wholly foreign to chemistry." In the same spirit a physiologist wishing to dismiss philosophy might reason thus: "Physiology does indeed proceed from the presupposition that there is a something we must call life, since in living organisms there is a law-governed system of vital phenomena; but since physiology essentially confines itself to investigating the phenomena, without penetrating into what lies hidden behind them, it can by no means be granted that there is here any proper problem of life. What life is in itself — a principle, a force, a product of the peculiar cooperation of mechanical and chemical forces — no matter! Let each form whatever individual representation of the essence of life he pleases; so long as he is not thereby kept from seeing the validity of the laws and demonstrating the physical causes, physiology is satisfied."

The difficulties which the natural influence of differing standpoints on the conception of a scientific problem always brings with it are considerably heightened by the ease with which something human — that is to say, something one-sided, accidental, arbitrary, passionate — mixes itself into the investigations. And since philosophy, for all its ideality, is like every other science subject to human conditions, it becomes important in this connection to take the matter from the purely human side as well, and so to note the obligations and rights which the sciences have to uphold in their mutual conflicts.

That the common problems in question must look otherwise in philosophy than in the subject-science is a matter of course. The problem of change, even if the chemist will acknowledge it in his own way, is in chemistry by no means what it is in philosophy; likewise the problem of life, even when the physiologist grants its reality, is one thing in physiology and another in psychology. The philosopher who knows the problem in the form of otherness, and is aware that the same is nevertheless not the same, must therefore make it his chief business to clarify the conditions of the development and to make the transformation plain. Anyone who would seek guidance in a particular science must of course make himself at home in its way of thinking. If the philosopher neglects this — if he misunderstands the character of the science in question either as a whole or in essential particulars, whether by ascribing to it an aim it does not have and cannot have, or by tearing its doctrines out of their proper connection and so distorting them, or by overlooking the peculiar presuppositions and conditions under which a given concept with its corresponding designation was formed, thereby laying a false meaning under the concept and in this way coming to force upon the science views, results and consequences it cannot acknowledge — then the specialist is within his rights in speaking out. This right becomes a formal duty to the truth when such blunders on philosophy's part, as has not seldom been the case (Propæd. pp. 170–171), are combined into the bargain with the conceit that it is philosophy which from its exalted standpoint must feel called upon to reform the subject-sciences. But further than a rejection of philosophical encroachments the specialist cannot and should not go. If he goes further — so that, for example, he is not content to secure to the concept the meaning it has in his science, but regards it as belonging to his subject to such a degree that he either flatly forbids the philosopher to touch it, or at any rate takes it upon himself to prescribe how far, in what manner and in what connections philosophy may develop the borrowed concept graciously made over to it by the subject-science — then there is to be seen in this not merely a presumption but a ludicrous misunderstanding.

“Confined to the several compartments of consciousness, the concepts gained are still so grown together with finite constituents, still so weighed down with particular presuppositions, that they find themselves as it were in a state of unfreedom. They are held here as the prisoners of their matter within a determinate horizon; they are, so to speak, the subject’s bondsmen, who dare not leave the appointed place of work.” (Propædeutik p. 113.)

Now in setting out to free the real concepts from what is accidental in their delimitation, to unfold the consequences and to bring the schematic forms into dialectical flow, philosophy must of course — once it has drawn the concepts into its own horizon, posed the problems in its own way, and thus taken the whole responsibility of the treatment upon itself — have an indisputable right within its own province to apply its own measure, to introduce its own designations and usage, and to carry out the development in its own spirit. If it stands firm that every subject-science must in essentials contain its own correctives, then philosophy too must surely contain its correctives. A judgment upon the errors and blunders a philosopher may possibly have committed in carrying through the dialectical analyses belongs in the last instance not before the forum of the subject-science but before that of philosophy.

To deny philosophy the full scientific right to treat in a free philosophical manner what it has appropriated from the subject-sciences is to deny it a development of concepts of its own, and to resort to a dictum of authority. If it is to go by dicta of authority, then a specialist may settle philosophy’s fate as easily as that caliph settled the fate of the Alexandrian library: “Either it contains only what stands in the Koran, and then it is superfluous, or it contains something else, and then it is false”; — “either philosophy contains only what stands in the subject-science, and then it is superfluous, or it contains something else, and then it is false.” It is a Saracen dilemma; but a Saracen dilemma cannot make philosophy give up its right.

A.
Philosophy's Relation to Mathematics: The Problem of Infinity.

"A philosophical fox or marten has stolen in upon mathematics' fat poultry to fill his schematic belly; he means to suck out their warm blood, so that nothing is left but the dead carcasses, void of sap and strength, which are then to be presented as mathematics' vigorous, practical, lively concepts" (Professor Steen's "Afregning," p. 7).†

These boding — indeed ill-boding — words may well seem at first glance to have something particularly forbidding about them, especially when set beside a declaration such as this: *"In the first place there is nothing that one may name 'the problem of infinity' in mathematics"* (cf. Begrebet: Nøiagtighed, p. 38). But when it is thereupon conceded that there is *"a mathematical concept 'infinite' which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature"*, the forbidding element disappears and something inviting shows itself. For if the mathematical concept "infinite" which arises and comes to be applied is, on the mathematician's own admission, *"extremely important and moreover difficult, because it is of a mixed nature"*, then this contains, so far as philosophy is concerned, a well-founded invitation to investigate whether this concept of infinity, in mathematics so *extremely important and moreover difficult*, may not after all be connected in one way or another — perhaps without the mathematician's knowing it — with a problem running through the whole of mathematics; whether philosophy may not have every reason to acknowledge, in this pervasive problem, its own age-old problem of infinity; and whether mathematics, without a mathematician's always quite thinking of it, may not be said to furnish essential and indispensable contributions toward that problem's solution.

a) The Concept of Infinity.

That the concept "infinite" which in mathematics "arises and comes to be applied" is "extremely important" we may grant at once; but we must most definitely deny that the

† *Translator's note.* This whole section is a reply in an ongoing quarrel. In 1860 the mathematician Adolph Steen attacked Nielsen's *Philosophie og Mathematik* in a pamphlet called *Afregning* ("Settling of Accounts"), and Nielsen answered with two pieces of his own — *Hr. Professor Steens "Afregning": Hermed en specificeret Opgjørelse til Revision* and *Om Begrebet Nøiagtighed med særligt Hensyn paa Hr. Prof. Steens "Bidrag"*, the latter a postscript to *Philosophie og Mathematik.* The short titles Nielsen cites here and below — *Afregning*, *Begrebet: Nøiagtighed*, *Om Begr. Nøiagt.* — are to those pieces, and the words he puts in quotation marks are his opponent's objections as he reports them there. Without this the pages read as an argument with nobody.

concept of infinity, taken as a concept by itself, is either difficult or of a mixed nature. Of the infinite in general only this can be said: that it is a negation of the finite, the not-finite. If any difficulty lay hidden in such a negation, the difficulty could not stem immediately from the negation itself, but would have to be referred essentially to what the negation cancels. So if one is to find difficulties in the mathematical "infinite," one must find corresponding difficulties in the mathematical "finite"; and if one raises no objection to the latter, it betrays mere thoughtlessness to raise objections to the former.[†] If the concept of infinity is difficult "because it is of a mixed nature," would not the concept of finitude, when the finite is specially emphasized, be of a still more mixed nature? If it be objected that the mathematical "finite," which holds only of magnitudes and not of things, cannot possibly be of a mixed nature, then one will also be obliged to grant that the negations in which such a "finite" disappears can as little be so.

Concepts of a mixed nature always have a questionable admixture of the conceptless, and as mixture-concepts they stand on the lowest rung in the order of concepts. How does a mixture-concept come about? By parts of unlike kind, brought together in an external and accidental way, being subsumed under a collective unity, which in consequence comes to contain marks of unlike kind accidentally brought together. Thus the concepts "topsoil," "gravel," "marl" are genuine mixture-concepts; but a concept of gravel and a mathematical concept of infinity — what a difference!

But, someone may perhaps ask, if the concept "infinite," in however many and various connections it occurs and however many transformations it undergoes, always preserves one and the same unalterable fundamental meaning — namely that of being "the negation of finitude" — how does it come about that so pure, so negative a logical concept can in a mathematician's eyes take on the appearance of being "of a mixed nature," or, what is the same, of belonging among the mixture-concepts? The answer is not difficult: *one mixes the accidental with the essential; by this mixing one turns up a number of meanings of differing kinds; one does not derive these separate meanings from the fundamental meaning, does not exhibit the necessary transition of the one into the other, but sets them up externally side by side, makes them an object of enumeration — and behold, the thing goes as of itself.* If one puts such meanings as "negation in positive form," "impossible," "transition involving something qualitative," and so on, externally together, one gets at once an impression of the mixed. How is one to distinguish a negation in positive form from a negation in negative form, except by the former's expressing the negative mediately

[†] *Translator's note.* Danish *gjøre Ophævelser over* is an idiom, "to make a fuss about"; but *Ophævelse* is also the technical word for sublation, and Nielsen has just used its verb (*ophævet*, "cancelled") two sentences earlier. The pun does not survive.

and the latter immediately? But if now the finite is set against the infinite, it is not the infinite that expresses the negation in positive form but on the contrary the finite, in so far as the finite, though it has a positive form, is yet mediately determined as the not-infinite. To take negation's own immediately negative expression for its positive form betrays a comical confusion. If someone, to display his acumen in matters logical, were to maintain that the proposition "a is not something" was straightforwardly negative, whereas the proposition "a is nothing" was affirmative, the negation being lodged in the word "nothing," one would surely agree without difficulty to assign such trials of acumen — formulated in all clumsiness after a formal logic's S is $Non - P$ — if not outright to the pedantic, then at any rate to that class of immature attempts which betrays that the acute man has not yet learned to distinguish between fruitful and unfruitful distinctions. If one now grants, so as not to be saddled with too much logical mixed goods, that the proposition "a is nothing" is just as directly negative as "a is not something," then one will also grant that the proposition "a is zero" is just as directly negative as "a is nothing." But if it stands firm that the sign 0 is just as direct and immediate an expression of negation as the word "nothing," then it takes a quite abnormal acumen to see that, while the proposition "between a and a there is no difference" is straightforwardly negative, the equation $a - a = 0$ should on the contrary be so nailed fast in the positive that only the screw of a paraphrase could bore through to the negation. (Cf. Om Begrebet: Nøiagtighed, p. 39).

The logical construing of mathematical equations is a peculiar business; the more weight is laid here on the conceptual side, the more evident it becomes that philosophy has a word to say here too. On the natural mathematical way of thinking, the equation $\frac{a}{0} = \infty$ shows us the well-known relation between divisor and quotient in its extreme consequence: if, on the assumption of a constant dividend, the divisor goes on shrinking to infinity, the quotient must go on growing to infinity. There is nothing in this relation to recall the impossible; everything points on the contrary to the possible: to the smallest possible divisor there answers the greatest possible quotient. If, on the other hand, in order that the infinite may come to signify the impossible, one chooses another point of view and construes the equation thus — "When it is asked how many times 0 must be made greater in order to give a, the answer in words is: that cannot be produced by multiplying 0; but in signs: $\frac{a}{0} = \infty$, 0 must be made infinitely many times greater in order to give a" (Om Begr. Nøiagt. p. 40) — then such a construal is well fitted to promote all manner of admixtures of an illogical miscellany. If the equation $\frac{a}{0} = \infty$ were really and seriously designed to answer the in itself absurd question "how many times must

0, the absolute 0, pure nothing, be made greater in order to give a?", then notations such as $\frac{a}{0} = 0$, $\frac{a}{0} = a$ would have been far more adequate. For from such expressions the impossible, the absurdity in the demand, is seen instantly, whereas the equation $\frac{a}{0} = \infty$, by its ironic pointing to the infinite, only mimically hints at what is absurd in the question, and then — quite rightly — transfixes the questioner by "a negation in positive form," because he is asking not as a mathematical thinker but as a logical fool. If one gives up the logical foolery of trying to reckon how many times pure zero must be made greater in order to produce a; if, in order that the equation may yield any sense, one lets 0 denote the smallest possible divisor while ∞ denotes the corresponding greatest possible quotient — then it is seen from every side how the impossible gives way to the possible. From the equation $\frac{a}{0} = \infty$ there follows directly, in virtue of the concept of the relation in question, $a = \infty\frac{a}{\infty} = \infty.0$; likewise from $\frac{b}{0} = \infty$, $\frac{c}{0} = \infty$, $\frac{d}{0} = \infty \ldots$ there follow $b = \infty.\frac{b}{\infty} = \infty.0$, $c = \infty\frac{c}{\infty} = \infty.0$, $d = \infty.\frac{d}{\infty} = \infty.0 \ldots$ Instead of the impossibility of $\infty.0$'s being able to give a, we see on the contrary the possibility of $\infty.0$'s giving not only a but b, c, $d \ldots$ — in short, as many different magnitudes as you please. Instead of the curt, dismissive "impossible," we get possibility upon possibility, possibilities in abundance. This result is of course so far from surprising the *thinking* mathematician that it does not even surprise ordinary sound human understanding. Everyone understands that any finite magnitude whatever may — and this is a possibility — be said to consist of infinitely many infinitely small parts, hence $a = \infty.\frac{a}{\infty} = \infty.0$, and that it may therefore also — and this is a possibility — be divided into infinitely many infinitely small parts, hence $\frac{a}{\infty} = 0$.

The fixed idea that the concept "infinite," as a "negation in positive form," should be better fitted to denote "the impossible" than the concept "finite" is easy to clear up. For "the impossible" is in and for itself as indifferent to "infinite" as to "finite"; the impossible shows itself every time a demand that has been set up annihilates itself, and such self-annihilating demands may just as well presuppose the finite as the infinite. It is, to stay with the infinite, impossible to *enumerate* all the terms of an infinite series; it is, to pass to the finite,

impossible for a human being to *count up to* a quadrillion; and it is — here we must lean at once on both the finite and the infinite — impossible that a sphere with an infinite radius should be a sphere, or a plane a spherical surface with a finite radius; and just as impossible that the equation $a = 2a = \infty.a = \frac{0}{a}$ should have any reality. If in the language of mathematical signs one wants a more direct notation for something impossible in itself, then it is not 0 and ∞ but $\sqrt{-1}$, $\sqrt[2m]{-a}$ and the like that one has to keep to. (The

"Afregning" as revised by R. N. Appendix, pp. 26–27).

In logical generality it is easily seen that the concept "infinite" does not as a concept take up constituents of unlike kind, and so cannot be reckoned among the mixture-concepts. That the concept nevertheless occurs in several meanings, shows itself in a different light, and to that extent may, when treated superficially, acquire a tinge of being "of a mixed nature," has its ground in something quite different: to think the concept through is to unfold the problem.

b) The Problem of Infinity. Suppose someone were to open an objection to a philosophical reference to the way in which the problem of finitude finds its solution in mathematics with the following declaration: "In the first place there is nothing that one may name the problem of finitude in mathematics, but rather a mathematical concept 'finite' which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature." One would surely find that a strange thing to say. What a finite magnitude is, is soon said; but the problems concerning the mutual relations of finite magnitudes are countless. Instead of saying that there is no problem of finitude in mathematics, one would rather have to say that there are countless problems of finitude in mathematics. If these problems fell apart from one another in loose multiplicity, one might indeed deny that there was any general problem running through them all; but one would then at the same stroke have to deny that there was any mathematical method — in other words, that mathematics was a science. If mathematics is a science, if there is a method by which the various problems of finitude are framed and solved in systematic connection, then there is also a general problem running through them all, a fundamental problem embracing and determining the method in all its ramifications, a problem whose all-round solution demands nothing less than the whole of "the analysis of the finite."

But what holds so incontestably of "the problem of finitude" in its relation to "the analysis of the finite" — should this not hold also of the relation between "the problem of infinity" and "the analysis of the infinite"? What holds so incontestably of the concept "finite" in its relation to "the problem of finitude" — namely that the difficulty lies not in the concept but in the problem — should this not, on closer inspection, be found to hold also of the concept "infinite" in its relation to "the problem of infinity"? —

α) Concept and Problem. As the distinguishing mark of mathematical concepts — however much they may otherwise resemble the general logical ones — it must be expressly emphasized (Mathem. og Dial. pp. 21–27) that they are operational concepts. From this it follows that, while in their generality they present no difficulty, analysing thought will soon discover that their proper development must always take place in the closest connection with the operations whose logical essence they are meant to clarify. What a logarithm is can easily be stated in a definition exact enough for sound understanding to hold the concept fast; but the manifold and characteristic consequences in which the concept's content first really shows itself come forth only in the course of the operations. What is meant by dividing a line in extreme and mean ratio — or, as it is also called, by golden-sectioning the line — is soon said and easily grasped; the concept "golden-section"† presents no difficulty. The difficulty, if there is one, is to solve the problem of the golden section and to develop all the consequences contained in it. What holds of the particular operational concepts holds of the general ones too: only from the systematic solution of the problems of finitude does it come to light what mathematically lies in the concept "finite"; only a carrying-through of "the analysis of the infinite" can open us a scientific insight into the mathematical concept of infinity. From the consequences of the operations it is further seen that the infinite falls not only outside but also inside the finite (cf. Phil. og Mathem. pp. 20–24; Mathem. og Dialekt. pp. 28–38), and that it develops in manifold ways out of finitude itself. Had a pervasive problem of infinity not originally stirred beneath "the analysis of the finite" and declared itself in many separate flashes, the differential and integral calculus would never have been developed. When Leibniz solved "the problem of tangents," what was essentially at stake was not the tangent and the secant (cf. Phil. og Mathem. p. 64) but the articulating reciprocity of finite and infinite: in "the problem of tangents" "the problem of infinity" is to be seen.

β) The Mathematical Conception. That thinking mathematicians, however much weight they may otherwise lay on the satisfactory results of the operational method, have nevertheless never been able to separate satisfactorily the concept they find inconceivable from the method that rests upon that concept, is shown by the history of mathematics from Leibniz and Newton (cf. Mathem. og Dial. pp. 90–91, p. 95 and elsewhere) down to the present time. "In the first place, when the principle is unknown we cannot know whether all the consequences following from it can be admitted: we go as if blindfold, and it might well happen that after laborious effort we should at last strike upon something which exhibited precisely the nature of the principle — that is to say, upon something we could not conceive. In the second place: is it not a shameful degradation of reason to make it follow a concatenation of terms, a series whose beginning and end

† *Translator's note.* The Danish verb is *høidele,* literally "to high-divide," the nineteenth-century Danish term for division in extreme and mean ratio, i.e. the golden section. English has no corresponding verb, so "golden-section" does duty as one here and in the next sentence.

it cannot see through? It is a great evil that is born of that kind of routine: accustomed to operating without seeing the ground, the mind loses all practice in thinking, and ends by carrying everything out quite mechanically." (Mathem. og Dial. p. III). These misgivings of Savrien's sufficiently explain the unrest and tension in which the analysis of the infinite has constantly held mathematical reflection[*)] (Mathem. og Dial. pp. 55–56), whose problem — as a prize question set by the Berlin Academy in the year 1784 characteristically puts it, "wie aus einer widersprechenden Voraussetzung so viele wahren Sätze haben können hergeleitet werden" [how so many true propositions could have been derived from a contradictory presupposition] — will not cease to return, even though "routine," the mechanical facility, will at all times no doubt afford "*habiles Calculateurs*" satisfaction so ample that it may easily escape this or that reckoning-master's notice that there is in mathematics something one may name "the problem of infinity."

γ) The Philosophical Conception. "To make the problem of infinity recognizable, reflection must lean upon the operation; to solve the problem, it must appropriate the intelligence expressed in the operation. That there can be a finite ratio between infinite magnitudes; that a sum of infinitely many terms may in some cases make up a finite and in others an infinite magnitude; that a series whose terms are so related that the limit of the ratio of two successive terms gives the quotient 1 may in some cases be convergent and in others divergent, and so on — such things will everywhere escape the thinking that does not, at the operation's hint, catch sight of the exact determinations. That these determinations must in philosophical form become dialectical, while in mathematical form they neither are nor can be so, shows a characteristic difference between mathematics and philosophy, but by no means any conflict. No concept, however exactly it be defined, can maintain itself in a being indifferent to the other concepts: the validity of concepts is seen in their transitions; but the transitions which exact analysis makes by the operation, the thinking that penetrates the operational concepts makes by dialectic" (Mathem. og Dial. p. 57. Cf. pp. 59–60). If the finite is set immediately against the infinite, sound understanding involuntarily gets the impression that it is the finite which, in its multiplicity of terms and relational determinations, contains the wealth of existence, whereas the infinite is a bottomless nothing, waste and empty as empty space. And since thought can determine only what is in itself determinable, and only that is determinable which admits in some way of being defined, and only that admits of being defined which admits of being bounded, and the bounded is a finite — it must

[*)] *See Carnot: Reflexions sur la Métaphysique du Calcul Infinitésimal. Paris 1797* (übersetzt von Hauf. Frankf. a. M. 1800). Versuch einer Philosophie der Mathematik — mit einer Kritik der Aufstellungen Hegel's über den Zweck und die Natur der höheren Analysis — von H. Schwartz. Halle 1853.

constantly appear to the understanding as though only the finite were conceivable, while the infinite, the unbounded, were really not to be conceived at all, but, like the ancients' hyle (τὸ ἄπειρον), to be known by a kind of spurious thinking (λογισμὸς νόθος).[†] Where this misunderstanding has taken hold there can of course be no question of entering more closely into the problem of infinity; to see the problem's philosophical significance one must absolutely have at least an inkling of the dialectical reciprocity of the finite and the infinite. "An angle of 90^0 can be divided into infinitely many infinitely small angles, and so can an angle of 1^0; a square mile can be divided into infinitely many infinitely small squares, and so can a square inch: what follows from this? If the emphasis is laid on the resolution of the finite into the infinite, the consequence does indeed seem to be that there is no longer any difference between the angle of 90^0 and the angle of 1^0, that the part becomes equal to the whole — that the boundary, in other words, is effaced. But if the emphasis is laid instead on the infinite's existing in and through the finite, then there is not only a difference between finite and finite, between the part and the whole, but at the same stroke a definite difference shows itself between infinite and infinite, with respect to the infinitely small as well as to the infinitely great. The one infinite number of infinitely small angles may then be as different from the other as 1^0 from 90^0; the one infinite number of infinitely small squares as different from the other as a square inch from a square mile. That the boundary is sublated in the finite means that it is posited in the infinite; that the infinite takes the boundary up into itself means that the finite is posited. The dialectical unity shows itself in this, that the boundary belongs as much to the infinite as to the finite" (Mathm. og Dial. p. 33). The fundamental thought that the infinite not only effaces the boundaries of finitude but also takes the sublated boundary-determinations up as articulating moments in its own development of concepts is, from the standpoint of the operations, sufficiently designated by saying that the infinite is analysed.

b). The Analysis of the Infinite. In the task of "analysing the infinite," then, mathematics and philosophy have a comprehensive common problem. That the problem must, however, undergo a transformation in being taken up by philosophy, and that the philosophical analysis neither can nor should be carried through in the same way as the mathematical, is a matter of course: for it is one thing to develop a concept for use in the operations that fall under it, and another to reflect the operations through in order to make plain the concept at work in them. "Where several sciences treat the same

[†] *Translator's note.* The phrase is Plato's, *Timaeus* 52b, where the receptacle of becoming is said to be apprehended by a "bastard reasoning" — *λογισμῷ τινι νόθῳ* — and hardly to be believed in. Nielsen pairs it with *hyle*, matter, and with the Aristotelian *τὸ ἄπειρον*, the unlimited.

thing, each will make its own clear; but they will never make the same thing clear in the same way. What one development must by its aim determine with the greatest exactness, another may by its aim only hint at in fleeting, hovering strokes, precisely because it has to let the emphasis of exactness fall on other points." (Begreb. Nøiagt. pp. 51–52).

Since mathematics as a subject-science looks directly to its own, it cannot as a rule be presupposed that mathematicians as specialists should have either any particular interest in, or any special facility for, following philosophy's dialecticizing analyses. But it may be presupposed — indeed, in the name of scientific standards, plainly demanded — that the specialist who feels called upon to pass judgment on what is distinctive in a philosophical procedure should at least grant the matter enough attention to hold fast the most essential starting points, follow the train of thought, and know what is being talked about. When, to cite a couple of examples, it is expressly to be brought out what difference it makes whether the categories opposed to one another are *division* and *composition* or *piecing-together* and *motion*, and this is then illustrated by a comparison between the broken and the curved line thus: "Of the broken line a composition holds, for its parts are finite; it is a piece-work belonging in each and all to finitude. Of the curved line no finite composition whatever holds, no piecing-together, not even the very finest; it owes its coming-to-be to motion alone, for it belongs to infinity" (Phil. og Mathm. p. 76) — would not anyone who grasps what is being talked about see at once the impossibility of characterizing the category of *piecing-together* without presupposing the immediately given existence of the finite parts? If we are to reflect here on how all the lines out of which the broken line is composed — the parts out of which a whole comes to be — themselves in turn have their becoming, then we are looking at becoming alone, not at piecing-together. Here too, of course, in so far as it is not a matter of clarifying a thought but only of lodging "a critique," one may on the subject's behalf lodge the following: "the straight and the broken line owe their coming-to-be to motion neither less nor more than the curved line does; the curved line is mathematically composed of finite lengths, just as the broken and the straight are." But what does science gain by such a critique? To let everything run together — formation by piecing-together and coming-to-be by smooth transition — to let everything come to the same thing: discreteness and continuity, finite and infinite, broken and unbroken, curved and straight, straight and crooked — that is surely as little mathematical as it is philosophical!

Ignorance and thoughtlessness are by no means always the same: one can be ignorant without being thoughtless, and one can be knowing in a finite sense and yet thoughtless to a striking degree. The examples are manifold. In a philosophical investigation the question is raised how the line's origin is really to be conceived in relation to the point; for that one always has the line once one has the surface, and the surface once one has the body, goes without saying. "In relation to the line as the positive, the point, the line's boundary, is the negative; when the line is a representation, the point is a thought. In relation to the surface as the positive, the line, the surface's boundary, is the negative. The surface is represented when the line is thought." (Phil. og Math. pp. 65–67). When, then, the question is how the line's origin is really to be thought — since letting the line arise by "composition of finite lengths" is merely to presuppose what was first to be shown, the finite lengths being themselves lines, and since letting the line's coming-to-be rest on surfaces will not do either, the surface's own origin necessarily presupposing the line — when this is expressly the question, and a man of the subject then, an expert who knows all the cases, with lines and with surfaces, with intersection and without, comes forward with the information that "*mathematically, curved lines also arise from the mutual intersection of surfaces*": then one receives not merely a piece of mathematical information but an aesthetic encouragement into the bargain.

α) Boundaries. The difference between exact and dialectical thinking is recognizable above all in a different use of the boundary. "Either something or nothing," "either positive or negative," "either finite or infinite," "either attainable or unattainable": such judgments of separation presuppose a sharp stress on what divides in the boundary. The sharply dividing boundary — the boundary that holds what it separates so entirely outside one another that the one must be denied when the other is affirmed — is *qualitative.* But the more the contradictory opposition is sharpened, the more exclusively the qualitative holds, the more conspicuous it becomes that the boundary, by being what divides, is precisely what unites. If one posits "either $+a$ or $-a$," the positive is indeed thereby held outside the negative; but since the division is so sharp that there is nothing whatever between, the positive and the negative are brought into immediate contact precisely in the dividing boundary, the zero point. The zero point is therefore the point of union of the positive and the negative just because it is their point of separation. If we now put the finite for the positive and the infinite for the negative — "either finite or infinite" — the same thing repeats itself. By staring at the negation, at what divides in the boundary, one gets the impression that the finite and the infinite are held apart by a yawning abyss: "either finite or infinite! One may therefore go on with the finite as long as one likes; a finite is and remains a finite, never an infinite." But by bethinking oneself of what unites in the boundary one easily sees that things so divided must nevertheless lie infinitely near one another: "either finite or infinite! So when one does not have the finite, one has straightway and without more ado the infinite." Where the finite ceases the infinite begins; the boundary, the negation, is therefore itself the logical point of contact, the point at which what is divided meets: here there is nothing between. "But if the finite and the infinite are in this way each other's boundaries, the determination of boundary must become dialectical. For if the infinite has its boundary in the finite, then it has a boundary of finitude, a finite boundary, and is itself finite. If conversely the finite first reaches its boundary in the infinite, then it has no finite boundary and is consequently infinite." (Phil. og Math. p. 22). The infinite is an unbounded; how, then, can that which has no boundaries at all be itself a boundary? The finite, which everywhere has boundaries in so far as one finite bounds another, may rightly be said to be something bounded in itself; but how can what is bounded in itself have an unbounded for its boundary? Or are there perhaps three kinds of boundary here: a boundary that holds for the finite and not for the infinite, a boundary that holds for the infinite and not for the finite, and a boundary common to finite and infinite? But if this is assumed, how does each of these boundaries stand to that whose boundary it is?

What, in respect of the mathematical mode of expression, easily gives the concept of boundary an ambiguous tinge is the circumstance that both magnitudes and negations of magnitude are posited as boundaries, and that 0 and ∞ cancel one another. By 0 the finite boundary is posited; by ∞ every finite boundary is annulled. Closer consideration will then easily make it evident that both negations belong to a complete grasp of the quantitative boundary. Without quality no quantity, without a zero-boundary no magnitude; but without a quantifying which in turn annuls the immediate zero-boundary and transforms the qualitative into a moment vanishing under increase and decrease, there is no magnitude either[*)]. To see this one need only set 0, 1 and ∞ side by side; for while 0 denotes the posited boundary, the abstractly qualitative, and ∞ the annulling of all possible boundaries, pure quantifying, the numerical unit itself is conditioned by both. If we say that a magnitude x falls, say, between the boundaries 0 and 1, then it is not really the magnitude 1 but the zero point at which 1 is passed that constitutes the boundary, namely the terminal boundary. If we call the terminal boundary 0_2 and the initial boundary 0_1, then 1 denotes that the distance between the two boundaries is posited as a magnitude determinate in full generality — that is, bounded — as a unit of distance. That x, in so far as it falls between 0 and 1, can with an unchanged initial boundary vary through countless terminal boundaries, or, as it is put, have infinitely many values, may then be expressed thus: $\frac{1}{\infty} = 0$, only by dividing the unit to infinity is

[*)] "Everything that can be thought of as increased or diminished, and that can be resolved into parts of like kind, is called a magnitude (*quantum*) and is said to possess quantity (*quantitas*)." Ramus.

the real boundary reached; $\frac{1}{0} = \infty$, within every finite unit there falls an infinite number of boundaries; or $1 = \infty.0$, the finite unit arises through the boundary's being posited and annulled infinitely many times.

The determinate quantum presupposes a quantifying within qualitatively determining zero-boundaries. In virtue of the determining negations the finite magnitudes differ so far from one another that the difference can not only be stated but stated exactly: it is not enough that a is greater than b; the question here is, how much greater? But since the difference of magnitude rests solely on fixed boundaries, and no finite boundary of magnitude stands so fast that it cannot — as ∞ reminds us — be annulled again, the finite magnitudes can be altered as the boundaries are altered: every magnitude can as magnitude be increased and diminished to infinity.

The dialectical unity of 0 and ∞, which has its subsistence in the finite magnitude, $1 = \infty.0$, develops exactly through a constant alternation between the fixing of boundaries and the passing of boundaries; and so a decisive opposition forms between *fixed* and *fluid*, *being* and *becoming* boundaries. The immediate zero-boundary has no becoming; it comes to be in being posited. But since every boundary can be passed, and the alteration of boundary can continue so that the one boundary falls incessantly either inside or outside the other, the boundary can find itself in a becoming. The boundary of the boundary's becoming, the terminal boundary, may in turn be regarded partly as posited and partly as annulled: posited in so far as the magnitude is finite, annulled in so far as a finite magnitude may in its generality be as great and as small as one pleases. In positing a determinate terminal boundary, countless preceding terminal boundaries have already been passed, and so far annulled.

β) Transitions. In the boundary the bounded is reflected. To think the boundary without implying something bounded by it is as impossible as to grasp the meaning of a denial without thinking of what is thereby denied. The immediate zero-boundary is therefore posited straightway only on the presupposition that there is in intuition something given whose boundary it is. That such boundaries as points, lines and surfaces, although in and for themselves pure negations, are nevertheless *intuited* in positive form — the point as the line's smallest part, the line as the surface's edge, the surface as a side of the body — is a striking proof that the boundary goes together with the bounded. But since the boundary nevertheless distinguishes itself from the bounded as negation from the positive, and since this distinction is equally valid for thought and for intuition, there is contained in this the task of following what is to be bounded up to the boundary. How, then, does one get from boundary to boundary? By a motion! Here there is originally no other way out, no other means. If one tries in thought to mark off on a line, say, two boundary points *A* and *B* at a certain distance, one must, if unconsciously, perform a motion, in that, to intuit the arbitrarily chosen distance, one involuntarily lets the eye run through the way from *A* to *B*. Let the eye fasten where it will, on its way it meets only boundary, for every point in the line is in and for itself a boundary point. As the eye moves along the line it looks as though it were merely following the moving boundary, the moving point; on this rests the mathematical explanation that the line arises by the motion of the point. Since this explanation rests essentially on an optical deception, it is no wonder that it turns out somewhat ambiguous. How can the pure point, the boundary, the abstractly negative, move? For that intuition, which in its immediacy presupposes the positive, can, once thought posits the negation, pass every boundary and go from boundary to boundary, is by no means yet sufficient to explain the mathematical miracle that the point, pure negation, should be able to move, to change place in space although it occupies no space, or as it were to stretch itself and form something so positive as a unit of length, a line. If the line could arise by a motion of the pure abstract zero point, then it would also have to be conceivable as composed of nothing but zero points, of $0 + 0 \ldots. + 0$[*)]. To set boundary immediately beside boundary is impossible, for in abstracting from the bounded the two boundaries coincide; Herbart's "die starre Linie" is as much a logical as a mathematical monstrosity[**)]; only in reflection of a positive can the negative come outside itself; only by means of something bounded can boundary follow upon boundary.

Between two successive boundaries, then, there is something bounded. If this bounded coincided immediately with the boundaries, so that there were nothing, nothing whatever, between, then the boundaries themselves would have to coincide and could not follow upon one another. If the bounded fell as an immediately existing thing between boundaries different from it, then there would be a finite distance, and hence countless boundaries between two immediately successive boundaries, which is a contradiction: all motion is grounded in the dialectical unity of the boundary and the bounded. By means of the boundary, motion is transition; the pure boundary is to that extent always a form of transition. It does indeed make an essential difference whether the transition is intuited in space or merely represented as an arithmetical transition from finite to 0, from finite to ∞; but this stands firm: without reflexes of space and time no motion,

[*)]That this proposition by no means conflicts with the equation $\frac{a}{\infty} = 0$ taken as an expression of possibility becomes evident as soon as one recalls that the possibility depends precisely on 0 and ∞ being taken, not as empty negations, but as symbols of the greatest and smallest possible magnitude.

[**)]Cf. Den Herbartske Philosophie by P. S. V. Heegaard. Copenhagen 1861. pp. 21–45.

without motion no quantifying, without quantifying no quantum, without quantum no mathematics. To exclude the category of motion from pure mathematics and still keep the categories "boundary," "transition," "approximation" is a contradiction. How the categories are chosen, stressed and applied in mathematics, so far as it is technically a question of the most suitable execution of exact operations, must of course be decided within mathematics itself; but the logical connection and dialectical transitions of the operational concepts are investigated in philosophy. If it then happens — for happen it may — that a specialist, a routineer whose mechanics of concepts has long since become one with his mechanics of operation, grows loud on such an occasion, and by a loud application of the Saracen dilemma expressly demands that philosophy give up dialectic and pay unconditional homage to the mechanics of concepts: such an occurrence cannot much interrupt the dialectical development, but rather hasten it.

If someone, in order to prove the proposition that "the transition from finite to finite coincides with the transition from finite to infinite," had appealed to an equation such as $\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm \varepsilon) = 0$, where ε denotes a point of finitude, then a mathematician would certainly have had occasion here to point out the mistake, and to set the true equation against the false one: $\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm \varepsilon) = \infty$. But when in a philosophical analysis (cf. Phil. og Math. pp. 23–24) there is no question at all of any special determination of the difference between $\operatorname{tg} 90^0$ and $\operatorname{tg}(90^0 \pm \varepsilon)$, but only of the well-known fact that $\operatorname{tg} x$ passes from being finite to being infinite as x passes from a value less than 90^0 to 90^0 — then where does the mathematician (cf. Begr. Nøiagt. p. 47; "Afregning," revised, p. 7) mean to go with his $\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm \varepsilon)$?. If ε is of a finite value, however small, and hence a

point of finitude[*)], then $\operatorname{tg}(90^0 \pm \varepsilon)$ is finite, and the transition from finite (90^0) through a point of finitude (ε) to finite ($90^0 \pm \varepsilon$) therefore coincides with a transition from infinite ($\operatorname{tg} 90^0$) to finite ($\operatorname{tg}(90^0 \pm \varepsilon)$). If ε is infinitely small, and hence a point

[*)]One has, say, $\operatorname{tg} 90^0 = \infty$, and one has $\operatorname{tg}(90^0 \pm \varepsilon)$ finite; how does one now get from $\operatorname{tg} 90^0$ to $\operatorname{tg}(90^0 \pm \varepsilon)$? Plainly by ε, as part of the arc, being traversed continuously; and in the course of this motion it happens that the tangent passes from being finite to being infinite. How, then, is such a transition really to be thought? "It can," it is maintained, "happen only by a leap." What, then, is leapt over here? "A finite!" But then there is only a finite distance between finite and infinite. If nevertheless the transition from finite to infinite is to happen by a leap, then the transition from finite to finite must also, where a finite distance is concerned, happen by a leap; but if the transition through the finite can happen only by a leap, then all continuous transition, and with it every possible motion — indeed every possible transition — is denied. "An infinite!" But to arrive at the infinite by leaping over it is surely an intolerable and senseless contradiction. "Neither a finite nor an infinite!" But what sort of magnitude is it that is neither finite nor infinite? The greatest possible finite magnitude it cannot possibly be, for the greatest possible finite magnitude has precisely the property that by being *neither* finite *nor yet* infinite it is precisely at one and the same time *both* finite and infinite. "Nothing!" But what sort of leap is it, then, by which nothing is leapt over? (Cf. Begr. Nøiagtighed p. 45. The revised "Afregning" p. 7).

of infinity, then tg $(90^0 \pm \varepsilon)$ is infinite, and the transition from finite through a point of infinity to finite (for $90^0 \pm \varepsilon$ is surely of a finite value just as much as 90^0) therefore coincides with the transition from infinite (tg $90^0 = \infty$) to infinite (tg $(90^0 \pm \varepsilon) = \infty$). How great the difference between ∞ and ∞ may otherwise be does not concern the transition as transition[*)]. The objection that the expression "coincide with" denotes merely "an accidental concurrence" is unwarranted; for changes of tangent do not merely occur *simultaneously* with, but are *necessary consequences* of, corresponding changes of angle: to confuse the consequence's issuing from the presupposition with an accidental concurrence of something simultaneous betrays mere thoughtlessness.

[*)]How infinitely dialectical the concept of magnitude is becomes evident when one asks, say: how great is the greatest possible finite magnitude? or: how small is the smallest possible finite magnitude? The demand expressed in these questions contains an essential contradiction. There is no finite magnitude so great that its boundary cannot be passed, so that a still greater is always possible; nor is there any finite magnitude so small that a smaller is not always possible. The greatest possible finite magnitude is, in other words, the infinitely great finite, and the smallest possible finite the infinitely small finite; here too it shows itself that the infinite lies, not in an impossibility, but in a possibility. How, then, can the contradiction be resolved? By seeing what is two-sided in the boundary of magnitude, that is, by taking the boundary partly as being and partly as becoming. The magnitude with a being boundary, the being magnitude, be it as great and as small as it will, is always finite; with a becoming boundary the magnitude, the becoming magnitude, moves either toward the greater, and then presents itself under the *infinite* becoming as the infinitely great, or toward the smaller, and is then grasped under the *infinite* becoming as the infinitely small: the infinitely great and the infinitely small are therefore magnitudes whose being boundary is resolved into becoming, as into unceasing growth and unceasing diminution. What is the notation for the greatest possible finite magnitude. Lim $x = \infty$! And for the smallest possible? lim $x = 0$. The boundary that separates the finite from ∞ and from 0 is, so to speak, not a hair's breadth wider than the one that separates the finite from the finite. "If the one point of finitude lies infinitely near the other, then the same may now be said of 0 and finite, and of ∞ and finite." (Philos. og Math. p. 24). If one rejects dialectic and tries by the critical separation "either a finite or an infinite" to remove the contradiction, it only then becomes seriously insistent; and if one then tries by all manner of technical turns, leaps and artifices to veil the contradiction lurking everywhere, one may easily, by venturing now and then into the logical, get into several comical embarrassments.

If one holds fast to the dilemma "either ε is finite or it is 0" one-sidedly and critically, that is, undialectically — hence "$\varepsilon - 0$ either finite or 0" — then one may as little say of 0 as of ∞ that it lies infinitely near the finite. If the mathematician will answer to this: "quite right! 0 does not lie infinitely near the finite either," then one must ask again: what, then, lies between 0 and finite? The infinitely small the mathematician dare not adduce, for he has himself expressly said: "either ε is 0 or it is finite." Nothing, then, lies between 0 and finite — and yet 0 is supposed not to lie infinitely near the finite! The case stands correspondingly with ∞ and finite. Undialectically one holds fast to the dilemma: either x is finite or it is infinite, *tertium non datur*; hence $\infty - x$ is either ∞ or finite or 0; and one may then as little say of ∞ as of 0 that it lies infinitely near the finite. But what, then, lies between finite and ∞? If the mathematician will answer "there lies ∞!" he answers precisely the opposite of what he really meant. For he meant that, by the equation $\infty - a = \infty$, for $x = a$, there must absolutely be an infinite distance between the finite and ∞ — and behold, the supposed distance, the *what-lies-between*, turns into precisely that which was to be reached across the distance, namely the infinite itself. In other words: the outermost boundary of the finite cannot possibly be passed without the infinite being reached. What was to be kept infinitely far from the finite by "the infinite distance" thus comes infinitely near that finite after all. "Either finite or infinite!" But then nothing, nothing at all, lies between finite and infinite; and yet $\infty - a = \infty$, tg $90^0 -$ tg $(90^0 \pm \varepsilon) = \frac{1}{\cos 90^0} = \infty$ is supposed to *be a proof, a fully valid mathematical proof, that between finite and infinite there is an infinite distance!* If a mathematical critic should really win adherents by blustering with such logic, then on logic's behalf one would indeed have to apply the old saying: *un sot trouve toujours un plus sot, qui l'admire.*

By $\frac{1}{a-x} = y$ the changes in y are point for point dependent on the changes in x; and when y, as $x = a$, passes from being finite to being infinite, who will see in such a transition a merely external accidental concurrence with the value assigned to x? If $\varepsilon = a - x$ is the smallest possible finite difference, then the turning point at which y passes from finite to infinite is also infinitely near; but from this it by no means follows that the difference between finite and ∞ can be parcelled out into either finite or infinite subtraction-differences. If the questions of continuity are to be decided by proofs from *immediate* subtraction-differences, the proofs will indeed turn out accordingly. The equation $\operatorname{tg} 90^0 \div \operatorname{tg}(90^0 \pm \varepsilon) = \frac{1}{\cos 90^0} = \infty$ holds also for $\varepsilon = 0$. By making the immediate subtraction-difference the measure of the continuous one arrives at the — in a logical sense — exceedingly beautiful result that, since $\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm 0) = \frac{1}{\cos 90^0} = \infty$, there must necessarily be an infinite distance between $\operatorname{tg} 90^0$ and $\operatorname{tg} 90^0$, or, what here comes to the same, $\operatorname{tg} 90^0$ must be infinitely far from itself*). That ∞, the in itself unbounded, can be a boundary rests essentially on 0, the determining negation, the boundary itself. That 0 lies infinitely near the finite means, in other words, that the boundary lies infinitely near the bounded; and what holds, with respect to the attainable boundary, of $a - x = 0$ in an equation such as $\frac{1}{a-x} = y$, holds also of $y = \infty$. The boundary ∞ is therefore, seen from this standpoint, as attainable as the boundary 0. From this it is in turn explained that, while no finite magnitude can denote any transition from + to −, ∞ can denote

*)The evasion that ∞ is not a magnitude says nothing; if ∞ were not a magnitude, how could one have $\infty - \infty = n$, a determinate *difference of magnitude* between two *non-magnitudes*? What it means to say that ∞ is a magnitude, namely a becoming magnitude, is seen for instance from $\frac{1}{1-3} = 1+3+3^2+3^3 \ldots. +3^n+\frac{3^{n+1}}{1-3} = \infty - \infty = -\frac{1}{2}$, for $n = \infty$. The evasion that $\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm \varepsilon)$ is really a *function*-difference does not help either; for if the function-difference is to acquire significance as against the immediate subtraction-difference, ε must be transformed into an element, and thereby becoming, continuous change, transition — motion, that is — must be what decides.

One further remark for consideration. A proposition is set up, illuminated from several sides (cf. Phil. og Math. pp. 23–24, Begrebet Nøiagtighed pp. 45–47, "Afregn." revised p. 7, Appendix p. 29), whose fundamental thought is this: in order to think the continuous transition from finite to ∞ it is necessary to look at the motion; if one abstracts from the motion and keeps to immediate subtraction-differences, the continuous transition becomes inconceivable. Against this a mathematician who is also an extraordinary logician comes forward with the announcement that the proposition set up is false, fundamentally false. How does the mathematical logician set about proving his claim? He demonstrates that the proposition's mathematical error is "infinitely great," in that by the equation

$$\operatorname{tg} 90^0 - \operatorname{tg}(90^0 \pm \varepsilon) = \frac{\sin(\mp \varepsilon)}{\cos 90^0 \ \cos(90^0 \pm \varepsilon)} = \frac{1}{\cos 90^0} = \infty$$

he expressly abstracts from the motion, and by abstracting from the motion proves clearly and incontestably what the assailed proposition precisely affirms: namely that the proof of a continuous transition must necessarily fail when one abstracts from the motion. The attempted refutation thereby turns into an express confirmation of what was to be refuted. So: one sets out to demonstrate a proposition's "infinite mathematical error," and behold, by a — of course, it goes without saying, insignificant — mathematico-logical blunder one comes precisely to prove the proposition's mathematical validity: ordinary mathematics, extraordinary logic!

this transition just as well as 0, precisely because 0 can denote it. "Since a transition, as regards the opposition of positive and negative, acquires significance only through the positive's passing over into the negative and conversely the negative into the positive — that is, through the positive and the negative both being in the transition and therefore belonging to the concept of the transition — such equations as $\sin 0 = 0$, cot. $0 = \infty$ ought strictly to be written $\sin(\pm 0) = \pm 0$, $\cot(\pm 0) = \pm\infty$, and so on. By this separation of + and − in 0 and ∞ there arises within these boundary notations themselves an inner difference which is dialectical in the highest degree. The notation ± 0 expresses, on closer inspection, that the one zero has divided itself into three different zeros: a +0, a −0, and the pure neutral 0, the sharp negation of + and ÷. The same separation repeats itself with $\pm\infty$: here too there is a $+\infty$, a $-\infty$, and a neutral ∞ in which + and ÷ have vanished. When, therefore, it is said that 0 and ∞ are notations of transition, this cannot hold of the pure 0 or the neutral ∞, but only of ± 0 and $\pm\infty$*). The way in which the so-called indeterminate forms $\frac{0}{0}$, $0.\ \infty$, 0^0, ∞^0 and 1^∞ are treated in mathematics shows plainly enough what forms of transition mean." (Math. og Dialekt. p. 72).

γ) Articulations. As forms of transition, then, 0 and ∞ denote the boundary's dialectical unity with the bounded, and thereby motion: transition presupposes motion. In so far as the finite magnitudes are determined by given signs, they can, without any motion being noticed, be grasped by representation and held fast in their finitude — even to such a degree that it looks as though the infinite were only an empty abstraction, an external negation hovering above. But in so far as the finite magnitudes are determined by relations and grasped as concepts, the negation shows itself as dwelling within finitude, and the infinite is articulated. "So long as representation involuntarily clings to the sign 1, it looks as though the unit were an immediate something, an existing thing one could intuit and point straight at; this immediacy of representation disappears when one begins to grasp the unit as a relation: $\frac{2}{2} = \frac{3}{3} = \frac{4}{4}$." (Mathem. og Dial. p. 79). What, then, is the unit? In and for itself every magnitude is the unit: $a = a$, $b = b$, $c = c$ ....; as many different magnitudes, so many different units! But none of the different units is the unit itself; how, then, do we arrive at the unit that determines all units? By the equality expressed in equal magnitudes! From $\frac{a}{a} = 1$, $\frac{b}{b} = 1$, $\frac{c}{c} = 1$ .... it is expressly seen that the unit is precisely every magnitude's equality with itself; and by specially bringing out the equality that holds for each magnitude, and positing this in turn as a magnitude, we have

*) Only by regarding zero from 3 points of view — 1) as the starting point of the positive, 2) as the starting point of the negative, and 3) as the dividing point of unity of both — and letting these 3 determinations reflect themselves in one fundamental determination, does one think the concept of zero, the pure concept of boundary, as a concept of transition.

the general unit. As the determinate expression of equality, as the quotient of equal terms of a ratio, the unit is independent of the immediate difference of the terms; so long as each term merely continues to be equal to itself, the terms can be altered, can grow and diminish to infinity, without such alterations in the least altering the unit. If therefore one has, by means of the bounded ($a = a$, $b = b$ ....): $\frac{a}{a} + \frac{b}{b} + \frac{c}{c} + \ldots = 1 + 1 + 1 + \ldots$, then one has also, by means of the boundless $\left(\text{Lim.}\ \frac{x}{x} = \frac{\infty}{\infty}\right)$: $\frac{\infty}{\infty} + \frac{\infty}{\infty} + \frac{\infty}{\infty} + \ldots = 1 + 1 + 1 + \ldots$ and, by means of the boundary $\left(\text{lim.}\ \frac{x}{x} = \frac{0}{0}\right)$: $\frac{0}{0} + \frac{0}{0} + \frac{0}{0} + \ldots = 1 + 1 + 1 + \ldots$ Out of these streams of change the unit rises forth as the unchangeable, as quantifying's pure quantum, the immovable within motion.

On the unity of the general and the particular unit, the dialectical unity of the magnitude's concept and the magnitude's value, the function depends; every magnitude is a function of itself and the unit: $x = f(x) = 1.x$ (cf. Math. og Dial. pp. 6–7). From $\frac{f(x)}{x} = 1$, the expression for the magnitude's unity with itself in general, the *general unit* ($f(x) = x$), there follows $\frac{\int_{f(x)=0}^{f(x)=1.a} df(x)}{\int_{x=0}^{x=a} dx} = \frac{a}{a} = 1$, i.e. the magnitude's unity with itself as a particular magnitude, the *particular unit* ($f(a) = a$).

If the transition is presented, with $d.f(x) = dx$, by $f(x) = \int_{x=0}^{x=\infty} dx = \infty$, it must above all be held fast that there can here be no thought whatever of a direct summation, but only of becoming, coming-to-be, intellectual motion. Nothing has from the beginning done more to displace the decisive point of view and force back the natural development of the concept than the circumstance that the mechanics of operation has far too much treated the differential now as an immediate atom of magnitude, now as a mystical zero, instead of grasping it as a moment of transition, and has correspondingly regarded integration merely as a kind of summation (Math. og Dial. pp. 99–104) instead of determining it as a restitution of the finite through the infinite by means of the motion that holds for quantifying. How impossible it is to represent $\int dx = dx + dx \ldots. = \infty\, dx$ as a sum proper[*)] is seen directly from the fact that dx, as a dialectical intermediate determination between 0 and x, cannot possibly be an addend. To be an addend, dx would have to have immediate finite existence — thus not only a but $\frac{a}{10000}$ can be an addend; but taken as existing, dx is a highly ambiguous magnitude, too great to be nothing, too small to be something. This ambiguity turns into a two-sidedness within the concept when dx is grasped as the determination of transition on which all motion essentially depends. As a determination of transition, dx cannot at any instant stand still; its being is becoming, it alters and becomes either 0 or x. If someone objects that one cannot get from dx to x by such a leap, but must first have $2dx$, $3dx$, ... up to ndx,

[*)] x can indeed, seen under the aspect of infinite discreteness, be *thought as resolved* into $dx + dx +$ ....; but it is one thing to *think* infinite discreteness, another to *add* dx to dx — to infinity.

for $n = \infty$, it is not difficult to point out the conceptual confusion hidden beneath this.

For in order to articulate the motion it becomes necessary to distinguish between a single and a double becoming. In the single becoming, n in ndx is either finite or infinite. In so far as n is finite, ndx is an element, and as an element an ordinary dx; if ndx is to become a finite magnitude, n must be ∞. So long as the question here is still only of what is generally essential in the transition, we are looking throughout at the dialectical difference-unity of dx and x, and not at the difference between dx and ndx; in the course of the motion every ndx is, as an element, on the point of becoming a finite, and as a finite of growing by an element. How great the element is cannot, in this comparison — the comparison being immediate — come into consideration, nor how small the finite magnitude is; the transition depends precisely on the greatest possible element and the smallest possible finite magnitude being reciprocal determinations. In the single becoming the motion goes from 0 to x not through myriads of dx but through dx. By representing the task contained in $\int dx = x$ as solved thus — $0 + dx + dx + ... + dx = ndx = x$ — one entangles oneself in the fixed idea that the series must contain some particular dx whose place marked precisely the being boundary at which the sum of the elements first gave a finite magnitude. The difference between addition and motion is here striking. By adding dx to dx one never reaches x: $x - dx = x - 2dx = x - 3dx = x - ndx$, so long as n is finite, and under addition n is always finite; by motion, on the other hand, one passes immediately and instantaneously from dx to x, x being the smallest possible *finite* magnitude. The x into which motion transforms the dialectical dx is in turn dialectical, like motion itself; what is dialectical in x follows directly from the fact that every motion must begin before it can end. With the beginning of motion dx has immediately become an x, for motion in its immediacy is itself relation, reflection; but since the motion is still only at its beginning, x is the smallest possible. If the motion now stops just as what has begun was to be completed, then the finite falls back again into the element: the smallest possible x is then itself a dx. If on the other hand the beginning motion is completed, then the result is reached: the smallest possible x grows out beyond a smallest possible, takes root in the finite, and thereby becomes qualitatively different from its element. The x that arises from $\int dx = x$ therefore has, however small it may be (Math. og Dial. pp. 111–112), the completed motion for its presupposition; it is of a determinate finite value, can as a finite be divided into infinitely many infinitely small parts, and by this possibility produces the semblance of being a sum of countless dx.

The Eleatic objections to the possibility of motion*) may in the present connection be traced to an undialectical identification of beginning and completion. For a motion to be completed it must of course have begun; if now the beginning motion is posited as identical with the completed one, then we have, as the cardinal point of the contradictions, this admittedly absurd proposition: "for a motion to be completed it must already beforehand have been completed." One demands, in other words, an absolute beginning; but to demand an absolute beginning is a self-contradiction, for no beginning is or can be absolute. In the course of motion every beginning is a continuation, and every continuation a new beginning. The dialectical unity of beginning and continuation is motion itself. Under the aspect of discreteness every motion therefore also shows itself as a sum of infinitely many motions. Seen from this standpoint there undeniably lie between x and dx infinitely many dx, and likewise between dx and 0 infinitely many Lim. $\frac{dx}{x}$; but to see how such discreteness is resolved into continuity we must consider particularly the doubled becoming, the becoming in which motion determines motion.

If we have $f(x) = y = ax$, and hence (Math. og Dial. pp. 84–85) $dy = adx$, then the relation between dy and dx points to a double becoming: a becoming of x by the increment $dx = dx$, and a becoming determined thereby, under which the y dependent on x grows by the increment $dy = adx$. Although dy is an element, and as an element an infinitely small value, vanishing in comparison with any finite magnitude whatever, just as dx is, it is nevertheless seen from this that the one element can be a times greater than the other; if for example $a = 2$, then $dy = 2dx$ and $dx = \frac{1}{2}dy$; if $a = 3$, then $dy = 3dx$ and $dx = \frac{1}{3}dy$, and so forth. That no immediate comparison can, however, be made between the elements themselves, and that nothing can here be formed out of such elements either by piecing-together or by parcelling-out, is easy to see. Instead of deriving $dy - dx = dx$ from $dy = 2dx$, one gives up subtraction as an unfruitful symbolism, applies division, and obtains $\frac{dy}{dx} = 2$, the differential coefficient, the factor by which dx must be multiplied in order to give dy (Math. og Dial. p. 81). What decisive significance it has for the articulation of the infinite by the finite that multiplication and division take the place of addition and subtraction can be shown without difficulty. The magnitude of a finite unit of time can be stated immediately, and so can the magnitude of the space traversed with a certain finite velocity in a finite time; and since the magnitudes are each immediately given, the comparison between them can also be made in so immediate a way that the terms themselves preserve a certain independence over against the relation in which they stand to one another. If the velocity on one track is 4 miles and on another

*) Cf. Math. og Dialekt. p. 131 and elsewhere.

8 miles an hour, then the subtraction-difference thus given falls out as of itself. It is otherwise when elements of time and space take the place of the finite units of time and space. How long, after all, is an instant? That cannot be stated immediately, and just as little can it be stated immediately how long a stretch of road can be traversed at a given finite velocity in a single instant. Instead of remaining in an external relation, the elements must transform themselves into moments in a pervasive reflection; and for this multiplication and division, product and quotient, are needed. In product and quotient the terms have already given up their external existence, indifferent to one another, and transformed themselves into reflected moments of relation. So long as product and quotient are formed only from finite magnitudes, however, the corresponding reflection is still limited by the merely representational grasp[*)]; but when the elements take the place of the finite magnitudes, the last remnant of representation's immediacy must vanish, and the concept — the pure concept made plain by the articulation — must hold. If we let $s = 4t$, $ds = 4dt$, $\frac{s}{t} = \frac{ds}{dt} = 4$ be different expressions for one and the same velocity, namely 4 miles an hour, then it is evident that, although the velocity is the same, the expression of velocity $\frac{ds}{dt}$ stands in quite another relation to thinking than the expression $\frac{s}{t}$. The quotient $\frac{s}{t}$ still keeps within the relation supported by representation, but in $\frac{ds}{dt}$ the whole insight rests on pervasive reflection; in the relation determined by them, ds and dt are nothing, nothing whatever in and for themselves, but reflexes of one another, and in this transparency determined by one another, so that they afford representation no finite foothold at all.

The becoming doubled within itself is articulated still more richly when the velocity answering to the motion changes at every instant. For a motion may, as regards velocity, be subject to infinitely many changes and yet, provided none of these changes occurs suddenly, show itself continuous. The continuity is conditioned by all the changes being subject to one and the same higher law and thus standing in a relation of mutual dependence.

"In so far as s is a space already traversed in the time t, one cannot yet know whether this space has been traversed with variable or with invariable velocity. To determine the velocity one must give the time an increment, that is to say: provide time for continued motion. If the increment to the time is called h, then, since $s = f(t)$, one has the series:

$$f(t+h) = s + \frac{ds}{dt}\frac{h}{1} + \frac{d^2s}{dt^2}\frac{h^2}{1.2} + \frac{d^3s}{dt^3}\frac{h^3}{1.2.3} + \ldots.$$

[*)]To finite reflection, to merely representational thinking, it does indeed look as though the sum $a + b = (a + b)$ were just as reflected as the product $a.b = ab$, and the difference $a - b = c$ just as reflected as the quotient $\frac{a}{b} = c$.

If the increment h is a finite magnitude and the velocity constant, the whole series will reduce to $f(t+h) = s + \frac{ds}{dt}\frac{h}{1}$; if h is infinitely small, equal to dt, the same will hold even when the velocity is variable. When h is finite, on the other hand, the terms of the series can be increased to infinity. Seen from the standpoint of the concept, the series states a totality of objective determinations of reflection. The individual terms may, where s means the space, v the velocity and φ the force, be rendered by different corresponding expressions:

$$f(t+h) = s + \frac{ds}{dt}\frac{h}{1} + \frac{d^2s}{dt^2}\frac{h^2}{1.2} + \frac{d^3s}{dt^3}\frac{h^3}{1.2.3} + \ldots.$$
$$= .. + v\frac{h}{1} + \frac{dv}{dt}\frac{h^2}{1.2} + \frac{d^2v}{dt^2}\frac{h^3}{1.2.3} + \ldots.$$
$$= .. + \ldots. + \varphi\frac{h^2}{1.2} + \frac{d.\varphi}{dt}\frac{h^3}{1.2.3} + \ldots.$$

The categories *motion*, *velocity*, *force* and *change in force* are thus determined by objective reflection*). Immediately s denotes a length of space; but in the equation $s = f(t)$ this length of space is seen in the reflection of a length of time; it owes its existence to time, in so far as it is traversed in a given time. Immediately v denotes a velocity, but in the expression $\frac{ds}{dt}$ the concept of velocity is determined by the objective reflection of space and time in the given relation. Immediately force is denoted by φ, next by the reflection of velocity and time in the determinate relation $\frac{dv}{dt}$, and finally, as $v = \frac{ds}{dt}$ is resolved into the reflection

of space and time, by a doubled reflection in the relation $\frac{d^2s}{dt^2}$. In this way the changes of force can, with respect to the relation of space and time, be determined as reflections of reflections to infinity." (Phil. Propæd. pp. 165–167).

*)To what degree an otherwise developed reflection may fail in developing an operational concept when the hints that exact analysis gives are entirely overlooked, Hegel has shown in his speculative interpretation of the motion of falling bodies (cf. Phil. Propæd. pp. 161–167). The well-known proposition that the spaces traversed with uniformly increasing velocity are as the squares of the times employed, Hegel interprets so that the quotient $\frac{s}{t^2}$ becomes a dialectical consequence of the abstractly *a priori* relation of the concepts of space and time: "Die der Einheit, als der Form der Zeit, entgegengesetzte Form des Außereinander des Raums, und zwar ohne daß irgend eine andere Bestimmtheit sich einmischt, ist das Qvadrat: die Größe außer sich kommend in eine zweite Dimension sich setzend, sich somit vermehrend. aber nach *keiner andern als ihrer eignen* Bestimmtheit — diesem Erweitern sich selbst zur Grenze machend, und in ihrem Anderswerden so sich nur auf sich beziehend" [the form of the asunderness of space, opposed to unity as the form of time — and that without any other determinacy mixing itself in — is the square: magnitude coming outside itself, positing itself in a second dimension, thus increasing itself, but according to no determinacy other than its own — making this expansion its own limit, and in its becoming-other thus relating only to itself]. It escapes Hegel's attention in this case that, even if motion in full generality can be deduced from the relation between pure space and pure time — a deduction which must always be conditioned by supporting subsidiary representations — the square of the time cannot possibly arise without a varying velocity, and a varying velocity not possibly without a corresponding cause: here a constant force acting on matter moving in space according to the law of inertia.

The dialectical reciprocal determination of the concepts of continuity and discreteness now comes to light in the way the continuous motion is articulated, in that the elements themselves can not only be divided and divided again, but under given conditions even be resolved into elements of higher and ever higher order[*)]. If the space s could not, say, be divided according to possibility into nothing but $ds = ds$, as the time t into nothing but $dt = dt$, then continuity itself, and consequently every continuous motion, would be inconceivable, and the relation between rest and motion, between faster and slower motion, entirely unintelligible.

"The difference between slow and fast motion must plainly rest on the first having more, the second less rest in it, and so on the first being nearer to rest than the second; a motion with an infinitely small velocity is precisely so infinitely near rest that it may be regarded as rest[*)]. Nonetheless the opposition between rest and motion is qualitative: what rests does not move, what moves does not rest. If we now reject infinite magnitudes, the difference of velocity can be explained only by pure motion's being interrupted at finite intervals by rest. That the body on A traverses only s in the unit of time t, while the body on B traverses $s_1 = 8s$, must then be explained by saying that the former body moves only for $\frac{1}{8}t$ and rests for $\frac{7}{8}t$. So long as the unit of time in which such an alternation takes place is an hour, the alternation of rest with motion is conspicuous; if the unit of time is 1 second, the alternation is less striking and looks like a jerky motion; in a unit of time of $\frac{1}{100}$ second the alternation will not be noticed at all, and the phenomenon will show itself as a smooth motion … If the motion on A is to be continuous, time and space

[*)]To reason thus — "the magnitude y may be thought of as divided into infinitely many dy, granted! but since dy thereby becomes itself the smallest possible part of a magnitude, dy cannot possibly be itself divisible, since by dividing dy one would get parts smaller than the smallest possible" — will not do; for $dy = a\,dx$, $dx = \frac{dy}{a}$, and, if we choose the function $y = f(x) = x^2$, $dy = 2x\,dx$, $dx = \frac{dy}{2x}$, shows plainly enough what it means to say that dy is divisible. But if an element of the first order is not absolutely indivisible, although as an element it is *infinitely small*, then it follows of itself that elements of the second order d^2y, the third order d^3y, the 4th order d^4y, the n^{th} order d^ny cannot on account of their *infinite smallness* be absolutely indivisible either. One may indeed in full generality set $\frac{d^2y}{dx} = 0$, $\frac{d^3y}{dx^2} = 0$, $\frac{d^4y}{dx^3} = 0$, $\frac{d^ny}{dx^{n-1}} = 0$, and, what is equivalent to this, $\frac{dx}{d^2y} = \infty$, $\frac{dx^2}{d^3y} = \infty$, $\frac{dx^3}{d^4y} = \infty$, $\frac{dx^{n-1}}{d^ny} = \infty$, denoting by 0 and ∞ that elements of different order are "*incomparable*" (cf. Math. og Dialekt. p. 96); but from this it of course by no means follows that $\frac{d^2y}{dx^2}$ should be smaller than $\frac{dy}{dx}$, or $\frac{d^ny}{dx^n}$ smaller than $\frac{d^{n-1}y}{dx^{n-1}}$; on the contrary! The relations may even entail $\frac{d^4y}{dx^4} > \frac{d^3y}{dx^3}$, $\frac{d^3y}{dx^3} > \frac{d^2y}{dx^2}$, $\frac{d^2y}{dx^2} > \frac{dy}{dx}$, $\frac{d^ny}{dx^n} > \frac{d^{n-1}y}{dx^{n-1}}$. The greater the quotient, into so many the more parts must the dividend be divided: the character of the dividends is determined by the measures chosen as divisors. Between the elements dy and d^2y, d^2y and d^3y and so on, in the quotients cited, there is here no immediately determined relation; the relation determined by $y = f(x)$ first arises through the measures dx^2, dx^3, dx^4 ... all being determined by $dx = dx$. Thus if $\frac{dy}{dx}$ is constant, say $= a$, then also $\frac{dy}{dx} = \frac{\int dy}{\int dx} = \frac{y}{x} = a$, and conversely. If $\frac{d^2y}{dx^2}$ is constant, say $= 2a$, then also $\frac{\int\int d^2y}{\int\int dx^2} = \frac{\int dy}{\int x\,dx} = \frac{y}{\frac{1}{2}x^2} = 2a$, and conversely; if $\frac{d^3y}{dx^3}$ is constant, say $= 6a$, then $\frac{\int\int\int d^3y}{\int\int\int dx^3} = \frac{\int\int d^2y}{\int\int x\,dx^2} = \frac{\int dy}{\int \frac{1}{2}x^2\,dx} = \frac{y}{\frac{1}{6}x^3} = 6a$, and conversely. Such relations may be continued to infinity.

[*)]Cf. Math. og Dialekt. pp. 55–56.

must be divisible to infinity; on this point Aristotle saw rightly. But what Aristotle did not see and could not see, because he lacked the fulcrum that only the analysis of the infinite can give, is that by this infinite divisibility of time and space there are formed precisely infinitely small magnitudes differing from one another in such a way that they enter into determinate finite ratios with one another." (Math. og Dialekt. pp. 144–45). To the constant velocity $v = a = \frac{ds}{dt}$ there answers a single continuity — a continuity which, since it is marked precisely by the differential coefficient of the first order being constant, we shall here call *the continuity of the first order.*

If the single continuity can be conceived only through a corresponding single discreteness going to infinity, then the continuity doubled within itself cannot be conceived either without a corresponding doubling of discreteness. If the velocity changed at every instant gives us the series $v_1, v_2, \dots\, v_{n-1}, v_n$ in n successive instants, so that $v_p - v_{p-1} = dv$, whereby the increment of space ds in each dt is successively transformed into ds_1, $ds_2 \dots ds_n$, then here, since $v_n \gtrless v_{n-1}$ and $ds_n \gtrless ds_{n-1}$, there is in these two series, in so far as they are regarded *immediately*, no continuity to be discovered. Continuity, which demands that the same must repeat itself without interruption at every instant, shows itself only when we discover, in the velocity as in the space, the possibility of an element unchanging under the changes. Thus if it is given that the force φ is constant, then the change of velocity dv effected by it at each instant, $dt = dt$, is constant too; but the constant element of space answering to this is not ds; it is, by $dv\,dt = d^2s$, precisely the element d^2s*) The continuity articulated by the functional law $s = at^2$, which presupposes that the differential coefficient of the second order is constant, we shall accordingly call a *continuity of the second order.*

If the doubled continuity demands a doubled discreteness, then in like manner the threefold continuity — that, namely, which answers to $s = at^3$ — demands a threefold discreteness, an element of space of the third order. For here it is not the force itself, φ, but the cause of the change of force, ψ, that is constant. The change of force, $d\varphi$, is then of course constant too. The constant $d\varphi$ now effects at each instant, dt, a constant

*)By applying the summation symbolism one is then also led, through an immediate summation of d^2s, to the expression for s as a function of the time.
In the 1st dt the velocity is dv, and effects the change of space d^2s
in the 2nd dt .... $2dv$ .... $2d^2s$
in the 3rd dt .... $3dv$ .... $3d^2s$

in the n^{th} dt .... ndv .... nd^2s
After the lapse of the time t there has here been traversed, since $t = ndt$, $n = \frac{t}{dt}$, a space: $(1 + 2 + 3 \dots + n)d^2s = \frac{1}{2}n(n+1)d^2s = \frac{t}{dt}\left(1 + \frac{t}{dt}\right)\frac{d^2s}{2}$, which, since $\frac{t}{dt}$ is infinitely small in comparison with $\left(\frac{t}{dt}\right)^2$, is equal to $\frac{t^2}{2}\frac{d^2s}{dt^2} = \frac{gt^2}{2}$, the usual expression for the law of falling bodies.

change in the increment of velocity, which in turn effects a constant change, $d^3 s = d^3 s$, in the increment of the increment of space; $d^3 s$ is therefore, in space, the element of discreteness answering to this law of continuity. It is the third differential coefficient, $\frac{d^3 s}{dt^3}$, that is here constant. *The continuity is of the third order.*

From the articulations thus exhibited it is seen that continuity is by no means interrupted by more and ever more changes being drawn under the law of continuity. If continuity were ever to burst under the weight of the multitude of changes, then an insurmountable barrier would at last have to show itself to the rising order of the elements, without this proceeding from any constant differential coefficient; but since this is a contradiction, the order of the elements being able to rise and rise to infinity, the continuity of the n^{th} order answering to a constant n^{th} differential coefficient, $\frac{d^n s}{dt^n} = C$, must hold for $n = \infty$ as well[*)].

If we gather the results here won into a single survey, the following main points are especially to be brought out: 1) the circumstance that between dx and x there may lie countless dx, and, in so far as x is a dependent variable, between 0 and dx countless d^2x, between 0 and d^2x countless d^3x and so on, does indeed indicate that between the corresponding and simultaneously vanishing increments of the dependent and the independent variable there are varying ratios, from which differentials of higher order necessarily follow; but by no means that 0, dx and x (the smallest finite) are, as far as the concept of transition in its generality is concerned, separated by countless intermediate terms. When dx is to represent the smallest possible magnitude in an immediacy answering to independence, it cannot possibly become smaller without becoming 0, nor greater without becoming x. In its *immediacy* the element is an outermost boundary of discreteness, an atom of magnitude, and as such indivisible; reflected in its relations to other elements it may on the contrary show itself infinitely divisible. Motion depends on the dialectic of the element indivisible-divisible in itself. 2) The beginning of motion takes place immediately neither with 0, nor with x by itself, nor with dx, but with a negative-concrete unity of all three moments. "In the absolute Now nothing is and nothing happens. In the relative Now the infinite is posited as the beginning of the finite: the beginning of the finite therefore coincides with, is in the mathematical instant simultaneous with, the beginning of the infinite. In the empirical instant the finite is posited as completed; but in and with this completion of the finite an infinity is itself completed: the completion

[*)]If s is a fractional, irrational or transcendent function of t, say $s = \cos t$, then there is no constant differential coefficient here; but this by no means means that continuity is broken — on the contrary, that it is so infinitely articulated that one cannot exhibit in isolation the atom of discreteness answering to such a continuity.

of the finite therefore coincides with, is in every actual instant simultaneous with, the completion of an infinity.” (Math. og Dialekt. pp. 134–35). 3) Although motions of every kind must always presuppose at least subsidiary representations, reflexes, of time and space, motion is nevertheless in the strict mathematical sense to be confined to pure quantifying. And since a quantifying as necessarily presupposes a quantifying quantum as motion presupposes a something that can move, while yet the abstract quantum which furnishes quantifying’s immediate presupposition owes its existence precisely to the quantifying, and thereby comes to presuppose that of which it was itself to be the presupposition — it is seen from this that quantity as an abstract concept can first acquire its true significance in and through an existing thing whose quantity it is.

Of pure mathematics it is said that it finds “application” to actual objects (in mechanics, astronomy, optics and the like) Essentially understood, such an “application” presupposes that the mathematical is originally realized in the objects to which mathematics is applied. An application of mathematics to actual objects would be, if not a sheer impossibility, at any rate a subjective arbitrariness, if the mathematical did not itself belong with inner objective necessity to the essence of actuality; but as a moment in the essence of actuality the mathematical belongs also to philosophy.

B.

Philosophy’s Relation to Chemistry: The Problem of Change.

In order to present philosophy’s relation to chemistry, it will in the present connection be necessary: a) to characterize chemistry as a distinct science; b) to show how the problem of change arises in the course of chemical analysis; and finally c) how philosophy appropriates the problem developed within chemistry.

That philosophy, however deeply it immerses itself in an other, nevertheless relates itself through this other to itself, and so far moves within its own domain, will in the course of the exposition be recognizable in this: that nothing of the nature of material is taken up immediately from outside, but all positive constituents are so appropriated that they yield, each in its own way, ideal contributions to the knowledge of essence and serve the dialectic of the development of concepts.

a) Chemistry as a Distinct Science.

“Since science has its root in human life, where the ideals of imagination go before those of thought, it is natural that the various sciences should from the first come into the world in the swaddling-bands of fancy and be encumbered with unscientific admixtures. Thus astronomy was long grown together with astrology, chemistry with alchemy, philosophy with mythology. It does indeed lie in science’s own drive to throw off the chrysalis, to break its husk and purge away the foreign constituents. But cultivation costs time and purging costs effort; the hindrances the human spirit has here to overcome are of various kinds.” (Phil. Propæd. p. 70).

It is science’s task to penetrate into the interior of things and through the phenomenon to discover the essence; but recognized as this demand is in abstract generality, its fulfilment is correspondingly difficult when attention turns to the particular, the concrete, the actual. What phenomenon is better known or more conspicuous than the phenomenon of fire — and yet, what is the inner in this outer, what is the essence of flame, what is fire? Most people will at least have heard in passing that in recent times a number of metallic substances formerly unknown have been isolated by art; among these substances is potassium, a tin-white, brightly lustrous, very soft and pliable metal. If a piece of this metal is thrown into water it does not sink to the bottom but moves about on the surface, takes fire and burns with a violet flame; after which it seems to disappear entirely. What has happened here? Is it the potassium that has burnt the water, or the water that has burnt the potassium? *What, in the phenomenon of combustion, is that which burns, and what is that which is combustible?* To answer such a question science has had to prepare itself for some 3000 years — if, to begin at the beginning, we look for our first chemists among the priests of Egypt.

α) The Origin of Chemistry[*)].

Among the Egyptians the cultivation of chemistry seems to have formed part of the mysteries hidden in the temples; among the Greeks the elements of chemical knowledge passed, like other empirical knowledge, into philosophy; in the Middle Ages chemistry as alchemy was a direction for fathoming the riddle of nature and finding the philosophers’ stone (*lapis philosophorum*).

Since insight into the connection of natural forces is of no less practical than theoretical interest, and depends as much on rich experience as on sustained reflection, it is

[*)] *Leçons sur la philosophie chimique par Dumas, recueillies par Bineau. Bruxelles* 1839. Geschichte der Chemie von Dr. H. Kopp. 4 Bd. Braunschweig 1843–47.

intelligible that the study of chemistry should long have been grown together with various alien endeavours, and that this branch of science should long have served as a means before it could be set free to be its own end and be developed in a purely scientific spirit. *Mystical speculation, groping after method, and the discovery of method* are then the stages that here mark the path of the development. *αα) Mystical Speculation: Antiquity and the Middle Ages.* Without experience, no thinking; but when visible things are only fleetingly surveyed, because the art of observation is not yet developed, abstracting thought unites with imagination and forms out of a few accidental and unreliable experiences a system of concepts whose content, so far as existing things are concerned, belongs more to fancy than to actuality. Conceptual forms can indeed, as Greek philosophy sufficiently shows from so many sides, be developed in abstract directions with clarity and sharp consistency although the content of experience is lacking; but if concepts of existence, the real concepts proper, are in question, it is self-evident that the development of form must everywhere be conditioned by a corresponding development of content.

The way in which Aristotle grounds the existence of four elements shows precisely how impossible it is to determine a thing's essence without knowing the thing. The elementary qualities — hot, cold, dry and moist — are brought out in such a way that each of them is posited as an essentiality in its own right, a substantial form determining matter; thus a simultaneous union of dryness and heat gives *the element of fire*, of heat and moisture *the element of air*, of moisture and cold *the element of water*, of cold and dryness *the element of earth.* It is a remarkable confusion of effect with cause, of the derived with the original, that we meet here. The thing itself, the existent, does not after all arise through the union of qualities, and so not through the union of fundamental qualities either; on the contrary, the union of qualities — and hence of fundamental qualities too — is posited in and with and by means of the thing. What we have here is not a system of general qualities subsisting on their own and acting upon things, but a system of things acting reciprocally upon one another through their qualities; it is not, for example, the quality dryness that dries the thing, any more than it is the quality heat that warms it; the cause of heat and of dryness is to be sought, not in a general essentiality, but in a conditioned action. What influence such a misdirection, inherited from philosophy and especially from Aristotle, must have exercised on the Middle Ages — fermenting as they were with deep intimations and obscure but bold endeavours — is not, so far as chemistry in particular is concerned, difficult to imagine.

Material substances such as quicksilver and sulphur must, in so far as they belong among externally actual things, show themselves as objects of experience calling for corresponding observation; but since speculation prefers to see in such substances only a sensuous embodiment of certain non-sensuous essentialities stamped upon matter, it came about by degrees that the alchemists' *Mercurius* and *Sulphur* denoted, not primarily the quicksilver and sulphur actually existing, but above all these substances' metaphysical characters. Once the thing, which by its nature ought to be an object of observation, has been transformed into a speculative essentiality, it lies near to regard the sensuous as a fleeting image of the supersensuous, the natural as a symbol of the spiritual. The black powder that arises when sulphur and quicksilver are ground together passes on ignition into a red mass, namely cinnabar; in one and the same substance red and black, the symbols of light and darkness, of the good and the evil principle, are therefore visible. On the presupposition that in all metals there was to be found *Mercurius*, the character of the permanent and the indissoluble, shining in the metallic lustre, and *Sulphur*, the combustible, volatile and changeable, and that the mutual difference of the metals depended chiefly on a peculiar mixture and "fixation" of these natures, the alchemists were carried ever further in their endeavour to transform, transmute and ennoble the metals.

"What was then regarded as the magicians' highest scientific striving aimed at purifying the divinely-homogeneous which was found in sinful nature, bound up with resisting elements, so that what lay hidden as the divine and sustaining might work freely in every inward form. This is alchemy — no accidental, arbitrarily contrived thing, but a wholly essential constituent of the prevailing physics. All physicists sought the philosophers' stone, for there was then no other physics, and no other could arise. The bringing forth of this noblest kernel of all existence was as much of a religious kind as it was a physical experiment, and this universally prevailing endeavour gives the most striking proof of the spirit's bondage to the earthly. What was thus purified, in which the original creative force concentrated itself, must, when applied to the macrocosm, bring forth the noblest materials, precious stones and above all gold; but applied to the microcosm it must on the same ground — the original sustaining principle in both being the same — promote health. and prolong life." (H. Steffens)[*)].

Mystical, marvellous and enigmatic that substance must indeed have been which by mere contact was to subdue *Sulphur*, enchant *Mercurius*, *Metallorum Draco*, and transmute every metal, even the basest, into pure gold. The philosophers' stone, the great elixir, *Magisterium magnum*, the red tincture, which has the power to ennoble the metals on so prodigious a scale — *mare tingerem, si mercurius esset*, says Raimundus

[*)]Cf. "*Naturvidenskabens Forhold til Tidsaldere og deres Philosophie*", a critical review by H. C. Ørsted. Aanden i Naturen. 2. pp. 161–62.

Lullus — and to heal all diseases, is accordingly not formed as an aggregate but grows from within, organically, like a living being. No wonder that the discovery of such a secret came at last (Basilius Valentinus) to appear as much a religious as a scientific task. The stone, which shone with a light of grace as well as of nature, spread a magical gleam over the whole of existence; earthly life with its sorrows and sufferings was then, seen in this light, a "digestion," a purification by "fermentation," the grave the place where a "putrefaction" should work which destroyed only the base constituents of the human body, while what is immortal in the soul emerged as by a "sublimation" of the nobler essence, a *Quinta Essentia.*

ββ) Groping after Method: Qualitative Analysis. The alchemists' reveries pointed in two directions: the *metallurgical* and the *medicinal.* That the philosophers' stone, once discovered, would heal all diseases was of course a fancy that could not possibly yield a scientific task; yet beneath this fancy there smouldered a spark of something rational, a thought which, though in fantastic form, broke through in so-called iatrochemistry, medical chemistry: the thought that the organic functions depend on chemical processes. The chemists of the sixteenth and seventeenth centuries — Paracelsus, Van Helmont, Sylvius — were of course by no means yet able to form any clear concept either of the chemical processes or of the organic functions, and so not able either to devise any definite, reliable or consistent procedure. What a Sylvius explains about the abnormal predominance of acids and alkalis in the body's fluids is at bottom as untrustworthy as what a Paracelsus reveals to us in the proclamation that every part of the human body has its own peculiar sulphur, its own mercury and its own salt; that a preponderance of sulphur gives fever and plague, a preponderance of quicksilver melancholy and paralysis, an excess of salt dropsy, and so forth. Yet the new direction nonetheless gave the investigations a new impetus. To draw the "quintessence" out of medicinal plants, to prepare the many kinds of tinctures, essences, extracts, decoctions and juices — these were endeavours well fitted to sharpen the spirit of observation, to multiply experiments, and to teach the individual investigator to exercise his eye and his judgment. It lies in the nature of science that it can become conscious of its end only in the same degree as it can become conscious of the procedure answering to that end, the method, and conversely. The end of medical chemistry, though in and for itself rational, must always show itself from a more or less fantastic side so long as research is unable to provide the means, so long as the road of investigation that leads to the goal has not yet been found; but the dialectic of progress is that on the path of error too there are truths to be found — among others the truth that here is an error. By the time iatrochemistry had been carried far enough for its hollowness and charlatanry to show themselves properly, chemistry itself, following the other track, the ancient metallurgical one, had come to the turning point at which it could at last stand forth independently and begin to operate on its own account.

Chemistry's emancipation is recognizable in the altered task; for the task is now in the main neither to transmute the metals into gold nor to find the arcanum of medicine, a universal remedy against all diseases, but to investigate the conditions and laws governing the separation and combination of substances, and above all to answer the question: on what does combustion depend? But scientific, and distinctively chemical, as this task presents itself, it was impossible that the first attempts at its solution should turn out other than somewhat doubtful. The presupposition inherited from the alchemists, that in all metals there was a hypothetical elementary substance conditioning combustion, still continued to hold; nor could the prejudice easily be shaken off that combustion must absolutely be a destruction, a dissolution, a separation, so that something departed from the burning body, namely what went up in the flame. In this, however, the problem of combustion differs essentially from the iatrochemical fancies: that however much of the fantastic mingles with it at the outset, it lies so near to research that by its own rational character it can gradually point the way and lead into the method.

The changes that all combustible bodies, organic as well as inorganic, undergo in fire, Stahl[*)] brought together under a single generally valid point of view, deriving combustion from a substance dwelling in all bodies, a substance of combustibility, namely phlogiston. Thus was formed the doctrine of phlogiston characteristic of the period from the latter half of the seventeenth to near the close of the eighteenth century — Stahl, Priestley — the phlogistic theory. By testing the analogies, bringing out the general, drawing theoretical inferences from the observation of phenomena, and applying more comprehensive rational points of view, science gathered its forces and so rose step by step into its own distinctive shape.

But despite the long series of experiments that the question of combustion alone must call forth, despite the multitude of discussions that the hypothetical substance phlogiston must occasion — whether it was sulphur, which Stahl denied; whether it was not rather a matter of light; or whether, as others found plausible, it might perhaps be hydrogen[†] — discussions which for all their hovering were always distinguished

[*)] Georg Ernst Stahl, born at Anspach in 1660, was in 1687 appointed physician in ordinary to the Duke of Weimar, and, at Friedr. Hofmann's instance, in 1694 professor of medicine at the University of Halle.

[†] *Translator's note.* Danish *Brint* for hydrogen and *Ilt* for oxygen (below, p. 91, with *Metalilter* for metallic oxides) are native coinages of H. C. Ørsted, who appears in the footnote on p. 86; they replaced the international *Hydrogen* and *Oxygen* in Danish scientific prose and are still the ordinary Danish words.

from empty speculation by the fact that they all turned on actual matters of fact and corresponding presuppositions: the investigations were nevertheless still one-sided and imperfect so long as emphasis was laid merely on the quality of the given constituents, without the quantitative in the combinations coming particularly into consideration. So, despite the many and even connected special investigations, things remained on the whole with the old abstraction: that since like effects must have like causes, there must underlie the prominent properties of the various bodies, the ruling qualities, a corresponding common element of property or quality — in all acids a quality of acidity, an original acid (*acidum primitivum, primordiale, catholicum*); in all earths an earthy quality, an original earth; in all caustic alkalis a caustic quality, a caustic principle; in all combustible bodies a principle of combustibility, a phlogiston.

γγ) The Fixing of Method: Quantitative Analysis. Already in antiquity attention was drawn to the difference the metals display when tested in fire, some — such as gold — keeping their properties unchanged, while others, such as copper and iron, are transformed into earthy bodies. What remains when the metal has been burnt, the Greeks and Romans denoted by the same name as holds for burnt wood, namely ash (σποδός, *cinis*); but when the Arabs later brought out the resemblance between metallic ash and burnt lime, there arose in the West such expressions as *Calces*, metallic calxes (answering to what are now called metallic oxides), and *Calcinatio*, calcination, the condition into which the metal comes by combustion.

How closely the phlogistic theory attaches to the view that the metals are composite substances, and how the metallic calx accordingly stands to phlogiston and to the pure metal, may be shown briefly by an example. The more phlogiston a body contains, the more combustible it is, and the less ash remains after combustion; carbon, which leaves very little ash, is therefore particularly rich in phlogiston. If now rusted — that is, burnt — iron, the calx of iron, is heated with charcoal, the carbon will give up its phlogiston to the calx; there will thus arise calx of iron with phlogiston, and dephlogisticated charcoal; and the calx of iron in its union with phlogiston will constitute the reduced metal, pure iron, a composite substance.

What is distinctive in this emerges still more clearly when the antiphlogistic theory is set sharply against the phlogistic. The metals — iron, for example — are, according to the antiphlogistic theory, not composite but simple substances; combustion takes place not by something departing but on the contrary by something being added; it consists, in other words, not in a separating but in a combining. Pure iron burns in uniting with

Nothing in the argument turns on them, but the reader should know that Nielsen is using a deliberately Danish chemical vocabulary here.

oxygen; oxidized iron is reduced, not by its taking up phlogiston — for phlogiston exists only in fancy — but by its losing again the oxygen taken up, and being presented in its simplicity as an element.

Now that the two theories, each according to its fundamental thought, have been made plain by their mutual opposition, the question presses itself upon us of how we are to convince ourselves of the truth of the one and the untruth of the other — a question on which so much the greater weight must be laid, since an insight into what really determines chemistry's method depends on its being answered. If one weighs the iron before and after its combustion, one easily convinces oneself that the burnt mass weighs more than the unburnt: with the recognition of the significance lying in such a change of weight, quantitative analysis begins; and once quantitative analysis is combined with qualitative, it is all over with the phlogistians, for now, but only now, can an all-round examination of the phenomena begin.

What, after all, is phlogiston? If phlogiston is the combustible, what then is that which feeds the fire? When a body increases in weight by combustion, how is it thinkable that combustion can depend on a secretion of phlogiston? How can a body become heavier by losing something of its mass? How does it come about that, when a body burns in atmospheric air, not only is a constituent of this air absorbed, but the burnt body has gained precisely an increase in weight answering to the weight of the absorbed air? When the adherents of phlogiston came to answer these and similar questions exactly, punctually and clearly, they entangled themselves in so many contradictions that Lavoisier, who wrote on his balance *in hoc signo vinces*, won the completest victory.

In the particular the general reflects itself: the grounding of a true theory of combustion coincides on the whole with a grounding of the chemical method. That matter is constantly maintained through all operations so that no creation and no annihilation takes place; that the weight of every chemical product is always equal to the weight of its constituents; that every increase of weight accordingly denotes a combination and every decrease a separation; and that it thus becomes possible to sense the otherwise non-sensuous and to follow matter's exchanges along otherwise invisible paths — these are now the fundamental principles on which the method thus won will henceforth know how to build. "For centuries," says Liebig, "the physicists measured; it is only fifty years since they began to weigh. Lavoisier has the balance to thank for all his great discoveries — the balance, that incomparable instrument which holds fast all observations and discoveries, which conquers doubt and draws truth forth into the light … With the balance Aristotle's realm came to an end; his method, of making the explanation of a natural phenomenon a game for the mind, gave place to genuine investigation of nature."

β) Chemical Forces.

Natural scientists are accustomed to lay down[*)] that "the ultimate causes of new forms and phenomena are the chemical forces, which are distinguished from all others in that we notice their existence in their manifestations only upon immediate contact between bodies: this class of phenomena bounds the domain of chemistry. Gravity, electric and magnetic force, light and heat do indeed exert an influence on the chemical actions; but as forces acting in a wider sense they also condition motions, changes of place, natural phenomena generally; the investigation of their nature and laws therefore belongs to physics in the narrower sense."

That the natural scientist should on the whole conceive "the ultimate causes" in this way — not as principles but as forces — is in the highest degree characteristic. To make plain the concept of what is distinctive in the chemical forces, we must in this connection treat particularly of the affinities that hold in chemistry, together with their corresponding conditions and proportions.

αα) Affinities. Already in Hippocrates there occurs the thought that when two bodies unite it must be on account of an essence common to both — that, in other words, like seeks like; and on this view rests the later term affinity, *affinitas.* What Albertus Magnus says of sulphur in his book *de rebus metallicis — sulphur propter affinitatem naturæ metalla adurit* — is in full agreement with the proposition that like seeks like; for since, on the alchemical view, there was sulphur in all metals, the thought lay near that it must be the sulphureous essence dwelling in the metals and ordinary sulphur that sympathized on account of the affinity: sympathy and antipathy were the decisive categories in the medieval view of nature[*)].

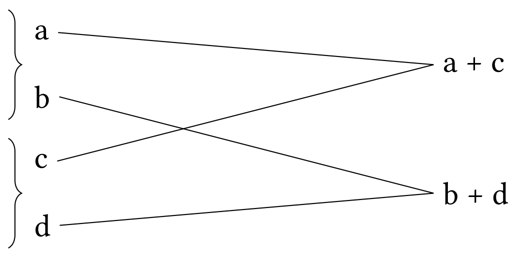
[*)]The following exposition cannot, given its particular aim, take account of the divergent views by which new textbooks indicate new paths — paths that lead not so much through the regions of actuality as through those of possibility.

[*)]"Since the Idea of the organism was the instinctive ground of all their (the medieval physicists') views, and therefore could not become an object of reflection, that principle too — that one homogeneous thing seeks another — presented itself, not as a merely dead attraction, but as sympathy and antipathy. For those qualities mentioned above were the outward forms of things; the deeper ground, on the other hand, from which this attraction sprang, arose for them out of the inward, really living forms.... When the conjunction of the stars seemed peculiarly favourable at a person's birth, when rubbed amber attracted light bodies, or the magnet iron, when stones moved in dissolving liquids, when human beings acted upon others by a glance, by a word, or by secret influence generally, all these effects sprang from the same principle. Antipathy, the disturbing element, sprang from the homogeneous thing's inclination to union. But they assumed a great division, an original contradiction in existence; it formed the keynote of all their knowing, and was wholly of a religious kind: it was the prevailing view of God and the Devil. The former was the principle of what is everywhere and always homogeneous, of the sustaining, of the pure; the latter was the

As late as the seventeenth century it was assumed that two substances were the more intimately akin the more they resembled one another in their properties; but in the eighteenth century Boerhaave already — considering how acids dissolve metals, how gold is dissolved in aqua regia, and setting this dissolution down to affinity — became aware that it is precisely not bodies alike but on the contrary bodies unlike, differing in their properties, that unite: a fact which in more recent times has found confirmation from many sides. Without being properly able to account for the mode of action, Boerhaave confined himself to denoting the uniting force as *amor, amicitia*; but the question how chemical attraction stood to mechanical could not be suppressed, the less so as the concept of affinity, on account of the multiplicity of existential relations, was gradually articulated into a system of stronger and weaker affinities, and thereby brought with it the concept of the so-called elective affinities (*attractio electiva simplex, duplex, multiplex*).

What such elective affinities mean chemistry can easily teach us: "When two substances *a* and *b* are combined with one another into a compound body, this can be separated (decomposed) by a third substance *c*, provided *c* has a greater affinity for *b* than *a* has for *b*; for then *b* and *c* come together into a compound body, and *a* is set free (simple decomposition)." If a compound of quicksilver and sulphur is ignited in a crucible with iron, the sulphur will separate from the quicksilver and unite with the iron. For iron and quicksilver both have an affinity for sulphur, but the iron's affinity is greater than the quicksilver's.

"Two compound bodies can also decompose one another reciprocally, so that by an exchange of the constituents two new compound bodies arise (double decomposition). Let *a* and *b* be combined into one compound body, *c* and *d* into another; then *a* and *c* may come together, and *b* and *d* likewise come together, and thus two new bodies are formed." Such a chemical process is usually presented thus:



If, to take one more concrete example, hydrogen sulphide is passed over heated silver oxide, silver sulphide and water will be formed. Why? Because the sulphur, which can indeed unite with the hydrogen — though only with great difficulty when it has to

principle of disturbance and of the hostile." (H. Steffens). See "*Aanden i Naturen*" by Ørsted. 2. pp. 160–61.

happen directly — has a stronger attraction for, and so, figuratively speaking, a preference for, the silver. As the sulphur and the silver thus prefer one another, the oxygen and the hydrogen are set free, and their combination now takes place all the more easily since the oxygen in turn by its nature prefers the hydrogen to the silver, and the hydrogen likewise the oxygen to the sulphur:

Silver — Silver sulphide.
Oxygen — Water.
Sulphur — Silver sulphide.
Hydrogen — Water.

It goes without saying that by the expression *prefer* no mystical sympathy can here be meant, any more than any arbitrariness. — After men had long disputed about what is operative in the degrees of affinity, Humphry Davy and Berzelius at last succeeded in shedding a new light on what had hitherto been so enigmatic. When water is decomposed by the voltaic pile, the oxygen makes for the positive pole and the hydrogen for the negative; the oxygen must therefore be negatively, the hydrogen positively electric. Berzelius, who in the year 1819 set up his fully developed electrochemical theory[*)], then arrived at the result that what we call chemical affinity, with all its variations, is nothing other than *the electrical polarity in the smallest parts of the various bodies*, and that electricity is therefore the “first cause” of all chemical effects.

ββ) Conditions. The reason why the natural scientist conceives all causes, even “the first causes,” as forces — that is, as causes belonging to things — plainly lies in this, that natural effects can be observed, and determined by observation, only in so far as they can be referred to material starting points: only material existence is the immediate object of sensation. When Lavoisier defined combustion as the union a combustible body enters into with oxygen, he certainly held any combustion without oxygen to be an impossibility. At first glance, then, it looks as though the old abstraction, that like effects must always have the same cause, were being renewed: if the chemists of the seventeenth century had held to the proposition *ubi ignis et calor, ibi sulphur*, it now looked almost as though the antiphlogistians would regard the proposition *ubi ignis, ibi oxygenium* as without restriction. But this was only a passing misunderstanding; even if the true method cannot directly keep errors out, it can always offer means for their discovery and correction. It was soon proved that oxygen could not be the exclusive

[*)] The fundamental thought of the theory was already stated in 1807 by H. C. Ørsted. Cf. Samlede Skrifter. 5. pp. 17–18.

cause of fire, since metals can burn both with sulphur and with chlorine without any cooperation of oxygen. Instead of unconditionally referring all like effects to one and the same cause, men now gradually learned what it means that not only can like effects have unlike causes, but apparently like causes even most unlike effects — learning that between the cause and its effect there falls an intermediate determination on which particular weight must be laid in the world of things, namely *the condition.*

Every element, every chemical product or educt, carries in itself its own system of forces determined by its properties: oxygen a system of oxygen-forces, sulphur a system of sulphur-forces, and so on. Whether, how far and in what way oxygen's forces are to manifest themselves and act depends not on the oxygen alone, but also on the substance with which the oxygen is brought into contact, and on a whole set of external relations — the circumstances, namely, under which the contact takes place. The forces dwelling in the substances therefore have a being according to possibility, so that they are roused to activity only under certain conditions. How this is to be understood the affinities, conditioned as they are in their action, can clearly show. Affinity in and for itself is only a possibility, if a real one; for the possibility to be actualized the necessary conditions must be provided.

The relation between real possibility and its conditions is everywhere a remarkable one. Only with respect to the required actuality is the possibility real; so long as a condition is still lacking, the possibility cannot become actuality and is so far impossible; but when all the conditions are present, the possibility is no longer possibility either, for then it is actuality. To this corresponds the Megarians' paradox that there is no possibility at all (Phil. Prop. p. 141): either the possibility lacks one or more conditions, and then it is an impossibility, or the possibility has all its conditions, and then it is an actuality; hence: either impossibility or actuality. Without dwelling on the Megarians' peculiar manner of proof, we shall in this connection test the paradox by considering the conditions that hold for affinity.

Is it, for instance, possible for iron to combine with sulphur? According to the paradox one must deny the possibility. See, here is a piece of iron, there a piece of sulphur: why do they not unite? "Because it is impossible." What if we now divide each of the pieces into very small parts, so that on the one side we get iron filings and on the other powdered sulphur, and then lay the two masses quite close to one another, yet without any contact taking place? "They still do not unite, because it is still impossible!" But if by mixing the masses we bring the parts into contact with one another, should their union not then be possible? "No, now the union is more than possible, for now it is necessary; hence:

either impossible or necessary!" It is easy to see, however, how what is over-subtle in the paradox rests on a confusion of absolute with relative impossibility, and at the same time on a misconception of the conditions and of the changing relations of things. For while absolute impossibility annihilates the possibility, relative impossibility must sustain it: absolute impossibility is necessity's own negative expression, relative impossibility possibility's. Where one case is possible, its opposite is possible too: real possibility has alternating terms. If, for instance, it is possible for iron to combine with sulphur, because the affinity is given once for all, but only possible so long as the conditions are not yet fulfilled, then the opposite — that no combination comes about — is possible too; but so long as both cases are thus merely possible, both are relatively impossible, for the one becomes impossible precisely when the other becomes actual: the combination is impossible so long as the non-combination is actual, and conversely. From this it is also seen that the necessity which coincides with actuality — the actuality that is possibility's completion — is not the absolute but the conditioned necessity.

Conditioned necessity, akin precisely thereby to cases and to the accidental, is so far from conflicting with possibility's reality that it shows itself on the contrary as the dialectical unity of possibility and actuality; for the possible is, just as much as the actual, always a consequence of given presuppositions, always law-governed, and so far an expression of conditioned necessity. If the paradox of impossibility were to be defended, it would have to be proved either that the conditions stood in so intimate a connection that where one was found all were found, and where one was lacking all were lacking, or — what, seen from another side, comes to the same — that because a condition was lacking at one given moment it must be lacking for ever; but how little such a view accords with the alternations of the relations of things is conspicuous.

As regards the operation of the chemical affinities in general, the following two fundamental conditions are especially to be brought out: 1) the substances must be brought into immediate contact with one another, and 2) the reciprocal attraction at the moment of union must be strong enough to overcome the hindering forces, if such are present; but in what degree the particular conditions, such as heat, light, electricity and so on, can act upon the fundamental conditions must be for the subject-science to demonstrate.

γγ) Proportions. How intimately the quantitative in all chemical combinations is united with the qualitative, and what significance quantitative analysis must have in this respect, a chemist explains to us in the following way: "To decompose 117 parts of cinnabar, a compound of sulphur and quicksilver, I use 27 parts of iron; I obtain 101 parts of quicksilver and 43 parts of sulphide of iron. Accordingly, 16 parts of sulphur are found combined with 101 parts of quicksilver, the latter being separated out and replaced by 27 parts of iron; 27 parts of iron have therefore combined with 16 parts of sulphur." Should anyone suppose that it is precisely the sulphur which by its attraction to the iron turns the scale in the given case, continued observation can easily show what is essential here and what is inessential and so far accidental. "For if the quicksilver in one of its compounds with oxygen, chlorine, bromine and so on is separated out and replaced by iron, then I invariably and without exception need 27 parts of iron for every 101 parts of quicksilver." Should anyone suppose that it is perhaps the number 16 on which the emphasis must fall, and that cases in which 101 parts of quicksilver have combined, not with 16 but, say, with 8 parts by weight of another substance, will make an exception to the rule laid down, then continued observations can likewise show that we have here not a rule with exceptions but a law. For experience confirms "that when 101 parts of quicksilver enter into any combination whatever with a body whose weight may be called a, then this body's weight a also enters into a combination with 27 parts of iron. Since 16 of sulphur combine with 8 of oxygen and with 1 of hydrogen, one also observes that wherever hydrogen in a compound is replaced by oxygen, or oxygen by hydrogen, 8 parts by weight of oxygen take the place of 1 part by weight of hydrogen, and conversely."

Should anyone suppose that concepts abstracted from the phenomena of experience are altogether unfit to yield universally valid determinations of essence (Hume; cf. Propædeut. pp. 15–17), then, so far as the chemical proportions are concerned, the inferences drawn from the facts can convince him of the opposite. How intimately the rational is grown together with the empirical is seen precisely in the present connection. That aA, bB, cC, dD must in their possible mutual combination correspond to one another when they, each term severally, correspond to eE — where a, b, c, d, e denote the parts by weight, the quantitative, and A, B, C, D, E the substances, the qualitative — is a consequence of concepts grounded upon experience. For one finds "that when the weight is known in which one body combines with two or more others, these parts by weight also express the quantities in which these various bodies combine with one another: 16 of sulphur combine with 8 of oxygen, 1 of hydrogen, 101 of quicksilver, 27 of iron; but 8 of oxygen combine exactly with 101 of quicksilver, and with 27 of iron, to give oxides of quicksilver, of iron, and so on.[*)] If, then, one knows the proportion in which

[*)] "To bring about a chemical process I need, for one purpose or another, 8 of oxygen; and when instead of the oxygen I can and will use sulphur, I always need 16 of sulphur; these combining weights express equal values."

one body, no matter which, combines with all others, one also knows the quantities in which all bodies, when they possess any affinity for one another at all, combine among themselves."

It must further be specially brought out that these proportions do not change either in the cases where one body forms more than one compound with another, or with a third, and so on. "Thus 14 of nitrogen combine with 8 of oxygen to give nitrous oxide, the so-called laughing gas; there is another compound, a colourless gas: 14 of nitrogen with 16 (twice 8) of oxygen, which forms red fumes in air; there is a third which contains 24 (three times 8), a fourth which contains 32 (four times 8), a fifth, nitric acid, which contains 40 (five times 8) of oxygen, always to 14 of nitrogen."

The composition of all chemical compounds can then without exception be expressed by fixed numbers, which for that very reason have been called combining weights or equivalents, because they really express the quantities in which bodies enter into "mixtures," that is, *combinations*, or in which they produce equal effects.

From the doctrine of equivalents and of conditions it then emerges what it means that different causes can have the same effect, and the same cause different effects. As an essential *condition*, heat is at the same time a *cause*; and if heat is posited in full generality as a causality identical with itself, how different are its effects! Increased heat can cause combinations to be entered into: oxygen and hydrogen do not combine at ordinary temperature, but do at a red heat; yet increased heat can also cause combinations to be dissolved: at ordinary temperature lime and carbonic acid have a strong attraction, but when the lime is ignited the carbonic acid departs. If the chemical affinity that binds 27 parts of iron is regarded as the expression of one and the same effect, while the oxygen and the sulphur denote two mutually different causes, and the quantities that must be used for the purpose — 8 for the oxygen and 16 for the sulphur — the corresponding conditions, then we have here an example which shows how *the conditions equalize the difference of cause*, so that *different causes* produce *the same effect*. If finally, to see the matter from this side too, the affinity of oxygen and nitrogen is taken as the one and the same possible cause of all nitrogen's oxygen compounds, then it is seen from this not only that a reciprocal action is itself a cause, but also how *the changing conditions* can bring it about that *the same cause* gives *different effects*; for among nitrogen's 5 oxygen compounds the difference is characteristic. The dialectic of causality thus indicated rests essentially on this, that a condition, by its very concept, is both different from and yet one with the operative cause.

γ. Chemical Combinations.

That chemistry is in the strictest sense of the word a distinct science, and that what in this science "is founded upon reason" by no means "depends upon philosophy," as the Hegelian misdirection suggests, must by now be sufficiently evident from what has already been treated. In considering the course of chemistry's development we saw what efforts it cost before the investigating spirit passed from speculation's mystical essentialities to a punctual knowledge of the forces operative in matter; by then considering the chemical forces and investigating the relation between possibility, cause, condition and effect, we arrived at the result that the possible cause must in each case grow together in its own peculiar way with its conditions in order to yield a peculiar product. The concept *chemical combination* is a concept of concretion; but a concept of concretion has many punctual determinations. For a closer insight into that concept it is therefore necessary to understand the *sign-language**) by which chemists express "both a compound's composition and the way in which they conceive the constituents to be arranged in such a composition"; for this sign-language answers precisely, in its distinctiveness, to the punctual analysis of the distinctive concept that finds its expression in it. We shall use this sign-language as we now pass to treating of compounds of different orders. "If two simple substances combine into a compound substance, a compound of the first order arises, $[a + b]$; if two substances of the first order combine with one another, a compound of the second order arises, $[(a + b) + (c + d)]$; if a substance of the second order combines with one of the first or with one of the second, a compound of the third order arises, $[\{(a + b) + (c + d)\} + \{(e + f) + (g + h)\}]$, and so forth."

αα) The First Order. The electrochemical theory, which teaches us that what is operative in electricity and in affinity is essentially the same, states a great and comprehensive truth. How rich in content and how pervasive this fundamental truth is can be fully recognized only when the dialectic of causes and conditions

has been properly made plain. If the dialectical is overlooked, one will constantly be tempted, now too one-sidedly to set chemical attraction, as the manifestation of a force

*)If, to keep to an example already cited, we denote by N 14 parts by weight of nitrogen and by O 8 parts by weight of oxygen, then the 5 nitrogen compounds may be expressed by $N + O$, $N + O_2$, $N + O_3$, $N + O_4$, $N + O_5$, or, putting a dot over N for each O, by $\dot{N}$, $\ddot{N}$, $\dddot{N}$, $\ddddot{N}$, $\overset{.....}{N}$. From such a manner of notation it will easily be understood that it is in and for itself quite indifferent at how many parts by weight, that is, units of weight, N is set, provided only that the number corresponds to the quantity of weight represented by O. If, for example, one lets O denote, not 8, but 100.00 parts by weight, the proportion suffers no essential change when N is reckoned, not at 14, or more exactly not at 14.2, but at 175.00; and by then bringing N and O under the aspect of elementary parts, $\dot{N}$ will denote that one elementary part, or rather one double elementary part, of nitrogen is combined with one, $\ddot{N}$ that it is combined with two, $\dddot{N}$ that it is combined with three elementary parts of oxygen, and so on.

subsisting on its own, over against the other natural forces more or less akin to it, now just as one-sidedly to identify the chemical force either with the mechanical, or with the electrical, or with the force of light, or with the force of heat, and so on; whereas the truth undoubtedly is that now this, now that of the particular forces can, according to the corresponding conditions, acquire a decisive influence on the chemical manifestation of force, precisely because the elementary forces, despite the difference reflected in the outward, are identical[*)]. The type theory, proceeding from Dumas and now already developed in several directions, has indeed pointed out not a few phenomena conflicting with the presuppositions of electrochemistry, and has also, in so far as electrical polarity is made without more ado the immediate cause of chemical effects, broken the system as set up; but when once the turning point is reached at which the concept of affinity enters a new stage, it will doubtless appear that the unitary type theory has by no means simply displaced the electrochemical dualism, but has rather developed, corrected, explained and in essential points confirmed it: the concept of affinity is a concept of concretion reflected in manifold conditions.

In abstract logical and mathematical form the opposition of positive and negative is formal; in the electrochemical form it is real. But what the concept, the true concept, demands, existence fulfils. It lies in the concept that every existing something must be integrated by its other, its opposite; it is seen in existence that no thing is anything by itself alone, but that a thing is in itself only what it is in its system of relations to other things. What, after all, are such substances as oxygen (O), sulphur (S), phosphorus (P), carbon (C), silicon (Si), calcium (Ca), magnesium (Mg), aluminium (Al), sodium (Na), potassium (Ka), when each is considered by itself, absolutely in and for itself? That is impossible to say, and for the good reason that an absolute in-and-for-itself does not exist. If one gathers under a single survey the 60 or 70 substances which science at its present stage must regard as elements, and divides them into metals and metalloids thus — the metals, to which Ca (the lime-metal), Mg, Al (the clay-metal), Na and Ka belong, are distinguished by a peculiar lustre, are opaque and moreover good conductors of electricity and heat; the metalloids, such as O, S, P, C, Si and others, do not have these properties — then, since a property cannot possibly manifest itself except in relations, it is conspicuous how every attempt at a characterization must rest on nothing but

[*)]"The steady progress of the natural sciences has brought physics and chemistry into an ever more intimate connection, and everything indicates that before long these two branches of science will entwine themselves into one. Manifold investigations have shown that matter's physical and chemical properties are most closely bound up with one another, so that the former cannot be known in their true connection except in conjunction with the latter, and conversely." Tidsskrift for Physik og Chemie, ed. A. Thomsen and J. Thomsen. 1st year, 1st number. Copenhagen 1862. p. 1.

determinations of relation.

In polar opposition, with its tension and its equalization, the original determination of relation is expressed in a distinctive way. Positive electricity does not attract the positive, for what it immediately is itself it has no need to seek outside itself; but it attracts the negative, for the opposite — the unlike within the like, the unhomogeneous within the homogeneous, what it is not itself — is precisely what it needs in its existence. Electrical tension, however, presupposes, so far as the phenomenon is concerned, that the opposed terms — think of glass and sealing-wax — hold themselves as masses with a certain outward independence over against one another, so that the equalization yields only a formal result, in so far as the mutually opposed activities are by their union freed from one-sidedness and come to rest. Chemical combination, by contrast, has this distinctive feature, that the masses, through whose smallest parts the polar tension is conceived to act, give up their outward independence at every point, so that the equalization is productive, existing in a product of its own.

When sulphur unites with hydrogen to form hydrogen sulphide, $S + H = SH = \dot{H}$, the tension is conditioned by the sulphur's furnishing the negative and the hydrogen the positive term; but in the hydrogen sulphide the sulphur is not sulphur and the hydrogen not hydrogen: in the union itself, that is, in the equalization of the tension, the several terms have given up their independence and formed a new body. If sulphur unites with oxygen, say to form sulphuric acid $S + O_3 = SO_3 = \dddot{S}$, the tension is conditioned by the sulphur's being the positive and the oxygen the negative; but the result of the union, the new product, is a realized relation in which the original terms have become moments.

That the product arising from the union is not a mixture of unlike constituents but a new body, an existing unity, becomes especially conspicuous when the united substances display a strong mutual opposition; for then the product will by its characteristic properties diverge most of all from both. "Thus the combinations between members of the same family have the easily recognizable family virtues and vices undiminished, often intensified; but when two quite opposite stocks ally themselves, there always arises from it a new body in which the parents cannot be recognized. Thus iron and quicksilver (two metals) stand by their pedigrees far nearer to one another than iron and sulphur, or quicksilver and sulphur (a metal and a metalloid). In a combination between the first two one sees the origin at once; but who would suspect in cinnabar the fluid, silver-white metal, the yellow combustible sulphur?" (Liebig).

ββ) The Second Order. For compounds of the first order to be able to combine again, one term must, according to the electrochemical theory, likewise be electropositive when the other is electronegative. If the compounds $\overset{\cdot\cdot\cdot\cdot\cdot}{\mathrm{P}}$, $\dddot{\mathrm{S}}\mathrm{i}$, $\ddot{\mathrm{C}}$ are set on the one side, and $\dddot{\mathrm{A}}\mathrm{l}$, $\dot{\mathrm{N}}\mathrm{a}$, $\dot{\mathrm{K}}\mathrm{a}$ on the other, the former are negative and are called acids: $\overset{\cdot\cdot\cdot\cdot\cdot}{\mathrm{P}}$, phosphoric acid, $\dddot{\mathrm{S}}\mathrm{i}$ silicic acid, $\ddot{\mathrm{C}}$ carbonic acid; the latter are positive and are called bases: $\dddot{\mathrm{A}}\mathrm{l}$ (oxide of aluminium) alumina, $\dot{\mathrm{N}}\mathrm{a}$ (oxide of sodium) soda, $\dot{\mathrm{K}}\mathrm{a}$ (oxide of potassium) potash. These acids and bases can now combine, for example, thus: phosphoric acid with potash to give $\overset{\cdot\cdot\cdot\cdot\cdot}{\mathrm{P}}\,\dot{\mathrm{K}}\mathrm{a}$, phosphate of potash; carbonic acid with potash to give $\ddot{\mathrm{C}}\dot{\mathrm{K}}\mathrm{a}$, carbonate of potash; silicic acid with alumina to give $\dddot{\mathrm{S}}\mathrm{i}\ \dddot{\mathrm{A}}\mathrm{l}$, silicate of alumina, and so on. What holds in general of the concepts positive and negative — namely that they presuppose and reflect one another — holds also in a special sense of the concepts acid and base: being an acid acquires significance only through being a base.

When the chemists of the seventeenth century became properly aware of the opposition between "acids" and "alkalis" (bases), it was natural that they should seek in their own way to characterize this opposition. Thus if in the acid one had "a hot and burning nature," in the alkali one had "a cold nature"; if the acids represented "the active and masculine," then the alkalis must of course represent the "passive and feminine." With respect to possible productions and formations, the alkali was on the whole conceived as the soluble, in itself indeterminate but yet determinable substance, as chaos, and the acid as the self-determining and forming, a *Spiritus imprægnans.* The first to state scientific determinations of the concepts was Robert Boyle; this acute chemist characterized acids and bases by their opposed modes of action: the acid turns blue litmus red, but the base turns red litmus blue; what is dissolved by the acid is precipitated by the base, what is precipitated by the acid is dissolved by the base; acid and base cancel by their combination one another's salient properties. Further than this one could not get on the phlogistic presupposition.

With Lavoisier the new research begins; but Lavoisier, who saw in combustion everywhere only an effect of the oxygen-cause, now also took oxygen for the general cause of acidity, the substance which in its capacity as »*Oxygene*« (*Oxygenium*, acid-stuff) gave all acids their sour property. It was not long, however, before it was discovered that hydrogen sulphide, hydrogen cyanide (prussic acid), hydrogen chloride and others could also act as acids; and finally Berzelius showed that it depended, not on the oxygen by itself, but on the radical with which the oxygen was combined, whether in a given case an acid or a base was to be formed — that oxygen, for example, which in its combination with sulphur forms a strong acid, forms in its combination with potassium a strong base. And now it was clearly seen that oxygen could not possibly be the unconditionally acid-forming substance (*principium aciditatis*).

Seen from the standpoint of the concept this is a very great advance. Force, when its manifestation is grasped immediately, is a cause belonging to a thing; but as the conditioned thing-causes begin to act, the effect itself becomes cause and the cause effect: as the oxygen is cause and attracts the sulphur, so the sulphur in turn is cause and attracts the oxygen. The force of affinity is essentially two forces, an identity of opposed causes. Although oxygen in its activity is of course always oxygen, it acts on sulphur otherwise than it acts on potassium. Why? Because it is acted upon by sulphur otherwise than it is acted upon by potassium. Not the substance as substance, not the thing as thing, but the conditioned reciprocal action is therefore from now on to be recognized as the true, self-reflected cause of all chemical formations[*)].

[*)]Dr. R. H. Lotze (Allgemeine Physiologie des körperlichen Lebens. Leipzig 1851. pp. 84–94: Von der Natur der Kräfte) has already noticed this categorial relation from his own standpoint, save that the otherwise so acute thinker has not properly distinguished between the ground and the thing-cause. "Wir nun freilich, gewöhnt an die Analogie unseres eigenen Handelns mit seinem Unterschied zwischen Absicht und Ausführung, pflegen, wenn die Wirkung eingetreten ist, in die Dinge, von denen sie anschaulich ausging, nicht nur die vollständigen Bedingungen ihrer Möglichkeit, sondern sogar eine beständige Tendenz zu ihrer Hervorbringung zu verlegen, und sprechen von Kräften, die ausdrücklich auf einen bestimmten Erfolg gerichtet, beständig den Dingen inhäriren, selbst wo gar keine äusseren Umstände vorhanden sind, welche eine Ausführung ihrer Tendenz gestatteten. Solche fertige Kräfte sind nun so wenig vorhanden, als in der Natur der Dinge allein der Grund ihrer Wirkungen liegt. Die Möglichkeit eines Ereignisses setzt nicht nur die eigenthümlichen Eigenschaften mehrerer zu seiner Verwirklichung beitragender Substanzen voraus, sondern ebenso nothwendig eine bestimmte Beziehungsweise, in welche diese zu einander treten. Wollen wir daher mit dem Namen Kraft den vollen und zureichenden Grund für das Eintreten einer Wirkung bezeichnen, so müssen wir weiter hinzufügen, daß eine solche Kraft niemals fertig in irgend einer Substanz präformirt liegt, sondern daß sie stets nur einem bestimmten Verhältnisse mehrerer Substanzen, einem Bruchstücke des Naturlaufs inhärirt. Wo wir von der Kraft einer Substanz sprechen, ist genau genommen die Vorstellung der Kraft auf ein falsches und unvollständiges Subject bezogen, und nur in vorsichtig bestimmten Gränzen kann dieser übliche Sprachgebrauch ohne Verwirrung der Begriffe beibehalten werden. Wie es nun keine Kräfte gibt, die einer einzigen Substanz inhäriren, so gibt es zweitens noch weniger deren, die einer solchen *beständig* inwohnten, und ihrer wesentlichen Natur nach von andern zu unterscheiden wären, welche derselben Substanz nur bedingungsweise zukämen. Wollen wir vielmehr jenem Sprachgebrauche folgend, die Fähigkeit zur Erzeugung einer Wirkung, die einer Substanz nur unter Beihilfe gewisser Umstände zukommt, auf sie allein als ihre Kraft übertragen, so müssen wir dann auch behaupten, daß keine Substanz beständige, sondern jede nur erworbene Kräfte besitzt. Je nachdem die Beziehungen wechseln, in denen sie zu der übrigen Welt steht, wachsen ihr nach allgemeinen Gesetzen Fähigkeiten bald zu dieser, bald zu jener Leistung zu; beständig erwirbt sie Kräfte und büßt deren andere ein; niemals aber bleibt ihr neben diesen veränderlichen Vermögen eine Summe ihr absolut angehöriger Kräfte übrig, die auch in Augenblicken, wo sie nicht wirkten, dennoch existirten." [We, accustomed as we are to the analogy of our own action with its distinction between intention and execution, are apt, once the effect has occurred, to place in the things from which it visibly proceeded not only the complete conditions of its possibility but even a constant tendency to bring it about; and we speak of forces which, expressly directed to a determinate result, inhere constantly in things, even where no external circumstances whatever are present that would permit their tendency to be carried out. Such ready-made forces exist as little as the ground of their effects lies in the nature of things alone. The possibility of an event presupposes not only the peculiar properties of several substances contributing to its realization, but just as necessarily a determinate mode of relation into which these enter with one another. If, therefore, we wish to designate by the name force the full and sufficient ground for the occurrence of an effect, we must further add that such a force never lies ready-made, preformed in any one substance, but always inheres only in a determinate relation of several substances, in a fragment of

The union of acid and base gives, according to a term at length fixed in chemistry, a salt; but what really is a salt? That the concept salt could not be fixed by a definition as the concept ellipse or the concept logarithm could, is conspicuous; one must first arrive at such a real concept by analysis's various roundabout ways, and by analysis develop it further again. But when in formal logic — as sometimes happens — a plainly external and undialectical separation is made between the analysis by which we arrive at the concept and analysis of the concept proper, and this in turn is divided into various kinds of more and less proper conceptual analyses, then logic by so naive a division has given the signal for an equally naive confusion. Were we to follow this direction here, the concept of salt would certainly become a rather motley mixture-concept; for what is a salt?

The substance characteristic by its peculiar taste and its solubility, common salt, together with every other substance having a predominant resemblance to it, has of course furnished the starting point for the determination and development of the concept of salt. The resemblance between common salt, soda and potash was from the first conspicuous enough; and since these substances are not consumed by fire, incombustibility as a third mark was at length so strongly emphasized that the alchemists ended by making salt the third fundamental element of the metals. From all three starting points analogies now operated in ever wider compass, so that not only substances like alum, saltpetre and the vitriols, but acids, alkalis and their neutral products generally, were reckoned among the salts. Among so many and such various respects (*diversi respectus*) emphasis could now be laid arbitrarily, now on the one, now on the other, now on the general "principle of salt," now on the particular saline substances, now on the taste, now on the solubility; and the crossing, conflicting, cancelling analogies could long hold the concept in suspense. How unsettled the concept of salt has been is seen, among other things, from Newton's reckoning water too among the salts in his Optics: *Aqua, quæ est sal*

the course of nature. Where we speak of the force of a substance, the representation of force is, strictly taken, referred to a false and incomplete subject, and only within cautiously determined limits can this customary usage be retained without confusion of concepts. As there are no forces inhering in a single substance, so, secondly, there are still fewer that dwell in one constantly and would be distinguishable by their essential nature from others belonging to the same substance only conditionally. If rather, following that usage, we transfer to a substance alone, as its force, the capacity to produce an effect which belongs to it only with the aid of certain circumstances, then we must also maintain that no substance possesses constant, but every one only acquired forces. According as the relations change in which it stands to the rest of the world, capacities for now this, now that performance accrue to it by general laws; it constantly acquires forces and forfeits others; but never does there remain to it, beside these variable capacities, a sum of forces absolutely belonging to it, which would exist even in moments when they were not acting.] — From the whole of natural science, with its manifold and mutually different circles of investigation, scarcely an example could be singled out which confirms and illustrates more strikingly the proposition that a critique of concepts carried through with understanding must consistently lead to dialectic.

admodum fluidus et saporis expers. Some way into the eighteenth century Boerhaave still gives us the following definition: *Sal vocatur corpus, quod aqua potest dilui, igne autem fundi, si non avolat prius in auras, quodque gustum humanum afficere valet eo sensu, quem saporem appellant*; whereupon he divides the salts into *salia alcalina*, *salia acida*, *sic dicta jam neutra* and *salia composita.*

These investigations do not exactly lack phenomena; but what help are all the phenomena of experience when the decisive point of view is lacking, when thought is unable to separate what is essential in the phenomena from what is inessential, the necessary from the accidental, the universally valid from the particular? Here one begins with the particular, with the peculiar substance which is called salt precisely after its peculiar taste; here one begins with a peculiar reciprocal determination of the subjective and the objective. For what is subjective in the taste has its objective presupposition in the solubility, since what cannot be dissolved yields no taste either. By the sense of taste, however, one easily convinces oneself that the sour, *sal acidum*, and the sharp or bitter, *sal alcalinum*, are after all only one-sided statements of what is properly saline, and that the salty taste arises — as van Helmont already observed — when an acid is united with a body opposed to it and forms a product that can be tasted. And if one then sees how the subjective difference between the sour and the sharp can be made objectively recognizable, can, as Boyle showed, be made evident by the change of colour that acid and base each in its own way bring about — what else is it then, on these presuppositions, than a direct consequence to let *sal neutrum*, the product in which acid and base have neutralized one another, be the typical expression of salt? And when so objective and comprehensive a foundation for the concept has once been won, what else is it again than a direct consequence that the first objective but restricted and in itself unsteady mark of character, namely solubility, must fall away as inessential? The fundamental concept thus won by an articulating reciprocal action of sensation and thinking, observation and inference, can then also be further analysed only by a corresponding reciprocal action.

To recognize the difference between oxygen salts and sulphur salts one must first have recognized how sulphur can play the part of oxygen both in the acid (sulphide) and in the base (sulphuret); but once this has been recognized, one sees also that the compound sulphide of arsenic — sulphide of sodium expresses the concept salt just as essentially as the compound phosphate of potash. To ground the distinction between the so-called amphid salts (phosphate of potash, carbonate of soda and the like) and haloid salts (chloride of potassium, chloride of sodium and others), it was first necessary to discover how the hydracids formed by hydrogen's union with chlorine, bromine, iodine and fluorine (*corpora halogenia*) combine with the oxygen bases, the acid's hydrogen combining with the base's oxygen to form water while the base's metal and the acid's radical by their combination form the salt; but once this has been recognized, it can also be seen that, for example, hydrochlorate of soda gives a salt (chloride of sodium + water) which, despite the peculiar mode of formation, answers to the fundamental concept just as well as carbonate of soda does. The advance is always made by the characteristic marks which observation discovers and tests being transformed by reflection into essential determinations in the real concept concerned.

γγ) The Third Order. Since a salt can combine anew with another salt, as an acid does with a base, one would at first glance suppose that, so far as the oxygen salts are concerned, it must be the acid salt which by virtue of its acidity combined with the basic one into a neutral double salt; nevertheless such a conception would conflict with the mode of presentation current in chemistry, and a categorial awkwardness would easily arise: every science always has its own grounds, were they only historical grounds, why it prefers to use these and not other designations.

For the question how the electronegative (the acid) stands to the electropositive (the base) in one and the same salt must not be confused with the question how two different salts stand to one another with respect to negative and positive character. The first of these questions concerns the product as an existing whole, in so far as its chemical character reflects itself in the constituents from whose union it has arisen. On this question chemists have agreed to call every quantity of a metallic oxide which contains one oxygen equivalent an *equivalent of metallic oxide*[*)], without any regard to the equivalents of metal contained in it. Only on the presupposition of a *chosen* measure can anything be measured, only on condition of fixed starting points can relations determinate in themselves be exactly determined; and so, likewise by agreement, the quantity of acid that saturates one equivalent of base has been given the name of an *equivalent of acid.* What is arbitrary in the "agreement" is therefore nevertheless something rational and necessary in its own connection; and by this road we come to the categories *neutral salt, acid salt, basic salt*[*)]. — The second question, on the other

[*)]To separate out 10 parts of one base one uses, say, 15 parts of another, 25 parts of a third, and so on. Now when the 10 parts of the first base contain 5 parts of oxygen, it turns out that the 15 parts of the second and the 25 parts of the third, and so on, likewise contain neither more nor less than 5 parts of oxygen. The quantities of oxygen in the metallic bases that replace one another remain constantly the same; only the metals combined with the same quantities of oxygen replace one another according to these equivalents.

[*)]The salt in which there are as many equivalents of acid as there are oxygen equivalents in the base is called a neutral salt. Thus sulphate of alumina, $(\dddot{S})_3 + \dddot{A}l$, is a neutral salt, for in the base $\dddot{A}l = Al + O_3$ there are contained 3 equivalents of oxygen, answering to the three equivalents of acid, $(\dddot{S})_3 = 3\,\dddot{S}$; likewise hand, concerns not the product's relation to the constituents contained in it, but the quality of the chemical relation, the relation of affinity in which one compound substance stands to another, in so far as by entering into combination they are to produce a still more compound one. How independent the second of the two questions cited is of the first is then clearly seen from the fact that two "neutral salts" can, when the conditions are present, enter into such a relation of opposition to one another that the one acts in the character of the electropositive and so far, without being "basic," plays the part of base, while the other, by assuming the character of the electronegative, functions as acid without being "acid." Examples of this we have in the double salts: sulphate of potash and alumina, $\dddot{\mathrm{S}}\,\dot{\mathrm{K}}\mathrm{a} + (\dddot{\mathrm{S}})_3\;\dddot{\mathrm{A}}\,\mathrm{l}$, which with 24 parts of water (24 Aq = 24 $\dot{\mathrm{H}}$) gives alum; and silicate of potash and alumina, $\dddot{\mathrm{S}}\,\mathrm{i}\dot{\mathrm{K}}\mathrm{a} + (\dddot{\mathrm{S}}\,\mathrm{i})_3\;\dddot{\mathrm{A}}\,\mathrm{l}$, the type of the so-called felspars — a salt which in chemistry's language one might with special emphasis call "a salt of the earth," since as one of the chief constituents of granite, gneiss, syenite, porphyry and several other rocks it is found so generally distributed in the earth's crust. That besides the double salts, compounds of the third order, there occur in the mineral kingdom compounds of a higher order as well, namely double double salts, is in the present connection without influence on the development of the concept.

What it means that the concept chemical combination is a distinctive concept of concretion must now be sufficiently evident from the foregoing; but the concrete acquires a different illumination according to the different point of view from which it is seen. The concrete is in general determined by opposition to the abstract — this is the logical conception; but the concrete must, from a physical standpoint, also be determined as the composite in opposition to the simple, the uncompounded; and finally the concrete will, in virtue of an infinite reciprocal determination of the logical and the physical, be able in a dialectical-ontological sense to assume the shape of the simple as well as of the particular and the general, and thereby present itself as the really-rational.

In *logical concretion* there is a plurality of abstract determinations whose mutual connection can be exhibited only in so far as one of these determinations is completed by the other with inner necessity, that is, with inescapable consequences of analogy — the one abstract side being integrated by the other. When common salt (chloride of sodium)

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sulphate of potash, $\dddot{\mathrm{S}}\,\dot{\mathrm{K}}\mathrm{a} = \dddot{\mathrm{S}} + \dot{\mathrm{K}}\mathrm{a}$, is a neutral salt, since the one equivalent of oxygen in the base answers exactly to the one equivalent of acid). But now many neutral salts can again enter into combination with more of the oxide they contain as base; such salts, in which there is therefore more base than in the neutral salts, are called basic (in silicate of oxide of zinc, $\dddot{\mathrm{S}}\,\mathrm{i} + (\dot{\mathrm{Z}}\mathrm{n})_3$, there is for instance more base than in carbonate of oxide of zinc, $\ddot{\mathrm{C}} + \dot{\mathrm{Z}}\mathrm{n}$. Further, many neutral salts can combine with certain equivalents of the acid they already contain; such salts, which therefore contain more acid than the neutral ones, are called acid salts (bisulphate of potash, $\dddot{\mathrm{S}}\,\dot{\mathrm{K}}\mathrm{a} + \dddot{\mathrm{S}}\,\dot{\mathrm{H}}$, contains, when it has lost its water, $2\,\dddot{\mathrm{S}} + \dot{\mathrm{K}}\mathrm{a}$, *twice as much* acid as the neutral one).

and potash (carbonate of potash) are denoted by the same name, salt, it is plain that a good many differences are here abstracted from; but real differences do not fall away because one abstracts from them. It must therefore be possible to prove that the likeness holding despite the unlikeness — the analogy — really is so pervasive that the common name becomes categorial. To establish this, the mode of formation is now brought out; but here in the mode of formation the analogy betrays a questionable unlikeness, for while the oxygen acid enters into direct combination with its base, the hydracid on the contrary brings about, as was noted above, a peculiar separation and exchange. To remove this disturbing unlikeness one might then also attempt—to carry the analogy through from a new point of view, namely by cancelling the difference between oxygen acids and hydracids. “Some chemists thus regard all acids, the oxygen acids too, as being hydracids, assuming that every acid is an acid only when it contains 1 atom of water, whose hydrogen is then assumed to be in combination with the acid + the water’s oxygen, these last forming a compound radical — for example English sulphuric acid: $\dddot{S} + \dot{H} = (SO_4)H$. They further assume that when the acids enter into combination with the oxygen bases the same thing happens as with the real hydracids: the hydrogen goes to the base’s oxygen and forms water, and the base’s metal enters into combination with the compound radical” .. (Wöhler’s Chemie, ed. Groth, p. 97). Every consequence of analogy must in this way be pursued until it is bounded and cancelled by another consequence running in the opposite direction.

While logical analysis, ideal separation, is essentially conditioned by abstraction, the *physical* is on the contrary *a real separation*, which in chemistry takes place by decomposition. What confusion arises when abstraction tries to overfly the real analysis grounded on chemical separation and combination is seen from the alchemists’ errors. For a production of the philosophers’ stone it was held necessary to seek out the pure primal matter, *materia prima, cruda*; they sought it in the metals, in quicksilver, in vitriol, in salt, in the air — a *spiritus mundi* — in the virgin earth, *terra virginea*, *terra virgo*; they looked everywhere, but in vain! The contradiction is that the general, which can be held fast only in an abstraction, is nevertheless sought as a particular substance, an object of real analysis. Real analysis, which explains the composite by the simple, explains at the same stroke thing by thing, for the simple is a thing just as much as the composite; abstraction gives a metaphysical, real analysis a chemical element. It is on the difference between the metaphysical and the chemical element that the emphasis must fall in this connection — a difference brought to clear consciousness in the seventeenth century by R. Boyle. As a proof of the validity of the Aristotelian elements, he says, “later natural scientists have adduced that from combustion in particular one could see how a body resolved itself into earth (ash), fire, air (smoke and gas) and water"; but Boyle shows how unsteady these expressions are, that ash, for instance, is by no means a real earth; he shows further that if sulphur, salt and quicksilver were, on the alchemists' still more absurd view, the elements of things, then a host of other bodies could with the same right be called elements. Without troubling about the primal essence of matter one should then, so runs his direction, "chiefly turn attention to those constituents which really admit of being separated out and exhibited. Those constituents — that is, those substances — which chemistry at a given stage is unable to separate, one might well call elements; but one would do well to consider that the last limit of separation is not to be fixed *a priori*."

Although real analysis, considered by itself, is a sensuous analysis of things, it nevertheless leads consistently to the concept of things, in so far as it leads to insight into the reciprocal action of substances. By seeing how separation conditions combination and combination again separation, how synthesis presupposes analysis and again assists analysis, how differently bodies behave under heat, how differently their disposition to dissolve in liquids is, how two substances which otherwise do not unite yet enter into combination when they meet at the moment both are set free from an earlier combination (*in statu nascendi*[*)]), and so on, we gain an insight into the fundamental relations of existence, into the nature of things — an insight whose loss no other science could make good to us[**)].

How concretion, under a pervasive *reciprocal determination of the physical and the logical, the empirical and the intellectual, the thing and the concept*, stands to the *simple* and the *abstract* may be seen already from such judgments as "iron is a metal,"

[*)]Chlorine cannot enter directly into combination with oxygen; these 2 gases remain wholly without effect on one another, under whatever conditions of pressure or temperature one may try to make them act together; only when chlorine is brought into contact with oxygen at the very moment the latter leaves another combination do they enter into union. (Forchhammer. Chem. pp. 35–36).

[**)]Welcher Antheil von Wahrheit in jener Ansicht liegt, die von einem bedeutungsvollen Gedanken die Wirklichkeit durchdrungen und geordnet glaubt, dies besonders hervorzuheben ist kaum nöthig. Denn wie sehr auch nach so vielen mislungenen Versuchen, jenen in allen Dingen schlummernden Gedanken auf menschliche Ausdrücke zu bringen, unsere Zeit abgestumpft für diese Voraussetzung ist, auf einem tiefen und lebhaften Bedürfnisse des Gemüths beruht sie dennoch fest und sicher für den, der weder den vereinzelten Aussprüchen des Verstandes gestattet, sich als einzige Quelle der Wahrheit zu benehmen, noch das, was er als unantastbare Gewißheit voraussetzt, dennoch bis zur tastbarer Deutlichkeit ausführen zu müssen meint. [How much truth lies in that view which believes actuality to be penetrated and ordered by a significant thought, there is hardly need to bring out particularly. For however blunted our age may be to this presupposition, after so many failed attempts to bring that thought slumbering in all things into human expressions, it nevertheless rests firm and secure upon a deep and lively need of the heart — for him who neither permits the isolated pronouncements of the understanding to conduct themselves as the sole source of truth, nor thinks he must carry out what he presupposes as unassailable certainty to the point of tangible distinctness.] Lotze. Allgem. Physiol. p. 18.

"hydrogen chloride is an acid," "alum is a salt." In each of these judgments the subject displays a dialectical unity of the concrete and the simple, seen from the standpoint of the thing, while the predicates in like manner give a dialectic of the abstract and the concrete, seen from the standpoint of the concept. According to real analysis iron is a simple substance, while hydrogen chloride and alum are compound substances, the latter more compound than the former; but dialectically seen, the alum is precisely a simple because it is a concrete, for concretion, chemical combination, does not give us an aggregate of unlike constituents but an existing unity, a body of its own, a thing by itself. On the other side the iron is, precisely because it is a simple substance, concrete; for as a simple substance, as something existing by itself, it is a unity of thing, and as a unity of thing a concretion of various properties and forces. If we then consider the predicates metal, acid, salt, it follows of itself that each of these categories must within its circle be the general. In so far as being a metal is common to all metals, the category metal is in this generality abstract; but since being a metal is at the same time identical with being, say, iron — to such a degree indeed that iron as iron is neither more nor less than precisely a metal — there are in the category metal exactly as many determinations of concept as there are in the thing iron necessary determinations of existence. The category, the general, is therefore concrete not merely *although* but also *because* it is abstract. And since in every judgment the subject must reflect its predicate and the predicate its subject; since the subject with its concrete simplicity has its general in the predicate, and the predicate in its concrete generality has its simplicity in the subject (the hydrogen chloride in its concrete simplicity has its general in being an acid, the acid in its concrete generality its simplicity in being hydrogen chloride); since both, subject and predicate, meet in the particular (being alum with being a salt — for by being a salt of its own, namely a double salt, the alum passes beyond mere objective simplicity and becomes a particular, and by being a double salt, namely alum, the salt as salt likewise passes beyond the merely general and becomes a particular) — it follows from this that the form of particularity must be precisely the content-rich form of concretion: the concrete is a particular just because it is a simple and a general. But when the dialectical system of judgments and inferences that allows of being developed from this shows us, from the side of the concept, that the concretely rational which is the subject-science's essential content must always be capable of being appropriated by philosophy, then from the side of existence we need only hold fast the total impression of the present section in order rightly to grasp what it means that chemistry — without whose real analyses such categories as metal, acid, base, salt and the like would be only hovering, empty abstractions answering little to actuality — that chemistry, we say, with its own observations, its own aim and its own method, is and must be a distinct science.

b) The Problem of Change in Chemistry.

That substances, things, bodies are subject to changes, experience shows; but the task is to find out the laws of the changes, to know their character and essence, for the changes are of most various kinds. When clay and sand are stirred in water, a change takes place; when common salt is dissolved in water, a change takes place; when zinc is dissolved in dilute sulphuric acid, a change takes place; but these changes differ among themselves: the first is mechanical, the last chemical, the middle one stands at the transition. The change by which oxygen and hydrogen are combined into water, or by which water is separated into oxygen and hydrogen, is pervasive in another way than the change by which ice passes into water, water into steam. The ancients, who had no clear concept of the nature of chemical combination and consequently no clear concept of change either, took the formation of the new product for a kind of coming-to-be out of nothing: they did not see that the body which arose from the combination must, however striking the change might be, have the combined terms for its constituents; that every substance which issued from a combination must, however surprising its appearance, nevertheless have been contained beforehand as a constituent in the combination itself. Wherever chemists before the sixteenth century speak of a certain body being contained as a constituent in another, what is generally thought of is the mixture (ores contain the metals as constituents, common salt is a constituent of sea-water, and so on), not chemical combination. What is brought about by chemical combination is, on that view, more than a change: it is a mystical transformation, a marvel of nature giving existence to a non-being. Thus Paracelsus held that the copper which appeared out of “the blue vitriol” in which “metallic iron” had been laid arose, not (as it appears to us) by the separating out of the vitriol (sulphate of oxide of copper), of which it forms a constituent, but by a mode of formation standing in essential accord with the theological explanation of a *conservatio* answering to an *annihilatio rerum.* A clearer insight into the relation between the product and its constituents, between the educt and the combination from which it has been separated, first forms itself in the seventeenth century.

Attempts to make plain wherein chemical affinity and chemical combination really consist soon led to a closer determination of the general foundation, with the conditions necessary for the changes, and thereby to a philosophical-mechanical presupposition that early found its expression. “The Hellenes,” says Anaxagoras already, “wrongly assume coming to be and passing away, for no thing comes to be or passes away; rather, things are composed of subsisting things, and so they would more correctly call coming to be composition and passing away separation." The mechanical presupposition was already among the ancients (Leucippus, Democritus, Epicurus) distinctively developed in the doctrine of σώματα ἐλάχιστα, the eternal indivisible elementary parts, *the atoms*. Epicurus' atomic doctrine was revived in the seventeenth century by Gassendi and made the basis of a mathematical treatment of physics. As for chemistry, there are found already in van Helmont expressions such as *atomi gas ob nimiam exiguitatem invisibiles*; but R. Boyle is probably the first who really sought to explain chemical combination by the atoms, the particles of the various masses, ranging themselves closely up against one another. *Non videri absurdum concipere*, he says in »*Chemista scepticus*« (1661), *in prima mixtorum corporum productione materiam illam universalem, ex qua ea inter cæteras universi partes constabant, in exiguas particulas, diversis magnitudinibus et figuris instructas, varieque motas, actu divisas esse* ... The atomic doctrine could, however, become fruitful for chemistry only with the introduction of quantitative analysis; after several wavering attempts, J. Dalton's work *New System of Chemical Philosophy*, which appeared in the year 1808, at last turned the scale. Of Dalton's atomic doctrine we bring out here the following fundamental features. The atoms, the last products into which all bodies resolve themselves when the division is thought of as carried on to the utmost limit, are for one and the same body of equal weight and equal size. "Whether atoms of different elements have the same figure is uncertain; but since every atom must be assumed to be surrounded by a sphere of heat, so that immediate contact between two atoms, be they otherwise of like or unlike kind, never takes place, every atom with its sphere of heat may be regarded as spherical. Chemical combinations arise through the atoms of the various bodies forming, by their mutual arrangement, new atoms, which are always at a greater distance from one another than those by whose union they are themselves formed. If two elements, A and B, combine in different proportions, then the compounds thereby arising must be found in the following series:

1	atom	A	with	1	atom	B	to	1	atom	C_0
1	„	A	—	2	„	B	—	1	„	C_1
2	„	A	—	1	„	B	—	1	„	C_2
1	„	A	—	3	„	B	—	1	„	C_3
3	„	A	—	1	„	B	—	1	„	C_4

and so forth."

The chemical doctrine of change is then, on such a foundation, essentially transformed into an atomic doctrine; and in order now to investigate what consequences are contained in this, we shall in what follows treat of the categories: atom and equivalent, atomic constitution and crystal form, atom and substance.

α) Atom and Equivalent.

As a material point the atom is sensuous; as a point it is, on account of its extraordinary smallness, non-sensuous. The existence of such a non-sensuous sensuous must in the doctrine of experience be hypothetical. The atom's hypothetical existence acquires a scientific validity in that it furnishes a fulcrum for the exact interpretation of what is an object of immediate observation, and so far not hypothetical. How much an atom of water weighs by itself can as little be settled by immediate observation as it can be settled by immediate observation how large an atom of oxygen is, or what figure an atom of hydrogen has; but it can be established by observation and calculation that when, say, 9 pounds of water are separated into oxygen and hydrogen, one will obtain 8 pounds of oxygen and 1 pound of hydrogen. It can further be established that when the 8 pounds of oxygen at a determinate degree of heat and under a determinate pressure fill a cubic feet of space, the one pound of hydrogen will at the same degree of heat and under the same pressure fill $2a$ cubic feet; in other words, that the quantities by weight of oxygen and hydrogen will be as 8 to 1, while their volumes are as 1 to 2. What holds here of 9 pounds holds of course also of 9 drachms, of 9 ounces, and so on; one may then conceive the part of water so small that it becomes an "atom": always will every atom of water admit of being separated into oxygen and hydrogen so that the oxygen makes up $\frac{8}{9}$ and the hydrogen $\frac{1}{9}$ of its weight. If we only knew how much an atom of oxygen and how much an atom of hydrogen weighs, we could very easily determine the weight of the atom of water; but since no atom can be weighed as an atom, since the absolute weight of atoms is quite unknown to us, the atomic doctrine, if it is to be grounded on relations of weight, must begin with arbitrary presuppositions.

How arbitrary the presuppositions are may be seen from the example cited. Since the hydrogen furnishes only one unit of weight, it lies near to let this unit coincide with the atomic unit, so that the atom of hydrogen is for the present denoted by 1; but what are we now to judge about the oxygen's 8 units of weight? If one proceeds from the presupposition that all atoms, without regard to the mutual difference of the elements, are equally heavy, then it follows that the atom of water must contain 8 atoms of oxygen and 1 of hydrogen; if one proceeds on the other hand from the presupposition that an atom of water is composed of 1 atom of oxygen and 1 of hydrogen, then it follows that 1 atom of oxygen must be 8 times as heavy as 1 atom of hydrogen. Each of these assumptions is in the present connection in and for itself equally probable; and if the chemical atomic doctrine is nevertheless built on the latter, with rejection of the former, then there is in this something arbitrary. This arbitrary element can then further be combined with another presupposition, probable enough in many cases but by no means directly demonstrable. Berzelius, to whom the stoichiometric development of the atomic doctrine is chiefly due, proceeded, so far as the gaseous substances are concerned, from the presupposition that equal volumes must at equal pressure and equal temperature contain an equal number of atoms. According to such a “volume theory”[*)] it must therefore be, since 1 measure (volume) of oxygen is in 2 measures of aqueous vapour combined with 2 measures (volumes) of hydrogen, that the number of hydrogen atoms is twice as great as that of the oxygen atoms; but if the number of hydrogen atoms is twice as great as that of the oxygen atoms, so that in the atom of water there are 2 atoms of hydrogen to 1 of oxygen, then it is not the single atom of hydrogen, H, but the double one, Ħ = 2H, that weighs 1 when the atom of oxygen weighs 8; and the notation for the atom of water accordingly becomes, not H + O = Ḣ, but 2H + O = Ħ̇. If the unit of weight represented by H is now multiplied by 12.50, the quantity of weight represented by O becomes = 100.00; and now the relation between atom and equivalent, atomic number and equivalent number, is, so far as the leading thought goes, easy to grasp.

As a category the atom denotes the indivisible, by which there must here at the same time be understood something subsisting on its own, something absolute in itself: the one atom is so far no concern of the other. The term equivalent, by contrast, leads the mind to a mutual relation of dependence in which the one term outweighs the other, takes the place of the other and replaces the other. As atoms, the equivalents are absolute, mutually independent real points indifferent to one another; as equivalents, the atoms are relative, mutually dependent parts replacing one another. In the relations of combination and of replacement the atoms make known their qualitative difference in a quantitative way. The common constitution, the general quality, is weight: the atoms are all heavy. But since the atoms of the one element weigh more than the atoms of the other, since the difference of quality here coincides with the quantifying difference of weight, the common constitution, the general quality, becomes articulated into a system of particular determinations of difference stated by quantities, and these mutually

[*)] The volume theory is restricted by the fact that the volumes of compound gases stand indeed in a determinate, but yet a very different, relation to their equivalents.

answering, mutually equalizing determinations of difference are expressed in the so-called atomic and equivalent numbers. From the notation $\dot{H} = 2H + O$ it is seen that, since 2 atoms of hydrogen are combined with 1 atom of oxygen, the atomic number of hydrogen must be 6.25 when that of oxygen is 100; but since the atom of hydrogen is doubled, since 2 atoms of hydrogen are combined into 1 equivalent, the equivalent number of hydrogen must be twice as great as its atomic number, that is, 12.50. Wherever, on the other hand, the atom coincides with the equivalent without being doubled, the atomic number of course coincides with the equivalent number as well.

What is hypothetical in the atomic doctrine now acquires, by means of the doctrine of equivalents, significance through the punctuality with which the analyses resting upon it are carried out — a punctuality characteristic of the doctrine of change as a whole. However many formations, transformations, educts and products may arise from the combination and separation of substances, the swarm of changes does not lose itself in a chaotic indeterminacy according to which "everything can come of everything and become everything"; here there is something unchangeable in the changeable: here there are measure and boundary. The combination of substances is indeed not restricted to a single proportion; but even when the combinations take place by multiples of equivalents, these multiples are on the whole to be referred to simple and determinate proportions. Leaving the organic bodies out of account, it holds as a rule that 1 equivalent of the one substance enters into combination with 1, 2, 3 equivalents of the other, or 2 equivalents of the one with 3, 5, 7 equivalents of the other. How products formed from the same constituents acquire, precisely by measure and boundary, a character answering to the relations of equivalence, we can — since oxygen combines with all elements hitherto known except fluorine — best illustrate by the so-called degrees of oxidation. A substance combined with oxygen is an oxide; thus manganese, Mn, a grey-white, slightly lustrous and very brittle metal, combines with oxygen in 5 different proportions: $\dot{M}n$, $\dddot{M}\ n_2$, $\ddot{M}n$, $\dddot{M}\ n$, and $\overset{\cdots}{\overset{\cdot\cdot\cdot\cdot}{M}}\ n_2$, and so forms 5 different oxides of manganese. But why is it that substances standing in affinity cannot, as Berthollet in his day assumed, enter into combinations in all possible proportions? What does it mean that the combinations take place at intervals and as if by leaps? — Existence's need of measure and boundary! — Seen in the concept, existence everywhere demands a separation in which the quality mediated by quantifying turns the scale. In mere gradation the qualitative becomes a moment for the quantitative, so that the boundary can be passed again and again and the change go on to infinity; in measure (μέτρον), by contrast, the qualitative has so determinate a preponderance over the quantitative that a boundary is thereby set which, gradation notwithstanding, cannot be passed. The difference between the 5 degrees of oxidation cited is thus more than a difference of degree: it is a difference of character determined by measure.

The opposition of the electronegative and the electropositive is articulated by fixed measures in such a way that the acid compounds are again separated into acids of different kinds (thus $\overset{\cdots}{M}\,n$ is manganic acid, $\overset{\substack{\cdots\\ \cdots\cdot}}{M}\,n_2$ permanganic acid), the basic into bases of different kinds (thus $\dot{M}n$, manganous oxide, and $\overset{\cdots}{M}\,n_2$, manganic oxide, are both bases, but of different character); indeed, by opposition to the acid and the basic even the neutral becomes characteristic. But since measure and boundary are seen also in the fact that not all bodies have equally many degrees of oxidation, the articulation varies with the quality of the combined terms[*]).

The many-sided way in which the formations are individualized by measure and boundary shows plainly enough that the changes in question are not merely to maintain a stream of alternations but to uphold a law-governed order of things. The single and the peculiar must everywhere be resolved into the general; only on this presupposition can the general determine the particular at every single point, and the totality satisfy the concept. That the chemical affinities stand under the influence of heat expresses the general; but the reciprocal relation between the action of heat and the great multiplicity of chemical formations first shows itself articulated when we see, for example, how characteristically the specific heat of the elements answers, by the law discovered by Dulong and Petit, to their atomic weight. If to warm 200 pounds of sulphur by 1 degree one needs the same quantity of heat as to warm 350 pounds of iron or 1294.50 pounds of lead by 1 degree, then it is already understood from this how distinctively the particular manifestations of bodies’ mutual affinity must be determined by their relation to heat, since the manifestation is determined by the atomic weight, and the specific heat of the elements stands precisely in inverse proportion to their atomic weight[*]). A modification in this as yet only provisionally determined theory will of course modify, but not cancel, the articulation denoted by it.

[*])Different degrees of oxidation of acid bodies give: hyposulphurous, sulphurous, under-acid, acid, per-acid (hyposulphurous acid, sulphurous acid, sulphuric acid); different degrees of oxidation of basic bodies give: suboxide, protoxide, intermediate or double oxide, peroxide.

[*])By Regnault’s discovery it has further been established that the case stands the same way with bodies of the same atomic and similar chemical composition; that, for example, 1 atom of gypsum (sulphate of lime) and 1 atom of heavy spar (sulphate of baryta) have the same specific heat; and finally that a compound atom’s specific heat is equal to the sum of the quantities of heat sufficient to raise the temperature of its single atoms by 1 degree. — “*The atomic heat*, the product of the specific heat with the atomic number, is therefore the quantity of heat which one atom of the compound contains, and this is equal for compounds of like composition.” J. Thomsen. Kortfattet Læreb. i den uorganiske Chemie. Copenhagen 1850. p. 86.

β) Atomic Constitution and Crystal Form.

The atom has not only, as equivalent, a certain relative weight; the equivalent has also, as atom, a certain determinate size and figure. How the matter stands with the figure and size of atoms must then best be inferred from the form of delimitation a body assumes when the transition from the fluid to the solid state takes place in such a way that its atoms can arrange themselves in full accord with the mutual attraction they originally have for one another — that is, from the *crystal form*.

Haüy, the real founder of crystallography, proceeded at the beginning of this century from the presupposition that to every distinctive crystal form there must answer a distinctive chemical composition. His theory of the primitive forms determined by *molécules intégrantes*, and of the lawfulness with which the layers of the molecules altered — a lawfulness from which the derived forms allowed of so natural an explanation — grounded a view which at first seemed bound to find unconditional confirmation in the facts of experience. Haüy really did succeed, with the theory's help, in distinguishing minerals which had hitherto been held identical, and in predicting from the crystal form what analysis afterwards showed, a difference in their chemical composition; in this way he distinguished heavy spar from strontian earth, dioptase from emerald, epidote from actinolite, and so on. But Haüy's view was soon to undergo a critique and suffer a considerable restriction. For it was proved (Mitscherlich) both that substances of different kinds can crystallize in the same form (isomorphism), and one and the same substance in different forms (dimorphism, polymorphism).

To understand the discussions belonging here we must carefully note that it is not only the term "chemical composition" that is ambiguous, in so far as one may think one-sidedly now of the combined constituents themselves, now of the proportion in which they are combined, but that there is also in the talk of "the same form" and "different form" an ambiguity which must first be removed. Pure crystallography distinguishes between fundamental forms and derived forms, so that every distinctively laid out crystal system is again divided into as many relative systems as it has fundamental forms. Since the crystal form is made up of plane faces, such mathematical bodies as the cube, the rhombohedron, the square octahedron and the rhombic octahedron may be chosen as fundamental forms[*)]. The cube has 8 corners, 6 faces, 12 edges; if now "touching faces" are laid in the cube's 12 edges, a figure bounded by 12 rhombs is formed, the rhombic dodecahedron; if "touching faces" are next laid in the rhombic dodecahedron's 24 edges, a figure bounded by 24 trapezia is formed, the trapezoidal icositetrahedron, and so on. If

[*)] G. Forchhammer. Grundtrækkene af den rene Krystallographie. p. 7.

we start from the rhombohedron, a figure formed of 6 congruent rhombs, as fundamental form, and set it upright on one of its corners, its projection will form a regular hexagon; and if the regular hexagon is moved so that during the motion it remains parallel with its first position, a six-sided prism is formed. The six-sided prism, an infinite form whose delimitation rests essentially on a combination, therefore belongs among the derived forms of the rhombohedral system; and so forth.

When isomorphism now presupposes that two substances of different kinds crystallize in the same form, then by what is of different kind one thinks chiefly, though not exclusively, of the constituents, and by the same form of the system of forms answering to a determinate fundamental form. Likewise, when dimorphism is in question, one must by the different forms expressly think of forms belonging under two different systems. A substance is, for example, not dimorphous because it crystallizes now in the rhombic dodecahedron (the garnetohedron), now in the trapezoidal icositetrahedron (the leucitohedron), since both these forms belong under the same system; carbon, on the other hand, is dimorphous, since as diamond it crystallizes in the cubic system, but as graphite in the rhombohedral.

If we now understand by atomic constitution the way in which the elementary parts in a body are bounded, arranged and combined, it will also be understood that there must here be a distinctive connection between the atomic constitution and the crystal form. As for the single atoms answering to the elements, polymorphism may arise from the same atoms arranging themselves differently under several different crystallizations; on the other hand isomorphism allows of explanation partly — as is sometimes one-sidedly emphasized — directly from the atoms in two or more different elements being of equal size and equally shaped, and partly also from the atoms, though of unequal size, standing in figure and size in such a relation that a group of the smaller atoms may be conceived as congruent with the single larger one. As for the atomic compositions answering to compound bodies, the number of possibilities is the greater the more alternatives the compositions in question offer.

Only bodies of like crystal form can in the proper sense crystallize together[*)]. If two salts of different crystal form crystallize in one and the same liquid, the crystals of the one develop as a rule as perfectly as though the other were not present at all. “If Epsom salt and saltpetre are dissolved in some warm water, and the liquid saturated with the two salts is decanted, then, as the liquid gradually cools, crystals of Epsom salt and

[*)]“In those cases where two salts of different crystal form crystallize together, it always appears that the mixed crystal’s form is like that of one of the two salts, and that its composition is like that of this last.” Liebig.

crystals of saltpetre arrange themselves side by side; but the single crystals of saltpetre contain no Epsom salt, the single crystals of Epsom salt no saltpetre. It stands otherwise, however, with Epsom salt and nickel or zinc vitriol; when both crystallize in one and the same liquid, one observes no separation of zinc vitriol and Epsom salt; the crystals that arise contain zinc vitriol and Epsom salt together in all possible proportions, which follow the quantity in which both salts were present in the solution. But Epsom salt, zinc and nickel vitriol also possess the same crystal form. Thus the crystal of Epsom salt looks like white nickel vitriol, the nickel vitriol like green Epsom salt." To have the same crystal form is so far a condition; but the likeness between two bodies' crystal form is not without more ado a ground for their crystallizing together: here as a rule a likeness in chemical composition must also be presupposed. A crystal of sal ammoniac has, for instance, the same geometrical form as a crystal of alum; nevertheless they crystallize in one and the same liquid separated from one another: the crystals of alum contain no sal ammoniac, the crystals of sal ammoniac no alum — "plainly," says Liebig, "because the force which draws parts of alum to parts of alum, or parts of sal ammoniac to parts of sal ammoniac, is, despite the same form of the crystal atoms, far greater than the attractive force operative between the parts of sal ammoniac and of alum"; for by its constituents the atom of sal ammoniac has only two, the alum on the contrary thirty compound atoms: "a more unlike constitution cannot be conceived." — Even the finer modifications in the crystal form allow of being referred to modifications in the atomic constitution. Exact measurements of the crystals have established that the uniform compounds of isomorphous bodies, those which replace one another without the crystal form being altered, do not always show perfectly the same form, so that the angles which the facets make with one another are not always quite identical; such divergences all allow of being explained by considerations which rest in each and all upon the atomic theory.

"Acquaintance with the isomorphous substances," says Liebig, "puts beyond all doubt the fact that their mutual replacement in compounds, without their crystal form being altered, rests on their atoms possessing the same form and being of equal size; and when, as one substance replaces another, we see that the compound's crystal form becomes a different one, we must presuppose that this alteration rests on this other body's atoms either possessing a different form or not filling the same space in the compound. By this presupposition the specific gravity is explained in a very simple way. When, then, lead at an equal volume weighs more than iron, iron more than sulphur, iodine more than chlorine, this rests either on the atom of iodine being heavier than the atom of chlorine, or on there being in the same space a greater number of atoms of lead than, say, of iron.

But chlorine and iodine are mutually isomorphous; we presuppose that their atoms are of equal size and possess the same form; and when the same number of atoms of iodine and of chlorine is present in the same volume of iodine and of chlorine, then in truth their specific gravities must stand in the same relation to one another as their equivalent numbers or atomic weights*)."

"The atomic weights or equivalent numbers of isomorphous bodies must, when divided by the specific gravity, give one and the same quotient, because these bodies contain in the same volume the same number of atoms*); if the number is different, or if the atoms diverge in their form, shape and size, this divergence will also make itself known in these quotients." — Whatever modifications, extensions or restrictions the view thus stated may undergo, it is nevertheless seen from it that the reciprocal relation between atomic constitution and crystal form will always continue to show itself articulated.

γ) Atom and Substance.

Such words as matter, substance-stuff, substance and the like are often used indiscriminately in free exposition; but this is of course no obstacle to the same expressions being distinguished by very essential marks when their meaning is to be fixed categorially. If we set matter and stuff side by side and see in both an expression for the sensuous, it should not be overlooked that matter denotes chiefly the general, stuff the particular. This distinction has its interest in the history of chemistry in that, whereas matter in abstract generality (*materia prima*) cannot be observed at all, every assumed stuff on the contrary makes claim to be presented by itself — if not always immediately, then mediately (hypothetical radicals) — and thereby made an object of particular investigation. So long as phlogiston passed *in abstracto* for an *ens sui generis* it played a great part; but when as an actual stuff, a particular matter, it was to be exhibited and expressly demonstrated, the illusion was discovered.

*)"If we conceive in a cubic inch of space an equal number, say 1000 atoms of iodine or of chlorine, then the specific gravity plainly expresses the relation of weight between their atoms; if a cubic inch of iodine weighs 4948 grains, then a cubic inch of chlorine must weigh 1330 grains; $\frac{1}{1000}$ cubic inch of iodine, in which there is 1 atom of iodine, would thus weigh 4.948 grains, $\frac{1}{1000}$ cubic inch of chlorine, in which there is 1 atom of chlorine, 1.330 grains. But chlorine and iodine are mutually isomorphous; we presuppose that their atoms are of equal size and possess the same form. To separate out 4.948 grains of iodine from a compound and replace what was separated with chlorine, neither more nor less than 1.330 grains of chlorine would be necessary. The specific gravity of iodine stands to that of chlorine as 4.948 to 1.330, or, what is the same, as their equivalents 12.6 of iodine to 3.52 of chlorine." Liebig.

*)Chlorine's atomic weight 3.52 divided by its specific gravity 1.330, and iodine's atomic weight 12.6 divided by its specific gravity 4.948, are two quotients which, each by itself, give roughly the same number 2.5. — Cf. J. Thomsen. Læreb. i den uorg. Chem. Copenhagen 1850. pp. 83–84.

If matter and stuff are next set against substance, a still more conspicuous difference shows itself; for whereas matter and stuff have above all the suffering, the indifferent, the yielding, the passive for their common mark — matter in so far as it is opposed in an abstract way to force, stuff in so far as it stands with its accidental form ready to receive any other form whatever — substance on the contrary has the forceful, the effective, the active for its distinguishing mark; and whereas stuff represents the changeable, and so far the possible, substance represents the unchangeable, the permanent, what is actual in itself. That this difference between stuff and substance is nevertheless grounded in the identity of both is easy to show: substance is stuff-like, stuff is substantial. That stuff is substantial can be seen from any material. Wood, stone, iron undeniably allow of being worked for different uses, and so of being treated as stuff; but every such stuff reacts again against the working, so that the one cannot be treated like the other; and in this independence the substantial shows itself. As for the relation between the permanent and the changeable, the difference-unity of stuff and substance is seen most clearly in the thing. Ice, water and steam are at once the same stuff and the same substance, and yet not without more ado the same thing. That the substance is here a stuff is seen from its having been able in its changeability to assume three such different forms of state; that the stuff is a substance is seen from its having been able through the changes to maintain its subsistence, its chemical character, $\dot{H}$, and so to keep itself essentially unaltered. If ice, water and steam are brought under the category thing, then each of the three states is denoted by a set of corresponding properties; and with the alteration of the properties the thing must surely be altered: ice, water and steam must therefore be different things. But then again some of a thing's properties are fundamentally essential, others less essential, others inessential; with the less essential and the inessential properties, then, alterations must be able to take place without the thing itself being essentially altered. If, then, the chemical properties are posited as the fundamentally essential ones, and the alterations of property brought about by the forms of state as inessential, then ice, water and steam are after all at bottom the same thing. By maintaining its subsistence under an alteration of properties the thing points to the substantial; by being altered along with the alterable properties it points to the stuff-like. The thing, the existing stuff-substance, is as the restless unity of stuff and substance subject to the laws of change.

By a critical separation of the changeable and the unchangeable the finite relation between stuff-substances of different orders is to be more closely determined. As representing what is absolutely unchangeable in itself, the elementary atoms are then, from chemistry's standpoint, the first and proper substances, substances in the strictest sense of the word; as the masses consisting of such single substances, the elements, the chemical elements, are already composite and so far derived substances. How change forces its way at once into these composite “simple substances” is recognizable already from their assuming different forms of state; still more from their being able, in the solid state, to alternate between the amorphous or massive form, the crystalline form and the crystal form proper; and, as regards carbon for instance, in a still deeper sense from one and the same element being able, by the mode of crystallization, to divide into two substances — for the difference between diamond and graphite may rightly be called substantial. In existential immediacy substantiality seems to be recognizable directly by duration, by the endurance with which an existing thing defies disturbing influences and resists alterations. Seen from this standpoint not only would a rock crystal be more substantial than a piece of ice, but a piece of ice at the North Pole even more substantial than a piece of ice in the temperate zone. If the substantial is to be grasped in a concept, substantiality — essential independence — must on the contrary be determined by the existing thing’s originality, its lesser or greater independence of other existing things: the derived has, for example, less substantiality than that from which it is derived, the composite less than the constituents of which it is composed, and so on. Essential independence is then something other than immediate duration. For duration can be secured by circumstances, by conditions provided from without, so that what is in itself fragile can last a very long time; the greater essential independence, and so the higher degree of substantiality, must so far be recognized by the greater force and endurance with which one existing thing, under outwardly equal conditions, resists influences coming from without more than another does. If we look to the concept of substantiality, then it holds that the more composite, and so the more derived, a substance is, the more it belongs to the perishable. Thus the double salt, however hardened and durable it may show itself in its petrifaction, is by its concept less substantial than the single salts of which it consists, for these, as the more original, can subsist without it, while it cannot subsist without them; for the same reason acids and bases, though less substantial than the elements from which they are formed, have more substantiality than the single salts, which can arise only by their union.

From the most various sides the reciprocal relation of the changeable, the relatively permanent and the absolutely unchangeable allows of being referred to the figure, grouping and arrangement of elementary and composite atoms. If polymorphism can bring it about that out of one stuff (titanic acid, say) several substances are formed (rutile, anatase and brookite), isomorphism can in turn bring it about that different stuffs alternately form one and the same substance, the substantiality being essentially determined by a typical formula. Before isomorphism was discovered, there showed themselves in every attempt to order minerals according to their constituents and composition countless entanglements and difficulties; "the most conscientious chemists contradicted one another with respect to the composition of the best characterized minerals. Thus in the garnet from Arendal over 13 per cent of magnesia was found, which was wholly lacking in the garnet from Fahlun, from Vesuvius, and so on; in the noble garnet analysis made out 27 per cent of alumina, of which not the slightest trace could be found in the yellow one from Altenau. Which constituents, then, belong to the garnet? How is it really composed? All this has resolved itself in a very simple way: where the alumina was lacking, the isomorphous oxide of iron was present; where the magnesia was lacking, there was isomorphous lime. It turned out that the garnet contains alternating quantities of isomorphous oxides, such as oxide of iron and alumina, or protoxides of lime, of manganese and of iron, which are able to replace one another without the compound's form being altered." (Liebig).

The different mode of arrangement of the atoms can of course as little be an object of immediate observation as their size, figure or weight; but what substantial significance must be attached to the way in which the atoms are conceived to be grouped, the influence which the crystal form exercises on the determination of the physical properties can alone teach us. The crystal forms are determined chiefly by the axes, the central lines about which the whole formation is regularly laid out. That the axis really furnishes the essential fulcrum for the arrangement — which must therefore differ according as it takes place parallel with the axis or perpendicular to it — is confirmed from many sides. The determinate directions along which the crystal cleaves stand in relation to the axis; the same holds of the passage of light, which, since it rests on the vibrations of the parts of the ether, is precisely dependent on the arrangement of the atoms. Parallel with the axis tourmaline is, for example, almost opaque, whereas when it is viewed in a direction perpendicular to the axis it is transparent; dichroite, which is blue in the direction of the principal axis, is brownish-yellow perpendicular to it, and so on. What holds of light holds in like manner of heat. Crystals of the cubic system, "the spheroedric system," which has three kinds of principal axis, conduct heat equally well in all directions; whereas calcspar and quartz, belonging to the rhombohedral system, which has only one principal axis, show a conductivity which is greatest parallel with the principal axis and least perpendicular to it.

A still more substantial influence is acquired by the atoms' mode of arrangement in the so-called isomeric bodies, "substances which, with the same constituents in the same quantity, show different properties." How, then, can a product alter itself and become another body, another substance, although the constituents in both a qualitative and a quantitative respect continue to be the same? The substantial alteration, answers the atomist, is brought about by the transposition and the absolute quantity of the parts. — It is chiefly in organic chemistry that the isomeric formations frequently occur. As regards the mode of composition, chemists have on the occasion of the isomeric formations seen themselves called upon to distinguish between *empirical* and *rational* formulae — a distinction which contributes still further to articulating the reciprocal relation between experience and thinking. Let a given product be, say, $C_6H_{12}O_4$, thus consisting, by the empirical formula, of 6 atoms of carbon, 12 atoms of hydrogen, 4 atoms of oxygen: what substance is it now? There is in this stuff a possibility of two substances, a doubleness which finds its explanation in the empirical formula's being separated into two rational formulae: $C_4H_6O_3 + C_2H_6O$ — the substance is then acetate of oxide of methyl — and $C_2H_2O_3 + C_4H_{10}O$ — the substance is then formate of oxide of ethyl. The substantial difference rests on a rearrangement of the parts, for which reason such isomeric formations are also called specifically metameric. In like manner the alteration of substantiality is in certain cases derived from the compounds containing the same stuffs indeed in the same relative but not in the same absolute quantity, so that polymeric formations arise; thus $C_8H_{16}O_4$ gives the same percentage composition as $C_4H_8O_2$, but since the quantity of atoms in each of the elements is different, the difference between the two products may be substantial.

Under all alterations, all formations and transformations, the stuff-like alternates with the substantial, being stuff and being substance, the difference between essential and inessential everywhere resting on the oppositions that form themselves[*)]. Thus carbon

[*)] "One of the finest examples of the transposition of a nitrogen-free body on account of a disturbance that has set in is given by *aldehyde*. This colourless liquid, which can be mixed with water and is volatile to such a degree that it comes to boil already in the warm hand (at twenty-one degrees), possesses a suffocating smell and has the property of attracting oxygen from the air with great avidity, whereby it passes over into acetic acid. Brought into contact with caustic potash lye it condenses into a brown resin. These are quite salient properties, but they are in the highest degree unstable. For in its formation and preparation the aldehyde cannot be protected from contact with the air's oxygen. If we put it in a glass vessel which we close air-tight by fusion, one or more of its smallest particles is nevertheless on the point of taking up oxygen; they are in a state of action which must necessarily cease when the oxygen is shut out. The process of oxidation in the aldehyde does indeed thereby find a boundary, but the disturbance that has set in in the equilibrium of its elements' attractive force continues to propagate itself. By the motion that has set in in those particles of the aldehyde which are oxidized, the state of rest in the aldehyde's neighbouring atoms is disturbed, whereby its elements order themselves into a new group wholly different from the original one; the motion of these particles propagates itself again to the nearest, and at last to all the others, so that after some days or weeks one no longer has aldehyde, or a body which by its properties possesses even a distant resemblance to aldehyde. We now have in the flask a liquid with which water can in its difference from sulphur is substance, but this carbon-substance is in turn as a stuff when diamond is opposed to graphite; thus the isomorphous constituents of garnet and of alum are, in so far as they are qualitatively opposed to one another, substances of different kinds, but in so far as they vicariously satisfy the respective formulae of garnet and of alum they are only stuffs, subordinating themselves in the compounds to the substantiality of garnet and of alum. According to a critically carried-through atomism, then, the elementary atom is the only original substance subsisting absolutely in itself, the unchangeable lying at the ground of all changes.

c) The Problem of Change in Philosophy.

If the relation between the changeable and the unchangeable, as chemistry has on the whole hitherto conceived and determined it, is first given in all its fundamental features, then this relation is next to be investigated and developed from another side in philosophy. A further development of the problem of change must also be in good accord with the chemical way of seeing, since chemistry, which in its character of subject-science cannot enter upon "speculations," must itself regard its whole atomic theory as a hypothesis. "There must," says Liebig, "be a cause which makes the coming together of the elements in other proportions impossible, which sets an insuperable obstacle to their diminution or increase. The fixed proportions are manifestations of this cause; but with these the domain of research is bounded. The cause itself cannot be observed with the senses, and can become an object only for speculation, for the spiritual faculty of representation. Now in setting out," he adds, "to develop the view that has at present become prevalent about the cause of the chemical proportions, I must beg that it not be forgotten that its untruth or truth has not the least to do with the law itself; this last remains, as an expression of experience, always true and does not alter, however the representation of its ground may change." (Chemiske Breve, 2nd ed. in the Danish translation, 5th letter).

For philosophy, on the other hand, the question of the reality of atoms is a vital question. That the atomic doctrine, consistently carried through, must end in materialism is easy to see; for in positing the atom as the unconditioned first, one at the same stroke

no longer be mixed, but which swims on its surface like oil. This liquid possesses an agreeable, ethereal smell; its boiling point lies sixty degrees higher than the aldehyde's; it no longer condenses with caustic potash into a resin, no longer passes over into acetic acid; but despite this great difference this body is by its composition nevertheless aldehyde." (Liebig, Chem. Breve, pp. 124–125). — In the compound $C^4H^8O^2$ we have, then, a *substance* which, in opposition to any other compound whatever, say $C^4H^4O^4$, is always aldehyde, and yet a *stuff* which by its substantiality is now aldehyde, now again neither aldehyde nor "a body which by its properties possesses even a distant resemblance to aldehyde": can the essence of changeability in the elementary, can the dialectic of stuff-substance, be more clearly stated?

ascribes absolute originality to matter. And if matter really has an absolute priority, then spirit, the Idea, everything ideal, must necessarily be something derived, something merely subordinate. Since, however, it is not matter but spirit as spirit that thinks, and in thinking asks after the original, the eternal first, it must always revolt thought to have to ascribe unconditional validity to an essence foreign and contrary to itself. It is therefore intelligible enough that the metaphysical atomism which has appealed to chemistry and physics has been contested by philosophers who strove to conceive matter as the derived and spirit as the original. Against the atomistic-mechanical view Kant in particular set his dynamical mode of explanation: instead of deriving the forces from matter, he derives conversely matter's existence from two mutually opposed forces — the attractive force (*vis attractiva*), by which matter's parts dispersed in space strive to approach one another, and the expansive force (*vis expansiva*), which makes matter fill space. Without entering upon a closer explanation of particulars, Kant expressed in full generality the thought that the chemical changes which take place as bodies combine rest not on the figure and mechanical motion of the smallest parts but solely on matter's attractive and repulsive force. This fundamental thought, further developed by Schelling, found through "natural philosophy" an entrance with a few brilliant chemists, who now attempted a more idealistic interpretation of what essentially went on in the chemical process; but as soon as one undertook to explain the phenomena in their particularity and determinateness, it was easily discovered that the dynamical view — according to which the constituents in entering a chemical combination were wholly and entirely to interpenetrate one another — is nevertheless not suited to any punctual analysis.

In order, then, to investigate how far an idealism of actuality immersed in punctuality is able to reconcile itself with what is essential in the atomic doctrine, we shall in what follows determine the atom as point of discretion, as point of function, and as substantial moment.

α) The Atom as Point of Discretion.

" 'According to the moment of its "being-in-itself" matter would be the single point; according to the moment of its "being-outside-itself" it would be, most immediately, a multitude of mutually excluding atoms. But since these, in excluding one another, relate themselves to one another, the atom has no actuality, and the atomistic as well as absolute continuity or infinite divisibility is present in it only according to possibility.' In this Hegelian conception it does indeed look at first glance as though thought penetrated matter. By the logical opposition of "Insichseyn" and "Aussersichseyn" and its dialectical treatment one is indeed spared the difficulties which atomism — that stumbling-block of speculation — otherwise brings with it when it is to be explained in a way satisfactory to the experts; but the difficulty is only evaded, not removed: as the element of sensibility asserts its right, the relation between matter's discretion and continuity takes up something unthinkable, something impenetrable to reflection; for the sensuous is the barrier of thought." (Phil. Propæd. pp. 47–48).

On the presupposition that matter is something independent over against thought, the barrier of thought, one has then in the opposite direction, out of respect for actuality, let the matter of fact rule; that is to say, one has restricted logic but forgotten dialectic: "thought thinks its barrier; the unthinkable is after all thinkable"! From the standpoint of the matter of fact one usually defends the atomic doctrine by appealing to matter's discretion, in appealing to the phenomena that point to this discretion. By taking the matter from this side Fechner, for example (Atomenlehre. Leipzig 1855), has made the task fairly easy for himself, for to demonstrate material discretion is fairly easy; but discretion presupposes continuity, a discrete is possible only by means of a continuum: to appeal to material discretion without giving an account of material continuity is unscientific. The question therefore becomes: how does the atom's independence stand to the reciprocal relation of discretion and continuity determined not merely by thought but by matter?

To immediate intuition, earth, water and air — bodies in the solid, the liquid and the gaseous state — present themselves as different continua; and it is precisely against this immediate representation of continuity that natural science has so emphatically asserted discretion. "In the solid state the parts, though separated by small intervals, are so attracted to one another that they offer perceptible resistance to whatever would move them out of their position; in the drop-liquid state the parts move among one another so easily that a slight influence causes a displacement; in the gaseous state, finally, the parts' mutual cohesion is = 0, they strive away from one another in all directions: vapours expand. In a pound of liquid water the parts of water are plainly closer beside one another than in a pound of steam, which at ordinary pressure occupies a space 1700 times greater; we can," it is said, "compress a volume of air into a space a thousand times smaller," and so on. It begins with the coarser discretion: the sponge has openings, every body has pores; these pores now become, as observation is sharpened, finer and ever finer, so that finite discretion seems at last to lose itself in the infinite.

The small intervals by which a body's smallest parts are separated are not continua of empty space; they are, as the doctrine of light and heat shows, filled with ether, and the parts of the ether again separated by small intervals. "Under refraction the refracted white ray spreads into a narrow fan of colours; this phenomenon can be brought into mathematical accord with the vibration hypothesis only when the parts of the ether are assumed to be sundered by finite discretion. For the narrow fan of colours rests on the different rays having different refrangibility, and on the distance of the parts of the ether not being vanishing in comparison with the breadth of a wave of light." (Propæd. p. 64). The intervals of the vibrating parts of the ether may then in turn be filled by a still finer fluid, whose parts may again be separated by still smaller intervals, and so forth. Where is the boundary now? Sound understanding says that there must necessarily be a boundary here; but since the understanding is, precisely under the impression of the matters of fact, in the service of sensation, and so far somewhat constrained and unfree, it says so in a rather back-to-front way. The discretion whose parts can be sensed is of course always finite, however sharply the eye be armed. Since the eye nevertheless sees for itself how the discretion can constantly become finer and finer, the sense will readily allow sound understanding to take one step beyond the limits of the sensuous while still within the sensuous, namely by making the discretion so fine and its parts so small that they must, though in their essence sensuous, remain non-sensuous. But more than one such step the understanding, in the service of observation, is not easily permitted to take; for when the discretion goes so far that representation cannot follow, giddiness begins. To guard itself against the giddiness, therefore, the understanding led by sensation dictates a boundary of discretion, or, as it is generally expressed, a boundary to matter's divisibility.

What is absolutely true in this way of considering the matter is that the discretion really must have a boundary; if the discretion has no boundary, it goes to infinity, and an infinite discretion cancels itself and turns over into continuity. That the boundary should on the other hand exist in order to prevent the division from going beyond an absolutely fixed and determinate boundary is an illusion. It is by no means the parts' smallness that essentially determines the boundary — as is expressly seen, too, from the fact that atoms of one element may, on the hypothesis, be larger than atoms of another. What is essential is that the atom, which sets divisibility a real boundary, itself furnishes a material continuum: only material continua, not mathematical points, are able to sustain a real discretion. Every continuum, on the other hand, whether material or abstractly mathematical, is in and for itself always divisible to infinity. "If," it is said, "matter really consisted of infinitely small parts, then it would be without weight, and a whole milliard of such parts could not weigh more than a single infinitely small part"; but objections of this nature rest on a misunderstanding already illuminated under "the analysis of the infinite" in an earlier section. Infinite divisibility means essentially nothing but that the boundary in the continuum in question can be set wherever one pleases. How easily the chemical atoms — those small masses of the absolutely indispensable continuity — fall back again to discretion is quite characteristic of the atomic doctrine. If, for example, the atoms of iodine and of chlorine have as isomorphous substances the same figure and size, then there must necessarily be pores in the atom of chlorine, since it weighs so many times less than the atom of iodine; but a porous atom is a discrete continuum.

Once the relation between continuity and discretion grounded in matter's essence is logically evident, the significance of what is material in the relation, the non-logical, will also be logically evident. In abstract logical form continuity and discretion everywhere reflect one another so that the one of these concepts always shines through the other, in so far as the one can be determined only as the other's presupposition, cancellation and boundary. In matter, the sensuous, the non-logical, such a reflection is on the contrary impossible; here continuity stands outwardly against discretion, here the one immediate continuum exists outside and beside the other, here space is posited, here finite discretion is realized. Finite discretion by no means requires that the corresponding continua should in their singleness be absolute; on the contrary! From the atom as real point of discretion to the atom in absolute unchangeability there is an infinite leap. In water the atom of water is just as real a point of discretion as the atom of oxygen is in oxygen. The objection that the atom of water is only a relative, the atom of oxygen on the other hand an absolute continuum, has in this connection nothing to signify: the atom of water has for the mass of water the same validity as the atom of oxygen has for the mass of oxygen. In order to defend the in itself true fundamental thought that the compound atom, the concrete point of discretion, is just as much a real unity, a material continuum, as the single one, the dynamical school committed, however, the blunder of presupposing an interpenetration of the combined constituents; for such an interpenetration is as unnecessary as it is inexplicable. For the atom of water to exist as a material continuum it is not at all necessary that the two atoms of hydrogen and the one atom of oxygen should materially interpenetrate one another, but only that they should by mutual attraction be so altered that they let their distinctiveness — as experience precisely shows — be resolved into one and the same common mode of acting and of being.

"To the senses," it is objected, "the compound atoms are indeed of like kind; but we know that they are nevertheless of unlike kind, since they are compounded. So long as the composition subsists, the compound atoms of course keep themselves unaltered; but we know that they allow of being altered by decomposition, whereas the elementary atoms, the true continua, discretion's outermost boundaries, are unalterable." Under this way of considering the matter more than one error lies hidden. In representing the single atoms as existing in the compound atom in such a way that the combination determined by the attraction

has exercised no influence at all on their mutually unlike constitution, one confuses the possible with the actual: so long as the compound atom, the concrete but nevertheless like-kinded continuum, is actual, the corresponding single atoms are in their unlikeness of kind only possible; for the unlike single atoms to become actual again, the compound atom must be decomposed, that is, must go back to the merely possible. When the matter is seen from this standpoint it is so far from being the case that the elementary atoms keep themselves in absolute unalterability, that they on the contrary show themselves to be just as alterable as the compound ones: they are altered in giving up, under the combination, what is unlike in kind and forming a compound like-kinded thing; they are altered again in becoming unlike in kind anew by the separation of the compound. If material discretion is subject to an incessant and many-sided alteration, then the corresponding points of discretion, the material continua, the single as well as the compound, must be subject to a corresponding alteration.

To prove the immediate and material unalterability of atoms from the phenomena of isomorphism is impossible; for partly an atom may well be qualitatively altered although the form of delimitation remains unaltered, and partly — and this is here decisive — the atom as point of stuff may under certain conditions very well alter its figure, since figure answers precisely to the filling of space, and the filling of space to the stuff-like, the material, the alterable. "One can," says Liebig, "hardly conceive anything more wonderful than that out of carbon and nitrogen there should proceed a gaseous compound (cyanogen) in which metals burn with development of light and heat as in oxygen gas — a compound body which by its properties and its behaviour is a simple body, an element, whose smallest parts possess the same form as an element of chlorine, bromine, iodine, in that it replaces them in their compounds without the crystal form being altered." — But from the fact that cyanogen ($N_2C_2 = Cy_2$) is isomorphous with chlorine, bromine and iodine, no one will infer what eternally unalterable figure either the atom of carbon by itself or the atom of nitrogen by itself must absolutely possess.

And what, then, does dimorphism mean, when even an element can be dimorphous? By what are we really to recognize the eternally fixed, unalterable figure of the atoms of carbon? "By an eternally fixed, unalterable crystal form!" But the crystal form is precisely alterable. “What is alterable rests on the ordering of the parts!” But when there are crystal forms which alter with the mode of arrangement, which then are the in themselves absolutely unalterable figures? It is possible that the atoms of carbon are eternal cubes, so that graphite’s rhombohedral form is due to “an ordering of the cubes”; but it is just as possible that they are eternal rhombohedra, so that the diamond’s cubic form proceeds from “an ordering of the rhombohedra.” Or might not a crystal atom with a cubic figure just as well be formed of eternal rhombohedra — if the rhombohedra, mark you, were sufficiently small — as a crystal atom with a rhombohedral figure of eternal cubes, if the cubes were sufficiently small? But if both cases are equally possible, then it is again just as possible that the atoms of which the crystal figure is formed may in both cases have a form quite indifferent to the figure itself, spherical, say. From every side, then, we get as regards the original figure of the atom by no means a finished actuality but a real possibility — that possibility whose conditioned realization determines finite discretion.

β) The Atom as Point of Function.

To clarify the relation between what is alterable and what is unalterable in the atom, the reciprocal determination of suffering and acting, the passive and the active, must be specially brought out. Seen from the passive side every atom is a point of stuff, and as point of stuff alterable; seen from the active side every atom is a point of function, and as point of function unalterable in essence: the atom is, as a functioning point of stuff, an alterable unalterable. If it holds of the masses oxygen, hydrogen, sulphur, nitrogen, chlorine, carbon and so on that as elements they must express their existence in their acting — that every single substance is in and for itself only what it is in its reciprocal action with other substances — then the same must hold of the atoms of which the masses consist. If there can be in no single atom whatever a being, a point so petrified, closed off, isolated and at rest that it should fall indifferently outside all acting and being acted upon, then the elementary atoms must themselves have their existence in reciprocal action; and if such a reciprocal action is everywhere an alternation of suffering and acting, then it follows that the atoms, at once suffering and acting, can also display their unalterability only by themselves being altered.

As the constituents of things the atoms are themselves things, small invisible things: every atom is a thing-cause. The atoms of different kinds — the atom of oxygen, of sulphur, of hydrogen, of carbon and so on — therefore each furnish a centre from which a corresponding system of effects proceeds; for though they can replace one another in the "proportions," the one element cannot on that account be transformed into the other, oxygen not into sulphur, hydrogen not into carbon. What nature works through the atom of carbon it does not work in that way through any other atom whatever: in the atom of carbon the carbon-cause, in the atom of hydrogen the hydrogen-cause, in the atom of oxygen the oxygen-cause, and so forth, is punctualized. It is therefore not in this or that material state, but in the whole system of distinctive law-governed effects, that the atom asserts its unalterability.

It is a great misunderstanding to suppose that an atom of carbon, an atom of hydrogen — in short, any single atom whatever — should show what it is in its essence only when it is held fast in an isolated state, much as a grain of barley or a pellet of sparrow-shot can show itself in an isolated state; and that then, if only it were possible to obtain the requisite magnification, one would in all masses, compounds and products duly see the swarm of unaltered elementary atoms, as one sees the single grains in a heap of corn. The isolated state is so far from being the most essential state for the atom that from chemistry's standpoint it might rather be called accidental. What is essential, what constitutes the determinate atom's distinctive nature, is not of immediately sensuous constitution: it is something unalterable that reflects itself in a stream of alterations. The atom as point of stuff an armed eye could, if the magnification were sufficient, follow through all its combinations, and thereby also observe the manifold though always law-governed alterations answering to its nature which it underwent — how it now, in combination with others, lost itself in a composition, now again, separated out from the composition, came forward severally, yet modified, altered in relation to the circumstances and conditions under which the separation had been brought about. But the atom in its unalterability, the atom as the point of function identical with itself in the multiplicity of different modes of being and of acting, the eye would not be able to observe, though the microscope magnified billions of times.

The one-sidedly critical, undialectical separation of alterability and unalterability — a separation according to which what is in itself unalterable stands indifferent to all alterations — leads to a quite untenable separation of the attracting thing itself and its attraction. Chemical attraction is then, on this separation, to bring about no other alteration in the atoms than that which concerns place and position, place and position being external determinations indifferent to the atoms themselves. But to let the essence of atoms stand indifferent to their mutual attraction is not even possible in mechanics. If two masses can attract one another only in that the atoms in each of them attract and are attracted, how should two atoms be able to attract one another except in that all possible points inwardly as well as outwardly in each of them attract and are attracted — that is, in that the reciprocal action of attraction penetrates them both? And if the mechanical reciprocal action is already so energetic a relation that, instead of playing on the atoms' surface, it presses through the shell into the kernel: how much more must the chemical reciprocal action penetrate the atoms' existence!

To let the essential difference between mechanical and chemical attraction rest exclusively on the former's acting at a distance and the latter's only upon immediate contact is really far too superficial. In the first place, the mechanical attraction is only one-sidedly characterized by being said to act at a distance, since, in order to act as a force of cohesion, it must also be conditioned by "immediate contact"[*)]; in the second place, the so-called immediate contact in the chemical reciprocal action is, when we look more closely, not immediate after all, since the atoms, surrounded by a "sphere of heat," are constantly kept apart by small intervals filled with ether; in the third place, it is and remains inexplicable how two bodies, by entering into a chemical combination, can bring forth a new body differing in its properties from the original ones, if this does not happen through the atoms of the two bodies altering one another, by attraction and arrangement, so qualitatively that a new atom is formed which differs in its properties from the original ones.

That the atom is a point of function means, in other words: 1) that its distinctive existence is determined by its distinctive acting; 2) that by its activity it both brings forth and itself undergoes a series of law-governed alterations; 3) that under all alterations it keeps itself unaltered, not in material immediacy but in essential form — a reflected constant! By conceiving the atom as a constant of reflection, philosophy can appropriate chemistry's exact interpretation with all the punctuality belonging to it. Chemistry is therefore from its side entitled to presuppose something constant in the atoms' filling of space; but since this constant reflects itself precisely in the alterations it itself runs through, the mode of crystallization nevertheless rests essentially, not on ready-made, eternally fixed figures indifferent to every local transposition, but on active dispositions answering to the nature of the points of stuff and specially determined by special conditions. If the atoms as material parts are to fill space, their mutual reciprocal action must of course be conditioned by a particular distribution in space; and chemistry is in this

[*)] Ein Körper, der einen andern in Bewegung setzen soll, muß mit diesem in unmittelbare Berührung treten. Hobbes (*Physica*, 26,8) sagt: *nihil movet, nisi corpus motum et contiguum.* Auch Newton konnte sich (nach Humboldt) dieser Ueberzeugung nicht erwehren. [A body that is to set another in motion must come into immediate contact with it. Hobbes (Physica, 26,8) says: nothing moves unless it be a body moved and in contact. Newton too could not, according to Humboldt, resist this conviction.] *Dr.* A. Kortüm. Die Lebenskraft. Berlin. 1856. p. 2.

respect too entitled to lay particular weight, in all formations and transformations, on the ordering of the parts. But to try to explain everything by mere transpositions is superficial. Attempts have indeed been made to render evident the different ways in which single atoms can group themselves into compound ones; but the attempts belonging here are too meagre, arbitrary and insufficient even merely to explain a single one of the transformations which fall, for instance, under isomorphism. In metameric phenomena, such as $C_4H_6O_3 + C_2H_6O$ as against $C_2H_2O_3 + C_4H_{10}O$, each of the elementary atoms is indeed, despite the transposition, like itself, namely as point of function: $C = C$, $H = H$, $O = O$; but under the conditioned reciprocal action each single C in C_4 nevertheless acquires another character, another influence and significance by means of C_4H_6 than it can obtain in C_2 by means of C_2H_6, just as each single H in H_6 also plays another part by its relation to C_2 than by its relation to C_4.

Chemists have sought in an ingenious way to make the multiplicity of attractions in the compound atoms intelligible. "With the number of the elements, with the number of the atoms that are united into a group," says Liebig, "the directions of the attraction are multiplied; for which reason the strength of the attraction also decreases in the same degree as the multiplicity of the directions increases. A part of common salt, one of the smallest parts of cinnabar, presents a group of not more than two atoms; an atom of sugar, on the other hand, contains thirty-six, the smallest part of olive oil several hundred single atoms. In common salt the attraction manifests itself in only one, in the atom of sugar in thirty-six different directions. Without anything being added or taken away, we can conceive the thirty-six single atoms of the atom of sugar ordered in a thousand different ways; with any alteration in the position of a single one of these atoms the compound atom ceases to be an atom of sugar, for the properties distinctive of it vary with the way in which its atoms are arranged." — But it must above all not be overlooked that the way in which the atoms are arranged is at the same time determined by the atoms' own actual distinctiveness, which in every particular case is conditioned by, and in turn conditions, the function determined by the reciprocal action.

The mechanical does indeed show itself from many sides as a transition reflected in the chemical*), a transition which precisely confirms the dialectical unity of the mechanical and the chemical activities. "All the phenomena of fermentation taken

*)"The merely mechanical motion is sufficient to give the force of cohesion of crystallizing bodies a determinate direction, and to alter the affinity in chemical compounds .... The slightest friction, a blow, brings fulminating mercury and fulminating silver to explode; contact with a quill is sufficient to decompose ammoniacal oxide of silver and iodide of nitrogen. The merely supervening motion of the atoms alters in these cases the direction of the chemical attractive force, and they order themselves, as a consequence of the motion that has set in, into new groups; their elements come together into new products." Liebig. Chem. Breve. pp. 110—112.

together prove the fundamental principle set up long ago by Laplace and Berthollet: *that an atom (Molecule) which has been set in motion by any force whatever can communicate its own motion to another atom with which it is in contact.* This is a law in dynamics of the most general validity everywhere that the resistance (*force, vital force, affinity*) which opposes itself to the motion is not sufficient to cancel it. As a newly recognized cause of alteration in chemical compounds this law is the greatest and most lasting gain which the study of the process of fermentation has procured for science." (Liebig. Chem. Br. p. 150). — But however much weight must here be laid on the conditions, on the influences proceeding from other forces, we may in no way overlook that the chemical activity itself, in its dialectical unity with its conditions, is precisely something different from these conditions, something distinctive in itself. The difference between the mechanical and the chemical must, to stay with this point, in the last instance undoubtedly consist in this: that chemical attraction presupposes a qualitative difference in the mutually attracting atoms, whereas such a difference of quality is in mechanical attraction without significance. And if this is so, is it not a conspicuous absurdity that the distinctive character of the chemical attractions should — in absolute likeness with the mechanical — rest exclusively on difference of strength and of direction in space?

That the mode of aggregation has essential significance in the chemical process is evident; but evident too, then, is that the mechanical alteration of arrangement must go together with an alteration of substantiality penetrating the atoms: every modification in the quantitative presupposes and brings with it a corresponding modification in the qualitative. Thus carbon's action in all compounds is a distinctive manifestation, conditioned by the arrangement, of the carbon-quality; hydrogen's of the hydrogen-quality; oxygen's of the oxygen-quality. On the infinite reciprocal action of the qualitative with the quantitative rests also the validity of the polymeric formulae. If, for example, the compounds $C_8H_{16}O_4$ and $C_4H_8O_2$ constitute two "polymeric bodies" — that is, two bodies which, although they consist of the same substances in the same relative quantity, nevertheless display so qualitative a difference that they may on that ground be said to constitute two particular stuff-substances — then it is impossible to derive such an alteration of quality from an alteration in the quantitative alone. In the present investigation there is, mark you, no question of the reliability of a chemical analysis, a question about which philosophy has nothing to reason, but of a rational interpretation of rational formulae. In a rational interpretation the emphasis falls on concept and consequence of concept, and the task so far falls under the philosophical.

According to the mode of explanation prevailing in the subject-science, the qualitative difference between the two bodies is to proceed from the atoms in the one being larger than the atoms in the other; but if mere difference of size is able to bring about a difference of quality, then the elementary atoms themselves, C, H, O and so on, must on account of their qualitative difference likewise be regarded as compound, and the difference of quality be explained by their isomeric — be it metameric or polymeric — constitution. In other words: *if in some cases one derives the qualitative from the merely quantitative, then one ought to be consistent and do so in all cases.* But if one now assumes, with the ancients, that all the elementary atoms are in essence absolutely of like kind, so that the difference can at most rest only on figure and size — how will one then explain to oneself the wealth of qualities and alterations of quality which the compounds bring with them? Even if, on a presupposition so contradictory to the total impression of the chemical processes as that all atoms are originally of the same quality, he were able to invent formulae for the formation of an atom of carbon, an atom of oxygen and so on, one would still — and this is here decisive — be unable to invent, let alone demonstrate in the matter of fact, the causes and grounds by which the realization of such formulae could be determined: to appeal to chance (τύχη, κίνησις κατὰ παλμόν) will surely not do.

If, on the other hand, we hold fast to chemistry's own original and natural presupposition — that quality, wherever substantial alterations take place, acts as one with quantity; that the determination of quality therefore has as original a validity as the determination of quantity; that a modification in the quantitative can therefore produce a modification in the qualitative only in so far as it is itself conditioned by a modification of quality — then the consequence of concept will hold for polymerism as well.

If the same elementary atoms (C, H, O) really. form in the one case the atom $C_8H_{16}O_4$ and in the other the atom $C_4H_8O_2$, then each of these formations must have its sufficient ground. The sufficient ground can in such cases be determined only by the same affinities having, under particular conditions, to act otherwise in the one case than in the other; but affinity is qualitative: every manifestation of affinity is a quantitatively determined manifestation of quality. In the compound $C_8H_{16}O_4$ the elementary atoms therefore have after all another quality than in the compound $C_4H_8O_2$.

That such alterations of quality determined by quantity may for the rest be now greater, now smaller — indeed that by running through a series of differential effects they may form imperceptible, often continuous transitions — accords in equal degree with the consequence of concept and with the matters of fact. Only in that the atomic actions are quantitatively determined manifestations of quality can the constants of reflection functioning in each group form a new unity qualified by "the absolute quantities"; only so can they as points of function concentrate their contributions into a new and distinctively concrete point of function.

γ) The Atom as the Substantial Moment.

As point of function, at once unalterable and yet alterable, independent and yet dependent, absolute and yet relative, the atom leads a highly dialectical existence. It is precisely what is dialectical in this existence that is so decisive for the determination of matter's essence, of the reality of stuff-substances, and thereby on the whole for a closer determination of the relation between change and transformation, ground and existence, possibility and actuality, material and ideal cause. If what is independent in the atom is volatilized, matter loses its original punctuality, sinks as the in itself indeterminate general-sensuous below actuality, and then keeps itself at the level of possibility as the stuff out of which form — the generally operative in all particular determinations — makes the different, the actual in existence. If the atom is on the other hand hardened into an eternally fixed material unalterability, then matter in its original discretion becomes the absolutely valid; motion, change, coming-to-be, destruction — in short, the possibility of visible things — then comes to rest on the unconditioned actuality of the material points, on the forces dwelling in the atoms. Both ways of seeing stand sharply against one another: the former falls one-sidedly under the analysis of abstraction, the latter one-sidedly under the analysis of things; the former is abstractly metaphysical, the latter abstractly physical; the former gives grounds upon grounds but never reaches the thing, the latter explains thing by thing but never reaches the ground.

That the abstractly physical view is really as far from the concretely rational as the abstractly metaphysical is easy to show from what has gone before. Water is decomposed, and oxygen and hydrogen arise; oxygen and hydrogen combine, and water arises: how, now, is this "arising" really to be conceived? — The one-sided metaphysician sees in such transpositions only a proof of form's decided superiority over stuff. If the question is of what is material in matter, there is no difference at all between the oxygen and the hydrogen; the difference is form's work alone: the same stuff which by means of form can become oxygen can by means of form become hydrogen too, or carbon, or sulphur, and so on. To what degree stuff as stuff is subject to form is seen precisely as the water arises; for where has the oxygen now gone? What has become of the hydrogen? They have vanished, and they had to vanish, for their fleeting existence rested on form alone. By the self-altering form they have been transformed, their difference is cancelled in the water-form: both have become pure water. And what, then, is the ground of oxygen and hydrogen arising again when the water is separated? Form, of course! Form's regress must be precisely in reflection of its progress: just because the water-form resulted from the forms oxygen and hydrogen, these must result again from the water-form. And what, then, is the ground of form's not immediately, without any circuit, making water directly out of the general stuff (*materia prima*), but first using oxygen and hydrogen as intermediate terms? Form! for form is in the mediation. The general stuff does not exist at all in its generality; by being over against the particular stuffs it becomes itself, precisely by means of form, a particular. Thus the activity of form must everywhere consist in a mediation 1). of the general (stuff) with the single (water) through the particular (oxygen and hydrogen) — composition; 2) of the particular (oxygen and hydrogen) with the general (stuff) through the single (water). — decomposition; 3) of the single (water) through the general (stuff) with the particular (oxygen and hydrogen) — mediation of composition with decomposition! And what, finally, is the ground of form's always using oxygen and hydrogen as mediating terms when it would make water? Why could the form that mediates itself with itself through everything not just as well use for its formations of water now sulphur and arsenic, now carbon and hydrogen? Why does it set itself such fixed barriers, why does it bind itself to determinate intermediate terms? Form! for form itself, the ideal causality, demands precisely a consistently carried-through, in itself unalterable system of mediating forms of transition; only in such a system is nature reason, and reason nature.

The emptiness of this formalism is conspicuous; but if we now go to the opposite extreme and let the atomist, entangled in abstractly physical representations, explain the formation of water to us, will his explanation be any more satisfactory? From eternity, then, or something like it, the two gases oxygen and hydrogen are not merely opposed to one another, but by the absolute independence of the atoms of unlike kind so absolutely opposed that their material difference can never possibly either vanish or be equalized in any existing unity whatever. In a certain way we are indeed compelled to treat the product arising from their combination,- the water, as a new stuff, a new thing; but this new alterable thing with its alterable properties is nevertheless, we must confess, quite unlike the atoms of oxygen and hydrogen, those eternal, unalterable little things with unalterable properties. Water is, strictly speaking, an illusory unity, and the phenomenon of water as an immediate sense-thing a semblance without essence. Instead of an essence answering to the phenomenon, there shows itself to thought an actuality of unlike kind with the phenomenon. It is the atoms with their distinctive arrangement and attraction that here constitute what is actual in itself; and that this actual in itself looks to our imperfect senses as though it were something quite else, in so far as it looks like water, is no concern of the existent. The existent we call water is therefore, properly speaking, not water after all; water as water is not present in actuality but only in our imagination. It is not the water that is the thing (the water is at most only an apparent, improper thing); no, the groups of atoms are the thing: the elementary atoms are the things of things. Instead of a ground of things the atomist therefore sees everywhere things of things, things inside things, things behind things, the proper things behind the improper ones; and since the proper things are non-sensuous sense-things, philosophical abstraction-physics comes at last, out of sheer sensuousness, into conflict with the senses.

What is one-sided in each of the extremes set up makes itself known at once when it is brought into relation with the problem of change. In formalism, the abstractly metaphysical one-sidedness, the otherness that holds in change lacks its due emphasis; by the difference of form the one does indeed stand against the other — oxygen against hydrogen, against carbon, against sulphur, and so on; but by form's decisive relation to itself through all forms the difference and the opposition are cancelled as easily as they are posited and presupposed: the change becomes so absolutely pervasive that it must be called a transformation. On the atomistic representation of things, the abstractly-physical way of seeing, the otherness is petrified to such a degree that an essential transition from the one to the other is quite unthinkable. Since all change rests on a transposition of the in themselves unalterable parts, the change of the apparent things falls in a superficial way outside the proper things. In metaphysical abstraction-formalism change is volatilized into transformation, because the otherness is too weak and therefore too strong: here *finitude* is wanting, here *existence* is wanting. In physical abstraction-realism change is volatilized into local transposition, because the otherness is too strong and therefore too weak: here *essentiality* is lacking, existence lacks a *ground.*

As a constant of reflection the functioning atom can give up and yet assert its independence, be reduced to a moment and yet preserve its substantiality; as the substantial moment the atom can bear the change. By holding one-sidedly to the atom's independence and to local transposition it is impossible to dismiss the concept of things' arising and passing away; by means of the pure doctrine of transformation it is impossible to bring such a concept into accord with the conditions of actuality. Out of nothing, pure nothing, no thing in nature can arise; into nothing, pure nothing, no thing in nature can pass. But transposition — the merely local transposition of eternal, in themselves unalterable parts — explains nothing either. When Paracelsus assumes that the copper which arises out of “the blue vitriol” arises as by a mystical transformation, or as by a natural creation out of nothing, the error is easily seen. But if one now goes to the opposite extreme and maintains that the copper which arises out of the vitriol already was, before it arose, contained as existing copper in the vitriol itself: would not such a claim, when we look to concept and essence, betray just as great an error? If the atoms of copper (Cu) are contained in material unalterability in the vitriol ($\dddot{S}\,\dot{Cu} + 5\,Aq$) then the same must hold of the atoms of sulphur, of oxygen and of water (Aq); the vitriol therefore does not have the unity of a thing, is not one thing but a swarm of atoms, countless things — is in other words not vitriol. In the blue vitriol the copper is so far from having actual existence that on the contrary it has gone to the ground — if one likes, even twice over: the first time when the oxide of copper arose, the second when the oxide of copper united with the sulphuric acid and the 5 equivalents of water, and the vitriol arose. When the copper now arises again out of the vitriol, there is here a real transition from being-in-the-ground to existential existence. At bottom the copper is already in the copper vitriol, for it has gone to the ground in the combination — and must so far be there — at the ground; but what has gone to the ground is not present, that is, it is present only in possibility, not in actuality: real arising is a transition from possibility to actuality. That the thing must have its ground means that its own inner presupposition cannot itself be a thing or a set of things, but must be a self-reflected unity, a system of operative relations, a system of essence in which the thing-causes are reduced to moments of cause. Thus oxygen, for instance, cannot be the ground of water, any more than hydrogen can. Oxygen and hydrogen are, as the combination is entered into, cooperating causes, and their conditioned reciprocal action is therefore water’s proximate cause; in this respect it holds that every thing must have its cause. But cause is one thing and ground another. The ground of the formation of water — the real ground, namely — is the transformation which at the moment of union takes place with the atoms of oxygen and of hydrogen. Without such a transformation the coming-to-be of water is, as we have seen — when it is to be real water, mark you, and not merely water apparent to “our imperfect senses” — impossible. Out of mere thing-causes no thing can arise; the thing-cause realizes the concept of cause only in a quite external, finite and one-sided way. The thing is indeed cause, in so far as it is the material unity of forces, the existential centre from which effects proceed; but the cause itself is nevertheless not a thing. The thing passes away, but the force, that which causes, does not pass away[*)];

[*)] “That the production of heat by heat, or of mechanical quantities of work by mechanical quantities of work, and so on, is really only a communication of activity from one system of material parts to another and not any new production — and likewise that the body to which it is communicated can at most it acts, only in another connection and in another way, identical with itself as another force, when the thing has gone to the ground.

The cause exists; as thing-cause it has its reality in immediate, sensuously-positive form. The ground does not exist,

for what exists is a thing; and yet the ground has reality — it has, like possibility, its reality in negative form. Reality in negative form is ideal; it is the ideal that is the true ground of the material, the outwardly real. Things alter and must alter, for existence presupposes becoming. Therefore: the existent must be grounded. That the thing must have its ground means, then: 1) that no thing exists which has not arisen through another thing's having gone to the ground; 2) that no existing thing has so absolute a subsistence that it must not one day go to the ground in order that another thing may arise; 3) that arising and passing away is an infinite alternation of the ideal with the real, a dialectic of existence which transforms all thing-causes into moments of cause, into atomistic points of passage for the ground-cause, the constituting principle.

Result.

The metaphysical unity of thinking and being is in actuality itself an energetic unity of thinking and acting: the actual is in its natural immediacy to be determined originally as a sensuous-thinkable.

Speculation — the Hegelian-dialectical as much as the Aristotelian-dogmatic — misses the goal of real knowledge in missing the method. The fault is by no means that the unity of thinking and being is held fast; the fundamental fault is that the moment of sensibility necessary to everything really existing is overlooked by thought and volatilized by abstraction. As the moment of sensibility is volatilized, attention to the factual, the special, must of course be weakened, observation's eye for the distinctive blunted, and thinking, which should appropriate the content-rich real concepts, contents itself with abstract general forms, metaphysical essentialities: in all acids with "a quality of acidity,"

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receive an increase in activity of the same magnitude as that which the communicating body loses — these are generally acknowledged truths. It has hitherto been unclear, on the other hand, how the relation stands in general between the acting and the produced forces, since one has rather stopped short at the obscure consideration that the activities have played out their part when certain material results have been produced .... This way of considering the matter has always seemed to me in the highest degree uncomfortable, and I have therefore never been able to give myself over to such a thought. The one natural view seems to me on the contrary to be that which I have developed earlier, namely: *that forces can never vanish in the corporeal, and that it must consequently be a general law of nature that forces without exception undergo only a change of form when they seem to vanish, and thereupon come forward again as acting causes, but in altered forms.*" L. A. Colding. Om de almindelige Naturkræfter og deres gjensidige Afhængighed. (Vidensk. Selsk. Skr. 5th series. Naturv. og math. Afd. 2nd vol. Copenhagen 1851. pp. 169—170).

an “original acid”; in all caustic alkalis with “a caustic principle”; and so on. (Cf. Phil. Propæd. pp. 45–52).

In proportion as empirical inquiry — whose methodical security rests precisely on the art of observation, on experiments and connected research — advances with the advancing subject-science, the moment of sensibility forced back in speculation comes into its full right. Instead of the abstractly general the special is now brought out, instead of metaphysical essentialities actual matters of fact, instead of *a priori* forms of reflection empirical concepts of facticity. Thus in the course of chemistry’s historical development the categories metal and metalloid, chemical affinity, acid and base, salt, double salt and so on have, as we have already seen, been gradually formed, tested, altered and determined as forms of the existential, as concepts of matter of fact, for whose reliability science will, on given presuppositions and under given conditions, vouch.

But although the categories of facticity are at bottom as much due to reflection as to observation, and so far rest on a tacitly presupposed unity of thinking and acting, of concept and actuality, this unity cannot be made plain so long as thinking is still conceived as a merely subjective-human thinking, and this subjective-human thinking is lost in the matter of fact and oppressed by sensation. From the empirical-critical standpoint it constantly looks as though the thing were one and the thing’s concept another; in the finite sense one must constantly distinguish between the stuff and the concept of the stuff, the force and the concept of the force, the acid and the concept of the acid, and so on. This finite divorce between concept and actuality is on the other hand cancelled in philosophy, in so far as philosophy, by preserving the difference in the dialectic of the unity, frees reflection from the one-sidedly subjective and lets the moment of thinking objectified in actuality come into its right.

The fundamental thought on which the dialectical analysis of actuality rests as a whole, and in particular the analysis of the chemical categories — the fundamental thought whose all-round development philosophy sets out from — may here be stated in the following two propositions: 1) all acting is, essentially understood, an energetic reflecting; 2) true actuality is an energy of reflection in which the unalterable shines through the alterable in the most various ways.

1) *That all acting is at bottom an energetic reflecting and consequently one with an objective thinking* will readily be evident from what has gone before. No substance can in and for itself be absolutely either acid or base; the substance *a* is acid only in so far as it has a base in the substance *b*, and conversely: the concept of acid is as surely in dialectical unity with that of base as the concept of ground is with that of consequent, or

of cause with that of effect. In so far as the concepts acid and base not only mutually presuppose one another but also shine through and ground one another, reflection is the true expression of their existence. So long as pure concepts are in question, the reflection is indeed still formal, and looks in its formal generality as though it rested only on our way of considering things, as though it took place only in our subjective thinking. But if we integrate the abstraction, if we note the inner inescapable necessity with which what has been abstracted — the pure concept — appropriates again and point for point reclaims the determinations of concretion, the distinctive moments of actuality which the abstraction removed: then we see at the same stroke how it really comes about that formal reflecting transforms itself and becomes a real acting bent back into itself.

That what we call acid must have its base, and what we call base its acid, is — even if one were to reject these designations and introduce others — in reality not a subjective invention but an objective matter of fact. What such a matter of fact signifies is seen from the concept *affinity*. The substance α attracts the substance β on account of the affinity, the acid *a* the base *b* on account of the affinity, the salt *A* the salt *B* likewise on account of the affinity: the attraction is here explained by the affinity, the affinity again by the attraction, and both by force.

If one now makes, in accordance with the usual way of considering the matter, a finite separation between stuff and force, then it undeniably looks as though the mutually attracting substances were each really something by itself, *outside* or at least *alongside* the force with which they attract one another. Indeed, it looks almost as though the attraction rested on fine, superfine clasps, on quite extraordinarily small hooks with which the substances catch into one another upon contact, so that the atoms whose hooks do not fit together are not in affinity either, while those whose hooks fit into one another are akin and display attractive force. That such hooks and clasps involve great difficulties — when one considers how the combined substances can be separated again, and so torn from one another, and that without the hooks breaking (for the separated can be united afresh); when one considers the great multiplicity of arrangements, groupings and rearrangements which the isomeric, and especially the metameric and polymeric, formations presuppose — no one will deny. To escape from all this quackery with clasps, hooks, threads of attraction and the like, one may of course give the matter the turn that the attraction takes place by a force, but that we human beings cannot possibly know what force is, since we cannot see force, not even under the microscope. But even this evasion has its quackery; for even if we do not — and for good reasons — know what force is in and for itself, we do know that there is an attraction here, and that an attraction rests on force.

We know, then, that "what we call *force*" is something operative and belongs among the actual. To say something about force's relation to the actual, one might here adduce what is so often repeated, that force is a manifestation of things' properties; but what is really said by this? What is really said is what must absolutely be said here to begin with: that the substances which attract one another display, precisely by this attraction, their distinctive constitution, their chemical nature. The substance α may indeed find itself in outward separation from β, the acid *a* from the base *b*, and it may look as though the separately existing things had nothing to do with one another but were quite indifferent to one another. If, however, we consider these indifferent things from the chemical standpoint, we easily discover that the indifference is only a mask. Scarcely have these apparently so indifferent things been brought into contact under given conditions before the mask falls: the affinity acts, the union takes place, and it appears — as is the truth — that the one of these substances has chemical reality only by means of the other, the acid only by means of the base, the base only by means of the acid. And if it really is the acid that makes the base a base, and that in the very moment in which it is the base that makes the acid an acid, then the base is also, not only in our subjective thinking but in nature's own objective acting, reflected in the acid, and the acid in the base. The attraction, the conditioned reciprocal action, is an acting bent back into itself through the acting terms, a real reflecting, a natural identity of thinking and acting.

2) *True actuality is an energy of reflection in which the unalterable shines through the alterable in the most various ways.* From the dialectic of the mathematical categories there emerged the result that the positive is at bottom as much a reflected thing as the negative, the finite as much reflected as the infinite; the concepts thus developed then receive a more concrete content, and thereby a richer articulation, in the analysis of the chemical categories. The immediately positive here becomes a material, the finite a stuff-like alterable, while the infinite presents itself as the unalterable within the alterations — the essence, the substantial, the imperishable. That the unalterable is as much a category of reflection as the alterable needs in this connection no closer grounding; whereas the reality of the immediate — the category of immediacy — may well require special illumination if it is to become properly clear to us what energy of reflection really means.

Critically considered, the matter stands thus: the immediate is not reflected, the reflected is not immediate; it looks then as though the one fell straight outside the other. But if we consider more closely how immediacy itself is determined, it becomes at once evident that the immediate can be determined only by its opposition to the non-immediate. The non-immediate may indeed be conceived as a chain in which every link is again an immediate; but precisely in being determined in this way by relation to other immediates, the immediate becomes determined mediately, and so takes mediacy up into its own determination. If the immediate is posited as the unreflected, then the mediate must be posited as the reflected; and it is then seen that the unreflected — as the negative expression at once indicates — can be determined only by a reflected and as a reflected: that immediacy, in other words, has its ground in reflection.

The categories *ground* and *grounding* are *categories of reflection*; if nature in its products and immediate manifestations of force were not infinitely reflected, then there would be no forces in natural things, and in nature itself no ground and no grounding. But such natural products as water, alumina, felspar and so on have in their material existence so sensuously striking an immediacy that it must seem to the observer unprejudiced by theories as though in such natural objects there were not even a glimmer of reflection to be discovered. If everything in nature's acting is to be reflected, and the material existences are nevertheless to be opposed to the reflected in nature-forceful immediacy and, so to speak, in palpable fashion, then reflection must be able at once both to affirm and to deny itself; then it must be able, by positing and presupposing its opposite, to confirm itself. "But," asks the sober inquirer, "should so flat and palpable a contradiction not after all belong to the wholly unthinkable, and should the solution of such a contradiction not be reckoned among the absolutely impossible?" We answer: if actual nature had not originally such a fundamental contradiction to solve, then *space* and *time* would be without true objective validity.

Existence's conflict with the ground is a contradiction which finds its solution by means of the forms of outwardness, space and time. If we ask what water as a stuff essentially is, then water is an essence reflected in its ground; for water is as a product of oxygen and hydrogen reflected in its constituents, and these constituents conversely reflected in their product. The product thus reflected is moreover — at bottom — afflicted with the contradiction that it cannot exist in so far as its constituents have existence, and the constituents cannot exist in so far as the product has existence: product and constituents mutually presuppose and cancel one another. Further, this product reflected in its ground is afflicted with the contradiction that its form of aggregation is at once gaseous, drop-liquid and solid, while each of these forms mutually excludes the others. What it means that the ground is ground precisely by uniting what in existence is not unitable becomes evident when we consider that the ground is in inwardness, and precisely therefore must in inwardness — that is, in reflection — gather together what existence separates in outwardness, in sensuous, material immediacy. It is impossible to say what most characterizes water's essence, its formation out of oxygen and hydrogen or its separation into them; for at bottom the separation must be reflected in the formation, and the formation conversely in the separation: separation and formation are with their difference at bottom one. It is otherwise with existence. In immediate existence one and the same product cannot possibly be at once both formed and dissolved; but what cannot happen at once can happen successively — therefore we have time; and what cannot at the same time happen in one place can happen simultaneously in different places — therefore we have space.

In space and time reflection does indeed vanish for the consideration of the multiplicity of the given; but it is not therefore lost. The existing things reflect themselves in the laws; the multiplicity of the given constitutes a system of operative relations; the operative relations are existing forces, manifestations in existence of the ground's energetic reflection. The existing felspar does indeed, in so far as it exists at a given time in space, exclude all objective reflecting; hardness, weight, opacity, everything in such a stone is as opposed as possible to the ideality of reflection. And yet this stone is on every side in the power of the fundamental reflection, at every point subject to ideality. An elementary existence like the felspar falls, on account of its material immediacy, its outward facticity, quite outside the process of formation by which it arises and the process of dissolution by which it goes to the ground. If its nature is to be more closely investigated, however, its chemical character must be determined; and its chemical character is determined precisely by its being separated into constituents, and constituents of constituents, until we come to the elements that set boundaries to further separation. The question what the felspar is, analysis therefore answers by *letting it go to the ground.* But the elements in whose existence the felspar goes to the ground are precisely the elements whose existence must in turn go to the ground if the felspar is to exist. The question what the felspar is, synthesis therefore answers by showing *out of what it arises* and *how it forms itself.* Whether the existence falling between formation and dissolution lasts for millions of years or only for a single second is in this connection inessential; what is essential is that the existence which goes to the ground is reflected in its elements, and that the elements which go to the ground are in turn reflected in the existence resting on their destruction.

Pervasive reflection will always require that the terms reflected in one another be not only transformed into dependent moments, into pure reflexes, but also that under the transformation they show themselves in their difference and preserve a certain relative independence. And the greater the independence with which the terms come forward, the stronger must also be the reflection that is to transform them into moments and master them as reflexes: here we have a measure of reflection's energy. In merely intellectual reflection the reflected terms are, as images, representations and concepts, only fleeting semblances, dependent reflexes; and from this is explained pure — that is, abstract — thinking's impotence to bring forth anything existing whatever. In nature-forceful reflection the reflected terms have a prominent independence, and become precisely by this independence of theirs striking expressions of the power that marks their impotence and holds them in dependence. Reflection in its energetic and substantial essentiality is power. By power the moments reflected in the ground are posited as actual existences, as particular things: from this comes material discretion, from this the swarm of atoms, from this the existence of aggregates and masses in space. By power the actual existences, the particular things, are posited as moments and reflexes in the reflected reciprocal action: from this all union and dissolution, from this all arising and passing away, from this the multiplicity of changes with all transitions from the fleeting modification to the pervasive transformation! The power that posits and the power that cancels the independence of things is indeed different in the form of existence, but at bottom a power identical with itself. The power identical with itself comes into view in the stream of changes as the unalterable, the truly substantial, the eternal. According to its abstractly negative relation to the alterable, the unalterable is to be seen in the laws themselves; according to its concretely positive reflection in the alterable, it is to be seen in the fulfilment of the laws. Let all finite existences vanish as modifications, as semblances and impotent shadows in the self-identical sole power, and we have in this sole power an infinite substance, an unalterable; but an unalterable which in consuming all finite reality consumes itself[*)]. Let the infinite substance, so as not to shrink to a shadowy being, an empty unity of empty shadows, confirm its absolute power by an effusion reflected from the ground into material elements, into actual substances and relative substances; and we have reality in the unalterable itself, in that we have reality in the world of changes.

And what, then, is the actual stuff? A reflex of the operative force! And the force? A reflex of the stuff's actuality! And what are things? Reflexes of concretion of fulfilled laws! And the fulfilment of the laws? A thinking reflected in concrete acting! And the material causes with their conditions and conditioned effects? Reflexes of the

reciprocal action articulated in the system of functions! And the reciprocal action

[*)]Cf. Propæd. Logik by R. Nielsen. Copenhagen 1845. p. 178 ff.

itself? An energetic reflection of the One reflecting itself in all causes, things and conditions — the Principle.

C.

Philosophy's Relation to Physiology: The Problem of Life.

In so far as the particular sciences have a finite delimitation, they are subject-sciences. How peculiarly the different subject-sciences' view is formed becomes manifest when they, by treating one and the same subject, approach mutual contact. In the determination of soul, life and life-principle a thoroughgoing difference shows itself between the strictly empirical physiology, for which the organism is "a living machinery" that does not even need a life-force for "machine-master," and the psychology grounded on self-observation, which grasps the soul as constituting essence and lets life well up from "an inner source." (Phil. Propæd. pp. 94–95; cf. pp. 187–88).

If the relation between philosophy and a determinate subject-science could be settled by a definition, the boundaries separating physiology from psychology could easily be stated. "The activity of organisms or of organs is determined by their anatomical, physical and chemical properties, and by those derived from or cooperating with these. Special anatomy, the doctrine of the structure of the tissues, physics and chemistry are therefore the foundation-pillars on which the physiology of the human organism, the doctrine of the manifestations and laws of our material and materially-psychical life, rests. The presentation of the norms for purely spiritual activity belongs to philosophy, most nearly to logic and psychology." (Valentin. Physiologie, ed. Hannover. p. 1).

But when the question is how physiology, while as a subject-science pursuing its own goal by its own particular investigations, nevertheless step by step plays into philosophy's hands, if only mediately; when the question is how psychology, which without a physiological foundation must help itself out with abstractions and explanations of words, can in particulars appropriate the results of empirical research; when, in other words, the question is how the concrete common problem of physiology and philosophy comes to be with inner necessity — then definitions and remarks will not suffice. It will in this respect be necessary to direct attention both to what is historical in physiology's scientific development and to those presentations by which expert and talented specialists have sought to make the essential yield of the investigations accessible to general cultivation.

"By physiology, or the doctrine of life," says Eschricht (Tolv Foredrag. Copenhagen 1850, pp. 1–2), "is understood the doctrine which presents and seeks to explain the whole series of phenomena that distinguish plants and animals from the other natural bodies throughout their whole existence, from their coming-to-be to their death. This doctrine is usually intended only for physicians and natural scientists of the profession; but it is not easy to say why it should be a closed book to every other cultivated person[*)]. In this doctrine there are answered and illuminated questions which at many a moment must press themselves inescapably upon everyone who has an eye and a sense for nature, everyone who takes joy in life. The germ in the seed grows up into a flower or a tree, the germ in the egg becomes an animal — is there in this germ a peculiar force, almost to be called a magic force, exalted above all other natural forces? Or does what is fundamentally distinctive in the whole existence of plants and animals, what we call in one word *life*, arise only from their original peculiar structure, and this again from the peculiar chemical composition of their substances?" — Here, then, is the problem in its generality. To see how the physiological conception has itself moved from extreme to extreme, from the assumption of a magical force exalted above the general natural forces to the consistent denial of a distinctive vital force, we shall here bring out some moments belonging to the historical development[**)].

Paracelsus, who, as was already brought out in the preceding section, introduced medical chemistry, assumed that besides the chemical forces there must be in the body a being to whose governance the otherwise so mutually different functions were subject. And if he moreover saw, by virtue of his mystical speculation, spiritual beings everywhere in the macrocosm — sylphs in the air, undines in the water, pygmies in the earth, salamanders in the fire — what wonder that in the microcosm, the individual organism, he saw a demon of life as well, an "Archeus," who operated chiefly in the stomach, directed digestion, separated out the useless and harmful parts, transformed the nourishing into blood, and saw to it that every organ in the whole received its due nourishment. Archeus is, according to Paracelsus, an independent spirit whose acting does not depend on man's will. If Archeus grows weary and his forces slacken, then one notices how the substances in the body begin to ferment and to operate as though they found themselves outside the organism: putrefaction then sets in, and secondary diseases follow from it. Archeus

[*)] "No science shall be something standing alone; it shall knit itself together with all the others and with the whole of human life; without this the Idea of science cannot be realized in its whole extent." H. C. Ørsted. Naturlærens mechaniske Deel. Preface. p. IV.

[**)] Sprengel. Geschichte der Arzneikunde. 5 Theile. Halle. 1792–1803. Fortges. von Eble. Wien 1837–40.

is the physician's model; it is Archeus alone who can heal the disease. "He has head and hands, and is nothing other than *spiritus vitæ*, man's astral body, and besides him there is no *spiritus* in the body."

In "Archeus" the essence of life is personified into a living being within the living being. The living individual, man, then stands to his own life as to a foreign, in itself independent being, and in this relation becomes a stranger to himself: such a dualism of personification is intolerable to thought. By conceiving life as the living being's own essence the personal dualism is indeed removed; but in so far as life in its ideal independence is intuited as soul, and the soul is seen in *immediate* opposition to the body, a dualism nevertheless remains, if in a milder shape. That the opposition between soul and body is held fast immediately makes itself known especially in the question: where has the soul its seat? Taken physiologically, this question hangs most closely together with the question of the origin of the nerves. But the nerves are sensuous, the soul non-sensuous; the sensuous is the object of empirical inquiry, the non-sensuous of speculation: the immediate opposition alternates with an equally immediate identity, and the consequence is that the non-sensuous is continually treated as a sensuous and the sensuous as a non-sensuous, so that the empirical and speculative elements of the investigations are manifoldly mixed. — Since most anatomists in the sixteenth century, with the exception of the Peripatetics, derived the nerves from the brain, Cesalpini exerted himself in every way to defend Aristotle's assumption that the heart must be the origin of the nerves. In the living human body, so he argued, only one part could be the first and the seat of the soul, since there was only one soul. And since the first thing that shows itself in the *punctum saliens* of the fertilized egg is the heart, the heart is also the most important part of the body and the seat of the soul. But if this is assumed, then the heart must also be the seat of the sensations and hence the source of the nerves — something which the influence of the passions upon the heart plainly shows as well. Daily experience did indeed seem to speak for the nerve-force's having to proceed from the brain; but Cesalpini helped himself by assuming, with Aristotle, that the arteries first conducted the nerve-force from the heart to the brain; in the brain their cavities then collapsed, their walls sundered into threads, and so they became nerves.

Over against the Peripatetic's syllogisms, the iatrochemist's humours, and the "animal spirits," "vital spirits," "vital air" and the like of both, anatomy, now working its way forward, asserted the scientific demand for exact investigations. The great anatomists of the sixteenth century — an Andr. Vesalius, a Barth. Eustachius, a Gabr. Faloppia, a Fabricius ab Aqvapendente — had already got the body's inner structure seen in a clearer light in many parts and from many sides; but it was only at the beginning of the seventeenth century that the decisive turning point for physiology arrived, when William Harvey discovered the circulation of the blood. "Harvey's happiest thought," says Eschricht, "lay precisely in this, that the age-old belief in the arteries' containing air[*)] was false, and that they, as well as the veins, contain blood. As soon as this thought had ripened in that distinguished English physician, he soon found the direction of the blood-stream, guided partly by the discovery of the valves of the veins made at the same time by Fabricius ab Aqvapendente, partly by the experience that discharges of blood from a vein are stopped by a pressure below, from an artery conversely above the wound."

The disputes that Harvey's discovery called forth had for the scientific spirit as a whole the beneficial effect of shaking the authority of the ancients, awakening mistrust of *a priori* theories, and helping to lead the investigator onto the path of matters of fact and of induction. The doctrine of the circulation of the blood, to which Descartes from his philosophical standpoint also soon subscribed, brought with it involuntarily the view that the blood in the body's vascular system must move in a way similar to water in a hydraulic machine. And since in such a machine the relation between the quantity of water and the moving forces can be exactly determined by calculation, it lay near to institute similar calculations concerning the motion of the blood. Thus, especially under the influence of Cartesian philosophy, physiology was at last conceived as a part of applied mathematics, and the so-called iatromathematical school arose.

In Italy Galileo's spirit was at work; his foster-son and apologist, Benedict Castelli, was at Florence among those organizing the academy founded by the brother of Grand Duke Ferdinand II, Prince Leopold, which as an academy of *experiments* got the by-name *del Cimento.* In this academy was formed Joh. Alphons Borelli, who spent the last days of his life at Rome with Queen Christina, for whose instruction he composed his "immortal work" on the motion of animals (*De motu animalium*). In the first part of the work the motion of the muscles is, among other things, interpreted in a "quite new, uncommonly clear and thorough manner according to the laws of statics." Borelli, who regards the bones as levers and the muscles as the cords by which they are moved, communicates on this occasion "a quantity of apt information about the mechanism in the flight of birds, in the swimming of fishes, in the creeping of worms and so forth. The iatromathematical

[*)]The ancients were misled especially by the circumstance that the arteries are as a rule found empty of blood after death, and so assumed that as "arteries" they contained only air in life too. It was, by Galen's theory, in the cavities of the brain that the animal spirits were secreted. By means of the arteries the blood was mixed with vital spirit and then carried through the convolutions of the brain into those cavities, so that the animal spirits might be secreted from it.

endeavours were, in accordance with Newton's principles, still further advanced by men such as the Bernoullis (Johann and Daniel) applying the differential and integral calculus, together with the theory of curved lines, to clearing up the doctrine of the pulse, the movements of the muscles and other functions in the organism. According to the fundamental principles of hydrodynamics the circulation and secretion of the humours in the animal body were at last explained.

From the features cited it will, however, easily be seen that if iatrochemistry with its mystical profundity could not satisfy the demands of science, neither could the iatromathematical direction with its clear good sense. In the first place the mechanical conception did not go to the fundamental cause itself but kept to the merely phenomenal; and next, as regards the phenomenal, one was not even always sure of the first presuppositions of the calculation, since instead of observing nature one not seldom set up hypotheses for whose validity the formal consequences of the mathematical analyses yielded only a semblance of stringent proofs. If the iatrochemists had in their doctrine of the fermentations, effervescences and mixtures of substances only a highly insecure physiological foundation, then the iatromathematician, in his attempts to calculate what figures, angles and bendings had to be presupposed in the smallest parts of which the vessels and tissues were formed, could gain no reliable foothold either. If the adherents of iatrochemistry had, in order to veil their ignorance, not seldom to resort to turns of phrase and obscure expressions, then there was now no lack of iatromathematicians who not only — which is still intelligible — cured congestions by an application of centrifugal force, but even with astonishing success knew how to make the higher algebra press the remedy out of centripetal force.

The need for a more spiritualistic conception of the organism's essence — a need which had been forced back under the prevailing interest in the mathematical-mechanical — made itself felt again, and with renewed force, in the transition from the seventeenth to the eighteenth century, the period of Pietism. The Paracelsian doctrine of Archeus, to which J. B. van Helmont had already given a more rational turn, had toward the close of the seventeenth century found an entrance in the German schools, and the presuppositions for the theory of the soul developed by Stahl lay in the age. As a Pietist Stahl did indeed cherish a certain contempt for "ostentatious learning"; but he worked nevertheless in science's service, and, although his doctrine was not so original as he himself held it to be, asserted his place as first representative of the dynamical school. The theory's foundation may in a way be sought in the Cartesian conception of matter as the absolutely passive. The body has as body no force whatever to move itself; it must then, so Stahl concluded, be immaterial substances that set it in motion: every motion in matter has an immaterial cause[**)]. The marks by which the organism is distinguished from inorganic bodies, living individuals from lifeless things, were now brought out with the greatest care. “The organism has indeed on the whole, and according to its material constitution, a mechanical arrangement; but if we consider it formally and specifically, it is infinitely far from mechanism.” As an example Stahl adduces a clock. A clock is indeed a machine when one considers it according to its structure and composition, but an organ when it has been wound up and fulfils the determination of indicating the time. With the human body it stands in a similar way: it is a machine when we look to its single parts, but an organ when we look to the end for whose attainment the parts serve. An organism is therefore every body whose parts are formed into means acting together toward a common end. The immaterial substance from which all activities and motions in the organism proceed — activities by which the body’s maintenance, the integrity of its composition, and on the whole its various purposes are aimed at and preserved — is the soul. Where the effects are so alike among themselves there is, by Newton’s rule *principia non esse præter necessitatem mnltiplicanda*, no ground for presupposing a plurality of forces: the soul is, without the assistance of vital spirits, the sole operative cause in the body. It is the ancients’ “nature,” as etymology teaches, “for *Ψυχὴ* comes from *Φύσιν ἔχων*” (!). Of this principle one may say what the Hippocratic writer says of nature: “it does without instruction what it has to do,” after he has previously remarked: “it does it without deliberation (*οὐκ ἐκ διανοίης*[*)]).” This utterance Stahl now explains to mean that the involuntary motions in the body do indeed also proceed from the soul, but that this happens without deliberation and clear consciousness.

“They are,” he says, “operations that follow *ratione* or *λόγῳ*, but not *ratiocinio* or *λογισμῷ*.”

How impossible it must be, however, on the whole way of thinking of that time, to reconcile the organic with the mechanical may be seen among other things from the exchanges between Leibniz and Stahl. Against Stahl’s psychical theory Leibniz makes the following objections[*)]: 1) the soul cannot govern the body immediately and independently of the laws of mechanism; 2) the body’s laws are laws of motion, the

[**)]To deny matter every possible indwelling force, Stahl went so far as to deny it the capacity to fill space. From this his opponents took occasion to accuse him of atheism; for, says Hoffmann (*de differentia inter doctrinam mechanicam et Stahlii organicam. p. 36*), “if God were the first author of all corporeal motions, then God would have to fill the space of the world, and so be one with the world, as Spinoza has already said.” Sprengel. 5ter Theil. pp. 19–20.

[*)]*Hipp. Epidem. lib. V. 6. p. 1184. ἀπαίδευτος ἡ Φύσις ἐοῦσα καὶ οὐ μαθοῦσα τὰ δέοντα ποιέει.* Sprengel. 5ter Theil. p. 25.

[*)]*Leibnitii Opera omnia, ed. Dutens. Genev. 1768. vol. II. P. II. p. 156.*

soul's are moral laws; 3) the soul is incorporeal, and the body's first entelechy, but the body has besides yet another entelechy, namely the force of motion. In his reply to these objections Stahl ascribes to the soul extension and materiality, and expects immortality from God's grace alone.

Instead of clarifying the relation between the psychical and the corporeal, disputes of this nature could then only serve gradually to clear up the difference between the physical and the metaphysical way of considering things, and thereby to bring about a more determinate separation of physiology and psychology. But to think about the psychical without forming a concept of the corporeal is impossible, and just as impossible is it to investigate what is corporeal in the organism without forming a concept of the psychical. As the physiologist then comes from his standpoint into relation with the problem of life, he is undeniably led into peculiar entanglements. The physiologist is a physicist; his first and last causes are thing-causes; beyond the thing-causes and their conditioned reciprocal action he cannot as a physicist come. But life is metaphysical; it does not resemble a thing, nor a thing-cause. It looks in the organism — the physiologist cannot deny it — as though the physical phenomena of life nevertheless presupposed a distinctive force, a principle, an ideal causality of its own. If every cause is nevertheless originally a thing-cause, then an ideal causality is only a fancy, a mystical speculation; and the task becomes that of getting the apparent principle, the ideal causality, resolved into a system of thing-causes and conditioned reciprocal actions[**)].

If, in order to prove that an ideal causality distinct from the physical causes is at work in the organism, one expressly demands that phenomena be demonstrable, in a mechanical, a chemical or a morphological respect, which do not admit of explanation as consequences of natural laws — then a scientifically carried-through physiology must necessarily lead to the result that there is no vital force distinct from nature's general forces. That a system of mechanical tasks answering to life's requirements is solved in the organism is well known; and the more deeply science penetrates into the connection, the plainer it becomes that these tasks, in the parts as in the whole, in the smallest as in the greatest, are everywhere solved in a natural way by natural means. "The organism

[**)]Der Charakter der mathematischen Methode, die Dubois-Raimond so treffend geschildert hat, fordert, daß die Naturforschung, die für das Geschehen in der Natur Ursache und Wirkung voraussetzt, und den Grund jeder Bewegung, die Grundbewegung jeder Folgebewegung untersucht, das *Prius movens* jedes Gliedes ihrer Reihen aufzusuchen bemüht sei, so weit, als sie empirische Anknüpfungspunkte findet. Wo diese aufhören, da hört auch das Gebiet der Naturforschung auf. [The character of the mathematical method, which Dubois-Reymond has so aptly described, requires that natural science — which presupposes cause and effect for what happens in nature, and investigates the ground of every motion, the fundamental motion of every consequent motion — should endeavour to seek out the Prius movens of every member of its series, so far as it finds empirical points of connection. Where these cease, there the domain of natural science ceases too.] A. Kortüm. Anf. Skr. p. 3.

and its single parts have often been compared with a machine or a similar physico-chemical apparatus. The heart resembles a pumping-work furnished with suitable valves, the arteries elastic pipe-conduits, the eye a *Camera obscura*, the vocal organs a reed instrument; the bones behave as levers, the secreting organs partly as filtering apparatus," and so on. (Cf. Propæd. p. 187). On a preliminary consideration it might indeed seem as though the so-called organic substances pointed to a distinctive causality under whose influence they are formed. "We can," says Liebig, "compose a crystal of alum out of its elements, out of sulphur, oxygen, potassium and aluminium, because we can to a certain degree command their chemical affinity, the heat, and thereby the ordering; but a particle of sugar we cannot compose out of its elements, for in the organizing of the sugar-atom's distinctive form the vital force cooperated, over which our will cannot command in the same way as over heat, light, gravity and so on." The physiologist seems, then, to have to acknowledge a vital force after all.

Since, however, it is not the name but the thing that matters, everything turns on understanding what is really to be thought under the expression vital force in the connection cited. In the first place we must note that the so-called vital force is here set straightforwardly in line with the elementary forces, light, heat, electricity and so on. "Neither heat, electric force, nor the vital force is able to make the parts of two substances of unlike kind cohere in a group; the chemical force alone is capable of this. Everywhere in organic nature, in all compounds produced in the living animal or plant organism, we meet the same laws, we observe the same fixed and unvarying relations of combination as in the inorganic."

One might now imagine that the elementary forces lost their independence through the vital force and sank to being serving factors; but this assumption too must be given up. Thus the distinctiveness and relative independence of the chemical force over against the vital force is emphasized most strongly. "The substance of brain and muscle, constituents of the blood, milk, bile and so on are compound atoms whose formation and subsistence rest on the affinity operative between their smallest parts. It is affinity and no other force that brings about their coming together … But light, heat, the vital force and gravity exercise a quite decisive influence on the number of the single atoms which unite into a compound atom, as well as on the way in which they arrange themselves; they condition the form of the properties, the distinctiveness of the compounds, precisely because they have the capacity of communicating motion to resting atoms and of annihilating motion by resistance." Instead of acting as fundamental cause, independently forming and producing, the vital force is here on the contrary reduced to a merely cooperating cause, *a condition.* "On the coming together of the elements into a chemical compound the vital force has not the least influence; no element by itself alone is able to serve for the nourishment and development of a plant or of the animal organism. All the substances that take part in the process of life are lower groups of single atoms which by the vital force's influence come together into atoms of higher order. Under heat's dominion the chemical force conditions the form and properties of the least compound groups of atoms; the vital force conditions the form and properties of the higher, the organized atoms." (Chem. Breve, 12th letter). — So long as the "chemical activity of digestion" was either not known at all or known only in a highly unclear representation, one could readily credit "Archeus," "the soul," "the vital spirits" with the marvellous. If the vital forces acted in an independent way during digestion, why should savages not be able to eat earth and digest stones[*)]? "One need only recall that one ascribed to the animal organism the capacity to produce lime in the bones and eggshells of birds, phosphoric acid in the brain, iron in the red constituents of the blood, out of substances which one did not even know how to designate by name — whereas the existence of these materials in the organism is no longer any riddle, since they have been found as never-failing constituents in the nutriments." (Liebig).

If the vital force[*)] is set on a level with the elementary forces, then only a couple of

[*)]That the phenomenon of the "*clay-eaters*" is not, however, quite so straightforward to explain, Lewes has expressly brought out. "Travellers who see strange things are very positive in their assurances with respect to this subject. Humboldt, a man whose word rightly has European authority, confirms Gumilla's report that the Otomacs of South America live in the season of flood exclusively on a fat and ferruginous clay, of which every man eats a pound or more daily. The well-known botanist and investigator Martius has told me that the Indians on the Amazon eat a kind of clay even when there is abundance of other nourishment. Molina says that the Peruvians frequently eat a sweet-tasting clay; and Ehrenberg has analysed the edible clay sold in the markets of Bolivia, finding it composed of talc and mica. The inhabitants of Guiana mix clay in their bread; and the negroes in Jamaica are said to eat earth when there is a lack of other food. According to Labillardière the inhabitants of New Caledonia still their hunger with a white, friable earth which Vauquelin found composed of magnesia, silica, oxide of iron and chalk. The same author relates that in Java a cake is made of ferruginous clay which is much sought after by women during pregnancy. To make this list complete we must add Siam, Siberia and Kamchatka as lands in which clay-eaters are found." After enumerating this — as he expresses it — "rather surprising quantity of testimony, which we cannot reject all together, even if we assume that a wide play must be allowed to scepticism" — Lewes continues: "If we acknowledge it as a fact that certain kinds of earth are really nourishing — and it is difficult to escape such a conclusion — we are completely unable to give a satisfactory explanation of it. Only a little light is thrown on it by the in itself plausible assumption that this earth must contain organic constituents; for we must assume that in ten pounds of such earth there can scarcely be contained sufficient organic matter to satisfy a grown person's needs. Nor will it rid us of the difficulty to say that the earth only stills the hunger without nourishing the body; for in the first place we have Humboldt's testimony that the Otomacs live on this clay at times when they lack other food; and in the second place, even if the *local* sensation of hunger allows of being stilled by bringing something into the stomach, the more significant *general* sensation of hunger cannot be removed by this. We must therefore for the present content ourselves with acknowledging a fact which science may possibly explain later." Hverdagslivets Physiologie. First part. Trans. pp. 54–55.

[*)]*La vie n'est pour moi rien qui ressemble à l'archè suprême de Van Helmont, espèce de souverain paraissant* questions remain:

whence does this force proceed? On what does its activity rest? To let the vital force, like gravity, rest on a general property of matter will not do; for gravity acts everywhere, the vital force only in the living organism. To let the vital force proceed from a substance of its own, as the sulphur-force from sulphur and the oxygen-force from oxygen, will not do either, since a determinate vital substance of its own cannot be demonstrated. *The so-called vital force can therefore not possibly be a force existing on its own account, but must be regarded as the resultant of the distinctive cooperation of different forces — of gravity, of heat, and of magnetic, electric and chemical forces*[*)].

*avoir son individualité propre, qui siége dans le centre phrénique, et gouverne tant bien que mal une foule d'*archè *inférieurs domiciliés dans les diverses parties du corps, et à chaque instant en révolte contre leur chef. A mes yeux, la vie n'a pas davantage d'analogie avec* le principe vital *de Barthès, ou mieux de ses disciples, autre entité passablement confuse, sans demeure bien déterminée, mais qui veille avec anxiété au bon état du corps qui lui est confié, et fait souvent plus de mal que de bien en voulant réparer quelque léger dommage. Non, elle est tout simplement la cause inconnue d'un ensemble de phénomènes spéciaux et particuliers aux êtres vivans, de même que l'électricité est pour le physicien la cause inconnue des phénomènes que présentent les corps électrisés, de même que la chaleur est la cause également inconnue des phénomènes qui se produisent dans les corps cbauffés, de même enfin que la force physico-chimique générale, quel que soit le nom qu'on lui dounera, sera pour tout esprit sérieux la cause sans doute à jamais inconnue des phénomènes propres aux corps bruts. La vie n'est pas non plus une force tellement spéciale qu'elle soit de sa nature en opposition avec les forces qu'on vient de nommer. Sans doute, dans une foule de circonstances, elle modifie et contre-balance leur action; mais les forces physico-chimiques, mises simultanément en jeu, agissent bien souvent de même les unes sur les autres. La chaleur modifie l'action de l'élec-*

tricité, et toutes deux l'emportent, dans certains cas, sur la pesanteur, c'est-à-dire sur l'attraction, sur cette force, la plus universelle de toutes, et qu'on retrouve dans les corps bruts et les êtres vivans tout comme dans les soleils et les mondes. Ainsi entendue, l'idée de la vie ne saurait rien avoir qui répugne à l'esprit le plus rigoureusement scientifique. C'est tout simplement une force qui vient s'ajouter à d'autres forces déjà reconnues et universellement acceptées, et qui, comme elles, se constate par ses effets. A. de Quatrefages. Histoire naturelle de l'homme. Revue des deux mondes. 15. Décbr. 1860. p 819–20.

[*)]On this result the physiologists must surely at last cease adding the vital force, as an otherwise unknown Primum movens, to the other forces. "Jene Fähigkeit des Primum movens ist eine Kraft; sie ist die Kraft, auf welcher als auf seinem wesentlichen Moment das Leben beruht; denn die Systeme von constitutiven Elementen ohne diese Kraft leben nicht, haben kein Leben. Diese Kraft ob Eigenschaft, ob Bewegung, ist die *Lebenskraft* .... Die motorische Kraft des cerebrospinalen und die des sympathischen Nervensystems bilden also zusammen die *Lebenskraft* .... Lebenskraft ist daher die Summa aus der motorischen des cerebrospinalen, und der motorischen Kraft des sympathischen Nervensystems. Das Leben selbst aber, als die Fähigkeit des Substanzcomplexes, den wir Organismus nennen, durch ein inmitten seiner constitutiven Elemente befindliches Primum movens die ihnen proportionalen Bewegungen auszuführen, *ist die resultirende Größe der Leistung aus der Vexelwirkung des cerebrospinalen und des sympathischen Nervensystems mit den passiven Geweben der animalen und der organischen Sphäre, und mit den abäqvaten aüssern Lebensbedingungen*". [That capacity of the Primum movens is a force; it is the force on which, as on its essential moment, life rests; for the systems of constitutive elements do not live without this force, have no life. This force — be it property, be it motion — is the vital force .... The motor force of the cerebrospinal and that of the sympathetic nervous system therefore together constitute the vital force .... Vital force is accordingly the sum of the motor force of the cerebrospinal and the motor force of the sympathetic nervous system. But life itself, as the capacity of the complex of substances which we call organism to carry out, by means of a Primum movens situated in the midst of its constitutive elements, the motions proportional to them, is the resultant magnitude of the performance arising from the reciprocal action of the cerebrospinal and the sympathetic nervous system with the passive tissues of the animal and the organic sphere, and with the adequate outer conditions of life.] A. Kortüm. Anf. Skr. pp. 89–91. *C'est*

But on what, then, does this distinctive cooperation rest? On a system of conditions answering to the ordering of the parts and the arrangement of the organs — that is, on a morphological distinctiveness. Does not the morphological distinctiveness, the distinctive way in which the organs are formed and the shape comes to be, presuppose in its turn a distinctive causality diverging from the elementary natural causes? This assumption too proceeds from unacquaintance with the microscopic structure, that arrangement of the atoms by which the first formations, in the animal and the plant as in the crystal, are essentially conditioned.

Until a few years ago it was a general assumption — a presupposition that pressed itself involuntarily upon one — that plants and animals were in their innermost microscopic composition essentially different. "Whatever part of a plant one subjects to microscopic investigation, it is always found to consist of closed vesicles, filled with liquid or with firmer substances, and otherwise of various form.... There are indeed found, especially in the higher plants, tubes of various kinds; but these, it was known, are always formed by extension and especially by the fusion of the original little vesicles, or, as they are called, cells.... If one laid a little piece of another organ, of an animal body, under the microscope, one found the conditions quite different. Flesh, sinews and ligaments, for example, showed themselves to consist of nothing but fibres; cartilage of a dense mass, interspersed with many small oblong grains; the nervous system partly of threads, partly of globules." (Eschricht).

One originally proceeded, then, from the presupposition that there must necessarily be a thoroughgoing difference not only between the animal and the plant, but also between the plant and the mineral — a difference of principle between the living and the lifeless. Theology, which refers the difference between the three kingdoms of nature to particular acts of creation, confirmed what pressed itself as it were involuntarily upon sound sense; and physiology found the prevailing way of seeing in the best accord with the matters of fact, so long as the inner structure of bodies of different kinds still showed itself so highly different. But precisely because the presupposition was from the beginning so immediately positive, it bore in itself the disposition to negativity, to doubt and to dissolution[*)].

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tout simplement une force qui vient s'ajouter à d'autres forces .... Quatrefages.

[*)]„Sprechen wir ... von einem Naturreiche, so wollen wir damit sagen, daß das in ihm zusammengefaßte nicht bloß Product natürlicher Kräfte, sondern Naturproduct sei, d. h. daß es seine Existens nicht nur einem gesetzlichen aber planlosen Zusammenwirken einzelner Theile der Natur verdanke, sondern daß die Natur als Ganzes das Producirende sei, und sich der einzelnen natürlichen Kräfte nur als der Mittel nach einem vorbestimmten Plane bediene .... Sobald wir nun die unbestimmte Voraussetzung, die sich in dem Begriffe der Naturreiche verbarg, auf diesen bestimmteren Ausdruck gebracht haben, zeigt sich deutlicher die Schwierigkeit, sie zu begründen. .... Die Charaktere ausgebildeter thierischer und

For if it turned out on closer inspection that the alleged difference in the mode of structure of animal, plant and mineral at bottom proceeded from a confusion of the derived with the original — that it was by no means the unlikeness but on the contrary the likeness, the sameness of kind, that was the original — then it lay near to conclude the other way about: the animal body, the plant and the mineral consist of the same microscopic elementary parts; they must therefore from the first have been formed by the same natural activity; there can therefore be no difference of principle between animal, plant and mineral. Once the partition-wall between the animal and the vegetable kingdom had been broken down, the partition-wall between the vegetable and the mineral kingdom, and then in general between the living and the lifeless, had to fall as of itself. The fall was the more natural in that the primary building-blocks which, so to speak, made up the stones of the old wall dissolved and crumbled the more, the longer one regarded them under the microscope.

With the microscope observation becomes sharp, exact, penetrating. As the microscope was improved, one learned at the same time to use it in a far more strictly scientific

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pflanzlicher Organismen sind so deutlich verschieden, daß ihre Trennung in verschiedene Reiche keinem Bedenken unterliegt. Die niedersten Formen beider Reiche sind dagegen für unterscheidende Erkenntniß zu ähnlich einander, zu sehr mit ihren Eigenschaften eine in das Gebiet der andern übergreifend. Die Schwierigkeit, diese Wesen dem einen oder dem andern Reiche zuzuweisen, kann nun freilich als die Folge unserer beschränkten Beobachtung gedeutet werden, welche die Merkmale, die den beständig objectiv vorhandenen scharfen Unterschied zwischen Thieren und Pflanzen bestimmen, nicht in jeder äussern Form und Umhüllung wahrzunehmen vermag. Sie kann ferner daraus hergeleitet werden, daß eben jener scharfe Unterschied, den das unwissenschaftliche Vorurtheil voraussetzt, nicht vorhanden sei, und deshalb nie von uns durchgeführt werden könne; und zwar kann man diese Meinung beliebig als ein Verfliessen beider Reiche in einander, oder als die Annahme eines dritten eigenthümlichen Mittelreiches zwischen beiden ausführen, welches ebenso wie jene beiden seine Gattungsbegriffe habe, in dem Ganzen seines Gebietes aber einer Bildungstendenz folge, die eine Art Mitte zwischen dem Geiste des vegetabilischen und dem des animalischen Daseins halte ....“ [If we speak of a kingdom of nature, we mean by it that what is comprehended in it is not merely a product of natural forces but a product of nature — that is, that it owes its existence not only to a lawful but planless cooperation of single parts of nature, but that nature as a whole is what produces, and makes use of the single natural forces only as means according to a predetermined plan .... As soon as we have brought the indeterminate presupposition concealed in the concept of the kingdoms of nature to this more determinate expression, the difficulty of grounding it shows itself more plainly. .... The characters of developed animal and vegetable organisms are so plainly different that their separation into different kingdoms is open to no doubt. The lowest forms of both kingdoms, on the other hand, are for a distinguishing knowledge too like one another, encroaching too far with their properties the one upon the other’s domain. The difficulty of assigning these beings to the one kingdom or the other may indeed be interpreted as the consequence of our limited observation, which is unable to perceive in every outward form and covering the marks that determine the constantly objectively present sharp difference between animals and plants. It may further be derived from the fact that precisely that sharp difference which unscientific prejudice presupposes is not present, and can therefore never be carried through by us; and this opinion may be worked out at will either as a flowing together of both kingdoms into one another, or as the assumption of a third distinctive intermediate kingdom between them, which, just like those two, has its generic concepts, but in the whole of its domain follows a tendency of formation holding a kind of middle between the spirit of vegetable and that of animal existence ....] R. H. Lotze. Allgemeine Physiologie des körperlichen Lebens. Leipzig 1851. pp. 487–96.

way than before, particularly with regard to plant physiology and the investigation of the infusoria. "The distinguished English botanist Robert Brown had discovered that in every plant cell there sits as a rule a small firmer body somewhere or

other on its inner side; and since he found this little body relatively largest in the youngest cells, he already divined its significance as the oldest part in each single cell. He called it *the cell-nucleus.* A younger distinguished German botanist, Schleiden, became convinced that the cell-nucleus is as a rule really what is formed earliest, that the cell-wall comes later, while the cell-content is taken up from outside and undergoes various chemical changes. He then put to himself the question how, during the growth of plants, the extraordinary multiplication of the cells goes on — for in the same proportion as a part of a plant grows, the number of the cells increases, but not at all their size — and he thought he had got it answered by investigation thus: that the new cells are formed inside the older ones, 3–4 or more in each, and become free by the bursting of these mother-cells. Yet it turned out that in several cases the multiplication of cells also takes place by a cell's extending lengthwise and dividing by a constriction. In a similarly strict scientific way the microscope was from now on applied also to the investigation of animal parts, particularly by the German naturalist Theodor Schwann. He pointed out that several parts of the animal body quite plainly show the same microscopic structure as the parts of plants, particularly the feathers of birds and horny parts generally, such as the epidermis, the hairs and so on.... It turned out that the microscopic globules of which fat consists are cells in which the oil constitutes the cell-content, and the blood-corpuscles cells with blood-red; but, what was the most important, Schwann demonstrated that cartilage from the first — as it is found particularly in the tip of a young fish's gill-rays — consists of cells in which new cells are again formed, just as in the parts of plants; and that these cells are precisely the little grains known in the dense mass of the cartilage, while this dense mass itself is only to be set equal to what in plant anatomy is called intercellular substance; further, that when cartilage is transformed

into bone, it is these cells which partly fuse together in the form of tubes, the medullary tubes, partly are filled with phosphate of lime and become bone-grains.... Of flesh too, which consists of bundles of fibres, it turned out that every such bundle has been a single hollow thread formed by the fusion of a row of cells; connective tissue was found to consist of nothing but fine fibres arisen from original cells; the globules of the nervous system were found themselves to be equatable with cells, and its threads to be tubes arisen from cells." ... All plants and animals begin, then, on this account, in the form of a simple cell; "the cell itself is formed out of a mother-substance (cytoblastema) in this way, that several microscopic little grains gather into a group (the nucleus), around which there lay themselves other, less firm parts of the mother-substance and an outer thin membrane. In crystallization there are separated out of a solution — or, if one will, a mother-liquid — homogeneous firmer substances, and around these others homogeneous, the smallest precipitated parts constantly observing a certain position relative to one another, whereby there arises a regular figure differing according to the chemical nature of the substance. It is true that a certain difference between that formation of cells and this crystallization cannot be mistaken. The cells are round, soft, hollow and filled with a liquid; the crystals angular, firm, containing a liquid of crystallization only in certain cases and in relatively small quantity. But are these differences essential enough to ground upon them a bipartition of all natural bodies?" (Eschricht. Tolv Foredrag. pp. 30–36).

That there is a difference between organic and inorganic bodies, and even a very essential difference, no one will deny; but it is another question how far the difference is so thoroughgoing that organisms must in consequence be assumed to have a causality of their own, distinct from the elementary forces. The system of conditions under which the crystal is formed is indeed far simpler than that which the formation of an

organism requires; but the difference between the simple, which we can survey, grasp and so far master, and the intricate, which withdraws itself from our observations and calculations, cannot possibly entitle us to presuppose for the intricate a causality exalted above the thing-causes. If we were to assume a principle of organization of its own because the conditions of life are so manifoldly interwoven that we cannot see through what really goes on in the organism — should we not then also have to assume an independent principle of putrefaction of its own, when the conditions are so entangled that here too we cannot see through what really goes on? — "If we arrest," it is said in a sketch by Harting, "the functions of life suddenly and in such a way that the form — that is, the organism's structure — and likewise the chemical composition remains unaltered, and by the circumstances is compelled to remain unaltered, then, as soon as the old condition is restored, life will go its accustomed course; in other words: the changes of substance and of form which constitute the essence of life will begin again where for a time they were broken off. A seed which keeps its germinative power for centuries, a rotifer which lies whole years dried up on a glass plate, one can therefore call neither living nor dead. They are merely organic beings which under certain determinate conditions can become living; in other words: they have the capacity for life." It is an important lesson that allows of being drawn from the fact that both animal and plant organisms can, under certain circumstances even at astonishingly long intervals, find themselves in so remarkable a condition that, without being dead, they are literally not living.

What, now, becomes of the principle of life in such an intermediate condition? "It slumbers," one says; "here is a slumbering life." That is figurative speech; what slumbers must exist in order to be able to slumber: so long as life slumbers it is not yet extinguished, its inner, quiet acting not yet ceased. But in the condition here spoken of the activity of life is = 0; life is extinguished. The contradiction that life is extinguished without the capacity for life being lost can find its solution only in the presupposition that the life which is not present in actuality is nevertheless present in possibility. On what, then, does life's possibility rest? On its disposition! And the disposition? On the form, that is to say: "the structure of the organs and the chemical composition!" From its physical standpoint physiology is consistent when it denies the existence of a vital force independent in itself and closes its eye to the ideal causality[*)]. But because the ideal causality is nowhere to be discovered in physics, it does not yet follow that such a

[*)]Die Frage, ob es eine Lebenskraft giebt oder nicht, ist in der neuesten Zeit wieder lebhaft aufgenommen worden. Wie verschieden sie beantwortet wird, sagen uns am besten einige der Männer selbst, welche die Nothwendigkeit erkannten, ihr Sein oder Nichtsein auf wissenschaftlicher Grundlage zur Entscheidung zu bringen. *Lotze* will, daß „die Lebenskraft nicht als die Eine wirkende Kraft des Lebens, sondern als die resultirende Größe der Leistung betrachtet werde, die aus der Vereinigung unendlich vieler partieller Kräfte unter gewissen Bedingungen hervorgeht." *Donders*, an eine Stelle aus *Sanctorius* anknüpfend: „Corpus humanum cur vivit et non pùtrescit? Quia quotidie renovatur," fährt fort: „In diesem Wechsel, in diesem Verbrauch von Stoff erkennen wir jetzt die unfehlbare Bedingung, den Grund des Lebens. Die geheimnißvolle Lebenskraft hat hierin ihr Grab gefunden. Chemische Kraft ist es, welche sich in Lebenserscheinungen umsetzt." Nach *Büchner* „gehört der Begriff Lebenskraft unter jene mystischen, die Klarheit naturphilosophischer Anschauung verwirrenden Begriffe, welche eine an Naturkenntniß schwache Zeit ausgeheckt hat, und welche von der neueren exacten Naturforschung über Bord geworfen sind." *Virchow* endlich findet in der lebendigen Zelle die Kraft, welche er in folgenden Worten andeutet: „in der Analyse der Lebenserscheinungen werden wir immer genöthigt sein, neben der Wirkung der dem Stoffe immanenten Molecülarkräfte die Nachwirkung einer, von einer früheren Stoffcombination übertragenen, immerhin mechanischen Kraft zuzulassen." [The question whether there is a vital force or not has in the most recent time been vigorously taken up again. How differently it is answered is best told us by some of the men themselves who recognized the necessity of bringing its being or non-being to decision on a scientific foundation. Lotze would have it that "the vital force be regarded, not as the one operative force of life, but as the resultant magnitude of the performance which proceeds from the union of infinitely many partial forces under certain conditions." Donders, taking up a passage from Sanctorius — "Corpus humanum cur vivit et non putrescit? Quia quotidie renovatur" — continues: "In this exchange, in this consumption of substance, we now recognize the infallible condition, the ground of life. The mysterious vital force has herein found its grave. It is chemical force that converts itself into the phenomena of life." According to Büchner, "the concept of vital force belongs among those mystical concepts, confusing the clarity of the natural-philosophical view, which an age weak in knowledge of nature has hatched, and which have been thrown overboard by the newer exact investigation of nature." Virchow, finally, finds in the living cell the force which he indicates in the following words: "in the analysis of the phenomena of life we shall always be compelled to admit, alongside the action of the molecular forces immanent in the substance, the after-effect of a force — at all events mechanical — transmitted from an earlier combination of substance."] — A. Kortüm. Die Lebenskr. Vorrede.

causality has no

validity. The investigations belonging here we shall in what follows let move about the categorial points of view: *the essence of life, the stages of life, the relation between physical and psychical life.*

a) The Essence of Life.

To assume a principle of life holding for organic individuals without presupposing a fundamental principle holding for the whole of existence is thoughtless. But how are we to convince ourselves that the outwardly actual world, with its things, thing-causes, conditions and laws, has so ideal an origin that this whole is held together and borne by so spiritual a power, that this manifold is worked through by a single, omnipresent principle? Existence divides itself — for this was the consequence to which the investigation of "the problem of change" led us — into two mutually opposed and yet necessarily belonging totalities: the inner and the outer, the ideal totality of the ground and the material totality of existence. Things, which have their ground and have arisen out of the ground, must go to the ground; but what holds of things holds of thing-causes too: at bottom no thing is originally cause. In going to the ground the thing nevertheless yields, as a moment of cause, its contribution toward another thing's arising, another thing-cause's stepping out of the ground: the existing causes are all without distinction grounded and caused. The principle, the true cause of causes, is therefore not to be sought in existence by itself, for existence has only derived, caused causes; nor in the ground by itself, for the ground has only moments of cause, cancelled causes; but in the originally operative unity of both. *By virtue of the ground-unity of the moments of cause the principle is an ideal unity of cause, a ground-cause; and as ground-cause it has existence and the existing causes in its power.*

What is to act as ground-cause cannot possibly have immediate existence; but although existence's principle does not exist as a thing by itself, and so cannot anywhere be pointed to directly, it is so far from being an empty thought-entity that this entity is on the contrary infinite capacity and acting: the capacity that is the ground's power, the acting that is existence's force. How far such a conception really admits of being carried through must be settled by an analysis of concepts answering to the matters of fact. Suppose a system of celestial bodies is to be formed in space: what has the principle to do with such a formation? Does existence's principle perhaps produce of itself at one stroke both masses and forces, distribute the masses in space, and fix the laws by which the forces are to act? By no means! The principle here spoken of is not a principle of miracle creating out of nothing, but a principle of nature that presupposes its material. Can it then begin by presupposing the atoms and itself adding the forces? Impossible! Can it perhaps begin by presupposing the forces and then itself adding the atoms? Wholly impossible! The truth is that the material with its atoms and masses is not something original but on the contrary something derived.

In the outer, material form of existence the atoms do indeed assert a mechanical independence of their own; what is mechanical in this independence rests on the outwardness by which space is posited and filled. In this outwardness the atoms then take on the semblance of things that have sprung up, of *facta* without *fieri*, because the *fieri* is hidden in the ground — of eternal facts, *facta, quæ non facta sunt.* But the masses which were to be formed of such facts, and the system of celestial bodies which was in turn to be formed of such masses, would then have to be formed under a conceptless and therefore inconceivable concurrence of finite conditions accidental to one another. Such a system, formed by virtue of accidental necessity, would then be a whole rational in its phenomenon

but irrational in its origin, an intelligible consequence of blind, groundless presuppositions — a senseless contradiction. The blind presuppositions are not, however, removed by an omnipotent knowing being slipped in without more ado, any more than what is rational in nature is vindicated by an Anaxagorean *νοῦς* or Platonic *δημιουργός* taking over the principle's part. No: the blind presuppositions are removed when an express distinction is drawn between *being in immediacy's outer material state and acting in reflection's inner, ideally overreaching form*; what is rational in the ground is vindicated when matter's points, masses and forces — in whatever state and at whatever stage of development the outer existential totality be held fast — are known as moments and serving terms for the inner totality in which the ground-cause reflects its acting[*)].

[*)]There is a dilemma which allows of being presented thus: *either everything must originally rest on atoms and physical causes, and then philosophy's ground-causes, principles and ideal powers are only "speculations"; or else philosophy with its principles and rational causes must be right, and then physics' doctrine of atoms and physical causes is of merely subordinate significance. when the question is of the knowledge of truth.* — It will now readily be seen that a true knowledge of nature is not possible so long as one will with critical wilfulness hang upon such a dilemma; it is further evident that this in itself fundamentally false dilemma must give materialism an absolute preponderance over idealism, for it is an incontestable matter of fact that nature's activity goes at every point through the physical causes. Every tolerably sober attempt to remove this dilemma, so ruinous in its consequences, is therefore to be valued. "Man setzt im gemeinen Leben das, was den Sinnen unzugänglich, in der Materie Bewegung und Veränderung hervorbringt, ihr als geistiges Princip *entgegen*, — die Materie aber, so wie das Höhere in ihr Wirkende sind nur verschiedene Stufen geistiger Kraft. — Für unsere Sinne existirt nur die materielle Welt, für unsern Geist existiren unmittelbar nur die in der Materie wirksamen Geister. Jedes Atom ist ein für sich selbständig Bestehendes, von einer Seele Bewegtes — ist nur diese Seele niederster Art selbst. .... Die Thätigkeit des Weltgeistes hat sich aber nicht darauf beschränkt, eine unendliche Zahl von Atomen — Wesen der niedersten Art — zu schaffen, es sollte zu höhern Produkten seiner Wirksamkeit kommen.

Here it can no longer be asked whence the principle gets its material, for atoms and forces are at bottom moments of its own content; here it cannot be asked how force originally came into matter, and reason into the system of the celestial bodies, for matter and force are in their factual difference-unity effects in existence of the ground-cause, and reason the power with which it masters conditions and thing-causes; here it cannot be asked how what is in itself unconditioned came from the beginning to be entangled in the web of conditions, for what is in itself unconditioned is not a thing whose conditionedness marks its dependence, but an operative selfhood which, in being conditioned, subordinates the conditions to itself and asserts its originality. The form of the principial acting is in general characterized by the formation of a relatively closed whole whose ideal centre, systematic articulation and pervasive unity the principle actualizes. Everything here rests on the opposition between the ideal and the material. This opposition must in its utmost abstraction be determined thus: *the ideal is the general selfhood reflected in itself, pure rationality; the material*

on the contrary is the otherness immediately present in the outward single thing. And since the otherness, the derived, is nevertheless necessary — indeed originally necessary — for the selfhood that conditions its own development, the principle of actuality everywhere sets out to work the two operative things together. Under this working-together the opposition then assumes on the whole a threefold character.

1) It begins at the lowest stage with the rule of that which states the outwardness, which stamps and marks finitude, namely the material: the otherness, which selfhood originally presupposes as its moment, then comes forward, not as moment, but in so forceful an immediacy that the ideal, whose acting is in reflection, must keep itself hidden beneath it. In intuitive form this is expressed thus: atoms and masses float in space; it looks as though they were quite indifferent to one another, so strong is the otherness,

Es sollten in allmählich aufsteigenden Kategorien *geistige Wesen* entstehen, welche gleichsam Gedanken des Weltgeistes, an und für sich nicht fähig, *sinnlich, materiell wahrnehmbar* zu erscheinen, aber mit der Macht begabt sind, sich in die Stoffwelt zu versenken, diese in verschiedener Art zu beherrschen, aus ihr sich Leiber zu gestalten und mittelst dieser räumlich und zeitlich auszutreten." [In common life one sets that which, inaccessible to the senses, produces motion and change in matter over against matter as a spiritual principle — but matter, as well as what acts in it as the higher, are only different stages of spiritual force. — For our senses only the material world exists; for our spirit there exist immediately only the spirits operative in matter. Every atom is something subsisting independently for itself, moved by a soul — indeed is only this soul of the lowest kind itself. .... But the activity of the world-spirit has not confined itself to creating an infinite number of atoms, beings of the lowest kind; it was to come to higher products of its operation. In gradually ascending categories spiritual beings were to arise which, as it were thoughts of the world-spirit, are in and for themselves incapable of appearing sensuously, materially perceptible, but are endowed with the power to sink themselves into the world of matter, to master it in various ways, to shape bodies for themselves out of it, and by means of these to come forth in space and time.] M. Perty. Allgemeine Naturgeschichte als philosophische und Humanitätswissenschaft. Bern. 1843. 1ster Band. pp. 115–116.

so predominant the outwardness; it looks as though these material parts would have remained eternally in this state of indifference had they not by chance — an eternally inconceivable chance! — been furnished with forces, so that operative relations could be formed. It is inertia, impact and gravity, mechanical attraction, that here represent the active essentialities. But whence have these essentialities their origin? They are reflexes of the selfhood acting through the otherness[*)]. What is rational comes into view, to express it in the language of physics, in the form of eternal, unalterable laws; but what are the unalterable laws other than the abstract forms of the general powers, and through these of selfhood? By means of the laws the outwardly manifold is held under the dominion of the inwardly one. The fulfilment of the laws shows us the rationality that goes out from itself, is at one with itself, and returns to itself through everything — the selfhood mighty in the selfless. The totality answering to this is the astronomical one, that system of motions which *Laplace* has denoted by *mécanique céleste* (cf. Phil. Propæd. p. 172).

2) At the next principal stage the otherness is already altered; it has therefore after all not been able to harden itself against selfhood, has not been able to conceal that with its immediate independence it is essentially moment. Mechanical otherness is sharp and as it were petrified: the one can be beside the other, can touch, strike, press the other, can adhere to the other; but a more inward union — a contact by which the one, qualitatively different from the other, should alter its quality in the likeness of the other

[*)]Although the categories are thrown about among one another somewhat too freely, the following description nevertheless gives on the whole a true total impression .... Nach einem Grundgesetz der ganzen Natur, vermöge welchem auf niederen Stufen erscheinende Kräfte auf höheren sich potenzirt wiederholen, zeigen die Weltkörper statt der verschwindenden Kleinheit der Kräftepunkte riesenhafte Dimensionen .... Entfernung von Centralkörpern, mittlere Dichtigkeit, Geschwindigkeit der Bahnbewegung und Axendrehung sind nothwendige Bestimmungen, verschiedene Seiten des eigenthümlichen Wesens der Weltkörperseelen. All dieses hat sich z. B. zu einem Planetensysteme geordnet durch Wechselwirkung und Wechselstellung mit den allen Gliedern eines solchen eigenthümlichen Proportionen. Verhältnisse dieser Art ordnen sich auf ähnliche Weise, wie etwa jene der einzelnen Personen eines Familien- oder der Gewalten eines Staatswesens, oder wie sich verschieden schwere, unmischbare Flüssigkeiten unter einander schichten — nach den sich in das Gleichgewicht einer bestimmten Modalität setzenden Kräften, ohne daß hier von Vorstellung und individuellem Bewusstsein die Rede sein könnte. [By a fundamental law of the whole of nature, in virtue of which forces appearing at lower stages repeat themselves raised to a higher power at higher ones, the heavenly bodies show, instead of the vanishing smallness of the points of force, gigantic dimensions .... Distance from central bodies, mean density, velocity of orbital motion and of axial rotation are necessary determinations, different sides of the distinctive essence of the souls of the heavenly bodies. All this has, for example, ordered itself into a planetary system through reciprocal action and reciprocal position with the proportions distinctive of all the members of such a system. Relations of this kind order themselves in a similar way to those, say, of the single persons of a family or the powers of a body politic, or as fluids of different weight and incapable of mixing arrange themselves in layers among one another — according to the forces setting themselves in the equilibrium of a determinate modality, without there being any question here of representation and individual consciousness.] M. Perty. Anf. Skr. p. 120.

— is at the stage of mechanism impossible. Reason, which by mechanical parcelling-out is prevented from immersing itself in the ground and mirroring itself in inwardness, then asserts selfhood as the general essentiality, an objective power ruling in a system of

mechanical laws. A more inward union, the identity of the mutually opposed, is reached in the chemical. Under the chemical processes things go to the ground in order to be grounded anew; here the existing terms become moments, in order that the moments may again become existing terms: it is the rational itself whose essence is the affinity of substances. Under the inward alterations of bodies the primordially forceful, elementary selfhood is seen to immerse itself in the ground, to articulate matter, and to work its way through the otherness. The totality answering to this is, in earthly concretion, the geological[*)]. "Seen from the standpoint of geognosy the earth is only a pieced-together

[*)]Was einmal die Verhältnisse der Erde zu andern Weltkörpern betrifft, so tritt dasjenige, in welchem sie zu ihrem Centralkörper, der *Sonne* steht, ungemein überwiegend hervor.... Wie durch den Umlauf um die Sonne die Phasen des Jahres gegeben werden, so erzeugt die (vielleicht auf Elektromagnetismus beruhende) Axendrehung den Wechsel zwischen Tag- und Nachtleben, wobei nach einander alle Punkte dem nahen glänzenden Fixstern, zu dessen System die Erde gehört, bald zu bald von ihm abgewendet werden. Wie aber im Leben eines sekundären Organismus keine bestimmten Abschnitte vorhanden sind, sondern jenes als eine stätige Linie erscheint, in welcher die wunderbarsten Abwechslungen, ja scheinbar widersprechende Erscheinungen, unmerklich in einander verfließen, so auch in den Erscheinungen, welche die doppelte Bewegung der Erde zur Folge hat.... Dieses Verhältniß allein, wenn auch keine andern bekannt wären, würde denjenigen genügen, die Erde für einen Organismus zu halten, welche das in Rhythmus und Metamorphose begründete tiefere Wesen eines solchen begriffen haben.... Kann man auch der Ansicht einer Umwandlung der Stoffe, wobei durch organische Thätigkeit der Erdorgane, wie im thierischen Leibe, Festes, Flüssiges, Gasiges und alle Stoffe in einander umgewandelt, und in jedem Organ des Erdganzen eigenthümliche Produkte, Sekreta erzeugt werden, wonach auch die Gesteine aus schleimigen Substraten entstehen und fest werden, dann regelmässige Krystallformen annehmen, und endlich verwesen sollen, nicht unbedingt beistimmen, so muß man einen Athmungsproceß von Atmosphäre und Erdfeste unbedenklich zugeben. Er erfolgt *rhythmisch*, mit Ueberwiegen bald des einen, bald des andern Factors, durch aufsteigende und absteigende Ströme in der Atmosphäre, und wird schon durch die regelmässigen und unregelmässigen Barometerschwankungen angedeutet. .... Die Luftmassen, welche über Meer und Erde streichen, müssen Friktions-Elektrizität aufregen; die so verschiedenen Massen, aus welchen die Erdrinde besteht, durch ihr blosses Aufeinanderliegen Kontaktelektrizität. Auch die ungleiche Erwärmung von Luft, Meer und Erde wird elektrische Phänomene hervorrufen. Welche auch die nähern Ursachen der *Erdbeben* und des *Vulkanismus* sein mögen, ob elektrische Ausgleichungen, Schwankungen einer unterirdischen Atmosphäre, chemische Bindungen und Zersetzungen, das Centralfeuer, oder begleitende Erscheinung organischer Umbildungsprozesse — nie können diese furchtbaren Vorgänge als bloß physische oder chemische, sondern sie müssen als zum Kreis des Erdlebens gehörig betrachtet werden. — Wenn nun die Erde, wie man nicht bezweifeln kann, im Organismus ist, so muß sie im Ganzen, wie ihre Organe im Einzelnen, *veränderliche Stimmungen* annehmen können, sie muß *pathologische* Zustände zeigen, die in ihrem eignen Entwicklungsgang begründet sind, oder ihr von aussen herbeigeführt werden. In solchen Verhältnissen werden wir die Ursache abnormer Jahre, so wie des Charakters der Jahrgänge überhaupt, ausserordentlicher Vermehrung mancher Thiere und Pflanzen, und jener grossen Epidemien zu suchen haben, welche von Zeit zu Zeit die Völker heimsuchen. [As regards the earth's relations to other heavenly bodies, that in which it stands to its central body, the sun, comes forward with quite overwhelming preponderance.... As the phases of the year are given by the revolution about the sun, so the axial rotation (resting perhaps on electromagnetism) produces the alternation between the life of day and of night, whereby all points are turned in succession now toward, now away from the near shining fixed star to whose system the earth belongs. But as in the life of a secondary organism there are no determinate divisions, that life appearing rather as a continuous line in which the most wonderful alternations, indeed whole; from the standpoint of geology, on the other hand, the earth's body is to be conceived as an elementary organism, an essential whole whose parts have developed out of an inner disposition;" (Phil. Propæd. p. 186). The earthly whole, the elementary organism, is an image of selfhood: moved by an other, while yet it moves itself; magnetic, while it magnetizes itself; electric, while it electrifies itself; a laboratory which itself brings about, conducts and directs the chemical processes.

3) If the ideal is to form the material through and through in existence, and the otherness is to become that for which it is originally disposed, a moment for the ruling selfhood, then the ideality operative in the elementary organism must raise itself to a higher power in higher principial fulgurations and furnish points of selfhood for organic organisms.

From Ehrenberg's Mikrogeologie, for example, these fulgurations, these sparks of life in the dust, are to be seen. "Whoever looks through the great plates of this magnificent work will be astonished and overwhelmed by the numerous and wonderful forms of the infinite life whose traces are met with everywhere in and above the solid crust of the earth, wherever the investigator's well-armed eye is able to penetrate. — Everything seems to acquire life; nothing dead remains, nothing of purely inorganic origin. This fullness of life, running through all ages, surprises us and easily leads us to the erroneous view which Gleichen, one of the first microscopic investigators, already entertained — namely that the whole earth is formed of infusoria. We are touched in an uncanny and

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seemingly contradictory appearances, flow imperceptibly into one another, so too in the appearances which the earth's double motion has for its consequence.... This relation alone, even if no others were known, would suffice to make those hold the earth for an organism who have grasped the deeper essence of such a thing, grounded in rhythm and metamorphosis.... Even if one cannot unconditionally assent to the view of a transformation of substances whereby, through the organic activity of the earth's organs, as in the animal body, solid, liquid and gaseous and all substances are transformed into one another, and in every organ of the earthly whole distinctive products, secreta, are produced — according to which view the rocks too arise out of slimy substrata and become solid, then assume regular crystal forms, and finally are to decay — still one must without hesitation grant a process of respiration between atmosphere and solid earth. It takes place rhythmically, with a preponderance now of the one, now of the other factor, through ascending and descending currents in the atmosphere, and is already indicated by the regular and irregular fluctuations of the barometer. .... The masses of air which sweep over sea and land must excite frictional electricity; the so various masses of which the earth's crust consists, contact electricity by their mere lying one upon another. The unequal warming of air, sea and earth will also call forth electrical phenomena. Whatever the nearer causes of earthquakes and volcanism may be — electrical equalizations, fluctuations of a subterranean atmosphere, chemical combinations and decompositions, the central fire, or an accompanying appearance of organic processes of transformation — never can these fearful occurrences be regarded as merely physical or chemical; they must be regarded as belonging to the circle of the earth's life. — If, then, the earth is, as cannot be doubted, in the organism, it must be able to assume, as a whole and in its organs severally, variable moods; it must show pathological states grounded in its own course of development, or brought upon it from without. In such relations we shall have to seek the cause of abnormal years, as of the character of years in general, of the extraordinary multiplication of certain animals and plants, and of those great epidemics which from time to time visit the peoples.] M. Perty. Anf. Skr. pp. 436–40.

ghostly way by this world, in part long since buried but now risen again, which seems to penetrate everything: the ice of the poles and of the glaciers, the ooze of the sea and the névé and loose earth of the highest mountain peaks, the dust which the whirling hurricane sweeps along and the mote that trembles in the quiet sunbeam, the fertile soil of the deltas and the water of the rivers and the seas, the solid rock-strata of the earth's crust and even the glowing ashes of the volcanoes." (B. Cotta[*)]).

With the organic organism, the organism in the proper sense, selfhood comes to immediate existence; this is, naturally regarded, the origin of life. For a closer understanding we treat separately of *the organism, the organic type, the organic purpose.*

α) The Organism.

In the astronomical whole, the system of worlds that fills the space of the heavens, the weight of matter holds; here is the realm of masses. But the realm of masses is nevertheless also a realm of light: sensuous light is the symbol of spirit, a reflected splendour of the ideality operative in the ground of reason. Already in the body of the earth what is of the nature of mass is forced under articulating forms; with the organism it disappears entirely: the cell, insignificant in mass, is here the significant starting point. In the formation of the first cell the whole problem of life is compressed. What is distinctive in the problem of life is that two causes so unlike in kind, so mutually opposed, as self-cause and substance-cause act together in an immediately forceful unity; what is distinctive is that the principle penetrates the material to such a degree that it is itself materialized, that selfhood is embodied: an organism is embodied selfhood. But although matter is taken up by the ideal, it does not on that account cease to be material. The question therefore becomes how the same atoms which make up the constituents in dead masses can now also make up the constituents in the living organism. In this respect it will be our task to characterize the organism from a chemical, a morphological and a metaphysical standpoint.

For *chemical* analysis the organic compound is a product of affinity just as much as the inorganic one; the product is selfless, and therefore organic chemistry cannot possibly give us any direct information about what life really is. Mediately and indirectly, on the other hand, the chemistry of organic compounds throws a peculiar light on life's activity. For if we compare the small number of elements — carbon, hydrogen, oxygen and nitrogen — of which organic bodies on the whole consist, with the unsurveyable quantity of compounds that can be formed from them, we get an impression of matter's

[*)] Geologiske Billeder, ed. C. Fogh. Copenhagen. 1859. pp. 102—103.

alterability, its many-sided pliancy and willingness to let itself be formed. How many different bodies do not admit of being formed out of carbon, hydrogen and oxygen — indeed, how many out of carbon and hydrogen alone![*)] "The different kinds of starch, such as amylum, lichenin, inulin, cellulose and gum, have the common expression $C_{12}H_{10}O_{10}$, and quite different oils, such as oil of lemon, oil of cloves and oil of turpentine, have one and the same formula $C_{20}H_{16}$. One frequently helps oneself with the assumption that the same relative quantities of atoms in these *isomeric* bodies are ordered in different ways ... But," it is added (Physiol. by Valentin, ed. Hannover, p. 19), "this hypothesis gives no true information, and cannot smooth away the practically subsisting disproportion between what investigation requires and what for the moment it can do"[**)]. If, then, we presuppose what was shown in detail in the preceding section — that the atoms as points of stuff themselves undergo a material modification at every rearrangement — then the alterability becomes so pervasive that the chemical laws can be satisfied and the significance of the affinities vindicated without matter's being on that account able to

[*)] The great classes of volatile oils, to which oil of turpentine, oil of lemon, oil of copaiba balsam, oil of rosemary, oil of juniper and others belong — so different in their smell, their medicinal effects, boiling point and so on — contain carbon and oxygen in the same proportion, no more of the one constituent than of the other. In what wonderful simplicity does organic nature not show itself to us from this standpoint: with two equal proportions by weight of two constituents it brings forth an extraordinary multiplicity of the most remarkable compounds. Bodies have been discovered which, solid and volatile at ordinary temperature, like the crystallizing constituent of oil of roses, have the same composition as the gas that burns in our candle-flames, and besides with a dozen other bodies all highly different in their properties. Liebig. Chem. Breve. Copenhagen 1845. 11th letter.

[**)] "The reader may open almost any work on physiology and organic chemistry, and he will there find developments of the theory of nutrition and of the nutritive value of the various foods so exact and so decisive in their formulae that he will scarcely listen to me with patience when I maintain that the exactness is deceptive and the doctrines demonstrably incorrect. Nevertheless I hope, before we reach the end, to convince the reader that chemistry itself is in far too imperfect a state to give clear and satisfactory answers to its own

questions in this direction — as Mulder and Lehmann candidly admit — and further that chemistry, even if we assume it to be perfect, must constantly be unable to solve physiological problems ...... The philosophical poet warns us: "From higher court to lower
Bring not thy cause;"
and the appeal from physiology to chemistry is just such an appeal from higher to lower. The analysis of a nerve will never throw light on sensation; no array of chemical formulae will explain the cell's form and properties. One can take a mechanism to pieces and explain by physical laws the action and engagement of every wheel and every chain; but one cannot take an organism to pieces and explain its properties by chemical laws as the laboratory has shown them to us. If an overwhelming illustration of this conspicuous truth is needed, we find it in the animal egg: here is a microscopic globule, composed of substances well known to chemists, which potentially contains an animal, and which will reproduce not only the parents' shape, features, growth and particular peculiarities, but also many of their acquired habits, inclinations and oddities. Has chemistry in its whole extended domain anything answering to this? Can chemistry give us so much as an approach to an explanation of it? Chemical analysis can lead us to life's threshold, but at the threshold its guidance ceases. Here a new class of complications appears, a new series of laws must be discovered." Hverdagslivets Physiologie (after George Henry Lewes, »physiology of common life«). First part. pp. 63—65.

set the forming principle any essential hindrance, let alone any insuperable barrier[*)]. As points of stuff the atoms must always adapt their material distinctiveness to the state in which they find themselves, the connection in which they act. And since no atom, in so far as it is material, is self-active, but every atom must always be roused by influence to conditioned activity, it is understood from this that the atoms can, as activity decreases, sink into a state of rest and so make up the constituents of dead masses; but since the atoms are essentially points of function, it is seen at the same time that they must be able to be roused again to distinctive acting — indeed some of them, under particular conditions, even to take part in the activity of life. But further than this result chemical analysis cannot possibly lead.

On the *morphological* conception the organism's parts are products of organic activity, are as phenomena images of life's essence; morphology lays decisive weight on the difference between an organic and a merely chemical product. Here, then, one does not look at the form of delimitation alone; one looks also at the inner structure, the microscopic composition: anatomy must support morphology as much as chemistry does. "Before Bichat, who must be called the founder of philosophical anatomy, the body was held to be composed of different parts or organs; when one had enumerated these parts, one had finished one's description. Bichat lit a light for science when he showed that the organs themselves consisted of different tissues or fundamental formations, each of which preserved its distinctive properties in whatever part of the body it was found. The heart, for example, is an organ consisting of muscular tissue, connective tissue, nerve tissue and fatty tissue, each of these tissues showing the same properties in the heart as it shows in every other organ — quite in the same way[*)] as the different materials of which a ship is made: wood, hemp, copper, iron, tar and so on preserve their characteristic properties although the wood may be rudder, deck or mast, and the iron

[*)]On the proofs of the absolute likeness of composition with highly unlike properties Liebig adduces (in the letter cited, p. 104): "In cyanuric acid, hydrate of cyanic acid and cyamelide we have three such bodies. The first can be dissolved in water, is crystallizable, and able to form salts together with metallic oxides; hydrate of cyanic acid is a volatile liquid, corrosive in the highest degree, which cannot be brought together with water without being decomposed; cyamelide is a white, porcelain-like mass quite insoluble in water. In a hermetically sealed glass vessel cyanuric acid is transformed under the influence of a higher temperature into hydrate of cyanic acid, and this passes of itself at ordinary temperature into cyamelide, without a constituent's departing or a body's being taken up into it from outside. Cyamelide allows of being transformed at will into cyanuric acid or hydrate of cyanic acid."

[*)]By the words "quite in the same way" one must not, however, be led into overlooking the essential difference between a ship and an organism: namely that the different materials of which a ship is made are delivered to the builder as materials whose inner constitution he must use as something given, whereas the organizing workman on the contrary himself weaves the tissues of different kinds, and thereby determines their inner constitution. "The tissues are the fundamental constituents of the animal structure, and a particular branch of science bearing the name of histology or *general* anatomy is devoted to studying it. The organism is composed of organs, and the organs of tissues." (Hverdagsl. Physiol. 1. p. 4.)

anchor, nail or chain." (Lewes).

That a plurality of formations are simultaneous in space, that each of these formations is transformed with time, that whole systems of articulating formations subordinate themselves to a ruling unity — these are, from the formation of the first cell to the completion of the most various organs with their organ-parts, delimitation, shape and habit, pervasive traits of character. In the cell at its first stage of formation there is already to be seen, besides the nucleus if such there be, the cell-slime (protoplasm, cytoblastema), the primordial membrane (*utriculus primordialis*), which is nitrogenous, and the cell-membrane, which consists of cellulose. That these different formations, which mutually presuppose and furnish presuppositions for one another, are terms of passage for a forming one — manifestations, that is, of one essence — is interpreted in the phenomenon by the cell's living as an organized unity, by what thus forms itself being one living thing.

The living cell in turn undergoes transformations — indeed has its life in a series of transformations. In its first youth the cell (we are thinking here chiefly of plant organisms) is filled with evenly distributed protoplasm, which serves for its nourishment, since in the beginning it always grows very strongly; but the even distribution of the protoplasm soon gives way to an articulating separation by the small drops of cell-sap; at last the drops flow wholly together, and the currents of protoplasm that flowed between them disappear. The cell enters a new stage. Between the first and second stage, between the youngest and the next-youngest cell, the difference is characteristic. The next-youngest cell now differs in general from the youngest by greater volume, by thicker walls, by the vanishing protoplasm confined to parietal currents. In the young cell there is no chlorophyll; the chlorophyll (by Mulder's analysis $C_{18}H_{18}N_2O_3$) forms itself by degrees, and is with its amylum in turn to furnish nutritive matter for new cells. The cell ages, the grains of chlorophyll become sparser; it withers, and must at last, as its content is transformed, itself furnish nutritive matter[*)].

Organism, organ and organ-element are categories which in this connection pass fluidly over into one another. Although the formation of cells by division is the most frequent — all vegetative organs, indeed most reproductive organs too, having their origin from it — there nevertheless also takes place a free formation of cells, an organic

[*)] From the cell's liquid, however, solid substances, secretions, have separated out; these are found, before it has yet dried up, either swimming in the liquid or depositing themselves on the cell-membrane. And if the deposition now takes place little by little to such a thickness that the whole cavity of the cell thereby disappears, then the secretion is in most cases what one calls "wood." Woody matter, lignin, which on Wöhler's statement (Org. Chem. ed. Scharling pp. 59—60) is assumed to be $C_{35}H_{48}O_{20}$, makes up the chief mass of wood, flax, hemp, straw, linen and so on.

new formation, in the embryo-sac of phanerogamous plants. Such an embryo-sac, such a momentary organism, is originally only an organic moment, a larger cell which shows itself in the nucleus of the egg — before fertilization. In the upper end of the embryo-sac there soon appear, however, two or three, sometimes more, rounded bodies; these are cell-nuclei, in which one observes one or more small luminous points (nucleoli), organically distinct centres. Around these cell-nuclei, surrounded by the cell's remaining nascent content, a membrane now forms; and so we have the embryo-vesicle — an organism, an organ, an organ-element. Of the two or more embryo-vesicles thus formed, as by way of preparation, in the embryo-sac, one (or in some cases a few) is then — after fertilization — developed at the expense of the others; and now a formation of cells by division begins, whose first concluding result is the formation of the germ. But if we consider the germ, whose original part is the germ-axis, it is seen that the germ itself is more than a particular organ, that it is an organism in its own right, a plant in disposition. By this disposition of its the plant articulates itself into essentially different organs: axis and leaves. For the germ-axis is divided into an ascending part, the germ-stem (*Cauliculus*) with seed-leaves (*Cotyledones*) and germ-bud (*gemmula*), and a descending part, the root-bud. From the system of the first characteristic formations the morphological development can then further be followed through all the phenomena of transformations and stages of formation up to that turning point which, in furnishing a conclusion valid for the organic cycle, at the same time marks a new beginning. That in such a cycle of phenomena an inner unity of essence is mirrored is evident; but held captive by the phenomenal, morphology cannot immerse itself in this unity of essence, cannot reach the forming principle. If it is to be possible to reach the principle and know what life is, the knowledge must be *metaphysical.* Metaphysical knowledge is not won by mystical speculation, any more than by wavering hypotheses. A thinking which, in order to grasp what is otherwise ungraspable, steals in behind the matters of fact, behind consciousness, and at the same stroke past the difficulties, science cannot and will not acknowledge. Here there cannot and shall not be applied a biological seer's gaze which, without the least hint being given of how knowledge reaches life's source, is at once and without any trouble at the source, sees deep into the source, describes everything, defines everything, explicates everything — so that the whole at last looks like a veiled tale, a trivialized apocalypse of opinion. The metaphysical knowledge here in question must in all essentials attach itself so closely to the physiological matters of fact and their exact interpretation that the expert can point for point control both its empirical presuppositions and the consequences it draws. Its distinctiveness rests on its distinctive form; for its form is dialectic, a form essentially answering to life.

As an actual unity in an actual opposition of the ideal and the material, the essence of life is dialectical; the dialectic is determined 1) by the two-sided nature of the atoms as points of stuff and points of function; 2) by the two-sided reciprocal action of organizing — between the atoms themselves and between the atoms and the principle; 3) by the organic doubling of totality, in that the inner and the outer are, by means of the ideal and yet material character of the functions, articulated into an opposition of ideal and material organism.

1) *The two-sided nature of the atoms makes the boundary of the chemical dialectical.* By ascribing to chemical affinity independence over against heat, the vital force and so on, one easily comes from chemistry's standpoint to overlook the circumstance that affinity, separated from its conditions, is really only a possibility, not an actuality. As far as the essence is concerned, one may indeed set affinity independently against heat, for example; for if the substances in question were not in the requisite affinity, heat could not possibly force any union: the union is to be referred to the activity of affinity alone as to the essential cause. But since the essential cause, which by itself alone is a real possibility, acts only by means of the conditions, the activity of heat is — as chemistry expressly acknowledges too — an integrating moment in affinity's activity: the chemical compound has in this respect not two or more pieced-together causes, but at bottom one cause, namely affinity operative by means of heat[*)].

What holds of affinity's relation to heat holds in a far higher sense of its relation to the so-called "vital force." To let the "vital force's" relative independence over against independent affinity rest on this force's having, like light, heat and gravity, "the capacity of communicating motion to resting atoms and of annihilating motion by resistance," betrays far too external and superficial a way of representing things. What would chemical affinity signify if the mutual attraction, motion and arrangement of the parts rested on affinity only in part, only in some half-and-half way, or thereabouts — while now a ray of light, now a ray of heat, now a ray of life, and heaven knows what other twitchings, tremblings and rays could push themselves in between the parts, give the atoms a little shove, and turn the whole relation about, so that affinity, which was to be cause, became at last condition, and the conditioning force cause? To ascribe to heat and light independence over against the chemical affinity whose activity they condition will

[*)]When a raised temperature brings it about that, say, pyrophosphoric acid is formed out of metaphosphoric acid, and a still stronger ignition that metaphosphoric acid is formed again out of the pyrophosphoric, then the action of affinity does indeed alter with the heat; but it is nevertheless affinity itself and nothing else that acts chemically. Heat rouses affinity to activity, but for the rest adds nothing, forces nothing that does not lie in affinity itself.

pass well enough, in so far as these potencies are assumed to have a medium of their own and moreover to exercise an activity that may fall outside the chemical; but to make the vital force a “condition” — a condition standing in so outwardly finite a relation to chemical affinity that there could be talk of rousing a motion by impact and annihilating it by resistance — will not pass at all. Where, then, has “this force” its medium? Or what activity does the organizing “vital force” exercise outside, above, below or beside the organism’s chemical processes? No: if affinity is to assert its independence, if life’s chemism is really in earnest, then there is no particular force. to set up against the force of affinity, no separate vital force to guide it, set it right or tinker with it; then, in other words, organic activity is only a higher expression, answering to a system of conditions of its own, of the chemical.

But when the organic has in this way become in each and all chemical, *the chemical itself has become more than chemical.* For if the organic is a higher chemical, then the chemical is conversely a lower organic. In this way the gulf between the lifeless and the living seems indeed to be levelled; but the matter is not thereby settled. Instead of falling upon organic individuals, the opposition between life and death is now laid into the atoms themselves: it is the atoms that by their original nature belong to two spheres. There must therefore be, for the function of every atom in question, a nodal point through which the function passes from being elementary-chemical to being organic-chemical — the nodal point through which the atom itself passes from being lifeless to being living.

If the cell, the organism from the first beginning, is formed under a reciprocal action of living points of function, and if this reciprocal action is only chemism raised to a higher power, then life does not come into the corporeal as a foreign essence either, but springs from the nature of the substances, blazes up by the self-ignition of the atoms, a flame of ideal combustion. The beginning is then made, on this conception, not from an unconditioned unity but from a manifold conditioned in itself. As the manifold tremblings of the ether first yield a ray, and the countless coloured rays gather into unity in white light, so the countless points of life first yield, by their organic-chemical cooperation, the living unity. Yet scarcely has the unity of life gathered itself out of this manifold before it again concentrates itself into a unity of self to such a degree that it rises above the manifold and subordinates to itself the activity of the points of stuff. But that the unity thus results in the consequent *is precisely a proof that as principle it was already present in the presupposition.*

Under the dominion of the unity of self the two-sided nature of the atoms comes into view: they are points of life and yet points of stuff; they are called to life, their elementary function raised to the power of a function of life. But it is only for some moments that the single material point can endure the ideal tension in this way; its nature as stuff is against ideality, and so it grows weary, so it is pressed out of the circle of the points of life and returns again to the elementary. Should this not be the real ground of the incessant change of substance in the organism?

2) *The two-sided reciprocal action of organizing makes the determination of the morphological dialectical.* That between the processes which organic chemistry is able to interpret and the activity of life itself there must be an essential difference, a considerable distance, is evident from many sides. Science is also in this respect conscious of its boundary. The very circumstance that the same chemical products can be won under different conditions by different roads shows that chemistry might very well succeed in producing a series of organic compounds without its being thereby in the least cleared up how these compounds are really formed in the organism itself[*)]. For the emphasis falls, as regards the organic, not on the product that is formed but on the processes under which it is formed, not on what is brought forth but on the manner of bringing forth.

For chemical analysis the bringing-forth and what is brought forth, the process and the product, must fall apart: in so far as the process is, the product is not yet; in so far as the product is, the process is over. In organic activity, by contrast, the process is essentially reflected in its product, what is brought forth simultaneous with its bringing-forth[**)]: the vessels with which the organism operates are formed, shaped and maintained by the operations themselves. The difference between the chemism of organic compounds and the activity of life is here conspicuous. For if the activity we know from chemistry as chemism were the same as the activity of life, then it would also have to be able to accomplish the same; then chemical art would at last have to be

[*)]“If cyanate of oxide of ammonium is warmed, or its solution in water boiled, then it is altered, by a rearrangement of the constituents and without any qualitative change in its composition, into a quite other body: urea, $C_2N_2H_4O_2$ (which is therefore metameric with the cyanate of oxide of ammonium), which is not a salt but must be reckoned among the organic bases.” (Wöhler.) — But from this we do not learn how urea is formed in the living organism.

[**)]Im Chemismus sind die sich zu einander verhaltenden Materien zwar durch ihren Begriff auf einander bezogen (chemische Verwandtschaft) und erhalten somit an sich ihr Product, welches nicht schon durch das vorher Vorhandene ihm Gleiche sich erzeugt. Seine Hervorbringung aber ist keine Selbsterhaltung . . . Der organische Ernährungsproceß ist dagegen eine vollkommene Bestimmung der materiellen Vermehrung durch die innere schon existirende Form, welche, als das Subjective, oder als die einfache Form aller Theile sich zu sich selbst, oder jeder gegen die übrigen Theile als gegen ein Objectives verhält und nur mit sich im Proceß ist.“ [In chemism the materials standing in relation to one another are indeed referred to one another by their concept (chemical affinity) and so receive in themselves their product, which is not already produced by what was previously present and like it. But its bringing-forth is no self-preservation . . . The organic process of nutrition, by contrast, is a complete determination of the material increase by the inner, already existing form, which, as the subjective, or as the simple form of all the parts, relates itself to itself, or each part to the remaining parts as to something objective, and is in process only with itself.] Hegel: Philosophische Propädeutik. Berlin 1840. p. 167.

able to awaken life and to master its morphological phenomena. But there is surely no sober physiologist who owns to the view that organic chemistry should ever bring it so far that it could form either a stem or a leaf or a foot or an eye or any organ whatever of either the plant or the animal[*)]. Thus by life's chemism the chemical goes

together with the morphological; but in chemistry's chemism the organically morphological falls quite outside the chemical. Life's chemism is therefore essentially different from chemistry's.

That every constituent of the cell — its protoplasm, its vacuoles, its primordial membrane and so on — must form itself point for point by the reciprocal action of the atoms and molecules is incontestable; but it is nonetheless an illusion when, because

[*)]So soll denn auch die Unmöglichkeit, organische Formen hervorzubringen, nur darin ihren Grund haben, daß wir die dazu nöthigen Bedingungen nicht herbeiführen können. Gelänge uns dies, so wäre kein Zweifel: der Homunculus spränge aus dem Tigel. Gewiß: wenn alle Bedingungen einer Erscheinung vorhanden sind, tritt sie in die Wirklichkeit. So weit aber für jetzt unsre Wissenschaft reicht, *ist der organische Proceß selbst die nothwendige Bedingung, sollen organische Formen entstehen.* Seltsamer Weise wird nun bisweilen von den Vertretern der physikalischen Physiologie auf eine ganz unbefangene Weise eben diese Bedingung eingeführt, um die organischen Processen zu erklären. Die Verdauung sagt man z. B. ist nichts weiter, als ein chemischer Proceß. Der Magensaft vollbringt sie auch ausser dem Organismus, dieser selbst thut ja nichts weiter dazu, *als daß er eben jenen Saft zubereitet.* Oder: wenn der Organismus im Stande ist, organische Combinationen und Formen zu erzeugen, wir dagegen mit aller unserer Ueberlegung nicht die Spur einer solchen Form herausbringen, so hat dieß *nur darin* seinen Grund, daß er mit ganz andern Apparaten operirt als wir in unsern Laboratorien in unserer Gewalt haben; *daß er statt der Retorten und Tigel aus Metall und Glas mit vegetabilischen Zellen und thierischen Häuten arbeitet.* Ja freilich, diesem Besitz der organischen Form hat der Organismus seine Effecte zu verdanken. Sobald wir aber eben diese Bedingung den allgemeinen mechanischen und chemischen Gesetzen hinzufügen, sobald wir also behaupten: die organischen Erscheinungen seien nichts anderes als die nothwendigen Wirkungen jener Gesetze, jedoch modificirt durch den Umstand, daß sich diese Wirkungen an einen bereits vorhandenen organischen Keim anlegen, so wird damit die ganze physikalische Erklärung des Organismus zu einer Illusion. Es ist gerade so viel, als wenn wir sagten: alle Erscheinungen des Organismus sind das Produkt der allgemeinen unorganischen Kräfte, nur er selbst nicht. [The impossibility of bringing forth organic forms is thus supposed to have its ground only in our being unable to produce the conditions necessary for it. If we succeeded in this, there would be no doubt: the homunculus would spring out of the crucible. Certainly: when all the conditions of an appearance are present, it steps into actuality. But so far as our science at present reaches, the organic process itself is the necessary condition if organic forms are to arise. Strangely enough, this very condition is now sometimes introduced in a quite unembarrassed way by the representatives of physical physiology in order to explain the organic processes. Digestion, one says for example, is nothing more than a chemical process. The gastric juice accomplishes it outside the organism too; the organism itself does nothing further toward it than prepare that juice. Or: if the organism is able to produce organic combinations and forms, while we for all our deliberation cannot bring out the trace of such a form, then this has its ground only in its operating with quite other apparatus than we have at our command in our laboratories — in its working, instead of with retorts and crucibles of metal and glass, with vegetable cells and animal membranes. Yes indeed: it is to this possession of the organic form that the organism owes its effects. But as soon as we add just this condition to the general mechanical and chemical laws — as soon, that is, as we maintain that the organic appearances are nothing other than the necessary effects of those laws, modified however by the circumstance that these effects attach themselves to an organic germ already present — the whole physical explanation of the organism thereby becomes an illusion. It is just as much as if we said: all the appearances of the organism are the product of the general inorganic forces, only the organism itself is not.] Leib und Seele von Julius Schaller. Weimar 1855. pp. 154–55.

these separate constituents allow of being held fast in a lifeless state as chemical products, one makes the cell itself a chemical product — when, confusing the functions of life with chemistry's elementary functions, one looks merely at the reciprocal action of the atoms and forgets the principle. It is the principle that makes the cell a cell; it is the activity of the principle, the double reciprocal action, a reciprocal action of stuff-like immediacy and pervasive reflection, by which the organic form is determined. The organic form must necessarily presuppose a stuff-like immediacy; for an abstractly ideal acting, a reflection whose terms and factors are reflexes without independence, will incessantly consume its own products and can give no outward, material and visible shape. But the organic form must with the same necessity presuppose something forming, whose acting transforms what is brought forth in matter into a mirroring of the bringing-forth, a reflected moment in the system of forms — an energy which, in that its acting returns through all conditioning intermediate terms to unity and selfhood, and so asserts the unconditionedness reflected in the conditions, assumes the character of principial reflection.

Such an organizing energy of reflection, answering to the nature of the principle, is of course only "a turn of phrase" and "wind" when one thoughtlessly overlooks the dialectical; but if one sees the dialectic of the energy of reflection, it shows itself in equal degree in accord with concept and with matter of fact. *Half-dialectically* "speculation" concludes thus: the more emphasis is laid on the principle's reality, the less validity must the physical causes have; now the principle of life has unconditioned reality; *ergo* the physical causes are only fleeting moments without essential significance. *Undialectically* "empirical inquiry" concludes thus: the greater the weight one lays on the conditions and the physical causes, the less the principle has to signify; that the physical causes with their conditions are operative at all points in the organism is evident: *vere scire est per causas scire*; *ergo* the principle of life is a fancy. *Dialectically* one concludes thus: the greater the energy which the principle — the unconditioned in its conditions — displays under given presuppositions, the greater the emphasis that must necessarily fall on the corresponding elementary causes; now the principle comes forward in the living organism with distinctive energy; *ergo* an essential, a principial emphasis falls here on the physical distinctiveness of the elementary causes and conditions.

The more the material, the stuff-like, is brought out, the more strongly does what is elementary in the forms come forward. It then looks as though the several organs, vessels, tissues, cells and groups of cells were only physical products of physical thing-causes acting under particular conditions; it looks as though the whole matter were settled by the apparatus composed of microscopic parts forming the "many small workshops whose activity engages into one another and is according to circumstances modified in manifold ways, since they are labile arrangements" (cf. Phil. Propæd. p. 187); it looks as though one had explained everything when one has merely pointed to "the whole set of necessary conditions," and so on.

How absurd it is to try to overlook the reciprocal action of atoms and molecules; how absurd it is to abstract from the system of conditions, to conceive the principle as a thing unconditioned in itself, a mystical thing-cause doing violence to the conditions, an architect who, however much it is asserted that he acts *from within*, nevertheless betrays on closer inspection that he has crept into the house *from without* — this physiology has excellently shown, and its arguments on this point are weighty and striking and strong. But how absurd it is to try to overlook the ground-cause, to close one's eye to the unity reflected in the multiplicity of the conditions, the unconditioned, and to explain everything by aggregations — this physiology is on the other hand, in its zeal to make plain "the favourable conditions," tempted in many ways to overlook. The arguments with which the principle of life is combated from this standpoint — one most often identifying, by a comical mistake, "the principle" with "the force," and so letting the former suffer for what may rightly be objected against the latter — are then also to a striking degree superficial, untenable, unscientific[*)] The more the principial reflection is

[*)] *The body is one; all its parts are subordinated, all are bound together into a higher unity.* Our consciousness *teaches us that our life is a unity.* — "This argument," says Lewes, "points to an important fact, but it is wrongly interpreted. There is unity, there is an agreement between all the parts of the organism. But this should in my opinion not be ascribed to a principle of life existing independently of the organism; it is due to the organic subordination; all the parts stand in relation to one another, all act in union with one another by means of the nervous system, just as all the parts of an army cooperate by means of officers and discipline … One can divide a polyp or a worm into several pieces, and all the pieces continue to live and to grow; but one cannot on that account assume that in such cases one has divided the principle of life into several principles. There is a life in the parts and a life belonging to the whole organism. Every microscopic cell possesses an independent existence, runs through its own course from its birth to its death, and the total sum of these single lives makes up what one calls the animal's life: the unity is an aggregate of forces, not a single governing force." (Hverdagsl. Physiol. 2. pp. 433–434.)

What is superficial in this manner of argument stands in a remarkable conflict with the energy of reflection and sustained thoroughness with which the author otherwise treats his tasks. It is said that the unity rests on the "organic subordination" of the parts; but when "every microscopic cell possesses an independent existence," what then does "the organic subordination" mean? If "the unity is an aggregate of forces," if there is here "no governing force," how does it then go with "the army"? What do "the officers" then accomplish? What does "discipline" mean?

"I have," the author continues, "dwelt so much the longer on this subject because, although the hypothesis of the principle of life" — he means the vital force — "has lost all repute, the metaphysical view lying at the ground of it is unfortunately still too general among physiologists; and since it is the purpose of this work to regard physiology even more with a view to general cultivation than as a branch of a special science, this discussion became necessary." But has the author not by this "discussion" overstepped the boundary he has himself prescribed in his Physiology: "*never to attempt the solution of the problems of one science by the order of ideas belonging to another*"? Suppose the cultivated reader really comes to agree with the brought out, the more clearly is it evident that "the many small workshops" are only points of passage for selfhood, "apparatus" for the one overreaching workman. Nucleus, plasma, primordial membrane, cell-membrane presuppose one another in the living cell so essentially, so originally, that the way in which each single one of such constituents is formed is precisely, by means of selfhood — the unity confirming itself through otherness, difference and plurality — determined by the way in which all the other parts, each by itself and in their mutual relation, are formed. What holds of the formations of different kinds in each cell at each stage of formation holds also of every system of cells, indeed of every system of systems with transitions, entanglements and fusions. The many formations and parts of different kinds are all together sides, moments and reflexes in the acting articulated by the reciprocal action of the atoms, which expresses selfhood's energy.

Chemical form-activity brings it only so far that what in a certain state, at a certain time, is an existing term, a thing by itself, can at another time, in another state, become a moment for another existing thing; but this is precisely the distinguishing mark of organic forming, that what is distinctively formed — this cell, this tissue, this vessel — is at once an immediately existing term in immediate shape and yet, with this shape and in this state, at the same time a moment for the whole, a shape-reflex of all the terms in the total shape.

3) *The organic doubling of totality makes the relation between the inner and the outer distinctively dialectical.* A dialectical opposition between the outer and the inner runs in wide generality through all actual existence. How could anything at all exist if it were neither an outer nor an inner? But if the existing thing is an outer, then as outer it can exist only by means of its inner; and if the existing thing is an inner, then as inner it can exist only by means of its outer. To exist as an outer is to be in immediacy's sensuous form; to exist as an inner is to be in reflection's essential form. The one of these forms can, so long as it is only a matter of the abstractly formal, be transformed again into the other — the inner thus itself becoming an outer, the outer an inner. There is in the

author, and with himself, that "the hypothesis of the principle of life has lost all repute," since "the single lives" are explained by "the favourable conditions" and "the total sum" of "these single lives," which makes up "the animal's life," by an "aggregate of forces": what is he then to judge, on this occasion, of the solemn pointing to "the great revelations of the inconceivable," to "the many mysteries that surround us"? If one really can explain the phenomena of life without a principle, then the materialist is perfectly right, then there is here nothing inconceivable in itself, no great mysteries. But if one cannot explain the phenomena of life without presupposing an ideal cause distinct from the physical forces — what right has one then to reject the principle of life? That "familiarity with the inorganic phenomena" only too easily "blunts the sense for their unspeakable mysteriousness," in this the author must be right; but, we ask, should being serving media for the all-operative reason, the ideal causality, the principle, not constitute precisely a chief trait in the "unspeakable mysteriousness" of the inorganic phenomena?

whole of actuality no determination so reflected that it cannot always, under one point of view or another, be seen immediately and so far reckoned among the outer; on the other side, not even the material itself has any immediacy so strong that it cannot always be resolved into some objective reflection and so far brought under a category of the inner. If the opposition is on the other hand to be determined *in concreto*, its differing character will of course depend on the distinctiveness of the existing things. It therefore makes in this respect an essential difference whether the existing thing presents itself as *a thing*, as *a whole* or as *an organism*. In *the thing* as thing the opposition between the inner and the outer has, although there is here a concrete existence, not yet raised itself above the abstractly formal. What is the thing's outer? This green, this solid, this heavy, this angular — in short, the set of its properties! And the inner? The one reflected in this multiplicity! — The thing itself, which has the properties, the thing in itself, which keeps behind the properties, is a hidden, an enigmatic essence; it is this essence which, as "the critique of reason" has taught us, knowledge can never reach, because the understanding, when it most exerts itself to penetrate into the thing as thing, entangles itself in the phenomenon and remains hanging in the properties. But with just as much right one might now turn the relation about and maintain that the thing as the immediate unity of the object is precisely the outer, what is manifest by itself, while the properties — each single one of which may demand a whole system of essential relations in order to reflect itself therein and by the reflection make itself known — on the contrary constitute the determinations in the proper inner. What sort of thing water is, we all know; but now the water is blue, now it is green, now it is warm, now cold, now liquid, now solid, now gaseous: how deep must one not penetrate into existence's inner before one can know what such properties as green and blue, warm and cold, solid and liquid essentially signify!

If we next conceive the existing thing as *a whole*, the tension between the outer and the inner will already make itself known. Over against the whole the parts have quite another independence than the properties have over against the thing. Every single part is itself something existing just as much as the whole is, whereas no single property can have independent existence, but is and remains a mode of existence, a constitution of the existing thing. The same existing thing which in a given specimen is heavy is also liquid, is also green, is also cold; thus: it is this heavy thing that is liquid, this heavy liquid that is green, and so on. In so far as the properties have no particular independence over against the thing, they have in the thing itself no particular independence over against one another either. It is otherwise with the parts! Precisely because the parts have a certain independence over against the whole, they also have a certain mutual independence over against one another: what, then, is in the wholeness itself the outer, and what the inner? Superficially regarded, both the parts and the whole seem to come forward as outer, so that the inner disappears altogether. That there must nevertheless be an inner here we notice as soon as the question is raised of the manner of connection, of the relation between the constitution of the parts and the determination of the whole. In the mechanical wholeness it looks almost as though the whole and the parts divided the outer and the inner between them, so that each part as part were partly an outer and partly an inner, and likewise the whole as whole partly an inner and partly an outer. Thus the parts of the watch — the hands, the dial and so on — belong to the outer, in so far as each of them presents itself immediately, but to the inner, in so far as their essential significance depends on the watch's arrangement and is seen in reflection of the wholeness's requirement. But the whole also presents itself as an outer, in so far as it immediately presents itself as the set of the outer parts; in so far as it on the other hand essentially consists in the connection of the parts and reflects itself in the manner of connection, the whole as wholeness is an inner.

How little such a partial separation of the inner and the outer stands a critical test, however, can easily be shown. What, now, really is the whole? "The connection of the parts!" But if the existence of the parts falls outside their connection, then the parts fall outside the whole; as the connection of the parts the whole becomes therefore only a side of the wholeness, not the wholeness itself. "No, the existence of the parts cannot fall outside the whole; in relation to the wholeness itself the parts are only real moments, reflected in the whole's own concrete connection." But when the existence of the parts vanishes, in so far as they become moments in the actual wholeness, do we not then get a whole that has consumed its own parts — a whole without parts? If the parts are on the other hand to retain their existence, do we not then get a whole parcelled out among the parts, a divided whole, a whole that is not whole? Seen in the opposition of inner and outer, the alternatives are these: either the whole must present itself, whole and undivided, as the outer, which has its inner in the parts transformed into real moments; or else the whole must itself vanish as an inner in the connection and constitution of the existing parts, into which it is really divided. In the first case it is the whole that swallows the parts, in the last it is the parts that swallow the whole. There is in such an actuality, when it is thought through, a logical revolution of the inner against the outer — a revolution which plainly shows that the mechanical whole by no means satisfies wholeness's essential requirement, does not answer to the concept of a true whole. The more wholeness raises itself above the merely mechanical, the more the parts alternate between being existing terms and being real moments, the nearer does the actual come to a fulfilment of the concept's laws.

The actuality in which the problem of wholeness is first solved, and the inner set in force by its outer, is the organism; for *the organism is precisely that whole which is able to let its parts be at once reflected moments and yet existing terms*[**)] The organic opposition

[**)] „Ueberall reicht das Leben so weit, als die eigenthümliche Form seines Daseins über die Stoffe der Natur eine unmittelbare Herrschaft ausübt, und in diesem Sinne schließt der Umkreis des lebendigen Körpers eine gewisse Summe physischer Elemente zu einem zusammengehörigen Ganzen ein; Unrecht dagegen würde man haben, wenn man alle Bestandtheile des Körpers lebendig nennen wollte." [Everywhere life reaches just so far as the distinctive form of its existence exercises an immediate dominion over the materials of nature; and in this sense the circumference of the living body encloses a certain sum of physical elements into a whole belonging together. One would be wrong, on the other hand, to call all the constituents of the body living.] Lotze. Allgem. Physiol. p. 592. This way of considering the matter can, in so far as emphasis is laid on the finite opposition between the inner and the outer, and on the elementary constituents' alternation between living and lifeless state, very well subsist together with the presupposition that the principial unity of life exercises a systematic dominion in the organism regarded as a whole. That something is immediately made living while something else partly dies off, partly serves as material, as substratum and so on, is in good accord with the mediacy with which the organic system of functions must work upon its material. „Es gibt Bestandtheile, und zu ihnen gehören die meisten Säfte, die in der That den Körper nur als benutzbare Materialien durchkreisen, und ein Stück nach innen gezogener Aussenwelt darstellend), niemals eine besondere Form eigenthümlicher Entwicklung erwerben; es gibt andere, die, wie die Mehrzahl der festen Theile, zwar eine dem äussern Naturlauf ungewöhnliche Mischung und Form, aber keine Fähigkeit zu Wirkungen erlangen, die den gewohnten Vorgängen in der äussern Natur fremd wären; zu ihnen rechnen wir das dienende Gerüst des Körpers, alle knöchernen, sehnigen, elastischen Substanzen; es gibt ferner zusammengesetztere Theile, die durch ihre Structnr selbst einer gewissen Reizbarkeit fähig sind, obgleich auch diese nie lange sich nach ihrer Abtrennung von dem ganzen Organismus erhält; zu ihnen gehören die Muskeln; endlich gibt es noch größere Theilganze, welche, wie die Zweige der Pflanze, die Form des Hauptganzen wiederholen und dadurch auch zu einem abgesonderten Leben sich zu entwickeln vermögen". [There are constituents — and to them belong most of the humours — which in fact only circulate through the body as usable materials, representing a piece of the outer world drawn inward, and never acquire a particular form of distinctive development; there are others which, like the majority of the solid parts, do indeed attain a mixture and form unusual in the outer course of nature, but no capacity for effects that would be foreign to the accustomed processes in outer nature — to these we reckon the serving framework of the body, all bony, sinewy, elastic substances; there are further more composite parts which by their structure are themselves capable of a certain irritability, although this too never maintains itself long after their separation from the whole organism — to these belong the muscles; finally there are still larger part-wholes which, like the branches of a plant, repeat the form of the principal whole and are thereby able to develop into a separate life.] Anfr. Skr. p. 594. What must be objected against such a way of seeing is essentially this, that here not merely is the principle of life abstracted from, but the principle is denied — that life is explained in a principleless way by the merely physical reciprocal action of the elementary constituents. „Wie wir … das Leben selbst als eine Verknüpfung physischer Processe betrachten, deren Eigenthümlichkeit weniger in der besondern Natur des Verbundenen, als in der specifischen Form der Verbindung besteht, so gelten uns auch alle Theile eines lebendigen Körpers zunächst nur als physische, vorhandene Massen, ausgestattet mit allen den molecularen Kräften, die ihrer qvalitativen Natur entsprechen, aber ledig aller inneren Heimlichkeiten einer latenten Lebendigkeit, durch die sie mehr wären, als ihre Verwandten, die noch ausserhalb des organischen Körpers sich im Weltall umhertreiben." [As we regard life itself as a conjunction of physical processes, whose distinctiveness consists less in the particular nature of what is conjoined than in the specific form of the conjunction, so all the parts of a living body count for us in the first instance only as physical, existing masses, endowed with all the molecular forces answering to their qualitative nature, but free of all inner secrets of a latent aliveness through which they would be more than their kindred of inner and outer is resolved into organic unity; for life is organizing selfhood, and selfhood the unity which through plurality, difference and otherness develops what is its own. For since all the organism's parts are moments, the whole totality is thereby set in the form of inwardness, of unity and of selfhood; the whole system of functions shows itself from an ideal side, and here in this inner there is a whole, complete, ideal organism. But since form-activity has its strength precisely in this, that the mastered real moments come forward at the same time as existing terms — since it is precisely these terms' material immediacy that furnishes the resistance on which the forming principle unceasingly tests its energy, the otherness through which selfhood incessantly confirms its ideality — the whole system of functions has at the same time a material character, and here in this outer there is a whole, complete, material organism: *two organisms in one organism.*

To let the inner organism, as something metaphysically finished, be the independent thing which by hyperphysical forces brings forth the outer, will not do; to let the outer organism, as something physically finished, be the independent thing whose elementary-physical forces bring forth the inner, will not do either. In assuming the first one distorts the truth that *life is a presupposition for the organism*; in assuming the last one distorts the truth that *the organism is a presupposition for life.* The first of these misunderstandings must be illuminated in physiology, the last in philosophy.

The first misunderstanding is at science's present stage easy to dismiss. "Life," says the physiologist, "must stand as the organization does. A single cell's life is *the sum of this cell's activity*, and is simple in the same proportion as the organization is simple. The life of an animal with a highly developed organization is *the sum of all the forces that stir in it*, and is complicated in the same proportion as the organism is complicated." In thus bringing out the identity of life and the forces operative in the organism the physiologist is from his standpoint entirely right; but the one-sidedness of the standpoint must not be overlooked either. The one-sidedness shows itself in this, that the physiologist, who in this way makes life "*a consequence of the organism*," forgets that the identity appealed to is grounded in an essential difference — that life is a consequence of the organism only in so far as the organism is itself *a consequence of life.*

The second misunderstanding is at the present stage of scientific standards a good

which still drift about in the universe outside the organic body.] Anfr. Skr. p. 593. — The fixed idea that the molecular forces „die ausserhalb des organischen Körpcrs sich im Weltall umhertreiben" should be in themselves independent, closed and completed existences which needed only to run together in a distinctive combination in order to bring forth a living thing — this the physiologist must really contrive to get out of his head if he is to gain any insight into the essence of life; for expressions such as "distinctive mode of aggregation," "specific form of combination," "favourable conditions" are only turns of phrase so long as one can form no concept of the original cause.

deal more difficult to combat. The chief difficulty is to make the dialectical reciprocal relation between the conditions and the unconditioned quite plain. To have something definite before us we choose the interpretation of *arrested life.* The grain of wheat which has lain for three thousand years in one of Egypt's tombs has through these three thousand years preserved a system of conditions of life and yet been without life: what has it lacked in these three thousand years? "The unconditioned!" answers a metaphysician; but the physicist corrects him with the remark that what is lacking here is so far from being a hyperphysical unconditioned that it is rather a set of well-known physical conditions, such as heat, light, moisture and so on. Which of the two is now right? Both, and yet neither. The metaphysician is right in so far as he sees in arrested life a stoppage of the activity that is due to an unconditioned; but he is wrong in so far as he implies that this unconditioned has departed, since on the contrary it rests in the organic form with which its ideal essentiality is identical. The physicist is therefore right if one interprets his declaration to mean that it is not the unconditioned that is lacking here; but he is wrong in the way in which he understands his own declaration. For his meaning is that one does not lack the unconditioned in an organism because "a principle of life is only a phrase which we use to conceal our ignorance," because the principle, the unconditioned, is something quite superfluous in itself once one has the conditions.

There lies in the abstractly physical way of seeing so dazzling a sophistry that anyone who does not take care must be beguiled. For it is quite true that when one conceives all life's fundamental conditions united in a seed-corn, one need only let the elementary conditions come in, and the activity of life will follow with inescapable consistency. What is sophistical lies in this, that one abstracts from the unconditioned while in a thought-experiment providing conditions which only the unconditioned itself can provide. By such a procedure it is an easy matter to reduce, not only the principle of life, but any other real principle whatever — indeed genius itself — to "a phrase which we use to conceal our ignorance." What, after all, does it take to write a Shakespearean tragedy? Of course the same set of "favourable conditions" as were at work in Shakespeare's consciousness, nothing else!

The fact that a grain of wheat can lie for three thousand years and still preserve its "germinative power" is, when it is seen in the principle, just as explicable as the fact that the functions of life can set in again after having completely ceased for three seconds. That the unconditioned is itself something conditioned is seen from the functions' going to rest when the requisite conditions are lacking; that what is thus conditioned is an unconditioned is seen from its instantly transforming the conditions that come in into moments for its own organic self-activity. This dialectical unity of the unconditioned with the conditions is a natural expression of reason’s validity, a thought of eternity that loses nothing in force by waiting three thousand years for conditions.

In the outer, the material organism divisibility is so preponderant that the unity of life separates into a plurality of relatively independent points of life, a “total sum of single lives”; in the inner, ideal organism indivisibility is so preponderant that the particular points of life show themselves as moments of articulation in life’s concrete unity. And if the opposition of the inner and the outer organism is grounded in the unity, and conversely the unity in the opposition, then it follows from this too that an alteration may be decisive for the organic total unity without being so in the same degree for all the single parts, and conversely decisive for single parts without being so in the same degree for the unity.

“The organism” can be dead and many organs still continue to live. “The interruption of the *relation* between the single parts of the organism which brought all the parts endowed with individual independence to a real unity by calling forth a reciprocal influence between the parts may have taken place, and the unity be cancelled; but the parts endowed with individual independence are not annihilated. For several hours after a man is dead his muscles live and can contract; his glands live and can secrete, his heart lives and can beat; his stomach lives and can digest”[*)]. Conversely the organism may in its inner unity of totality still be living after many of the parts have died. What is thought of here is not primarily the change of substance; for “molecular death” of course goes on incessantly: the organism “survives this death of the organs, just as a people continues to live although many human beings die daily.” No: what is thought of here is expressly the fact that even the higher organisms — those in which the totalized energy of unity is predominant — can lose their limbs and other essential parts without death setting in on that account. — If the phenomena cited have now helped to make evident the opposition grounded in the unity, then the following investigations shall, if possible, help to make intelligible the unity actualized in the opposition.

β. The Organic Type.

In so far as the inner organism stamps itself in the outer, the organic shape is doubled; here there shows itself a non-sensuous reflected in the sensuous with its transformations — a fundamental shape, a type. It is with the type and its impress as with the inner and

[*)]Lewes adds in a note — after Ch. Robin, *Mémoires de la société de biologie, v. 134*— — that digestion in certain fishes continues even 24 hours after death. Hverdagsl. Physiol. 2. p. 459.

the outer. The opposition is general, but in the organic form becomes quite distinctive. Everywhere that one and the same kind of formation repeats itself with alterations, what is constant in the alterations, and precisely thereby characteristic, can as regards the shape always be brought out as type. Thus elementary formations — even those which depend most on the concurrence of conditions and varying circumstances — may be said to have their type. How changeable is the appearance of the clouds,

and yet cloud-formation has, as is well known, its types: the feather-cloud (*cirrus*), the heap-cloud (*cumulus*), the layer-cloud (*stratus*) with corresponding modifications and transitions. However much may be accidental in the outward appearance of mountains, there is nevertheless a typical difference between such forms as mass-mountains, chain-mountains, plateau-mountains and alpine mountains; just as the geologist also makes, in the case of coral islands for example, a typical distinction between fringing reefs, barrier reefs, atolls and so on. Still more pervasive is the significance of the type of minerals, as it has been brought out and characterized (particularly by Mohs); but most of all, surely, that of the pure crystal type.

The opposition between the organic and the inorganic can therefore not consist in a type answering to the former and none to the latter; the opposition is determined rather by the way in which the organic type essentially differs from the inorganic. If we compare an organic type with the crystal type, the difference is conspicuous: the plane faces, the sharp edges, the mathematical character predominant throughout at once give the crystal an appearance of its own, contrasting with the organic. To this difference in the outward there answers a difference in the inner structure as well: while the organism's constituents are solid and liquid and gaseous alike, the parts of the crystal are all in the solid state; while the organism shows itself articulated by formations of different kinds, the crystal is in each and all homogeneous, consisting of nothing but parts of like kind. Still more essential is the opposition determined by the relation between being and becoming. The crystal is a being that has its becoming behind it, a subsisting thing which, in order to subsist, must exclude alterations; the organism on the other hand is a being that has its becoming in itself, its subsistence in a system of alterations. From this it follows that the crystal's actual shape must be an immediate expression of the type — whether the expression be adequate or not — whereas

the organism will always display a definite difference between the essential and the immediate shape, so that the type does not simply coincide with the outer but is reflected in it: the crystal type is a *schema*, the organic type an *archetype*.

The typical schema stands to the original archetype as substance-cause to self-cause, as physical reciprocal action to ideal causality*). To assume a

particular force of crystallization is still more unwarranted than to assume a particular vital force; by the expression "vital force" attention may at any rate be drawn to the principle, but for assuming a particular principle operative in crystallization there is surely no ground at all. "In a crystallizing medium," says Liebig, "one observes an incessant motion; as though the smallest parts were magnets, they repel one another in one direction while they attract one another in another, and range themselves side by side; they form themselves into a regular shape which under like conditions never alters." If after these hints there were still ground to presuppose a particular principle of crystallization, then there would surely be good ground also for assuming a particular

*)Von Individualität, oder was uns als dasselbe gilt, von Ganzheit, kann in der Natur zuerst bei den Pflanzen die Rede sein. Das Mineral ist selbst als Krystall kein Ganzes und zwar deswegen, weil jeder Krystall, in seine Mutterlauge gebracht, sich zu vergrößern vermag, also nach außen keine selbstbestimmte Abgrenzung hat, dann weil es völlig gleichgültig ist, ob man zur Bestimmung des Minerals ein großes oder kleines Stück und dies von der Oberfläche her oder aus dem Innern wählt, weil also der Stein in sich gleichförmig, unbegrenzt ist. Was aber unabgeschlossen und unterschiedslos ist, das ist kein Fertiges, sondern bloßes Moment im Werden eines Ganzen, und zwar hier der Erde, es ist kein Ganzes, sondern bloßer Theil in der Zusammensetzung größerer Massen. Daher wird das Mineral äußerlich gemessen, innerlich gewogen, nach Gestalt und Härte dort, nach chemischen Bestandtheilen und der Dichte da, — es wird ihm nur durch Vergleichung mit Anderem eine Bestimmtheit gegeben, weil es keine eigene hat. Das Mineral ist in seiner selbständigsten Form ein bloßes Beispiel für irgend ein mathematisches Gesetz.

Die Pflanze zwar vergrößert sich auch, ja sie vergrößert sich so lange sie lebt, und diese Vergrößerung ist daher ein Wesentliches bei ihr, während sie beim Minerale von dem zufälligen Dasein einer Mutterlauge abhängt, daher auch am Wesen des Minerals keinerlei Aenderung herbeiführt. Allein diese Vergrößerung der Pflanze, zu dem sie allerdings einer Materie bedarf, welche man der Mutterlauge vergleichen könnte, ist das Resultat eines Processes, in welchem die Nahrungsmutterlauge zu einem ganz Anderen und zwar innerhalb der Pflanze selbst verarbeitet wird. Daher tritt denn an der Pflanze sogleich ein Gegensatz von Drinnen und Draußen, eines Inhaltes und Umschlußes auf. Die Pflanze ist eine durch ihren Typus bestimmte und denselben festhaltende Zelle, und je vollkommener sie wird, um so präciser und allseitiger gestaltet sich der Unterschied zwischen Peripherie und Centrum. [Of individuality — or, what counts for us as the same, of wholeness — there can in nature first be talk in the case of plants. The mineral is not a whole even as crystal, and that for this reason: because every crystal, brought into its mother-liquor, is able to enlarge itself, and so has outwardly no self-determined delimitation; and then because it is wholly indifferent whether one chooses for the determination of the mineral a large or a small piece, and this from the surface or from within — because, that is, the stone is in itself uniform, unbounded. But what is unconcluded and undifferentiated is nothing finished, but a mere moment in the becoming of a whole — here of the earth; it is no whole, but a mere part in the composition of larger masses. Hence the mineral is measured outwardly, weighed inwardly, in the one case by shape and hardness, in the other by chemical constituents and density; a determinateness is given it only by comparison with something else, because it has none of its own. The mineral is in its most independent form a mere example of some mathematical law. — The plant too, it is true, enlarges itself; indeed it enlarges itself as long as it lives, and this enlargement is therefore something essential in it, whereas in the mineral it depends on the accidental presence of a mother-liquor and accordingly brings about no alteration whatever in the mineral's essence. But this enlargement of the plant — for which it does indeed require a matter that might be compared with the mother-liquor — is the result of a process in which the nourishing mother-liquor is worked up into something quite other, and that within the plant itself. Hence there appears in the plant at once an opposition of within and without, of a content and an enclosure. The plant is a cell determined by its type and holding fast to that type; and the more perfect it becomes, the more precisely and many-sidedly does the difference between periphery and centre take shape.] F. Th. Bratranek. Beiträge zu einer Aesthetik der Pflanzenwelt. Leipzig 1853. pp. 224—25.

principle holding for the amorphous formations; for amorphism is precisely the concept by which one denotes "a distinctive state opposed to crystallization." To the properties, forces, conditions and laws of stuff-substances amorphism stands in the same way as crystallization does. "When we force a liquid or gaseous body to become suddenly solid, when we do not give its small parts time to range themselves in the directions in which their attraction (force of cohesion) is strongest, then no crystals will form; such bodies will refract light in another way, have another colour, hardness and a different cohesion. Thus we know a red and a coal-black cinnabar, a solid, hard sulphur and a transparent, soft sulphur which allows of being drawn into long threads, glass in the state of an opaque, milk-white body so hard that

it gives sparks when struck against steel, and in its usual transparent state with a conchoidal fracture. These states, so unlike in their properties, are conditioned in the one case by a regular, in the other by an irregular arrangement of the atoms; the one body is amorphous, the other crystallizes." (Liebig. Chem. Breve. pp. 105—106).

If one were now to assume a particular *a priori* archetype for the formation of crystals — a παράδειγμα hypostatized on its own account, after which nature with its "plastic, drastic drive" composed the crystals — then we should, to be consistent, have to assume also an archetype hypostatized in itself, after which nature's plastic art formed the amorphous in every single case; but an *a priori* form for the amorphous is surely a manifest contradiction[*)].

It stands otherwise with the organism and its organisation[**)]. Since organisation

[*)]That the crystal form is substantial and characteristic, the amorphous form of delimitation on the other hand accidental, proves only that the stuff-substance displays its distinctiveness in crystallizing — not that a particular principle should be needed for it. Even if one adopts Hegel's brilliant interpretation: . . . „in den Krystallen aber ist die gestaltende Thätigkeit keine dem Objekt fremdartige, sondern eine thätige Form, die diesem Mineral seiner eigenen Natur nach angehört; es ist die freie Kraft der Materie selbst, welche durch immanente Thätigkeit sich formt, und nicht passiv ihre Bestimmtheit von Außen erhält." [in crystals, however, the shaping activity is not one foreign to the object, but an active form belonging to this mineral by its own nature; it is the free force of matter itself, which forms itself by immanent activity and does not passively receive its determinateness from without.] (Aesthetik. 1ster Band. p. 168) — the form-activity dwelling in matter cannot for all that be identified with a particular forming principle. It is an uncritical running-together of things of different kinds when Hegel continues: „In noch höherer konkreterer Weise zeigt sich die ähnliche Thätigkeit der immanenten Form in dem lebendigen Organismus" [in a still higher and more concrete way the like activity of the immanent form shows itself in the living organism] . . ., as though organic form-activity were only a modification of the inorganic.

[**)]To hold fast the difference of principle between the inorganic and the organic, between the crystal and the plant, one must impress upon oneself again and again the pervasive difference between the organic process and any inorganic process whatever, be it mechanical, physical, chemico-physical or chemical. The organic process rests, as was shown above, on a *doubled reciprocal action: between the atoms themselves and between the atoms and the ground-cause.* The reality of this doubling is secured by the atoms' being, in their elementary state, *as much real possibilities as actual existences.* In the doubled reciprocal action the atoms' disposition for life is developed; in the development of this disposition for life the activity presupposes an original self-cause, since the whole system of forms operative in the organism is determined by an ideal causality, the outer organism must necessarily stand to the inner as image to archetype. The organism's archetype is however, when it is seen in the principle, itself imageless. As little as the self-reflecting selfhood can have in the reflection itself any immediate shape or figure, so little can validity be ascribed

of the principle is seen. In Lotze's declaration „so gelten uns auch alle Theile eines lebendigen Körpers zunächst nur als physische, vorhandene Massen, ausgestattet mit allen den molecularen Kräften, die ihrer qualitativen Natur entsprechen, aber ledig aller inneren Heimlichkeiten einer latenten Lebendigkeit, durch die sie mehr wären, als ihre Verwandten, die noch ausserhalb des organischen Körpers sich im Weltall umhertreiben" (Allgm. Physiol. p. 593) one may of course take „die inneren Heimlichkeiten einer latenten Lebendigkeit" in so mystical a way that the natural scientist must feel called upon to reject the mystification. But in avoiding the mystification of concepts under one form one may, by overlooking the dialectical, easily entangle oneself in it under another — *incidit in Scyllam, qui vult evitare Charybdim.* To see the difference between "the molecular forces" with which substances and their atoms act in their physical, elementary existence, and the forces whose activity is specially conditioned by the organism, simply effaced — and that by an investigator who has himself, as was already noted, expressed himself on the existence of forces as follows: „Wollen wir . . . . die Fähigkeit zur Erzeugung einer Wirkung, die einer Substans nur unter Beihilfe gewißer Umstände zukommt, auf sie allein als ihre Kraft übertragen, so müssen wir dann auch behaupten, daß keine Substans beständige, sondern jede nur erworbene Kräfte besitzt" . . . — really does look most peculiar.

If one denies the atoms' disposition for the activity of life, dismissing all „innere Heimlichkeiten einer latenten Lebendigkeit", then it is of course quite consistent to deny the difference of principle between crystal and plant. — „Selbst die Blüthe mit all ihrem Reichthum zusammengehöriger und auf einander berechneter Organe ist in keinem andern Sinne ein Individuum als in dem, daß die allgemeine gesetzliche Norm der gegenseitigen Lagerungsverhältnisse der Theile in diesem besonderen Falle zugleich eine bestimmte Anzahl derselben festsetzt, durch welche eine geschlossene, der Erweiterung nicht fähige Gesellschaft entsteht. Diese Zusammengehörigkeit ist nicht größer als die der krystallinischen Gebilden, denen überhaupt die zusammengesetzten Pflanzen sich in dieser Hinsicht analog verhalten. Auch der Krystall gestattet an seinen Außenflächen neuen Ansatz, sobald dieser dem allgemeinen Lagerungsgesetze der Theilchen genügt, so wie die Pflanze an ihrer Axe Nebenaxen gestattet, die demselben Bildungsgesetze wie jene unterworfen sind. Auch der Krystall nöthigt den neuen Ansatz irgendwo Ecken zu bilden, in denen unveränderlich die Theilchen mit gleicher Anzahl von Kanten und unter denselben Winkeln zusammentreten; gleich dieser Spitze beschränkt die Blüthe den weitern Ansatz der vegetabilischen Masse, indem sie eine bestimmte Anzahl von Theilen in ein äußerstes auch arithmetisch bestimmtes Product der Bildung zusammentreten läßt. Ueberall sehen wir hier Kolonien von Theilen ohne inneres Band, aber allgemeinen Gesetzen der Formung unterworfen, die aus ihrer eignen Wechselwirkung entspringen, und nicht überall gleichgiltige ins Unendliche fortzusetzende Reihen, sondern oft auch geschlossene Vereinigungen mit dem Scheine der Individualität hervorbringen." [Even the blossom, with all its wealth of organs belonging together and calculated upon one another, is an individual in no other sense than this, that the general lawful norm of the mutual relations of position of the parts in this particular case at the same time fixes a determinate number of them, by which a closed society incapable of extension arises. This belonging-together is no greater than that of crystalline formations, to which indeed the composite plants behave analogously in this respect. The crystal too permits new accretion at its outer surfaces as soon as this satisfies the general law of position of the particles, just as the plant permits secondary axes on its axis which are subject to the same law of formation as it is. The crystal too compels the new accretion to form corners somewhere, in which the particles invariably come together with an equal number of edges and at the same angles; like this apex, the blossom limits the further accretion of the vegetable mass by letting a determinate number of parts come together into an outermost product of formation, arithmetically determined as well. Everywhere we see here colonies of parts without an inner bond, but subject to general laws of formation which spring from their own reciprocal action, and which do not everywhere produce indifferent series to be continued to infinity, but often also closed unions with the semblance of individuality.] Allgem. Physiol. pp. 597—98. — But where do we now arrive?

If certain superficial analogies have first brought the observer to forget the proposition otherwise well to a paradigm subsisting on its own account — a magically operating image of fancy hovering above the organisation. But how, then, can what is in itself imageless furnish an archetype? Indeed, how does it come about that what is in itself invisible can become visible, that what itself occupies no space can fill space? — The extensive magnitude is realized only by means of the intensive, and conversely; matter only by means of force, force only by means of matter. "Rational mechanics does not stop at the fact that finite masses attract one another in a finite way, but interprets the attraction so that every smallest part in every mass must, in attracting and being attracted, be at once both point of departure and point of application (*point d'application*) for an operative force . . . . From the analysis thus indicated there emerges as an incontestable result that matter presupposes force to infinity, as force in turn presupposes matter to infinity; that matter can therefore as little be without force as force without matter; that the existence of matter and of force is essentially one and the same being. Since force is possible only by means of matter, it is impossible to derive

matter from force; since matter conversely is possible only by means of force, it is impossible to derive force from matter. The *dynamical* view, which makes force the principle, is as untenable as the *materialistic*, which makes matter the principle: *the principle is neither force as force nor matter as matter, but the dialectical unity of matter and force.*" (cf. Mathem. og Dialekt. pp. 127—131).

If this is now applied to the filling of space, it is seen that matter by itself cannot possibly fill space, since pure extension without force is a phantom as empty as empty space; that neither can it be force by itself, an opposition of pure forces (force of attraction and of repulsion), that is what immediately fills space, follows directly from force's not possibly being able to have predicates of extension in common with space — from one's not being able to speak of broad, thick, round forces and so on. It is therefore by means of force, the non-space-filling, that matter fills space; or rather: it is the principle, the

known from the reciprocal action of the elements, *that effects resembling one another may under certain presuppositions proceed from essentially different causes*, and at the same stroke to overlook the difference of principle between the crystal's formation and the plant's growth, between elementary aggregation and the organic process — then it is mere inconsistency when he would make much of the original difference between plant and animal. If the plant organism can be formed by "molecular forces" without ideal causality, why should the animal organism not be able to be so too? . . . .

. . . „abgesehn von Eigenthümlichkeiten, die in den niedersten Anfängen des Thierreichs noch nicht hervortreten" — it is said of the animal's organism (Anfr. Skr. p. 598) — „finden wir wenigstens das Zusammengehörige stets durch die Herrschaft eines Willens zu einem ungetheilten Ganzen vereinigt." [apart from peculiarities which do not yet come forward in the lowest beginnings of the animal kingdom, we find what belongs together always united into an undivided whole by the dominion of a will.] — What one is really to understand by this „durch die Herrschaft eines Willens", when one may not think of an organizing principle, is not easy to see. But how one can see an organizing principle in the animal organism while the plant organism is to be without principle, is unintelligible.

dialectical unity of matter and force, which, without itself occupying any space, brings about the filling of space. The case stands correspondingly with the making-visible of what is in itself invisible, with the shape of what is in itself shapeless. What is to become visible and gain shape must by its essence have an other, must present itself to intuition so that by this its other it fills a space, with a delimitation of its own. And since space is essentially filled by what does not itself immediately fill any space, it is seen from this that *it must be precisely the invisible itself that makes itself visible, the in itself shapeless that gives itself shape — an inner, ideal shape — in so far as it gives its other, the real medium through which it lets itself be seen, an outer, material shape.*

The paradigm holding for the organism is at the moment of fertilization already complete, but in possibility's imageless form. For the imageless paradigm is nothing other than the system of ideal determinations which lies in selfhood's own distinctive essence and conditions its energy. The ideal determinations are in the beginning themselves only requirements which are only gradually to be fulfilled, possibilities which little by little are to be realized. As the functions advance, form-activity advances at the same stroke, the possibilities are realized, the requirements fulfilled: to the system of functions there then answers a system of forms, typical forms in which the material is point for point determined by the ideal, the outer by the inner. And since the inner shows itself at all points as the forming, the original, that which is grounded in selfhood's energy, while the outer everywhere bears the stamp of being something formed, something brought forth in time, something derived — the in itself imageless thereby assumes an image-form, and the sensuous organism presents itself as the image that presupposes its archetype. If, to illustrate this, we choose a determinate organism, a rose — *Rosa canina*, say — it is seen that such a paradigm must be presupposed to be original and constant if botany is to hold it fast in opposition even to paradigms standing so near as *R. rubiginosa*, *R. inodora*, *R. tomentosa* and so on. The paradigm's *a priori* validity is expressly presupposed when the course of formation and development is in all essentials assumed to be determined from the germ's very first beginning in such a way that nothing in this respect is left to circumstances and chance encounters. The archetype's concrete execution may indeed vary in relation to the varying conditions of the surroundings, but the proper marks of character nevertheless continue under all such alterations — at least through extraordinarily long periods — to stand forth with the stamp

of something original, something in itself unalterable.[*)] What is original in the

[*)]Only a development of the principle, in accord with the matters of fact, in its paradigmatic reciprocal action with the varying conditions can lead to scientific insight into the relation between the unalterable and the alterable; whereas a subjective opinion about what in this respect is most worthy of the Creator is archetype becomes still more conspicuous when we look to the change of substance. Under the incessant exchange of the parts and the unceasing transformations there would arise such instability in all formations, such accidentality in all forms, that every trace of typical features must at last be effaced, did not the system of functions determined by the principle hold the plan fast and give the fundamental form consistency.

Only by the fundamental form — the paradigm transforming itself into an archetype — can such categories as *disposition and preformation* also acquire their full significance. What, after all, is a disposition? Every disposition is in itself, as disposition, a real possibility; but not every real possibility is on that account a disposition. There is in the carbonate of lime that is precipitated a real possibility of calcspar, but there is

also a real possibility of aragonite; and precisely because there is here a real possibility of both, there is here no particular disposition to either of them.[*)] In the germ of *Syringa vulgaris* there is indeed a possibility of a repetition of the same form; but a *Syringa vulgaris* is, mark you, the only form to which there is a possibility in such a germ. If this possibility is realized, then the form for its realization is determined beforehand; *such a possibility, determined by preformed actuality, is a disposition.* What is to come of the merely real possibility when it is realized depends chiefly on conditions supervening from without; what is to come of the disposition, if it is carried out, rests on the other

without significance in science. — „Schriftsteller ersten Rangs scheinen vollkommen davon überzeugt zu sein, daß jede Art unabhängig erschaffen worden sey. Nach meiner Meinung stimmt es beßer mit den der Materie vom Schöpfer eingeprägten Gesetzen überein, daß Entstehen und Vergehen früherer und jetziger Bewohner der Erde, so wie der Tod des Einzelwesens durch sekundäre Ursachen veranlaßt werde. Wenn ich alle Wesen nicht als besondere Schöpfungen, sondern als lineare Nachkommen einiger weniger, schon lange vor der Ablagerung der silurischen Schichten vorhanden gewesener Vorfahren betrachte, so scheinen sie mir dadurch veredelt zu werden. Und aus der Vergangenheit schließend dürfen wir getrost annehmen, daß nicht eine der jetzt lebenden Arten ihr unverändertes Abbild auf eine ferne Zukunft übertragen wird.“ [Authors of the first rank appear to be perfectly convinced that every species has been independently created. In my opinion it accords better with the laws impressed upon matter by the Creator that the coming-to-be and passing-away of earlier and present inhabitants of the earth, like the death of the individual, should be brought about by secondary causes. When I regard all beings, not as special creations, but as lineal descendants of some few ancestors already existing long before the deposition of the Silurian strata, they seem to me thereby ennobled. And reasoning from the past we may confidently assume that not one of the species now living will transmit its unaltered image to a distant future.] — Charles Darwin. Ueber die Entstehung der Arten im Thier- und Pflanzen-Reich. Trans. by Dr. H. G. Bronn. Stuttgart 1860. p. 493.

[*)]If we consider the chemical elements under a single survey, they all have, in so far as they are conceived as factually given, *possibility*, but not in the stricter sense *disposition* to different compounds; if on the other hand we presuppose that the elements are originally developed out of a fundamental unity determinate in itself, so that as distinctive terms in this unity's total development they have each their properties answering to the requirements of the whole, then the possibility in each single one and in all of them is transformed into disposition. What *harmonia præstabilita* expresses in an outward and mechanical way, the disposition with its preformations expresses in an inward, dynamical way. The disposition — the content-rich boundary between actuality and its preformation — is at the same stroke the content-rich boundary between selfhood's inner and existence's outer: the disposition, which presupposes a principial activity *with which*, presupposes at the same time an existing thing *in which* it begins.

hand not on outward but on inward conditions enclosed within it. From this it is seen that only on the presupposition of an ideal causality can there in the stricter sense be talk of disposition.

Under the development of the system of functions and forms, formation must then, by the disposition's inner determinations, follow upon formation, transition upon transition, what follows always determined beforehand by what precedes, so that

the typical is seen in a series of successively unfolded but simultaneously preformed forms. The younger the various formations are, the more they resemble one another; but beneath the indeterminate, uniform, still ambiguous outer an inner difference, articulated in itself, is already operative. For the egg — we are still thinking chiefly of the plant egg — the germ (*embryon*) is what is essential; and this essential thing begins originally with a single cell (*cellula embryonalis*): so simple is the beginning, so great the uniformity in the outer. But this too is characteristic of disposition, that the inner wealth must contrast with an outer poverty. What holds of the germ holds correspondingly of the bud. Every bud is from the beginning a cell, then a plurality of cells, a cell-tissue which then differentiates itself physiologically. If we consider the somewhat developed part which one commonly calls a bud — the bud of a lilac, say, with its imbricated bud-scales — then such a bud is at once the execution of a disposition and itself a disposition. A whole system of typical forms — several series of internodes of unlike kind with their unlike leaves, bud-scales, several generations of anthoplastic leaves — must therefore be preformed at the bud's first formation, under the cell-tissue's first physiological differentiation.

The first disposition, the disposition in the beginning itself, by no means has the whole set of nascent forms existing in it; and yet every form to come — whether it appear earlier or later, in the middle or at the end — is already determined, originally determined with the first disposition, according to the stage of development to which it belongs. Here a careful distinction must be drawn between the formations determined immediately in the disposition and those determined from the beginning to follow upon a series of mediating intermediate determinations. With the lilac-bud's first cell-tissue there are predetermined not only the articulated bud, the bud with its finished

bud-scales, but also the flowers, which nevertheless first appear toward the close of the development. And yet, what a distance between the unfolded flower and the imbricated bud-scales — indeed, what a difference between these and the first cell-tissue! With respect to the reciprocal determination of disposition and execution an express distinction must here be drawn between the several formations' remoter, nearer and nearest dispositions. For since what is developed out of its nearest disposition is again a disposition, there are seen here in the course of the development not one but several dispositions, a system of dispositions. And if this system of dispositions, unfolded for intuition, is now led back to the disposition of dispositions — the beginning itself, the first disposition — then inwardness's wealth shows itself in ideal doublings, and the dialectical unity of the ideal doublings is reflected in *the paradigm's originality, in selfhood's energy, in preformation.*

With every new organism the paradigm must repeat itself, since every organism originally presupposes selfhood; and the paradigm thus repeating itself must in the execution further undergo the modifications answering to the opposition of the sexes. But under all repetitions and modifications the paradigm determined in its selfhood continues to be like itself, and shows itself thereby as an ideal concretion of the general. Reflected in concrete generality the paradigm is *species*, and the existing organism the *individual*[*)].

The old dispute about the relation between the individual, the species and the genus still continues to repeat itself, as though it were never to end. From philosophy it has passed over into natural science, from metaphysics into empirical inquiry, and has particularly in recent times, with respect to the difference between "the artificial" and "the natural systems," gained heightened interest. While some (Schleiden) ascribe reality only to individuals, others (Lindley) would conceive the species too, and yet others (Fries) even both the species and the genera, as nature's work; whereas most seem to

regard families, orders and so on as pure abstractions[*)].

[*)]Even in cases where the species' complete presentation requires a series of more or less one-sided forms of individuality, the preformation is law-governed and the development paradigmatic. — "If we once gather into one image the whole development through nursing generations, as it comes forward for us in the bell-polyps (*Campanularia*), the club-polyps (*Coryne*), the medusae, the salps, the vorticellae and the intestinal worms, it shows itself at once as a particular and therefore essential. feature in this course of development that the species (that is, the species in its development) is not completely represented by the single, full-grown, breeding individuals of both sexes and their development, but that there are further required individuals of one or more preceding generations which form as it were a supplement to those. The difference between this course of development and the one generally recognized in nature — under which the species is represented by the individual (of the two sexes) and its development — is therefore, on the individuals' side, a lack of complete individuality as representatives of the species, as species-individuality, if such an expression may be used. — If we are now agreed in regarding such an incompleteness in the individual as the essential thing in this development, then we shall grasp its significance in nature when we consider this course of development itself in its development, see how it begins and how it proceeds, so that we at last discover where it is going . . . . . Each member or generation surely brings its brood nearer to the intended perfection; but this calling-forth of something more perfect happens only by a nursing within particular beings, and is left to an organ's quiet and calm activity, without the nursing beings being themselves conscious of it; it is only a function, not a manifestation of will." Joh. Japetus Sm. Steenstrup. Om Forplantning og Udvikling gjennem vexlende Generationsrækker. Copenhagen, 1842, pp. 64—66.

[*)]*Schleiden säger (Morph. II. Ed. 2, p. 25)*: „Hier muß man festhalten, daß die Natur unserer wis-

What in general makes investigations of this nature partly so entangled, partly so unfruitful, is the circumstance that one only too easily runs together starting points of quite different kinds, confuses the categorial determinations, and finds oneself in much obscurity concerning reality and existence. In treating the question of individual and species it makes a difference of principle whether

one begins by presupposing without more ado the existing individuals and then, by grouping these according to resemblances, abstracts the concepts species, genus and so on; or whether one begins by investigating the presuppositions necessary for an individual's coming-to-be and is thereby led to principle and paradigm: in the first case the individual, in the last the species, must show itself as the original.

If one proceeds without more ado from the assumption that only what can be pointed to as an outwardly existing thing has reality, then it is of course thereby settled that the species cannot possibly have reality, for outside the individuals the species' existence cannot be pointed to. If on the other hand one sees that the categories existence and reality do not run together without difference, then one will also understand that the individual's existence is the species' reality. Against those who deny the species reality it has rightly been objected that if the category species had no objective validity grounded in nature, then the kingdom of nature would be without laws, without plan, without

senschaftlichen Betrachtung kein System überliefert, sondern Einzelwesen. . . . Die Anordnung in größere oder kleinere Gruppen, Arten, Geschlechter oder Familien tragen erst wir in die Menge der Einzelwesen hinein. Wir finden größere Uebereinstimmung unter einer gewissen Zahl von Individuen und stellen diese zusammen, nun erst suchen wir nach einem Ausdruck zur Charakterisirung dieser Gruppe. . . . ." [Schleiden says: Here one must hold fast that nature hands over to our scientific consideration no system but single beings. . . . The arrangement into larger or smaller groups, species, genera or families, we first carry into the multitude of single beings. We find greater agreement among a certain number of individuals and put these together; only then do we look for an expression to characterize this group. . . . .] *Hos Lindley heter det på ett ställe (Bot. Reg. Vol. XIII, fol. 1066): »All genera and indeed all the divisions of Naturalists are necessarily artificial; and when one genus is called natural, and another artificial, all that can be meant by such expressions is, that the species of the one are less artificially combined than those of the other: this we apprehend, is, at the present day an universally admitted principle, to the proof of which we need not proceed.« På ett annat ställe säger han (Nat. syst. of Bot. Ed. II. Pref., p. VII). »Our genera, Orders, Classes, and the like are mere contrivances to facilitate the arrangement of our ideas with regard to species. A Genus, Order or Class is therefore called natural, not because it exists in Nature, but because it comprehends species naturally resembling each other more than they resemble any thing else.« Fries afgör deremot frågan så, att arter och slägten äro naturens verk och icke vårt; de högre grupperne deremot äro konstens verk; de förra äro systemets concreta innehåll, de sednare äro framställningens form. Af de förra er slägtet det mest ursprungliga, oförgängliga och i naturen djupast grundade begreppet; arterne äro slägtets mångfaldiga uppträdande i verkligheten. J. G. Agardh. Vext-Systemets Methodologie. Lund. 1858. p. 1—2.* [The Swedish frame reads: In Lindley it is said in one place (the first English quotation above follows); in another place he says (the second English quotation follows). Fries, by contrast, settles the question thus: that species and genera are nature's work and not ours, whereas the higher groups are the work of art; the former are the system's concrete content, the latter the form of its presentation. Of the former the genus is the most original, imperishable concept, and the most deeply grounded in nature; the species are the genus's manifold appearance in actuality.]

any reason at all, and the whole edifice of natural science would have to collapse*). But even if one can in this way vindicate the species' reality, one cannot in the same way vindicate the genus's reality without more ado.

As the actualized unity of species and individual the fulfilled paradigm is entelechy. Entelechy, the actuality complete in itself, has then a two-sided reality, an ideal and a material one. In the existing individual — this plant (in so far as a specimen of a plant can represent an individual), this animal — entelechy has its material reality.

The material is not, as the mystics assume, a defilement for the spiritual, a burden for life, a troublesome barrier for the ideal; on the contrary! The material belongs in the opposition itself under the ideal, and is a barrier only in so far as the barrier conditions the ideal's self-development. "The full-grown tree," says Aristotle (cf. Prop. pp. 143—144), is the seed's entelechy." It was an essential advance that Aristotle saw how necessarily the stuff-like belongs with it, if the form operative as the species (τὸ εἶδος) is by forming its stuff through and through to attain complete actuality: the Platonic εἶδος (cf. Propæd. pp. 137—140) is in its pure essentiality, indifferent to matter, an empty form, not an ideal actuality. But when the material in each and all only serves the energy, and the energy is in itself ideal, then the whole emphasis nevertheless falls on the ideal reality, and the question presses itself involuntarily upon us: what becomes of the entelechy when the individual falls away, when the plant withers, when the animal dies?

Since no acting and no actuality can vanish absolutely without trace, it cannot be thought that entelechy in its ideal reality should become sheer nothing, Can an entelechy, once developed, be effaced again, then the paradigm has no originality, selfhood no energy,

life no principle, and we fall back again into the senseless. But although the ideal

*) *Vi anmärka till en början, att om formerne således utbildades regellöst, så måste man tänka sig hela naturen utan någon bestämd plan, och allt vårt systematicerande, ja! all naturforskning blefve väl en byggnad på lös sand, som, huru ofta den instörtade, alltid skulle återuppbyggas på samma osäkra grundval. Hvad arterna särskilt beträffar, så synes denna åsigt också vederläggas af de ständigt lika former, som ur frö årligen uppdragas i våra trädgårdar. Man tror sig också ha funnit, att exemplar i våra museer, insamlade för 3:ne sekler tilbaka, fullkomligen motsvara nu lefvande vexter och djur. Om således formernes utbildning till de bestämda typer, vi kalla Arter, icke är en redan tillräckligen bevisad sats, så måste detta antagande dock vara ett axiom, hvarpå hela naturforskningen i sjelfva verket ytterst hvilar. Agardh. Vext-Syst. Method. p. 2—3.* [We remark at the outset that if forms were thus developed without rule, one would have to conceive the whole of nature without any determinate plan, and all our systematizing — indeed all investigation of nature — would be a building on loose sand which, however often it collapsed, would always be built up again on the same insecure foundation. As regards species in particular, this view seems also to be refuted by the constantly like forms which are raised yearly from seed in our gardens. It is also believed to have been found that specimens in our museums, collected three centuries ago, correspond perfectly to plants and animals now living. If, therefore, the development of forms into the determinate types we call species is not a proposition already sufficiently proved, this assumption must nevertheless be an axiom on which the whole investigation of nature in fact ultimately rests.]

entelechy cannot be annihilated, neither can it maintain itself as an immediately existing thing when the material falls away; for existential immediacy rests precisely on the material. Indeed, could one set up a “pneumatic organism” which remained standing as an invisible though well-conserved shape when the material one collapsed — a Paracelsian “astral body” which held itself when the earthly fell to dust — then the apocalyptic would be attained; then it would be merely a matter of immersing oneself in “das Hellsehen,” relying on “ecstasy,” and consulting those who see with the pit of the heart. Without misjudging psychological hints that may perhaps be glimpsed in enigmatic phenomena, we must on science’s behalf impress that every insight into life’s essence is conditioned by a dialectic which acknowledges no immediate seeing or glimpsing of ideal real shapes subsisting on their own account and moving behind the sensuous curtain. When the material organism goes to the ground, the ideal one must go to the ground too; but *the ideal goes to the ground in another way than the material.* The material goes to the ground by being dissolved, dispersed, transformed and lost as moments in another material thing; selfhood, the ideal, the self-reflecting singleness, cannot be dissolved: entelechy goes to the ground in such a way that it is preserved in the ground, made inward in reflection. Only the immediate can pass away[*)]; what is

reflected in itself cannot pass away: entelechy cannot pass away, it can only give up immediacy and go back into the inwardness that enriches itself with every development, reflect itself in ideality’s depth, reason’s eternal remembrance.

As selfhood is, so must remembrance be too; for selfhood is the medium of remembrance. Just as space with its material content of existence is the medium of outwardness, so is selfhood, with its ground-content, its ideal content of remembrance, the medium of inwardness. That in such a medium of remembrance one cannot again point to things, thing-causes and “molecular forces” is surely self-evident, and can in no way be adduced as proof against its ideal reality. Nor can the assumption of this reality be said to rest on “a hypothesis”; for — let us reflect on it once more — what is here meant by the general selfhood at work everywhere and in everything, even in the selfless? The general selfhood is nothing other than existence’s substantial unity of reason — grasped, be it noted, not as an empty abstraction but according to its concept *in concreto.* For the substantial unity of reason can as little have the really manifold outside itself as it can

[*)]“It is only the *visible* beauty that is perishable, because it exists in transitory moments; and it is only this that Goethe speaks of in his well-known charming epigram: *Warum bin ich vergänglich, o Zeus? so fragte die Schönheit.*
Macht’ ich doch, sagte der Gott, nur das Vergängliche schön. And this finds its application to painting, because this art is precisely restricted to visible beauty. But this is perishable, transitory; therefore painting can present only the perishable, the transitory; but by granting it duration it rescues beauty from passing away.” J. L. Heiberg.

be a result of the manifold's aggregation into a collective whole. If, then, it stands fast that the substantial unity of reason must with inner necessity have it in its power to transform every element in the outer manifold into a real moment of its own rich content, then it stands fast also that

the substantial self-development of the unity of reason must go together with a substantial remembrance.

The earthly whole, the elementary organism, presupposes, as we have seen in the foregoing, a reflection of the elements in the inwardness of the kingdom of reason: the elementary principle of totality remembers the existing elements; but in the elements themselves, the selfless existences, there is no remembrance. With the organic organism the remembrance of reason is doubled. The organic principle, the selfhood articulating itself in the paradigm, comes — in virtue of the infinite energy of reflection belonging to the ground of reason — to be remembered in such a way that it appropriates the development's ideal-real result and preserves its content by remembering itself. To the organic disposition there answers the organic remembrance as presupposition to result, as the concrete beginning to the concrete conclusion. The organism's complete development in the system of temporal and material forms is only the conclusion's actual middle term (*terminus medius*), not the conclusion itself.

Physiology shows how "the seminal germs" develop in the organism of the male animal and the eggs in that of the female; but it is remarked at the same time that "one has hitherto not been in a position to undertake a satisfactory chemical investigation of the semen"[*)]. We may, however, imagine that the turning point had been reached at which the chemical, anatomical and morphological investigations were complete: what will the consequence then be? One of two cases: either one discovers in the semen of each particular animal species a manifold of peculiar, pervasive, properly characteristic marks, or else the likeness in everything sensuous is seen to be greater than the unlikeness. But in either of these possible cases the formation of semen will show itself as an inner total effect of the male animal's whole organic peculiarity, as the formation of the egg does of the female's[*)]. Were it possible to intuit in an immediate way what is originally at

[*)]"According to Frerichs, albumin is contained in the fluid provided with immature seminal bodies, but a mass akin to horn-substance in the mature semen. One might therefore see in the development of the seminal germs a chemical, but not a morphological, likeness to the formation of horn. There have also been demonstrated in the semen not inconsiderable quantities of fat and phosphate of lime, and, according to report, free phosphorus as well." Physiologie af Valentin ved Hannover. p. 702.

[*)]"If an animal's sex" — says Steenstrup (Undersøgelser over Hermaphroditismens Tilværelse i Naturen. Copenhagen, 1845, p. 8) — "really had its seat in the sexual organs alone, then one might well conceive *two sexes* gathered in *one animal*, were two such sexual organs placed side by side. But sex is not something that has its seat at a given place, or that expresses itself only through a specified organ; it works through the whole being and has developed itself in every point of it. In a male creature every part, even the work in such a seminal germ, what is absolutely essential in such an *ovulum*, then the whole corresponding organic type — an organism, that is, with just such teeth, such claws, such a spine, such a stomach and so on — would mirror itself in these "organic secretions," in this semen, in this germ. But the teeth, the claws, the spine, the stomach and so on have, physically regarded, no immediate share in the sexual secretions: the mediate co-operations of teeth, claws, spine and stomach are, in the sexual materials, co-operating remembrances. Only in a complete system of such remembrances can a total effect proceeding from the whole organism — a total effect which has not merely all essential features but even apparent accidents, everything down to the disposition of the horny parts and the colour of the hair, in the unity of its remembrances — be concentrated[*)]. Such a

total effect is not a result that can come about piecemeal and be altered piecemeal; the concentrated total effect is whole and undivided in the very first beginning of the disposition, in the first points, in every one of "the granular formations" (*Granula seminis*) with which the coming-to-be of the seminal germs is introduced, in every one of the germs secreted in the female organism's *Ovaria*, The peculiar reciprocal action of the mature sexual materials — the reciprocal action under which the new life's mid-point,

smallest, is male, be it never so like the corresponding part of a female creature; and in the latter, likewise, even the very smallest part is only female. A union of both sexual organs in one creature will therefore first make that creature two-sexed when the natures of both sexes can rule through the whole body and can assert themselves at every single point of it — something which, in virtue of the opposition of the two sexes, can express itself only as a vanishing of all sex in such a creature." — From such a physiological conception one may then form a rational notion of what it means that the whole individuality, with such claws, such teeth and so on, can be co-operative, as in a substantial remembrance, in the formation of the female and male sexual material. A view, on the other hand, like the one that forms the basis of a consideration set out by Johann Müller (Physiologie des Menschen. 2nd edition. Coblenz, 1835. Part I. pp. 817—18). — "Ob das Lebensprincip und das psychische Princip von dem Gehirne aus in einem latenten Zustande auf dem Wege der Nerven zum Samen oder Keime gelange, ob es im latenten Zustande im ganzen Körper verbreitet sei . . . ." [Whether the life-principle and the psychical principle pass from the brain in a latent state along the path of the nerves to the semen or germ, or whether they are diffused in a latent state throughout the whole body . . . .] — can do nothing but awaken confusion, since the individual's life in its inner ideal unity of totality is hereby altogether overlooked.

[*)]"*Hybrid formation* shows us that the semen from one determinate animal species can fertilize the egg from another. When one leaves doubtful cases out of account, or at least regards them as rare exceptions, hybrid formation is possible only between related species, for example between horse and ass, dog and wolf, canary and siskin. Rusconi fertilized frog's eggs with the semen of toads, but could only exceptionally carry it as far as an irregular cleavage of the yolk, and not to any complete development of the embryo. The hybrids themselves are as a rule imperfect in respect of sex and are wont to die out with the first generation, unless mating takes place with one of the two original species." Physiol. af Valentin ved Hannover. p. 720. — "Ich habe schon mehrmals angeführt, daß ich eine grosse Menge von Thatsachen gesammelt habe, welche zeigen, daß, wenn Pflanzen und Thiere aus ihren natürlichen Verhältnissen gerissen werden, es vorzugsweise die Fortpflanzungsorgane sind, welche dabei angegriffen werden." [I have already several times adduced that I have gathered a great mass of facts which show that, when plants and animals are torn out of their natural conditions, it is above all the reproductive organs which are thereby affected.] Charles Darwin, über die Entstehung der Arten. Übersetzt von Dr. H. G. Bronn. Stuttgart, 1860. p. 274.

the new selfhood, forms itself — gives in the embryo itself the energy of remembrance its organically positive expression[*)].

In its unconscious and so far negative form the metaphysical energy of remembrance forms an essential ground for the psychological remembrances of self-consciousness. — As an example of the share the cerebrum must have in the spiritual manifestations of life, Lewes adduces[**)] a report of a sailor who, after falling from the yardarm down onto the deck, had lain unconscious for several months.

"On examining the head . . . . . . Cline found a perceptible depression, and thirteen months and some days after the sailor had come to harm he was brought into the operating theatre and there trepanned . . . . . Four days later the man was able to get up and began to speak, and a few days after that he was able to tell us where he came from. He remembered that he had been press-ganged and dragged down to Plymouth or Falmouth; but from that moment until the operation was performed his mind had been sunk in a complete forgetting. He had, as it were, drunk of Lethe's cup; he had suffered death with respect to his bodily and spiritual faculties; but by the removal of a little piece of bone with the saw he recovered at one stroke the use of all his mental faculties and almost all his bodily ones."

But on what does this lethargy now rest — how has such a draught of "Lethe's cup" actually worked? The circumstance that the man's mind had been sunk in a complete forgetting cannot possibly ground the assumption that one half of life, namely the life of consciousness, had really been annihilated; for an annihilated consciousness cannot possibly come to be again, unless it were to happen by a miracle. Consciousness is only in virtue of remembrance. If the vegetative living does not rest just as much on a self-continuing remembrance as the animal and "relative" living does, then the phenomenon is absolutely inexplicable.

The complete forgetting is here to be grasped as a negative remembrance. Negative is this remembrance, for the light of the spirit is extinguished; self-consciousness goes to

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[*)]"It must be granted that many of the examples which have been collected to prove heredity have been allowed to pass unchallenged by criticism, and that many of them are without worth as proofs. But does Buckle intend to deny that men's inclinations and peculiarities rest on their organization? If he does not intend to deny this, his scepticism is illogical, since there can be no shadow of doubt that organization is inherited. He will not maintain that it is a matter of chance that preserves the different races of animals, that brings it about that the bulldog resembles the bulldog, and that the offspring of a bulldog and a pinscher resembles a bulldog and a pinscher. . . . . . If the parents' organization, peculiarities and excellences did not pass by inheritance to the offspring, there could be no such thing as races. The cur would have the same prospect of becoming valuable as the dog that issues from the finest race. The greyhound would be able to point at game, and the cart-horse to win the Derby." Lewes. Hverdagsl. Physiol. 2. pp. 387—88.

[**)]After *Astley Cooper. Lectures on surgery.* Cf. Hverdagsl. Physiol. 2. pp. 105—107.

the ground. But although self-consciousness goes to the ground, the self does not thereby become selfless: *the negation of remembrance is itself a remembrance.* The unconscious self continues, that is, to live in the system of the vegetative functions. The self can thus in this way lose itself without becoming selfless; it

can depart from itself by being sunk into itself, can forget itself — in negative remembrance. Could the self be dissolved into the selfless, were the negation of remembrance not itself a remembrance, then the lost consciousness would be irrecoverably lost; but here at bottom nothing is yet lost: the system of the functions of conscious life is not annihilated; it is only arrested, it has gone back, with the whole result of all preceding acts of consciousness, into its disposition, that ground of inwardness in which it was originally preformed.

γ) The Organic Purpose.

As the execution of what is preformed in the disposition, as the fulfilment of the paradigm's demands, the fully developed organism is a realized plan, an attained intention: life is purpose.[†] *"The purpose (the end)"* — says Kant — *"is the supersensible unity that contains the ground of the objective's actuality. . . . Outer purposiveness is, since it signifies only a thing's usefulness for something else, something merely relative. . . It is otherwise with inner purposiveness, which consists in this, that something is in itself both means and end, so that its end does not fall outside it. The organic product of nature is such a one, in which each of its parts is aim and at the same time means or instrument. In generation the natural product is brought forth as species; in growth the natural product is brought forth as individual; in formation every part of the individual brings forth itself. This organism of nature does not let itself be explained by merely mechanical causes; it must be explained teleologically."* (cf. Phil. Propæd. pp.118-124).

But however great Kant's merit in the development he gave the teleological in his "Kritik der Urtheilskraft," he has nevertheless, by his critical distinction between what concerns only a mode of apprehension necessary for the human mind and what holds for nature's objective, to us unknown essence — between what he calls a regulative and a constitutive principle — by his acuteness and authority again contributed to strength-

[†] *Translator's note.* Nielsen works here with a family of terms English tends to collapse. *Formaal* is the purpose or end at which something aims; *Hensigt* the intention with which an agent acts; *Øiemed* the particular aim held in view; *Hensigtsmæssighed* the purposiveness a thing shows in being fitted to an end (Kant's *Zweckmäßigkeit*); and *Hensigtsaarsag*, alongside the Latinate *Finalaarsag*, the final cause. "Purpose," "intention," "aim," "purposiveness" and "final cause" are used for these throughout, and the distinctions matter to the argument: the whole objection Nielsen is about to put in the naturalist's mouth turns on whether purposiveness can be had without an intending consciousness.

ening the misunderstanding that "purposiveness" should be a merely anthropomorphic category, of no concern to nature itself. "The earlier systems treated the concept of natural purposiveness dogmatically; they have either ascribed to nature as thing in and for itself this purposiveness (hylozoism) or denied it to her (fatalism). But we, who hold fast that teleology is only a regulative principle, decide nothing concerning the question whether inner purposiveness belongs to nature in and for itself or not. Our understanding thinks discursively, always proceeds from the parts, and grasps the whole as a result of its parts; it cannot therefore comprehend the organic products of nature — in which, conversely, the whole is the parts' ground of coming-to-be and their prius — in any other way than under the point of view of the concept of purpose. Were there, on the other hand, an intuitive understanding (*intellectus archetypus*) which cognized the particular in the general, the parts in the whole as already co-determined, then such an understanding would comprehend the whole of nature out of one principle, without needing the concept of purpose."

There is in the kingdom of nature no lack whatever of facts that invite a teleological interpretation[*)];

but what the investigator of nature lacks in high degree is clear and distinct insight into teleology's significance for the objective cognition of nature. It is not seldom, namely, that one hears an investigator of nature — especially when he philosophizes for the educated in popular lectures — appeal now to nature's aim, the purposive, now again to blind necessity; indeed it may even happen that the two mutually conflicting ways of seeing are allowed, in one and the same presentation, to step up so sharply against each other that it is after all neither purposiveness nor necessity but naked chance on which the whole view at last comes to rest[*)];

[*)]*Les* causes finales *sont l'expression philosophique la plus haute de nos sciences, et la plus douce. Il y a un plaisir d'un ordre supérieur à découvrir et à contempler cet assemblage merveilleux de tant de ressorts divers combinés dans des proportions si justes. Le spectacle d'une sagesse infinie donne du calme à l'esprit des hommes. »Ce n'est pas peu de chose, disait Leibnitz, que d'ètre content de Dieu et de l'univers«. Éloge historique de M. H. D. de Blainville par Flourens. Académie des sciences. Paris, 1854, p. 14.* [Final causes are the highest expression of our sciences, and the sweetest. There is a pleasure of a superior order in discovering and contemplating this marvellous assemblage of so many diverse springs combined in proportions so just. The spectacle of an infinite wisdom gives calm to the minds of men. It is no small thing, said Leibniz, to be content with God and with the universe.]

[*)]A quite telling example of such an improvised piece of naturalist's philosophy we have, among others, in a treatise by Dr. H. Burmeister: "Vergangenheit und Gegenwart des Thierreichs" ("Geologische Bilder". Vol. I. Leipzig. 1851). After the author has at once at the beginning (p. 146), as regards the earth's development, referred to "die unwandelbare Nothwendigkeit" [the immutable necessity]; has further let the animals' basic types (p. 150) belong "mit Nothwendigkeit zur organischen Natur der Erde," [with necessity to the organic nature of the earth] while these types' more or less perfect "Darstellungsart" [mode of presentation] is on the other hand "den Umständen preisgegeben," [abandoned to circumstances] — he then considers again, by way of a closer understanding of what it really means to be "den Umständen preisgegeben", still on the same page "die gesammten Phänomene der Genesis" [the collective

It must constantly appear to the investigator attentive to thing-causes and laws as though there were something highly unnatural

in presupposing natural, unconscious purposes. Where there is to be action according to purpose, there must be action with a determinate intention; but where there is to be action with a determinate intention, there must be a seer and a something toward which

phenomena of genesis] — "unter dem allein richtigen Gesichtspunkte der absoluten Nothwendigkeit." [under the sole correct point of view of absolute necessity] Under this sole correct point of view, *"die absolute Nothwendigkeit"*, the teleological way of consideration nevertheless raises itself now and then so involuntarily and with such insistence that it not seldom takes the word out of absolute necessity's mouth; thus (p. 207): "Alle beständigen Wasserbewohner, namentlich die im Meere lebenden, haben kurze Wirbelkörper mit concaven Berührungsflächen (— *warum*?), weil eine solche Anlage der Wirbelsäule die schlängelnden Bewegungen des Rumpfes beim Schwimmen am leichtesten ausführbar macht." [All constant water-dwellers, especially those living in the sea, have short vertebral bodies with concave contact surfaces (— why?), because such a disposition of the vertebral column makes the serpentine movements of the trunk in swimming most easily executable.] Further (p. 213), where the author, after touching on the likelihood that the primeval world's *Iguanodon* lived on plant food, continues: "Demnach ist es nicht unwahrscheinlich, daß die plumpen Dinosaurier auf eine entsprechende Nahrung angewiesen waren (— *warum*?), denn kaum möchte für so große Geschöpfe eine ausreichende Fleischnahrung in jener Zeit vorhanden gewesen sein, wenn sie nicht von Fischen leben sollten. Dagegen spricht aber ihr plumper Bau, der einen schwerfälligen Gang andeutet, womit stets Pflanzennahrung verbunden ist. Große Raubthiere müssen zwar kräftig, aber nicht schwerfällig gebaut sein (— *warum*?), weil zum Beutemachen Gewandtheit und Schnelligkeit ebenso sehr, wie Kraft, vonnöthen ist." [Hence it is not improbable that the clumsy dinosaurs were referred to a corresponding nourishment (— why?), for scarcely could a sufficient flesh-nourishment have been available for such large creatures at that time, unless they were to live on fish. Against this, however, speaks their clumsy build, which indicates a ponderous gait, with which plant nourishment is always bound up. Large beasts of prey must indeed be powerfully, but not ponderously built (— why?), because for the taking of prey agility and speed are as needful as strength.] It might well seem as though the author with all this inclined more to a grounding in virtue of "absolute necessity" than to a grounding in virtue of purposiveness's "Warum"; but the balance is restored in the finest manner, inasmuch as he, after a many-sided grounding, a manifold answering of a manifold "Warum", ends by breaking the staff over all scientific grounding and confiscating every "Warum" without distinction. By this turn, as surprising as it is striking, a most lively contrast is produced between the treatise's beginning and its end. It runs at the beginning (p. 150): "Eine jede schärfere wissenschaftliche Untersuchung weist unwiderleglich nach, daß überall nicht Zufälligkeiten oder Willkür die Phänomene hervorgerufen haben... Diese Wahrheit wird uns um so klarer, je vorurtheilsfreier wir die Thatsachen prüfen und die Gründe ihres Entstehens beurtheilen." [Every sharper scientific investigation demonstrates irrefutably that nowhere have accidents or arbitrariness called forth the phenomena... This truth becomes the clearer to us the more free of prejudice we test the facts and judge the grounds of their arising.] But at the end we are yet warned in the most emphatic way against, as regards "die Thatsachen", "die Gründe ihres Entstehens zu beurtheilen"; for it is declared (pp. 243—44) outright: "Alle wissenschaftliche Forschung lehrt nichts mehr als den Gang kennen, welchen die organische Entwickelung genommen hat; nicht das Warum desselben, nach dem so viele Neugierige und Unverständige täppisch fragen." [All scientific research teaches us to know nothing more than the course which organic development has taken; not the Why of it, after which so many curious and uncomprehending people gropingly ask.] And lest anyone here, out of misunderstood zeal, should exert himself to distinguish between a twofold "Warum" — one of the ground (διότι) and one of purposiveness (οὗ ἕνεκα) — the author strikes both "Warum" with one blow: "Das Warum liegt überall außer der Sphäre des Begreiflichen," [The Why lies everywhere outside the sphere of the comprehensible] and makes short work of it with the help of "die Thatsache." "Welcher Verständige will fragen, warum es Menschen gebe? — Diese Frage ist gleichbedeutend mit der: warum giebt es eine Erde, ein Sonnensystem, ein Weltganzes, ja endlich Materie überhaupt? Solche Fragen sind kindisch, ein Vernünftiger kann nicht darauf antworten. Es ist so! und damit müssen wir uns begnügen. Denn was ist, hat eine Nothwendigkeit der Existenz in sich; es ist berechtigt zu sein, ohne Anderen von seiner Existenz Rechenschaft ablegen zu müssen." [What understanding person will ask why there are human beings? — This question is equivalent he looks; this something is then the purpose, the expressly intended. If a mechanic will construct a clock, he first intuits the end in a model answering to the concept of a clock. Looking toward the model, the determinate aim, the master-workman then chooses the material serviceable for the aim's execution; only in the material's being chosen and worked in accordance with the aim's demands are the means procured. From this it is seen what doublings the teleological must require. The purpose comes to be literally twice, in two ways, in two different media: the first time in the workman's consciousness, that is, in ideal fashion in an ideal medium, as an anticipation answering to the beginning — the *intended* purpose; the second time, by way of the execution, in real fashion, in a material medium — the *attained* purpose. A similar doubling is seen in the means. In and for itself the material is only stuff, not means. With respect to a given purpose one kind of material may then be serviceable and another not, and among

the serviceable one thing again more serviceable than another. The usable, what is serviceable for the aim, must be determined in and by the consciousness that determines the aim, and the means must accordingly likewise have an ideal existence in the technical sense before it can be realized as means in the material. Since further the concept of purpose is not merely theoretical but at the same time practical — since the plan's execution rests not merely on the workman's insight but also on his will, drive and skill (*technica intensionalis*) — there must during the execution enter into the purpose something accidental from the objective side and something arbitrary from the subjective. But how is it now thinkable that doublings conditioned by a surveying glance, combinations resting on an ingenious interplay of something arbitrary with something accidental, should answer to nature itself in her own inner, objectively necessary activity?

"The final cause" — so the investigator of nature will still be able to continue — "presupposes that the inner can in inwardness raise itself above the outer; such a liberation is conditioned by self-consciousness; nature has no self-consciousness; therefore a production of nature, however much likeness it may otherwise have to a human production of art, cannot possibly be teleological. Consider the bone-structure in the human body; weigh what a manifold of mechanical problems have here found their solution, the most important planes of gravity being laid out so that a labile equilibrium is thereby attainable; go into particulars and see how, for example, the vertebral column unites in

to: why is there an earth, a solar system, a world whole, indeed finally matter at all? Such questions are childish; a rational person cannot answer them. It is so! and with that we must content ourselves. For what is, has a necessity of existence in itself; it is entitled to be, without having to render an account of its existence to others.] So once again the Megarians': either necessity or impossibility! "Every existence is what it is, and is because it is;" in this tautology of fact absolute necessity is one with pure chance — "und damit müssen wir uns begnügen" [and with that we must content ourselves].

itself its great bearing power with great mobility and considerable resistance to shock; how, besides, even the atmosphere's pressure is employed to hold the joints together, lest the burden otherwise incumbent on the muscles should become too great; how the different joints — gliding joints, pivot joints, hinge joints, ball joints — with their peculiar forms and

articulations are each in its own place suited precisely to the lesser and greater degree of mobility that the body in its different parts requires: then we get undeniably an impression as though the whole were laid out by a self-conscious wisdom, thought through by a divine understanding, executed by an invisible artist. But if it nevertheless stands fast that nature in her forming and bringing-forth is not a self-conscious being and therefore cannot intend anything, cannot act with a determinate intention, then we are surely most strictly called upon to give up the anthropomorphic conception and look about us for an interpretation more in accord with nature's mode of working."

"If it holds of every organism that the whole and its parts develop by and with each other reciprocally, then it is a natural matter of course that each part must accord with the whole and the different parts suit one another. So long as we keep to organisms of the very simplest forms — single-celled algae, sponges, small jellyfish and so on — surely no one will take it into his head to present the parts' agreement with the whole's demand as proof of a teleological production. Or should nature, who without intention brings forth the many so highly characteristic crystal forms, not be able just as naturally to bring forth such plant organisms as diatoms, in the form of needles, wheels, combs and so forth, or such animal organisms as sea-corks and sea-bladders? Indeed even when we take the vertebrate type before us, but let it be represented by so simple a form as, for example, "the fringe-mouth (*Branchiostoma lanceolatum*), a little fish two inches long, jelly-like, water-clear and transparent, ... whose skeleton consists merely of the notochord and the membranes going out from it,"[*)] it will fall out as of itself that such a whole must go together with its parts, and these therefore answer to one another. But if it is now, in accordance with the law of development holding for

organisms, natural that the parts and the whole answer to each other in the lower and simpler forms, then one must surely find it just as natural that the same is also the case in the higher and more composite ones; if nature can bring forth in "the fringe-mouth's" skeleton a notochord (*chorda dorsalis*) without any final causes or preceding speculations, why should she not also be able, without express final cause, to bring forth the vertebral column in a human skeleton? In consideration of this we then grant Kant that we may

[*)] Dyreriget, af Chr. Fr. Lütken. Part I. 1855. p. 290.

employ teleology only as a regulative principle, in so far as our discursive understanding is not quite capable of comprehending the organic products of nature.”

b) The Stages of Life.

α) Plant Life.

β) Animal Life.

γ. Human Life.

c) Soul and Body.

α) Metaphysical Theories.

β) Psychophysical Investigations.

γ) The Limits of Human Knowledge.

III.
Philosophy and Academic Study.

Contents.